

Chapter 1

Fundamentals of Fluid Mechanics

1.1. Fundamental Concepts

Aerodynamics is the science of predicting and controlling the forces and moments that act on an aircraft or other object moving through the atmosphere. Aerodynamics is traditionally included as a part of two more extensive fields of science, *fluid mechanics* and *aeronautics*. Fluid mechanics is the broad field of science that deals with fluids at rest and in motion. Aeronautics is the branch of science that deals with the design and operation of aircraft. Aerodynamics builds on the foundation of fluid mechanics, and it forms a foundation for the study of aeronautics.

An understanding of the fundamentals concepts of fluid mechanics is a necessary prerequisite to the study of aerodynamics. Thus, we begin our study of the aerodynamics of flight with an overview of some essential topics from fluid mechanics.

As with any engineering discipline, to quantitatively describe fluid flow we must first model the physical phenomena with a consistent set of mathematical equations. The physical phenomena associated with the flow of fluids are extremely complex, and thus, the mathematical equations that describe such phenomena can be complex as well. To aid in building an understanding of fluid mechanics and aerodynamics, we will start by examining some simple problems and gradually build toward the more complex.

The Concept of a fluid

The first question that must be answered is, “What is a fluid?” Most engineering students will be familiar with the concept of classifying matter into three states; solids, liquids, and gases. A piece of steel at room temperature is one common example of a solid, water is likely the first liquid that comes to mind, and air is probably the most familiar example of a gas. Engineering students should also have a broad understanding of what is meant by the word “fluid” as it is used to characterize a state of matter, and they will likely know that both liquids and gases are classified as fluids. However, a precise definition of a fluid may seem more elusive. Intuitively, we understand that a fluid is a substance that flows. However, defining a fluid to be a substance that flows is not a rigorous definition, unless we have a rigorous definition for the word “flows.”

In the study of classical engineering mechanics, matter is classified as being in one of two states, either a solid or a fluid. The technical distinction between a solid and a fluid is linked to the manner in which the substance responds to a shear stress.

*A **solid** is a substance that can resist a shear stress with a finite static deformation.*

*A **fluid** is a substance that deforms continuously under the action of a shear stress.*

In this engineering definition, no distinction is made between liquids and gases. This is because the fundamental principles of fluid mechanics apply equally to both liquids and gases. In fact, the usual distinction between a liquid and a gas only exists at pressures below the critical pressure.

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Consider a volume of some substance that has a rectangular shape in the absence of any shear stress, as shown in Fig. 1.1.1. If a constant force per unit area to the right is applied to the top of the substance and an equal force per unit area to the left is applied to the bottom of the substance, the substance will deform such that the upper surface is displaced to the right relative to the lower surface, similar to what is shown in Fig. 1.1.1. The tangential force per unit area τ is referred to as the *shear stress*. If the deforming substance is a solid, the upper surface will be displaced only a finite distance to the right relative to the lower surface, and then no further deformation will take place as long as the applied shear stress remains constant. For example, if the substance were aluminum with a constant shear stress of 5000 psi, the equilibrium angle of deformation would be about 0.07 degrees. On the other hand, if the experiment is repeated with a fluid, the upper surface will continue to move indefinitely to the right relative to the lower surface, no matter how small the shear stress τ . For instance, if the substance were air at standard temperature and pressure with a constant shear stress of 0.0001 psi, the deformation angle would increase indefinitely at a rate of about 2.2×10^6 degrees per second.

It is important to note that the general definition of a solid does not require stress to be a linear function of deformation. For an important class of materials the shear stress is very nearly linearly proportional to the angular deformation. Such materials are typically modeled as *elastic solids* using the familiar relation known as *Hooke's law*.

Similarly, the general definition of a fluid does not require the shear stress to be a linear function of the angular deformation rate. However, most fluids (including air and water) can be accurately modeled by assuming that the shear stress is linearly proportional to the rate of angular deformation. Fluids that exhibit this linear relation between stress and rate of deformation are called *Newtonian fluids*. The exact mathematical relations between stress and rate of deformation for a Newtonian fluid will be discussed in detail later in this textbook.

Because air at the altitudes normally encountered in flight can be accurately modeled as a Newtonian fluid, the analytical methods presented in this textbook will be limited to the assumption of a Newtonian fluid. However, an engineer should never lose sight of the fact that the Newtonian fluid is only a mathematical model used to approximate the behavior of real fluids. Many common fluids such as air and water are for all practical purposes Newtonian, at normally encountered temperatures and pressures. However, there are also a number of common fluids, including human blood and some lubricating

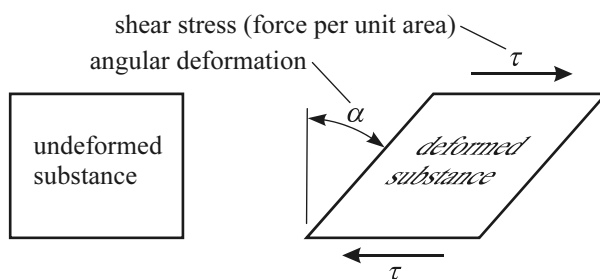


Figure 1.1.1. Deformation of a substance under the action of a shear stress.

oils, which cannot be accurately modeled as Newtonian fluids. Additionally, even fluids that are normally treated as Newtonian can exhibit non-Newtonian behavior at extremes in temperatures, pressures, and/or deformation rates.

In most cases it is quite easy to distinguish fluids from solids. However, in some instances this distinction is not so clear. Some substances, such as *glass* and *asphalt*, appear to be solids when subjected to a shear stress for short periods of time. However, when the shear stress is exerted over a period of many years, these substances are found to deform slowly, behaving as non-Newtonian fluids. The treatment of such materials is beyond the scope of this textbook.

Although the engineering definition of a fluid makes no distinction between liquids and gases, from an introductory course in thermodynamics, engineering students should be familiar with the distinction. Recall that a gas will always expand to occupy the entire volume of its container. In contrast, the volume of a liquid does not change substantially with pressure, so its volume does not depend on the volume of its container. When liquid is placed in a container having a volume greater than that of the liquid, a *free surface* forms, separating the liquid from its *vapor*. The words *gas* and *vapor* are often used synonymously. The word *vapor* usually implies the gaseous phase of some substance, which is not far from the condensation point. Any pure substance can exist as a solid, liquid, or gas depending on its temperature and pressure. The usual distinction between a liquid and a gas does not exist at pressures above the critical pressure.

The Concept of a Continuum

As mentioned previously, the methods used in classical engineering mechanics treat all matter as being either a solid or a fluid. Thus, classical engineering mechanics is traditionally divided into two disciplines, solid mechanics and fluid mechanics. This raises an important question, “Can all matter be treated as either a solid or a fluid in accordance with our engineering definitions?”

To answer this important question, consider an object moving through a stationary portion of the Earth’s atmosphere. Obviously, the air surrounding the object cannot resist a shear stress with a finite static deformation, so the air is not a solid. As the object moves through the atmosphere, the air immediately adjacent to the object is moved with the object, whereas the air far from the object remains stationary. Thus, the air directly above and below the object is subjected to shearing motion. The question remains, “Does the air satisfy the engineering definition of a fluid?” Strictly speaking, the answer to this question is always no.

To understand why air does not precisely satisfy the engineering definition of a fluid, we must examine the definition more closely. To satisfy this definition, a substance must “*deform continuously*” under the action of a shear stress. The key word here is *continuously*. Obviously, the air directly above and below the object deforms as the object passes by. But, is this deformation continuous?

According to our current understanding, stationary air is comprised of very small moving molecules separated by relatively vast amounts of empty space. The air is said to be “*stationary*” only because the statistical average velocity of the molecules is zero. When a particular molecule collides with a moving object, its velocity is changed. As many molecules collide with the moving object, the statistical average velocity of the

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molecules changes. This change in average velocity is manifested as a deformation of the air. However, strictly speaking, the deformation of the air is not continuous but comes as a series of discrete impulses. Nevertheless, in most commonly encountered situations such deformation can be approximated accurately as being continuous, because the number of collisions with the object per unit time is very large. Thus, air at the altitudes normally encountered in flight can be modeled accurately as an engineering fluid. Under such conditions, the air is said to be modeled as a *continuum*.

At very high altitudes or when considering flow about very small objects, air cannot be modeled accurately as a continuum and the mathematical formulations that are used in classical fluid mechanics do not apply. A dimensionless parameter that is commonly used to define the boundary of continuum flow is the Knudsen number K , which is defined to be the ratio of the molecular mean free path λ to the smallest significant length associated with the flow geometry d (i.e., $K \equiv \lambda/d$). The molecular mean free path is the mean distance that a molecule travels between collisions with neighboring molecules. For an object moving through the atmosphere, the accuracy of continuum theory also depends on the Mach number M , which is defined to be the ratio of the airspeed V to the speed of sound a (i.e., $M \equiv V/a$).

Strictly speaking, **continuum theory applies only in the limit as the Knudsen number approaches zero**. The practical limiting boundary for the application of continuum theory to fluid flow is a complex function of geometry, Knudsen number, and Mach number. However, **as a first approximation, the flow about an object moving through the atmosphere can usually be approximated as continuum flow if**

$$K < \sim 8 \times 10^{-3} [\sqrt{1+1/(2M)} - 1]^2 M \quad (1.1.1)$$

The approximate relation presented in Eq. (1.1.1) is shown graphically in Figs. 1.1.2 and 1.1.3. For low-speed flows the limiting Knudsen number shown in Fig. 1.1.2 is constant.

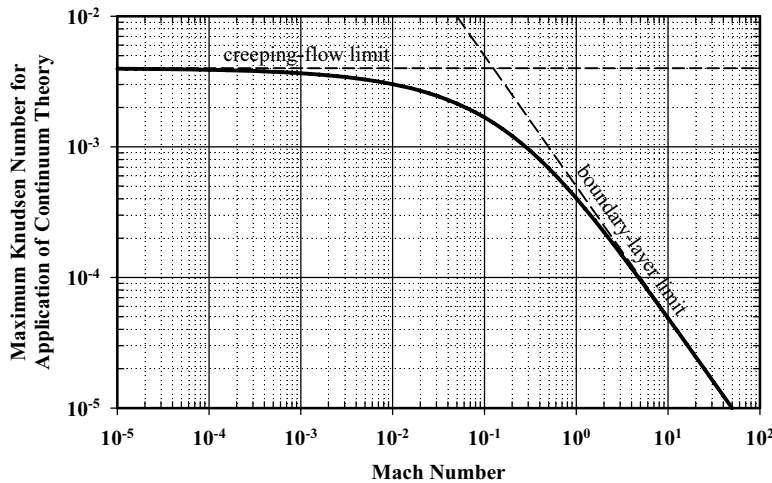


Figure 1.1.2. Approximate limits of continuum theory, from Phillips (2010).

This is called the **creeping flow limit**,

$$K < \sim 4 \times 10^{-3} \quad (1.1.2)$$

For high-speed flows the limiting Knudsen number shown in Fig. 1.1.2 is inversely proportional to the Mach number. This is called the **boundary layer limit**,

$$K < \sim 5 \times 10^{-4} / M \quad (1.1.3)$$

For objects as large as manned aircraft, continuum theory first breaks down near the leading edge of objects moving through the atmosphere at high altitudes and hypersonic airspeeds. As can be seen from Fig. 1.1.3, for atmospheric particles less than about 10 to 20 microns in size, continuum theory may not be applicable even for low airspeeds at sea level. The reader should keep in mind that results obtained from Eq. (1.1.1) are only approximate and continuum theory should be used with caution near these limits.

The molecular mean free path for air at sea level is approximately 2.1×10^{-7} ft. At higher altitudes the mean free path increases, and at 50,000 ft, $\lambda \cong 1.2 \times 10^{-6}$ ft. Even at an altitude of 200,000 ft, the mean free path is still only about 9.1×10^{-4} ft. Thus, Fig. 1.1.3 shows that **except for very high altitudes and Mach numbers, and/or for flow about very small objects, atmospheric air can be approximated as a continuum.**

Nevertheless, an engineer should never lose sight of the fact that the continuum theory of fluid mechanics is only a mathematical model used to approximate the behavior of a collection of molecules. There are important applications, such as spacecraft launch and reentry and flow about small aerosol particles, which cannot be analyzed accurately using continuum theory. At an altitude of 400,000 ft, the molecular mean free path for atmospheric air is nearly 24 ft. Thus, at this altitude, atmospheric air cannot be modeled

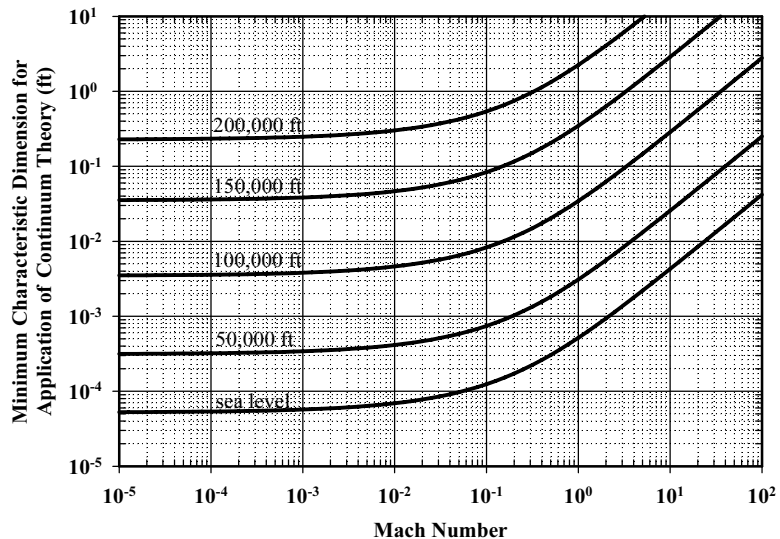


Figure 1.1.3. Size and altitude limits of continuum theory, from Phillips (2010).

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as a continuum. During spacecraft launch and reentry, aerospace vehicles must pass through a region of the atmosphere where continuum theory does not apply and yet the forces exerted on the vehicle by the surrounding air cannot be ignored. For further details on the limits of continuum aerodynamics the reader is referred to Phillips (2010).

Because continuum theory strictly applies only in the limit as the Knudsen number approaches zero, when the Knudsen number becomes significant, we must account for the fact that a gas is made up of individual molecules in a state of constant motion. In such cases, the macroscopic motion of the gas and the forces exerted on any bounding solid surface must be analyzed from a statistical consideration of the motion of the molecules. The related field of science that deals with the statistics of molecular motion in gases is called the *kinetic theory of gases*, or less commonly, *rarefied gas dynamics*.

The kinetic theory of gases is a well established scientific discipline, which dates back to Maxwell (1860, 1867). Readers who want to study kinetic theory could start with the classic work by Jeans (1940). Other useful textbooks on the topic include Present (1958), Vincenti and Kruger (1965), Brush (1966), Liboff (1998), and Struchtrup (2005).

In the kinetic-theory regime of finite Knudsen number, the aerodynamic force acting on an object moving through the atmosphere depends on the geometry, Knudsen number, and Mach number. To give the reader a better feel for how the Knudsen number can affect aerodynamic forces, Fig. 1.1.4 shows how the drag on a sphere moving through a gas at low Mach number depends on Knudsen number. Note that significant departure from continuum theory begins in the Knudsen number range 0.001–0.01 and produces a 20% reduction in drag at a Knudsen number of approximately 0.1. At a Knudsen number of 1.0, the drag is reduced by more than 75%.

The example shown in Fig. 1.1.4 demonstrates that the effects of departure from continuum theory can be very significant. Furthermore, there are many applications in

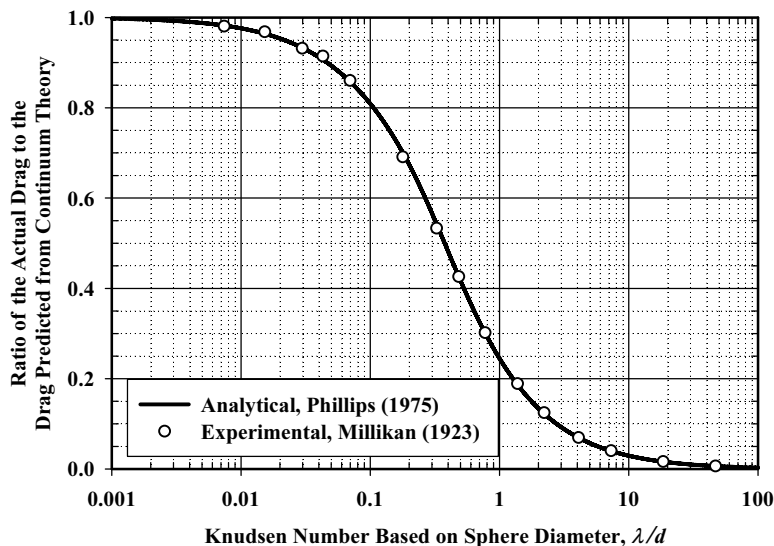


Figure 1.1.4. Knudsen-number effect for drag on a sphere moving in a gas at low Mach number.

engineering and science where the departure from continuum theory is very important. In aerospace engineering, one of the most important such examples is flight in the upper atmosphere. Nevertheless, in this textbook, we shall limit our study of aerodynamics to continuum theory. Although additional results obtained from the kinetic theory of gases will be presented occasionally without proof, a detailed presentation of kinetic theory is beyond the scope of this textbook.

1.2. Fluid Properties

The primary simplification introduced by the continuum theory of fluid mechanics is that it allows us to describe the macroscopic motion of the fluid and the forces exerted on the bounding surfaces in terms of only macroscopic fluid properties, without considering the statistics of molecular motion. From an introductory course in thermodynamics, engineering students should be familiar with the concept of thermodynamic properties like *pressure*, *temperature*, and *specific volume*. They should also be familiar with the concept of an *equation of state*, which relates one thermodynamic property to two other independent thermodynamic properties. Here we review some fundamentals from thermodynamics and discuss additional fluid properties that are useful in the study of fluid mechanics and aerodynamics.

Pressure

Engineering students should be familiar with the concept of thermodynamic pressure. In a gas, the thermodynamic pressure is a macroscopic manifestation of the momentum that is associated with the random motion of molecules at the microscopic level. More precisely, thermodynamic pressure is defined for a fluid in equilibrium. If $\delta\mathcal{S}$ is a small planar surface area, $\delta\mathcal{S}'$ is the smallest area over which the fluid can be considered to be a continuum, and δF_n is the outward normal component of force on $\delta\mathcal{S}$, then the pressure p is defined as

$$p \equiv \lim_{\delta\mathcal{S} \rightarrow \delta\mathcal{S}'} \left(-\frac{\delta F_n}{\delta\mathcal{S}} \right) \quad (1.2.1)$$

The thermodynamic pressure p is *isotropic*. In other words, the pressure at a point in a fluid in equilibrium is the same in all directions. When a fluid is deforming under the action of a shear stress, it is not in equilibrium and the normal stress is not necessarily isotropic. These considerations will be discussed in detail at a later point in this textbook. For now, it is sufficient to say that the thermodynamic pressure p is the isotropic inward normal component of the fluid stress tensor.

When solving engineering problems, the *units* used for pressure must be consistent with other units used in the problem. For the SI system of units, pressure is expressed in newtons per square meter (N/m^2). A newton per square meter is commonly referred to as a pascal (Pa), i.e., $1 \text{ Pa} \equiv 1 \text{ N/m}^2$. In the English system of units, the consistent units for pressure are pounds force per square foot (lbf/ft^2). However, in common practice, pressure is often stated in pounds force per square inch (lbf/in^2 or psi). Thus, students should remember the conversion $1 \text{ ft}^2 = 144 \text{ in}^2$.

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The thermodynamic pressure p is defined to be an *absolute pressure*. However, many pressure and vacuum gages read the difference between the absolute pressure and the atmospheric pressure existing at the gage. This pressure difference is called *gage pressure*. A barometer reads the absolute pressure of the atmosphere, which in this textbook will be denoted as p_∞ . Thus, the thermodynamic pressure is the sum of the barometric pressure and the gage pressure. In this textbook, p will always denote absolute pressure and gage pressure will be denoted as $p - p_\infty$.

Temperature

Most engineering students should also be familiar with the thermodynamic concept of temperature. In a gas, temperature is a macroscopic manifestation of the kinetic energy that is associated with the random motion of molecules at the microscopic level. There are four temperature scales in common use, the Celsius (C), Fahrenheit (F), Kelvin (K), and Rankine (R) scales. The Celsius scale is sometimes still referred to by its former name, the Centigrade scale. The Celsius and Fahrenheit scales are relative temperature scales having their zero points defined in terms of equilibrium with a particular pure substance at a known fixed-point temperature. The Kelvin and Rankine scales are absolute temperature scales having their zero points defined in terms of the hypothetical temperature at which an *ideal gas* would occupy zero volume.

The Celsius scale is defined in terms of a single fixed-point temperature, which is the triple point of water (0.01 C). The magnitude of the Celsius degree is defined in terms of the ideal-gas temperature scale, which is discussed in any undergraduate thermodynamic textbook. On the Celsius scale, the equilibrium temperature of a mixture of liquid water and ice at standard pressure (101,325 N/m²) is 0.000 C. The equilibrium temperature of a mixture of liquid water and its vapor at standard pressure is 100.000 C.

Relationships between the Celsius scale (°C) and the Kelvin (°K), Fahrenheit (°F), and Rankine (°R) scales are respectively given by

$$^{\circ}\text{K} \equiv ^{\circ}\text{C} + 273.150 \quad (1.2.2)$$

$$^{\circ}\text{F} \equiv 1.800(^{\circ}\text{C}) + 32.000 \quad (1.2.3)$$

$$^{\circ}\text{R} \equiv ^{\circ}\text{F} + 459.670 \quad (1.2.4)$$

Students should remember these conversions. Standard atmospheric temperature at sea level, which is 15.000 C, can also be expressed as 288.150 K, 59.000 F, or 518.670 R.

Density and Specific Volume

From an introductory course in thermodynamics, most engineering students are familiar with the thermodynamic property called *specific volume*. However, in the disciplines of fluid mechanics and aerodynamics, it is traditional to use the fluid property known as *density* rather than specific volume. **Density is the reciprocal of specific volume.** More precisely, if $\delta\mathcal{V}$ is a small volume, $\delta\mathcal{V}'$ is the smallest volume over which the fluid can be considered to be a continuum, and δm is the fluid mass within the volume $\delta\mathcal{V}$, then the fluid density ρ is defined as

$$\rho \equiv \lim_{\delta v \rightarrow \delta v'} \frac{\delta m}{\delta v} \quad (1.2.5)$$

When solving engineering problems, the units used for density must be consistent with other units used in the problem. For the SI system of units, density is expressed in kilograms per cubic meter (kg/m^3). Students should remember that the units of force and mass are related, i.e., $1 \text{ kg} \equiv 1 \text{ N}\cdot\text{s}^2/\text{m}$. In the English system of units, the consistent units for density are slugs per cubic foot (slugs/ft^3). However, in common practice, density is often stated in pounds mass per cubic foot (lbm/ft^3). Thus, students should remember some reasonable contraction of the conversion $1 \text{ slug} \equiv 1 \text{ lbf}\cdot\text{s}^2/\text{ft} = 32.17404855643 \text{ lbm}$.

A complete mathematical formulation of the laws of continuum fluid mechanics requires an *equation of state* relating fluid density to temperature and pressure. For gases at low Mach numbers as well as liquids, the problem is significantly simplified by assuming that the density of the fluid is a known constant. This simplification is commonly referred to as the *incompressible flow* approximation. For gases at higher Mach numbers, the ideal-gas equation of state is commonly used. The density of an ideal gas is related to temperature and pressure through the relation

$$\rho = \frac{p}{RT} \quad (1.2.6)$$

where p is absolute pressure, T is absolute temperature, and R is the ideal-gas constant ($R = 287.0528 \text{ N}\cdot\text{m/kg}\cdot\text{K} = 287.0528 \text{ m}^2/\text{s}^2\cdot\text{K}$ for standard atmospheric air). From their introductory course in thermodynamics, students may be more familiar with the ideal-gas equation of state written in terms of specific volume. However, Eq. (1.2.6) makes use of the fact that density is simply the reciprocal of specific volume.

Internal Energy, Enthalpy, and Specific Heat

Engineering students should be familiar with the thermodynamic property called *specific internal energy* (i.e., internal energy per unit mass), which is traditionally denoted as u . **For both incompressible flow and the flow of an ideal gas, specific internal energy can be expressed as a function of temperature only.** Thus, an infinitesimal change in specific internal energy can be written as

$$du = C_v dT \quad (1.2.7)$$

where C_v is called the *constant-volume specific heat* and is defined to be the change in specific internal energy with respect to temperature at constant volume,

$$C_v \equiv \left(\frac{\partial u}{\partial T} \right)_v \quad (1.2.8)$$

Another familiar thermodynamic property is the *enthalpy*, which is usually given the symbol h . By definition, enthalpy is related to the specific internal energy, pressure, and density through the relation

$$h \equiv u + \frac{p}{\rho} \quad (1.2.9)$$

For an incompressible substance, u is a function of temperature only and ρ is constant. Thus, employing the relation for specific internal energy that is given by Eq. (1.2.7), an infinitesimal change in enthalpy for an **incompressible flow** can be written as

$$dh = C_v dT + \frac{1}{\rho} dp \quad (1.2.10)$$

For an **ideal gas**, we can combine Eqs. (1.2.6) and (1.2.9) to yield

$$h = u + RT \quad (1.2.11)$$

and with the application of Eq. (1.2.7), we see that the enthalpy of an ideal gas is a function of temperature only. Thus, an infinitesimal change in enthalpy for an **ideal gas** can be written as

$$dh = C_p dT \quad (1.2.12)$$

where C_p is called the *constant-pressure specific heat* and is defined to be the change in enthalpy with respect to temperature at constant pressure,

$$C_p \equiv \left(\frac{\partial h}{\partial T} \right)_p \quad (1.2.13)$$

Combining Eqs. (1.2.7), (1.2.11), and (1.2.12), the constant-pressure specific heat for an **ideal gas** can be written in terms of the constant-volume specific heat and the ideal-gas constant

$$C_p = C_v + R \quad (1.2.14)$$

Another important thermodynamic property for compressible flow is the ratio of the constant-pressure specific heat to the constant-volume specific heat, which is traditionally denoted as γ . Using Eq. (1.2.14), for an **ideal-gas** the specific-heat ratio is

$$\gamma \equiv \frac{C_p}{C_v} = 1 + \frac{R}{C_v} \quad (1.2.15)$$

For standard atmospheric air, the specific-heat ratio is defined to be $\gamma = 1.4000$ and the ideal-gas constant is defined as $R = 287.0528 \text{ N}\cdot\text{m/kg}\cdot\text{K}$. From Eq. (1.2.15), this implies that the constant-volume specific heat for standard atmospheric air has a constant value of $C_v = 717.6320 \text{ N}\cdot\text{m/kg}\cdot\text{K}$, and after applying Eq. (1.2.14), $C_p = 1004.6848 \text{ N}\cdot\text{m/kg}\cdot\text{K}$. Students should remember that a newton-meter is a unit of energy commonly called a joule, i.e., $1 \text{ N}\cdot\text{m} \equiv 1 \text{ J}$.

Speed of Sound

As discussed in Sec. 1.1, the forces acting on an object moving through the atmosphere depend on the Mach number, which is defined to be the ratio of the airspeed to the speed of sound. The speed at which an infinitesimal pressure disturbance propagates through a fluid is designated as the speed of sound and is traditionally denoted as a . The speed of sound depends solely on properties of the fluid, so the speed of sound is itself a property of the fluid. For an **ideal gas** the speed of sound is given by

$$a = \sqrt{\gamma RT} \quad (1.2.16)$$

where γ is the specific-heat ratio, R is the gas constant, and T is absolute temperature. For atmospheric air at standard temperature (15 C), $\gamma = 1.4$, $R = 287.0528 \text{ N}\cdot\text{m}/\text{kg}\cdot\text{K}$, and $T = 288.15 \text{ K}$. Thus, using the definition $1 \text{ N} \equiv 1 \text{ kg}\cdot\text{m}/\text{s}^2$, the **speed of sound in air at standard temperature** is

$$a = \sqrt{1.4(287.0528 \text{ N}\cdot\text{m}/\text{kg}\cdot\text{K})(288.15 \text{ K})} = 340.29 \text{ m/s} = 1116.45 \text{ ft/s}$$

For standard atmospheric conditions below 11 kilometers, temperature and the speed of sound decrease with increasing altitude.

Viscosity

As discussed in Sec. 1.1, a fluid is a substance that deforms continuously under the action of a shear stress. For most fluids, the shear stress is linearly proportional to the rate of angular deformation. Fluids that exhibit this linear relation between shear stress and rate of angular deformation are called Newtonian fluids.

For example, consider the two-dimensional deformation of the fluid element shown in Fig. 1.2.1. At time t the fluid element is rectangular and at time $t + dt$ the fluid element is rhombic. From the geometry shown in Fig. 1.2.1, the rate of angular deformation is

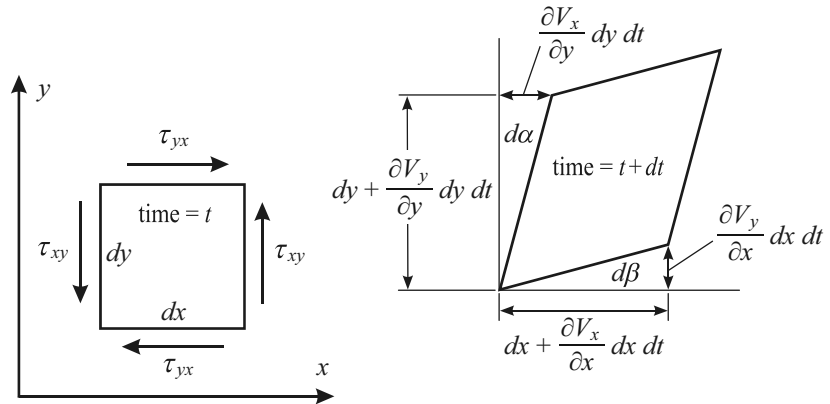


Figure 1.2.1. Deformation of a moving fluid element.

$$\text{rate of angular deformation} = \lim_{dt \rightarrow 0} \frac{d\alpha + d\beta}{dt} \quad (1.2.17)$$

In the limit as dt approaches zero, the angles $d\alpha$ and $d\beta$ can be written as

$$d\alpha = \lim_{dt \rightarrow 0} \frac{\frac{\partial V_x}{\partial y} dy dt}{dy + \frac{\partial V_y}{\partial y} dy dt} = \lim_{dt \rightarrow 0} \frac{\frac{\partial V_x}{\partial y} dt}{1 + \frac{\partial V_y}{\partial y} dt} = \frac{\partial V_x}{\partial y} dt \quad (1.2.18)$$

$$d\beta = \lim_{dt \rightarrow 0} \frac{\frac{\partial V_y}{\partial x} dx dt}{dx + \frac{\partial V_x}{\partial x} dx dt} = \lim_{dt \rightarrow 0} \frac{\frac{\partial V_y}{\partial x} dt}{1 + \frac{\partial V_x}{\partial x} dt} = \frac{\partial V_y}{\partial x} dt \quad (1.2.19)$$

where V_x and V_y are respectively the x and y components of the fluid velocity. Using Eqs. (1.2.18) and (1.2.19) in Eq. (1.2.17) yields

$$\text{rate of angular deformation} = \left(\frac{\partial V_x}{\partial y} dt + \frac{\partial V_y}{\partial x} dt \right) / dt = \frac{\partial V_x}{\partial y} + \frac{\partial V_y}{\partial x} \quad (1.2.20)$$

For a Newtonian fluid, the shear stress components, which are denoted in Fig. 1.2.1 as τ_{yx} and τ_{xy} , are linearly proportional to the rate of angular deformation. Thus, for **Newtonian fluids** we use the relation

$$\tau_{yx} = \tau_{xy} = \mu \left(\frac{\partial V_x}{\partial y} + \frac{\partial V_y}{\partial x} \right) \quad (1.2.21)$$

where the proportionality coefficient μ is called the *dynamic viscosity*. Like pressure, the shear stress is a force per unit area. In the SI system of units, shear stress is expressed in newtons per square meter (N/m^2), velocity is expressed in meters per second (m/s), and the coordinates x and y are expressed in meters (m). Thus, from Eq. (1.2.21), we see that the SI units for dynamic viscosity can be expressed as newton·seconds per square meter ($\text{N}\cdot\text{s/m}^2$). Equivalently, we can use the definition of a newton ($1 \text{ N} \equiv 1 \text{ kg}\cdot\text{m/s}^2$) to express dynamic viscosity in kilograms per second·meter ($\text{kg/s}\cdot\text{m}$).

In general, dynamic viscosity is a function of the fluid composition, temperature, and pressure. For gases at the temperatures and pressures encountered in the atmosphere, μ is typically independent of pressure. From the kinetic theory of gases, Sutherland (1893) obtained a relation that accurately describes the dynamic viscosity of many ideal gases,

$$\mu = \mu_0 \left(\frac{T}{T_0} \right)^{3/2} \frac{T_0 + T_S}{T + T_S} \quad (1.2.22)$$

where T is the absolute temperature and μ_0 , T_0 , and T_S are constants that depend on the molecular structure of the gas. Note that Eq. (1.2.22) predicts that the dynamic viscosity

of an ideal gas is independent of density. A similar but somewhat less accurate result was first obtained by Maxwell (1860), who found the result so surprising that he would not rest until he had confirmed it experimentally. At the temperatures and pressures encountered in the atmosphere, Sutherland's equation accurately describes **the dynamic viscosity of atmospheric air**,

$$\mu = 1.7153 \times 10^{-5} (\text{kg/s}\cdot\text{m}) \left(\frac{T}{273 \text{ K}} \right)^{3/2} \frac{383.4 \text{ K}}{T + 110.4 \text{ K}} \quad (1.2.23)$$

where T is the temperature in K.

The kinetic theory of gases also provides an analytical relation between the dynamic viscosity and the molecular mean free path for an **ideal gas**,

$$\mu = \sqrt{\frac{2RT}{\pi}} \rho \lambda \quad (1.2.24)$$

where R is the ideal gas constant, T is the absolute temperature, ρ is the density, and λ is the molecular mean free path. Because the mean free path is not an observable property of a gas, Eq. (1.2.24) is not a particularly useful relation for evaluating the dynamic viscosity. However, when combined with Eq. (1.2.22), the relation given by Eq. (1.2.24) does provide a means for evaluating the molecular mean free path for an ideal gas. Comparing Eqs. (1.2.22) and (1.2.24), we see that the product of ρ and λ depends only on the temperature.

In the mathematical formulations used in fluid mechanics and aerodynamics, the ratio μ/ρ appears frequently. This ratio is commonly called the *kinematic viscosity* and is traditionally given the symbol ν ,

$$\nu \equiv \frac{\mu}{\rho} \quad (1.2.25)$$

In the SI system of units, dynamic viscosity is expressed in kilograms per second-meter (kg/s·m) and density is expressed in kilograms per cubic meter (kg/m³). Thus, from Eq. (1.2.25), we see that the consistent SI units for kinematic viscosity are square meters per second (m²/s). In the English system of units, the consistent units for kinematic viscosity are square feet per second (ft²/s).

An important relation between the kinematic viscosity, the speed of sound, and the molecular mean free path is obtained by using Eqs. (1.2.16) and (1.2.24) in Eq. (1.2.25). This yields

$$\nu \equiv \frac{\mu}{\rho} = \sqrt{\frac{2RT}{\pi}} \lambda = \sqrt{\frac{2}{\pi\gamma}} \sqrt{\gamma RT} \lambda = \sqrt{\frac{2}{\pi\gamma}} a \lambda \quad (1.2.26)$$

where γ is the specific-heat ratio, a is the speed of sound, and λ is the molecular mean free path. Equation (1.2.26) is strictly valid only for a single-component ideal gas. However, it works well for air because the principal components, oxygen and nitrogen, are nearly identical diatomic molecules.

1.3. Important Dimensionless Parameters

In Sec. 1.1, we noted that the limiting boundary for the application of continuum theory to fluid flow is a function of Mach number, which is likely the most widely recognized dimensionless parameter in fluid mechanics. The Mach number M is defined to be the ratio of the characteristic velocity V to the speed of sound a ,

$$\text{Mach number} \equiv M \equiv \frac{V}{a} \quad (1.3.1)$$

Aerodynamic flows are commonly subdivided into several Mach-number regimes over which different analytical methods are applied. There are three major divisions; *subsonic flow* ($M < 1$), *sonic flow* ($M = 1$), and *supersonic flow* ($M > 1$). However, for the purpose of analysis, atmospheric flight is traditionally subdivided into five practical regimes; *incompressible subsonic flow* ($M < 0.3$), *compressible subsonic flow* ($0.3 < M < 0.8$), *transonic flow* ($0.8 < M < 1.2$), *supersonic flow* ($M > 1.2$), and *hypersonic flow* ($M > 5$).

In Sec. 1.1, we also observed that the limiting boundary for continuum theory is a function of Knudsen number, which is not nearly as widely known as the Mach number. The Knudsen number K is defined to be the ratio of the molecular mean free path λ to the minimum significant characteristic length associated with the flow geometry d ,

$$\text{Knudsen number} \equiv K \equiv \frac{\lambda}{d} \quad (1.3.2)$$

Strictly speaking, continuum theory applies only in the limit as the Knudsen number approaches zero. However, as a first approximation, the practical limit for the application of continuum theory to fluid flow can usually be estimated from Eq. (1.1.1). At Knudsen numbers greater than that specified by Eq. (1.1.1), the kinetic theory of gases should be considered as a means for analyzing aerodynamic flows.

In Sec. 1.2, we noted that most fluids can be approximated as Newtonian fluids, which exhibit a shear stress that is equal to the product of the angular deformation rate and the dynamic viscosity. The *Reynolds number* is a dimensionless parameter, which characterizes the relative importance of the effects of fluid inertia and viscosity. The Reynolds number R is defined to be the product of the fluid density ρ , the characteristic velocity V , and the characteristic length d , divided by the dynamic viscosity μ ,

$$\text{Reynolds number} \equiv R \equiv \frac{\rho V d}{\mu} \equiv \frac{V d}{\nu} \quad (1.3.3)$$

The reader should be careful not to confuse the Reynolds number with the ideal gas constant. Note that the symbol “ R ” used for the Reynolds number is not italic. The Reynolds number is a very important dimensionless parameter in all viscous flows. When possible variations in velocity and size are combined, the Reynolds number can vary widely in aerodynamic flows of practical interest. For example, in the flow about aerosol particles, such as small liquid droplets or dust particles, the Reynolds number can be less than 10^{-3} . On the other hand, in the flow about large high-speed transports, the Reynolds number could be greater than 10^9 . At very low Reynolds numbers, the viscous effects dominate to produce a smooth or *laminar* fluid motion known as *creeping flow*.

As the Reynolds number increases for a given geometry, inertia effects become important but the flow remains laminar. At still higher Reynolds numbers, laminar flow becomes unstable and the flow enters a *transition region*, which at even higher Reynolds numbers leads to a fluctuating fluid motion known as *turbulent flow*.

The aerodynamic forces and moments acting on an object moving through the atmosphere originate from only two sources,

1. The pressure distribution over the object's surface.
2. The shear stress distribution over the object's surface.

Both pressure and shear stress are traditionally nondimensionalized with respect to what is called the *freestream dynamic pressure*, which is a measure of fluid inertia defined to be one half the product of the freestream fluid density and the velocity squared,

$$\text{dynamic pressure} \equiv \frac{1}{2} \rho V^2 \quad (1.3.4)$$

For SI units, density is in kilograms per cubic meter (kg/m^3) and velocity is in meters per second (m/s). Thus, SI units for dynamic pressure are $(\text{kg/m}^3)(\text{m/s})^2 = \text{kg/m} \cdot \text{s}^2$, or after applying the definition $1 \text{ kg} \equiv 1 \text{ N} \cdot \text{s}^2/\text{m}$, the SI units for dynamic pressure are newtons per square meter (N/m^2). Similarly, in the English system of units, density is in slugs per cubic foot (slugs/ft^3), velocity is in feet per second (ft/s), and $1 \text{ slug} \equiv 1 \text{ lbf} \cdot \text{s}^2/\text{ft}$. Thus, the English units for dynamic pressure are pounds force per square foot (lbf/ft^2).

For dimensionless pressure, it is traditional to define the *pressure coefficient* to be the difference between the local fluid pressure p and the freestream fluid pressure p_∞ divided by the freestream dynamic pressure,

$$\text{pressure coefficient} \equiv C_p \equiv \frac{p - p_\infty}{\frac{1}{2} \rho V^2} \quad (1.3.5)$$

Similarly, the traditional *skin-friction coefficient* is defined to be the ratio of the surface or *wall* shear stress τ_w to the freestream dynamic pressure,

$$\text{skin-friction coefficient} \equiv C_f \equiv \frac{\tau_w}{\frac{1}{2} \rho V^2} \quad (1.3.6)$$

The resultant aerodynamic force acting on an object moving through the atmosphere is the net effects of the pressure and shear stress distributions integrated over the entire surface of the object. It is often convenient to split the resultant aerodynamic force into two components; the *lift*, which is defined to be the component of aerodynamic force perpendicular to the velocity vector, and the *drag*, which is defined to be the component of aerodynamic force parallel with the velocity vector. Both the lift L and the drag D are traditionally nondimensionalized with respect to the product of the freestream dynamic pressure and some reference area S . This yields the *lift coefficient* and *drag coefficient*

$$\text{lift coefficient} \equiv C_L \equiv \frac{L}{\frac{1}{2} \rho V^2 S} \quad (1.3.7)$$

$$\text{drag coefficient} \equiv C_D \equiv \frac{D}{\frac{1}{2}\rho V^2 S} \quad (1.3.8)$$

From continuum theory, the aerodynamic coefficients C_p , C_f , C_L , and C_D are typically found to be functions of flow geometry, Reynolds number, and Mach number. For very low density flows where continuum theory does not apply, the kinetic theory of gases predicts that these same aerodynamic coefficients are functions of flow geometry, Knudsen number, and Mach number. Because continuum theory is the limiting case for kinetic theory as the Knudsen number approaches zero, this suggests that there must be some relation between the Reynolds number, Mach number, and Knudsen number. This relation is exposed for an ideal gas by using Eq. (1.2.26) in Eq. (1.3.3), which yields

$$R \equiv \frac{Vd}{\nu} = \sqrt{\frac{\pi\gamma}{2}} \frac{Vd}{a\lambda} = \sqrt{\frac{\pi\gamma}{2}} \frac{M}{K} \quad \text{or} \quad K = \sqrt{\frac{\pi\gamma}{2}} \frac{M}{R} \quad (1.3.9)$$

Thus, the Knudsen number for an ideal gas can be computed directly from the Mach number and Reynolds number. Using Eq. (1.3.9) in Eq. (1.1.1) and rearranging, the approximate limiting boundary for the application of continuum theory can be expressed in terms of only the Mach number and Reynolds number, see Phillips (2010),

$$M < \sim \frac{4 \times 10^{-3} [2/(\pi\gamma)]^{1/2} R}{1 + 2 \{8 \times 10^{-3} [2/(\pi\gamma)]^{1/2} R\}^{1/2}} \quad (1.3.10)$$

This relation is shown graphically in Fig. 1.3.1. Note that continuum theory is valid at any altitude for subsonic flows with Reynolds numbers above 4,000.

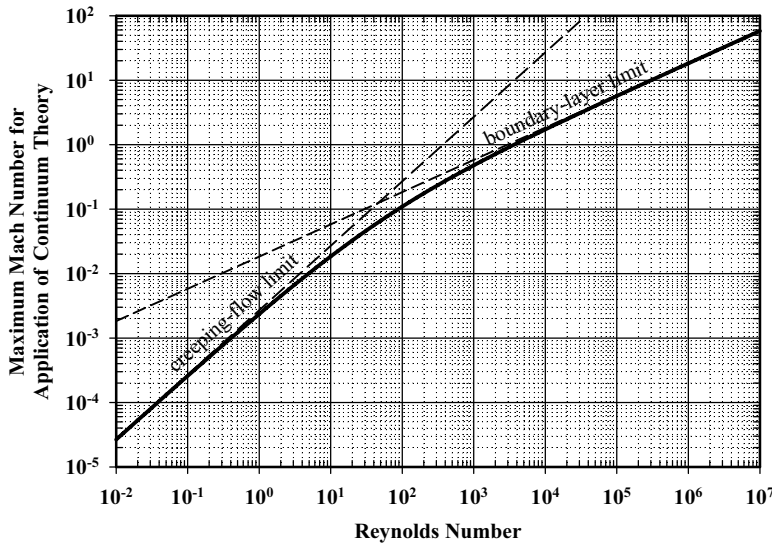


Figure 1.3.1. Mach-number and Reynolds-number limits of continuum theory, Phillips (2010).

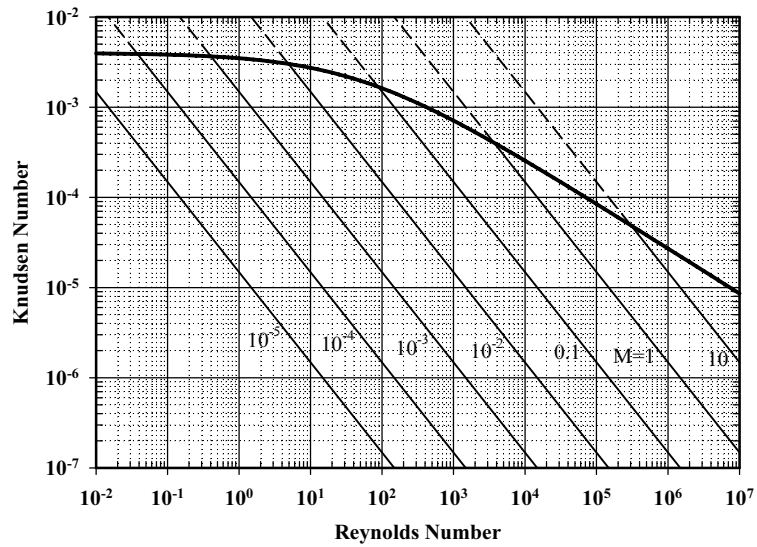


Figure 1.3.2. Knudsen-number and Reynolds-number limits of continuum theory, Phillips (2010).

The relation given by Eq. (1.3.9) can also be used to express the constraint given by Eq. (1.3.10) as a relation between the Knudsen number and Reynolds number,

$$K = \sqrt{\pi\gamma/2} \frac{M}{R} < \sim \frac{4 \times 10^{-3}}{1 + 2 \{8 \times 10^{-3} [2/(\pi\gamma)]^{1/2} R\}^{1/2}} \quad (1.3.11)$$

This relation is shown graphically in Fig. 1.3.2 compared with several lines of constant Mach number. These Mach number contours are shown as solid lines in the region where continuum theory is predicted to be valid, and they are shown as dashed lines in the region where kinetic theory should be considered. For a given geometry, the Knudsen number increases with altitude. Thus, the vertical axis in Fig. 1.3.2 could be viewed as a dimensionless altitude. It is important to recognize that, for fixed geometry and Mach number, the Reynolds number always decreases with increasing altitude.

EXAMPLE 1.3.1. At an altitude of 30,000 m (98,425 ft), the temperature and pressure of the air are -46.641°C and $1.1970 \times 10^3 \text{ N/m}^2$, respectively. A small object with a minimum significant characteristic length of 1 cm is moving through this air at 400 knots. Find the Mach number, Reynolds number, and Knudsen number. Do you expect continuum theory to be valid for this flow?

Solution. A knot is 1 nautical mile per hour and a nautical mile is 1852 m. Thus, the velocity in SI units is

$$V = \frac{400 \text{ naut. mi/hr}(1852 \text{ m/naut. mi})}{3600 \text{ s/hr}} = 205.78 \text{ m/s}$$

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The absolute temperature is $-46.641\text{ C} + 273.150 = 226.509\text{ K}$. From Eq. (1.2.16), the speed of sound is

$$\begin{aligned} a &= \sqrt{\gamma RT} = \sqrt{1.4(287.0528\text{ N}\cdot\text{m/kg}\cdot\text{K})(1\text{ kg}\cdot\text{m/s}^2\cdot\text{N})(226.509\text{ K})} \\ &= 301.71\text{ m/s} \end{aligned}$$

From Eq. (1.3.1), the Mach number is

$$M \equiv \frac{V}{a} = \frac{205.78\text{ m/s}}{301.71\text{ m/s}} = \underline{0.682}$$

From Eq. (1.2.6), the air density is

$$\rho = \frac{p}{RT} = \frac{1.1970 \times 10^3\text{ N/m}^2}{287.0528\text{ N}\cdot\text{m/kg}\cdot\text{K}(226.509\text{ K})} = 1.8410 \times 10^{-2}\text{ kg/m}^3$$

From Eq. (1.2.23), the dynamic viscosity is

$$\begin{aligned} \mu &= 1.7153 \times 10^{-5}\text{ (kg/s}\cdot\text{m)} \left(\frac{T}{273\text{ K}} \right)^{3/2} \frac{383.4\text{ K}}{T + 110.4\text{ K}} \\ &= 1.7153 \times 10^{-5}\text{ (kg/s}\cdot\text{m)} \left(\frac{226.509\text{ K}}{273\text{ K}} \right)^{3/2} \frac{383.4\text{ K}}{226.509\text{ K} + 110.4\text{ K}} \\ &= 1.4752 \times 10^{-5}\text{ kg/s}\cdot\text{m} \end{aligned}$$

From Eq. (1.2.25), the kinematic viscosity is

$$\nu \equiv \frac{\mu}{\rho} = \frac{1.4752 \times 10^{-5}\text{ kg/s}\cdot\text{m}}{1.8410 \times 10^{-2}\text{ kg/m}^3} = 8.0132 \times 10^{-4}\text{ m}^2/\text{s}$$

From Eq. (1.3.3), the Reynolds number is

$$R \equiv \frac{Vd}{\nu} = \frac{205.78\text{ m/s}(0.01\text{ m})}{8.0132 \times 10^{-4}\text{ m}^2/\text{s}} = \underline{2568}$$

From Eq. (1.3.9), the Knudsen number is

$$K = \sqrt{\frac{\pi\gamma}{2}} \frac{M}{R} = \sqrt{\frac{\pi(1.4)}{2}} \frac{0.682}{2568} = \underline{3.9 \times 10^{-4}}$$

From the right-hand side of Eq. (1.3.11), the maximum Knudsen number is

$$K_{\max} \cong \frac{4 \times 10^{-3}}{1 + 2 \{8 \times 10^{-3} [2/(\pi 1.4)]^{1/2} 2568\}^{1/2}} \cong 4.7 \times 10^{-4}$$

Because the actual Knudsen number is slightly less than the maximum allowable Knudsen number predicted from Eq. (1.3.11), we should expect continuum theory to give a reasonable first approximation for this flow. However, because the actual Knudsen number is very near the limit predicted from this approximate relation, results obtained from continuum theory, in this case, should be used with caution.

1.4. Review of Vector Mathematics

One of the fundamental laws of physics that forms the foundation for all of engineering mechanics, including fluid mechanics, is *Newton's second law*. Students should be very familiar with Newton's second law in the form

$$\mathbf{F} = m\mathbf{a} \quad (1.4.1)$$

where $\mathbf{F}$ is force, m is mass, and $\mathbf{a}$ is acceleration. Because both force and acceleration are vector quantities, Newton's second law is a vector equation, which has three scalar components. In this textbook all vectors will be denoted using bold un-italicized symbols like $\mathbf{F}$, $\mathbf{a}$, and $\mathbf{\Omega}$; and scalars will be denoted using italic symbols like m , a , and Ω . The acceleration vector in Newton's second law can be written as the total derivative of the velocity vector with respect to time,

$$\mathbf{F} = m \frac{d\mathbf{V}}{dt} \quad (1.4.2)$$

where $\mathbf{V}$ is the velocity vector and t denotes the scalar time.

Any vector can be written as the sum of its three components relative to a particular coordinate system. Each component can be written as a scalar magnitude multiplied by a unit vector in one of the three coordinate directions. For example, the force $\mathbf{F}$ can be written in Cartesian coordinates (x, y, z) as

$$\mathbf{F} = F_x \mathbf{i}_x + F_y \mathbf{i}_y + F_z \mathbf{i}_z \quad (1.4.3)$$

where F_x , F_y , and F_z , are the three scalar component magnitudes; and $\mathbf{i}_x$, $\mathbf{i}_y$, and $\mathbf{i}_z$ are the unit vectors in the x , y , and z directions, respectively. Hence, Eq. (1.4.2) can be written

$$F_x \mathbf{i}_x + F_y \mathbf{i}_y + F_z \mathbf{i}_z = m \frac{dV_x}{dt} \mathbf{i}_x + m \frac{dV_y}{dt} \mathbf{i}_y + m \frac{dV_z}{dt} \mathbf{i}_z \quad (1.4.4)$$

Because all three components of any vector equation must be satisfied independently, Eq. (1.4.2) can be viewed as shorthand notation for a system of three scalar equations, which can also be written in the notation of linear algebra as

$$\begin{Bmatrix} F_x \\ F_y \\ F_z \end{Bmatrix} = \begin{bmatrix} m & 0 & 0 \\ 0 & m & 0 \\ 0 & 0 & m \end{bmatrix} \begin{Bmatrix} dV_x/dt \\ dV_y/dt \\ dV_z/dt \end{Bmatrix} \quad (1.4.5)$$

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Because the vector formulation of Newton's second law forms the foundation of fluid mechanics, a thorough understanding of vector algebra and vector calculus is a prerequisite to the study of fluid mechanics and aerodynamics. In this section a concise review of vector algebra and vector calculus is presented. Most engineering students should be very familiar with this material. If you feel that your personal background in vector algebra and vector calculus is weak, you should revisit your mathematics textbooks and review the related chapters before proceeding further with the material in this textbook.

Vector Algebra

Let $\mathbf{A}$ be an arbitrary vector having the Cartesian components $\mathbf{A} = A_x \mathbf{i}_x + A_y \mathbf{i}_y + A_z \mathbf{i}_z$. The *magnitude* of this vector, denoted as $|\mathbf{A}|$, is a scalar quantity equal to its length

$$|\mathbf{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2} \quad (1.4.6)$$

and if the *unit vector* in the direction of $\mathbf{A}$ is denoted as $\mathbf{u}$, we have

$$\mathbf{u} = \mathbf{A}/|\mathbf{A}| \quad (1.4.7)$$

Let $\mathbf{B}$ represent another vector having the Cartesian components $\mathbf{B} = B_x \mathbf{i}_x + B_y \mathbf{i}_y + B_z \mathbf{i}_z$. *Vector addition and subtraction* are defined such that

$$\mathbf{A} + \mathbf{B} = (A_x + B_x)\mathbf{i}_x + (A_y + B_y)\mathbf{i}_y + (A_z + B_z)\mathbf{i}_z \quad (1.4.8)$$

$$\mathbf{A} - \mathbf{B} = (A_x - B_x)\mathbf{i}_x + (A_y - B_y)\mathbf{i}_y + (A_z - B_z)\mathbf{i}_z \quad (1.4.9)$$

There are two forms of vector multiplication; the *dot product* (also called the *scalar product* or the *inner product*), and the *cross product* (also called the *vector product*). The *dot product* of two vectors is a scalar defined according to

$$\mathbf{A} \cdot \mathbf{B} = A_x B_x + A_y B_y + A_z B_z \quad (1.4.10)$$

The dot product is also equal to the product of the two vector magnitudes multiplied by the cosine of the angle between the two vectors, i.e.,

$$\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}||\mathbf{B}| \cos \theta \quad (1.4.11)$$

where θ is the angle from $\mathbf{A}$ to $\mathbf{B}$. It should be noted that the dot product is commutative, i.e., $\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}$. The following rules apply to the dot product:

$$\mathbf{A} \cdot \mathbf{A} = |\mathbf{A}|^2 \quad (1.4.12)$$

$$\mathbf{C} \cdot (\mathbf{A} + \mathbf{B}) = (\mathbf{A} + \mathbf{B}) \cdot \mathbf{C} = \mathbf{A} \cdot \mathbf{C} + \mathbf{B} \cdot \mathbf{C} \quad (1.4.13)$$

$$\mathbf{i}_x \cdot \mathbf{i}_x = \mathbf{i}_y \cdot \mathbf{i}_y = \mathbf{i}_z \cdot \mathbf{i}_z = 1, \quad \mathbf{i}_x \cdot \mathbf{i}_y = \mathbf{i}_y \cdot \mathbf{i}_z = \mathbf{i}_z \cdot \mathbf{i}_x = 0 \quad (1.4.14)$$

The **cross product** of two vectors is a vector defined according to

$$\begin{aligned}\mathbf{A} \times \mathbf{B} &= \begin{vmatrix} \mathbf{i}_x & \mathbf{i}_y & \mathbf{i}_z \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} \\ &= (A_y B_z - A_z B_y) \mathbf{i}_x + (A_z B_x - A_x B_z) \mathbf{i}_y + (A_x B_y - A_y B_x) \mathbf{i}_z\end{aligned}\quad (1.4.15)$$

The cross product is also equal to the product of the two vector magnitudes multiplied by the sine of the angle between the two vectors and a unit vector normal to the plane that contains both $\mathbf{A}$ and $\mathbf{B}$, i.e.,

$$\mathbf{A} \times \mathbf{B} = |\mathbf{A}| |\mathbf{B}| \sin \theta \mathbf{n} \quad (1.4.16)$$

where θ is the angle in the plane containing $\mathbf{A}$ and $\mathbf{B}$ that is formed by rotating $\mathbf{A}$ into $\mathbf{B}$, and $\mathbf{n}$ is the unit vector that is normal to both $\mathbf{A}$ and $\mathbf{B}$ in the direction of the axis of a right-hand rotation from $\mathbf{A}$ to $\mathbf{B}$. In other words, if the fingers of your right hand curl in the direction of the rotation from $\mathbf{A}$ to $\mathbf{B}$, your right thumb will point in the direction of the unit vector $\mathbf{n}$. The following rules apply to the cross product:

$$\mathbf{A} \times \mathbf{B} = -\mathbf{B} \times \mathbf{A} \quad (1.4.17)$$

$$\mathbf{C} \times (\mathbf{A} + \mathbf{B}) = \mathbf{C} \times \mathbf{A} + \mathbf{C} \times \mathbf{B} \quad (1.4.18)$$

$$(\mathbf{A} + \mathbf{B}) \times \mathbf{C} = \mathbf{A} \times \mathbf{C} + \mathbf{B} \times \mathbf{C} \quad (1.4.19)$$

$$\mathbf{i}_x \times \mathbf{i}_x = \mathbf{i}_y \times \mathbf{i}_y = \mathbf{i}_z \times \mathbf{i}_z = 0 \quad (1.4.20)$$

$$\mathbf{i}_x \times \mathbf{i}_y = \mathbf{i}_z, \quad \mathbf{i}_y \times \mathbf{i}_z = \mathbf{i}_x, \quad \mathbf{i}_z \times \mathbf{i}_x = \mathbf{i}_y \quad (1.4.21)$$

It should be noted that because $\mathbf{A} \times \mathbf{B} = -\mathbf{B} \times \mathbf{A}$, the cross product is not commutative.

Vector Calculus

A scalar field is simply a scalar quantity having a value that varies with space and time, $\psi = \psi(x, y, z, t)$. Similarly, a vector field is simply a vector quantity having components with values that vary with space and time, $\mathbf{V} = V_x(x, y, z, t) \mathbf{i}_x + V_y(x, y, z, t) \mathbf{i}_y + V_z(x, y, z, t) \mathbf{i}_z$. The **gradient of a scalar field** expressed in **Cartesian coordinates** is the vector field

$$\nabla \psi = \frac{\partial \psi}{\partial x} \mathbf{i}_x + \frac{\partial \psi}{\partial y} \mathbf{i}_y + \frac{\partial \psi}{\partial z} \mathbf{i}_z \quad (1.4.22)$$

The **divergence of a vector field** expressed in **Cartesian coordinates** is the scalar field

$$\nabla \cdot \mathbf{V} = \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z} \quad (1.4.23)$$

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The **curl of a vector field** expressed in **Cartesian coordinates** is the vector field

$$\begin{aligned}\nabla \times \mathbf{V} &= \begin{vmatrix} \mathbf{i}_x & \mathbf{i}_y & \mathbf{i}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ V_x & V_y & V_z \end{vmatrix} \\ &= \left(\frac{\partial V_z}{\partial y} - \frac{\partial V_y}{\partial z} \right) \mathbf{i}_x + \left(\frac{\partial V_x}{\partial z} - \frac{\partial V_z}{\partial x} \right) \mathbf{i}_y + \left(\frac{\partial V_y}{\partial x} - \frac{\partial V_x}{\partial y} \right) \mathbf{i}_z\end{aligned}\quad (1.4.24)$$

The **Laplacian of a scalar field** expressed in **Cartesian coordinates** is the scalar field

$$\nabla^2 \psi = \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} \quad (1.4.25)$$

The **Laplacian of a vector field** expressed in **Cartesian coordinates** is the vector field

$$\nabla^2 \mathbf{V} = (\nabla^2 V_x) \mathbf{i}_x + (\nabla^2 V_y) \mathbf{i}_y + (\nabla^2 V_z) \mathbf{i}_z \quad (1.4.26)$$

Cylindrical Coordinates

Although Cartesian coordinates are useful for simple geometry, it is often convenient to use other coordinate systems in formulating commonly encountered fluid flow problems, such as flow in a circular pipe. The transformation between Cartesian coordinates (x, y, z) and cylindrical coordinates (r, θ, z) is specified by the relations $x = r \cos \theta$, $y = r \sin \theta$, and $z = z$. Hence, in cylindrical coordinates we obtain

$$\nabla \psi = \frac{\partial \psi}{\partial r} \mathbf{i}_r + \frac{1}{r} \frac{\partial \psi}{\partial \theta} \mathbf{i}_\theta + \frac{\partial \psi}{\partial z} \mathbf{i}_z \quad (1.4.27)$$

$$\nabla \cdot \mathbf{V} = \frac{1}{r} \frac{\partial(rV_r)}{\partial r} + \frac{1}{r} \frac{\partial V_\theta}{\partial \theta} + \frac{\partial V_z}{\partial z} = \frac{\partial V_r}{\partial r} + \frac{1}{r} \frac{\partial V_\theta}{\partial \theta} + \frac{\partial V_z}{\partial z} + \frac{V_r}{r} \quad (1.4.28)$$

$$\begin{aligned}\nabla \times \mathbf{V} &= \begin{vmatrix} (1/r)\mathbf{i}_r & \mathbf{i}_\theta & (1/r)\mathbf{i}_z \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial z} \\ V_r & rV_\theta & V_z \end{vmatrix} \\ &= \left(\frac{1}{r} \frac{\partial V_z}{\partial \theta} - \frac{\partial V_\theta}{\partial z} \right) \mathbf{i}_r + \left(\frac{\partial V_r}{\partial z} - \frac{\partial V_z}{\partial r} \right) \mathbf{i}_\theta + \left(\frac{\partial V_\theta}{\partial r} - \frac{1}{r} \frac{\partial V_r}{\partial \theta} + \frac{V_\theta}{r} \right) \mathbf{i}_z\end{aligned}\quad (1.4.29)$$

$$\nabla^2 \psi = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \psi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 \psi}{\partial \theta^2} + \frac{\partial^2 \psi}{\partial z^2} = \frac{\partial^2 \psi}{\partial r^2} + \frac{1}{r} \frac{\partial \psi}{\partial r} + \frac{1}{r^2} \frac{\partial^2 \psi}{\partial \theta^2} + \frac{\partial^2 \psi}{\partial z^2} \quad (1.4.30)$$

$$\nabla^2 \mathbf{V} = \left(\nabla^2 V_r - \frac{V_r}{r^2} - \frac{2}{r^2} \frac{\partial V_\theta}{\partial \theta} \right) \mathbf{i}_r + \left(\nabla^2 V_\theta + \frac{2}{r^2} \frac{\partial V_r}{\partial \theta} - \frac{V_\theta}{r^2} \right) \mathbf{i}_\theta + (\nabla^2 V_z) \mathbf{i}_z \quad (1.4.31)$$

The unit vectors in cylindrical coordinates are related to the Cartesian unit vectors by

$$\mathbf{i}_r = \cos \theta \mathbf{i}_x + \sin \theta \mathbf{i}_y, \quad \mathbf{i}_\theta = -\sin \theta \mathbf{i}_x + \cos \theta \mathbf{i}_y, \quad \mathbf{i}_z = \mathbf{i}_z \quad (1.4.32)$$

Spherical Coordinates

The transformation from Cartesian coordinates (x, y, z) to spherical coordinates (r, θ, φ) is defined by the three relations $x = r \sin \theta \cos \varphi$, $y = r \sin \theta \sin \varphi$, and $z = r \cos \theta$, where θ is the co-latitude and φ is the longitude. Hence, in spherical coordinates we obtain

$$\nabla \psi = \frac{\partial \psi}{\partial r} \mathbf{i}_r + \frac{1}{r} \frac{\partial \psi}{\partial \theta} \mathbf{i}_\theta + \frac{1}{r \sin \theta} \frac{\partial \psi}{\partial \varphi} \mathbf{i}_\varphi \quad (1.4.33)$$

$$\begin{aligned} \nabla \cdot \mathbf{V} &= \frac{1}{r^2} \frac{\partial(r^2 V_r)}{\partial r} + \frac{1}{r \sin \theta} \frac{\partial(V_\theta \sin \theta)}{\partial \theta} + \frac{1}{r \sin \theta} \frac{\partial V_\varphi}{\partial \varphi} \\ &= \frac{\partial V_r}{\partial r} + \frac{1}{r} \frac{\partial V_\theta}{\partial \theta} + \frac{1}{r \sin \theta} \frac{\partial V_\varphi}{\partial \varphi} + \frac{2V_r}{r} + \frac{V_\theta \cos \theta}{r \sin \theta} \end{aligned} \quad (1.4.34)$$

$$\begin{aligned} \nabla \times \mathbf{V} &= \begin{vmatrix} (1/r^2 \sin \theta) \mathbf{i}_r & (1/r \sin \theta) \mathbf{i}_\theta & (1/r) \mathbf{i}_\varphi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \varphi} \\ V_r & r V_\theta & r V_\varphi \sin \theta \end{vmatrix} \\ &= \left(\frac{1}{r} \frac{\partial V_\varphi}{\partial \theta} - \frac{1}{r \sin \theta} \frac{\partial V_\theta}{\partial \varphi} + \frac{V_\varphi \cos \theta}{r \sin \theta} \right) \mathbf{i}_r \\ &\quad + \left(\frac{1}{r \sin \theta} \frac{\partial V_r}{\partial \varphi} - \frac{\partial V_\varphi}{\partial r} - \frac{V_\varphi}{r} \right) \mathbf{i}_\theta \\ &\quad + \left(\frac{\partial V_\theta}{\partial r} - \frac{1}{r} \frac{\partial V_r}{\partial \theta} + \frac{V_\theta}{r} \right) \mathbf{i}_\varphi \end{aligned} \quad (1.4.35)$$

$$\begin{aligned} \nabla^2 \psi &= \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \psi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \psi}{\partial \varphi^2} \\ &= \frac{\partial^2 \psi}{\partial r^2} + \frac{2}{r} \frac{\partial \psi}{\partial r} + \frac{1}{r^2} \frac{\partial^2 \psi}{\partial \theta^2} + \frac{\cos \theta}{r^2 \sin \theta} \frac{\partial \psi}{\partial \theta} + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \psi}{\partial \varphi^2} \end{aligned} \quad (1.4.36)$$

$$\begin{aligned} \nabla^2 \mathbf{V} &= \left(\nabla^2 V_r - \frac{2V_r}{r^2} - \frac{2}{r^2} \frac{\partial V_\theta}{\partial \theta} - \frac{2V_\theta \cos \theta}{r^2 \sin \theta} - \frac{2}{r^2 \sin \theta} \frac{\partial V_\varphi}{\partial \varphi} \right) \mathbf{i}_r \\ &\quad + \left(\nabla^2 V_\theta + \frac{2}{r^2} \frac{\partial V_r}{\partial \theta} - \frac{V_\theta}{r^2 \sin^2 \theta} - \frac{2 \cos \theta}{r^2 \sin^2 \theta} \frac{\partial V_\varphi}{\partial \varphi} \right) \mathbf{i}_\theta \\ &\quad + \left(\nabla^2 V_\varphi + \frac{2}{r^2 \sin^2 \theta} \frac{\partial V_r}{\partial \varphi} - \frac{V_\varphi}{r^2 \sin^2 \theta} + \frac{2 \cos \theta}{r^2 \sin^2 \theta} \frac{\partial V_\theta}{\partial \varphi} \right) \mathbf{i}_\varphi \end{aligned} \quad (1.4.37)$$

The unit vectors in spherical coordinates are related to the Cartesian unit vectors by

$$\begin{aligned}\mathbf{i}_r &= \sin \theta \cos \varphi \mathbf{i}_x + \sin \theta \sin \varphi \mathbf{i}_y + \cos \theta \mathbf{i}_z \\ \mathbf{i}_\theta &= \cos \theta \cos \varphi \mathbf{i}_x + \cos \theta \sin \varphi \mathbf{i}_y - \sin \theta \mathbf{i}_z \\ \mathbf{i}_\varphi &= -\sin \varphi \mathbf{i}_x + \cos \varphi \mathbf{i}_y\end{aligned}\quad (1.4.38)$$

1.5. Fundamentals of Fluid Statics

The branch of fluid mechanics that deals with fluids that are either at rest or in rigid-body motion is called *fluid statics*. A fluid that is at rest or in rigid-body motion throughout will be referred to as a *static fluid*. From the engineering definition, *a fluid is a substance that deforms continuously under the action of a shear stress*. In fluids that are either at rest or in rigid-body motion, there is no relative motion of the fluid particles. Thus, in a static fluid there can be no fluid deformation. This leads to an important corollary of the engineering definition of a fluid, i.e., **there can be no shear stress in a static fluid**.

Newton's Second Law for a Static Fluid

Because there is no relative acceleration in a static fluid, Newton's second law requires that the summation of forces acting on a static fluid element must be zero. Because there can be no shear stress in a static fluid, the only forces acting on a static fluid element are the pressure forces acting over the surface of the element and the body forces acting over the volume of the element, which include any effects of rigid-body acceleration.

Consider the net pressure force $\mathbf{F}_p$ acting on the differential fluid element shown in Fig. 1.5.1. This element is centered at an arbitrary point having coordinates (x, y, z) and the element's lengths in the three coordinate directions are dx , dy , and dz , respectively. Because force is a vector, $\mathbf{F}_p$ has a scalar component in each of these three coordinate

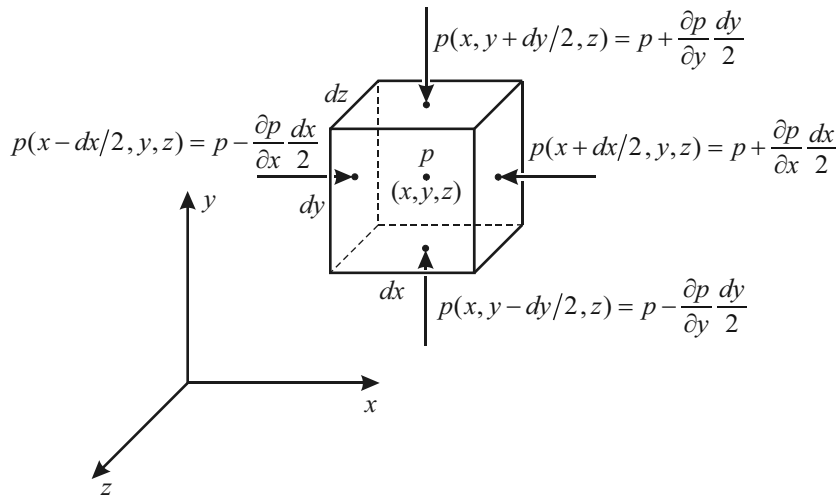


Figure 1.5.1. Pressure forces in the x and y directions acting on a differential fluid element.

directions, i.e., F_{px} , F_{py} , and F_{pz} . As shown in Fig. 1.5.1, the x component of $\mathbf{F}_p$ is the pressure acting on the left side of the element multiplied by the area of the left side, less the pressure acting on the right side multiplied by the area of the right side. This gives

$$\begin{aligned} F_{px} &= p(x - dx/2, y, z)dydz - p(x + dx/2, y, z)dydz \\ &= \left(p - \frac{\partial p}{\partial x} \frac{dx}{2} \right) dydz - \left(p + \frac{\partial p}{\partial x} \frac{dx}{2} \right) dydz \\ &= -\frac{\partial p}{\partial x} \frac{dx}{2} dydz - \frac{\partial p}{\partial x} \frac{dx}{2} dydz = -\frac{\partial p}{\partial x} dx dydz \end{aligned} \quad (1.5.1)$$

As also shown in Fig. 1.5.1, the y component of $\mathbf{F}_p$ is

$$\begin{aligned} F_{py} &= p(x, y - dy/2, z)dx dz - p(x, y + dy/2, z)dx dz \\ &= \left(p - \frac{\partial p}{\partial y} \frac{dy}{2} \right) dx dz - \left(p + \frac{\partial p}{\partial y} \frac{dy}{2} \right) dx dz \\ &= -\frac{\partial p}{\partial y} \frac{dy}{2} dx dz - \frac{\partial p}{\partial y} \frac{dy}{2} dx dz = -\frac{\partial p}{\partial y} dx dy dz \end{aligned} \quad (1.5.2)$$

Similarly, the z component of $\mathbf{F}_p$, which is not shown in Fig. 1.5.1, is found to be

$$F_{pz} = -\frac{\partial p}{\partial z} dx dy dz \quad (1.5.3)$$

The vector sum of Eqs. (1.5.1), (1.5.2), and (1.5.3) gives

$$\mathbf{F}_p = F_{px} \mathbf{i}_x + F_{py} \mathbf{i}_y + F_{pz} \mathbf{i}_z = -\left(\frac{\partial p}{\partial x} \mathbf{i}_x + \frac{\partial p}{\partial y} \mathbf{i}_y + \frac{\partial p}{\partial z} \mathbf{i}_z \right) dx dy dz \quad (1.5.4)$$

where $\mathbf{i}_x$, $\mathbf{i}_y$, and $\mathbf{i}_z$ are the unit vectors in the x , y , and z directions, respectively. Note that the three bracketed terms on the far right-hand side of Eq. (1.5.4) combine to form the vector, which is defined in Eq. (1.4.22) to be *gradient* of the pressure ∇p . We now define $\mathbf{f}_p$ to be the **pressure force vector per unit volume** acting on a differential fluid element. From Eq. (1.5.4), this gives

$$\mathbf{f}_p = -\nabla p \equiv -\frac{\partial p}{\partial x} \mathbf{i}_x - \frac{\partial p}{\partial y} \mathbf{i}_y - \frac{\partial p}{\partial z} \mathbf{i}_z \quad (1.5.5)$$

Since there can be no shear stress acting on a static fluid, the pressure force given by Eq. (1.5.5) is the total net force acting over the surface of a static fluid element. When this pressure force is added to any body forces that are acting over the volume of the fluid element, the result must sum to zero in a static fluid. If we define $\mathbf{f}_b$ to be the **vector sum of all body forces per unit volume** acting over the volume of a differential fluid element, then in view of Eq. (1.5.5), **Newton's second law for a static fluid requires**

$$-\nabla p + \mathbf{f}_b = 0 \quad (1.5.6)$$

Two familiar body forces that could be acting on a fluid element are the gravitational force and the electromagnetic force. However, if the coordinate system used to obtain Eq. (1.5.5) is accelerating, there will also be apparent body forces resulting from this coordinate-system acceleration. If the coordinate-system acceleration includes rotation, which is always the case for an Earth-fixed coordinate system, then an exact representation of Newton's second law would need to account for centripetal and Coriolis accelerations. The effects of coordinate-system acceleration can be included through the addition of appropriate body forces in our statement of Newton's second law. In certain cases the Coriolis force resulting from the Earth's rotation can be important for fluids in motion. For example, some oceanic and atmospheric flows are driven by the Coriolis force. However, in a great majority of practical engineering problems, all body forces except the gravitational force can be neglected. This is the important special case that will be considered first.

The magnitude of the gravitational body force acting on any substance is the mass of the substance multiplied by the scalar acceleration of gravity g . Thus, the magnitude of the gravitational force per unit volume is the product of the mass per unit volume and the scalar acceleration of gravity. Because mass per unit volume is simply the density, the magnitude of the gravitational body force per unit volume is

$$|\mathbf{f}_g| = \rho g \quad (1.5.7)$$

The direction of the force of gravity is always down. However, for a mathematical formulation we must define "down" mathematically. Many textbooks solve this problem by simply choosing the reference coordinate system so that the z axis points in a direction opposite to the force of gravity. In this particular coordinate system, the direction of the gravitational force is, by definition, $-\mathbf{i}_z$. However, it is often more convenient to choose a coordinate system that is not aligned with the gravitational force. Throughout this text, we shall use the symbol H to denote geometric altitude measured normal to the Earth's mean-sea-level surface. With this notation, the gravitational force acts in the direction opposite to the gradient of H , and the **gravitational body force per unit volume** is

$$\mathbf{f}_g = -\rho g \nabla H \quad (1.5.8)$$

Recall from Eq. (1.4.22) that the gradient of H is a vector having three components,

$$\nabla H = \frac{\partial H}{\partial x} \mathbf{i}_x + \frac{\partial H}{\partial y} \mathbf{i}_y + \frac{\partial H}{\partial z} \mathbf{i}_z \quad (1.5.9)$$

Because H is a linear distance measured in the same units as x , y , and z , the gradient of H is always a unit vector, i.e., $|\nabla H| = 1$.

The result given by Eq. (1.5.8) can be used with an arbitrary coordinate system. A Cartesian coordinate system that is commonly chosen for fluid statics problems has the z axis pointing vertically upward. In this particular coordinate system, we have

$$H = z, \quad \frac{\partial H}{\partial x} = 0, \quad \frac{\partial H}{\partial y} = 0, \quad \frac{\partial H}{\partial z} = 1, \quad \nabla H = \mathbf{i}_z$$

Combining Eqs. (1.5.6) and (1.5.8), yields a vector differential equation that allows us to relate the pressure distribution in a static fluid to the altitude, fluid density, and acceleration of gravity. **If the only body force is that associated with the acceleration of gravity, the pressure distribution in a static fluid can be found from**

$$\nabla p = -\rho g \nabla H \quad (1.5.10)$$

From Newton's law of gravitation, g varies with altitude according to the relation

$$g = g_o \left(\frac{R_E}{R_E + H} \right)^2 \quad (1.5.11)$$

where g_o is the standard acceleration of gravity at mean sea level, $g_o = 9.806645 \text{ m/s}^2$, and R_E is the radius of the Earth at mean sea level, $R_E = 6,356,766 \text{ m}$. While the acceleration of gravity does vary with altitude in a manner that agrees closely with that specified by Eq. (1.5.11), for many practical applications, changes in altitude are small compared with the radius of the Earth, and thus, g can often be approximated as a constant.

Equation (1.5.10) is a vector equation, which has three scalar components that must be satisfied individually. In Cartesian coordinates, the three components are

$$\begin{aligned} x \text{ component: } \frac{\partial p}{\partial x} &= -\rho g \frac{\partial H}{\partial x} \\ y \text{ component: } \frac{\partial p}{\partial y} &= -\rho g \frac{\partial H}{\partial y} \\ z \text{ component: } \frac{\partial p}{\partial z} &= -\rho g \frac{\partial H}{\partial z} \end{aligned} \quad (1.5.12)$$

Equations (1.5.12) specify how the thermodynamic pressure will vary in each of the three coordinate directions within a static fluid. For the special case where the z axis is chosen to point vertically up, the Cartesian component equations reduce to

$$\text{for } H = z: \quad \frac{\partial p}{\partial x} = 0, \quad \frac{\partial p}{\partial y} = 0, \quad \frac{\partial p}{\partial z} = -\rho g$$

For this special choice of coordinates, the pressure depends only on z .

In a more general manner, Eq. (1.5.10) can be expressed as an ordinary differential equation by taking the dot product with the vector $dx \mathbf{i}_x + dy \mathbf{i}_y + dz \mathbf{i}_z$. This yields

$$\frac{\partial p}{\partial x} dx + \frac{\partial p}{\partial y} dy + \frac{\partial p}{\partial z} dz = -\rho g \left(\frac{\partial H}{\partial x} dx + \frac{\partial H}{\partial y} dy + \frac{\partial H}{\partial z} dz \right) \quad (1.5.13)$$

Since the left-hand side and the bracketed expression on the right are the total derivatives of p and H , respectively, Eq. (1.5.13) reduces to an ordinary differential equation for the variation in pressure with altitude in a **static fluid, when the only body force is gravity**

$$\frac{dp}{dH} = -\rho g \quad (1.5.14)$$

Pressure Variation in a Static Fluid

If gravity provides the only body force, integration of Eq. (1.5.14) will produce a relation for the pressure in a static fluid as a function of altitude. However, before we can integrate Eq. (1.5.14), we must be able to express ρ and g in terms of p and H .

For most practical applications, Eq. (1.5.11) provides an adequate model for the variation in the acceleration of gravity with altitude. However, the gravitational variation predicted by Eq. (1.5.11) is quite small and can be neglected for many applications. For example, the Mariana Trench is the deepest part of the world's oceans, and has a maximum depth of about 10,911 meters below mean sea level. At 8,850 meters above mean sea level, Mount Everest is the highest mountain on Earth. Between these two altitudes, Eq. (1.5.11) predicts that the acceleration of gravity varies from 9.840396 m/s^2 to 9.779396 m/s^2 , which is a variation of about ± 0.3 percent. Thus, for most practical engineering applications near the Earth's surface, the acceleration of gravity can be assumed to be a constant, which is approximately $g = 9.81 \text{ m/s}^2 = 32.2 \text{ ft/s}^2$.

The density of most liquids varies only slightly with temperature and pressure. Thus, for computing the pressure variation in static liquids near the Earth's surface, we can usually approximate both the acceleration of gravity and the density of the liquid as being constant. With this approximation, we can easily integrate Eq. (1.5.14) from some given altitude H_0 , where the pressure is known to be p_0 , to some arbitrary altitude H , where the unknown pressure is p ,

$$\int_{p_0}^p dp = - \int_{H_0}^H \rho g dH \quad (1.5.15)$$

Thus, the variation in pressure with altitude in a **constant-density static fluid, when the only body force is that resulting from a constant gravitational acceleration is**

$$p = p_0 - \rho g (H - H_0) \quad (1.5.16)$$

where p_0 is the known pressure at the known altitude H_0 .

EXAMPLE 1.5.1. The surface of a fresh-water lake has an altitude equal to that at mean sea level. The surface temperature and pressure for both the air and the water are 15°C and $101,325 \text{ N/m}^2$, respectively. Neglecting all body forces except gravity, assuming that water has a constant density of 999.0 kg/m^3 , and assuming constant gravitational acceleration of 9.81 m/s^2 , estimate the pressure 10 m below the surface of the lake.

Solution. The approximations given in the problem statement are consistent with the approximations used to obtain Eq. (1.5.16). Thus, the pressure 10 m below the surface of the lake is estimated to be

$$\begin{aligned} p &= p_0 - \rho g (H - H_0) \\ &= 101,325 \text{ N/m}^2 - 999.0 \text{ kg/m}^3 (9.81 \text{ m/s}^2) [(-10.0 \text{ m}) - 0.0 \text{ m}] (1 \text{ N} \cdot \text{s}^2 / \text{kg} \cdot \text{m}) \\ &= 199,327 \text{ N/m}^2 = 4,163.03 \text{ lbf/ft}^2 = 28.9099 \text{ psi} \end{aligned}$$

Because the density of liquids varies only slightly with the temperature and pressure, Eq. (1.5.16) is commonly used to estimate pressure variations in a static liquid. On the other hand, the density of atmospheric air and other common gases can vary significantly with temperature and pressure. Air and many other common gases can be accurately modeled as ideal gases at the temperatures and pressures that are normally encountered in the atmosphere. The density of an ideal gas varies with temperature and pressure as specified by Eq. (1.2.6). Thus, for an **ideal gas**, Eq. (1.5.14) can be rearranged to give

$$\int_{p_0}^p \frac{dp}{p} = - \int_{H_0}^H \frac{g}{RT} dH \quad (1.5.17)$$

For an isothermal gas and constant gravitational acceleration, the integration yields

$$\ln(p/p_0) = -g(H - H_0)/(RT) \quad (1.5.18)$$

Thus, the variation in pressure with altitude in an **isothermal static ideal gas, when the only body force is that resulting from a constant gravitational acceleration is**

$$p = p_0 \exp[-g(H - H_0)/(RT)] \quad (1.5.19)$$

where p_0 is the known pressure at the known altitude H_0 .

EXAMPLE 1.5.2. The surface of a fresh-water lake has an altitude equal to that at mean sea level. The surface temperature and pressure for both the air and the water are 15 C and 101,325 N/m², respectively. Neglecting all body forces except gravity, using the ideal-gas constant $R = 287.0528 \text{ N}\cdot\text{m}/\text{kg}\cdot\text{K} = 287.0528 \text{ m}^2/\text{s}^2\cdot\text{K}$ for atmospheric air, assuming constant temperature in the air, and assuming a constant gravitational acceleration of 9.81 m/s², estimate the pressure 10 m above the surface of the lake. Also compute the density of the air at the surface of the lake and 10 m above the surface of the lake.

Solution. The given approximations are consistent with the approximations used to obtain Eq. (1.5.19). Thus, the pressure 10 m above the surface of the lake is

$$\begin{aligned} p &= p_0 \exp[-g(H - H_0)/(RT)] \\ &= 101,325 \text{ N/m}^2 \exp\left\{\frac{-9.81 \text{ m/s}^2(+10.0 \text{ m} - 0.0 \text{ m})}{287.0528 \text{ m}^2/\text{s}^2\cdot\text{K}(15 + 273.15)\text{K}}\right\} = 101,205 \text{ N/m}^2 \end{aligned}$$

From the ideal gas law, Eq. (1.2.6),

$$\text{at } H = 0 \text{ m, } \rho = \frac{p}{RT} = \frac{101,325 \text{ N/m}^2}{287.0528 \text{ N}\cdot\text{m}/\text{kg}\cdot\text{K}(288.15 \text{ K})} = 1.22500 \text{ kg/m}^3$$

$$\text{at } H = 10 \text{ m, } \rho = \frac{p}{RT} = \frac{101,205 \text{ N/m}^2}{287.0528 \text{ N}\cdot\text{m}/\text{kg}\cdot\text{K}(288.15 \text{ K})} = 1.22355 \text{ kg/m}^3$$

We see from Example 1.5.2, that over an altitude change of 10 m, there is very little change in the density of atmospheric air. For this reason, when estimating static pressure variations over small distances in a gas, an accurate approximation can be obtained by assuming constant gas density and applying Eq. (1.5.16).

EXAMPLE 1.5.3. The surface of a fresh-water lake has an altitude equal to that at mean sea level. The surface temperature and pressure for both the air and the water are 15 C and 101,325 N/m², respectively. Neglecting all body forces except gravity, assuming that air has a constant density of 1.225 kg/m³, and assuming constant gravitational acceleration of 9.81 m/s², estimate the pressure 10 m above the surface of the lake.

Solution. The approximations given in the problem statement are consistent with the approximations used to obtain Eq. (1.5.16). Thus, the pressure 10 m above the surface of the lake is estimated to be

$$\begin{aligned} p &= p_0 - \rho g (H - H_0) \\ &= 101,325 \text{ N/m}^2 - 1.225 \text{ kg/m}^3 (9.81 \text{ m/s}^2) [(+10.0 \text{ m}) - 0.0 \text{ m}] (1 \text{ N} \cdot \text{s}^2 / \text{kg} \cdot \text{m}) \\ &= 101,205 \text{ N/m}^2 = 2,113.71 \text{ lbf/ft}^2 = 14.6785 \text{ psi} \end{aligned}$$

Comparing results from Examples 1.5.2 and 1.5.3, we see that the prediction from Eq. (1.5.16) agrees with the more accurate result in Eq. (1.5.19) to within six significant digits. This is typical of results obtained for the static pressure variation in gases when the change in altitude is on the order of several meters or less. When we wish to predict static pressure variations in a gas for altitude changes of several hundred meters or more, for accurate results, we must account for the variation in gas density with altitude.

Fluids in Rigid-Body Acceleration

Fluids are commonly transported in vehicles that are capable of acceleration. Gasoline in the fuel tanks of an automobile or a general aviation airplane can be treated as a static fluid, if the vehicle is moving with a constant velocity relative to an inertial coordinate system. This is because *any coordinate system that is moving with a constant velocity relative to an inertial coordinate system is also an inertial coordinate system*. When a vehicle has been moving at constant velocity for some time, all of the fuel in its tanks will assume the same velocity as the vehicle. This is called *rigid-body motion*, because there is no motion of any fluid element relative to any other fluid element.

When a vehicle accelerates, there can be some initial sloshing of the fuel in its tanks. However, if the vehicle maintains constant acceleration for some time, eventually all of the fuel in its tanks will once again assume the same velocity as the vehicle, but now the fuel is also accelerating with the vehicle. This is called *rigid-body acceleration* because, although the fluid is accelerating relative to an inertial coordinate system, there is still no relative motion between fluid elements. A fluid in rigid-body acceleration can be treated as a static fluid; however, the surfaces of constant pressure are not necessarily horizontal or even planar.

Students should remember from their introductory class in *dynamics* that the forces acting on accelerating objects can be analyzed using an accelerating frame of reference that is fixed to an accelerating object. ***Newton's second law of motion can be written relative to an accelerating coordinate system by adding an apparent body force per unit mass that is equal to the negative of the coordinate-system-acceleration vector.***

For example, if a motionless object having a mass of 100 kg is placed on a scale in a fixed room, the contact force between the object and the scale must exactly balance the gravitational body force acting on the object, which is a downward force equal to the product of the mass and the acceleration of gravity, i.e., $(100 \text{ kg}) \times (9.81 \text{ m/s}^2) = 981 \text{ N}$. If the same object is placed on a scale in an elevator that is accelerating upward at 5 m/s^2 , the contact force between the object and the scale can be evaluated by treating the object as though it were in a fixed room where the total body force per unit mass is the vector sum of that due to gravity and the negative of the coordinate-system-acceleration vector, i.e., a total downward body force of $(100 \text{ kg}) \times (9.81 \text{ m/s}^2 + 5 \text{ m/s}^2) = 1481 \text{ N}$.

Because density is mass per unit volume, the body force per unit volume acting on a fluid is the product of the fluid density and the body force per unit mass. Using the relation given by Eq. (1.5.8) for the gravitational body force per unit volume, **the total body force per unit volume acting on a fluid in an accelerating reference frame is**

$$\mathbf{f}_b = -\rho(g\nabla H + \mathbf{a}) \quad (1.5.20)$$

where $\mathbf{a}$ is the coordinate-system-acceleration vector. Using Eq. (1.5.20) in Eq. (1.5.6), **Newton's second law for a static fluid in an accelerating reference frame requires**

$$\nabla p = -\rho(g\nabla H + \mathbf{a}) \quad (1.5.21)$$

The only requirement on Eq. (1.5.21) is that there can be no relative motion between fluid elements. Neither the acceleration of gravity g nor the coordinate-system-acceleration vector $\mathbf{a}$ need be constant throughout the fluid. For example, Eq. (1.5.21) can be applied to the rigid-body rotation of a fluid, in which case the acceleration vector $\mathbf{a}$ varies with the radial distance from the axis of rotation.

EXAMPLE 1.5.4. Consider an arbitrary mass of fluid in rigid-body rotation with a constant angular velocity of ω about a vertical axis. The pressure at a known point on the axis of rotation is p_0 . Assuming that the fluid density and acceleration of gravity are constant, find the pressure distribution in this static fluid.

Solution. For convenience, we can choose a cylindrical coordinate system rotating with the fluid. Its z axis is aligned with the vertical axis of rotation and the origin is located at the point where the fluid pressure is p_0 . The r coordinate is outward, normal to the axis of rotation. In this coordinate system we have

$$H = z, \quad \nabla H = \mathbf{i}_z \quad (1.5.22)$$

The magnitude of the centripetal acceleration for any rigid-body rotation at some constant angular velocity of ω and a distance r from the axis of rotation is $\omega^2 r$.

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The direction of this acceleration is radially inward toward the axis of rotation, which is the negative r direction. There is no acceleration of this reference frame in the θ or z directions. Hence, the coordinate-system-acceleration vector is

$$\mathbf{a} = -\omega^2 r \mathbf{i}_r \quad (1.5.23)$$

Using Eqs. (1.5.22) and (1.5.23) in Eq. (1.5.21) and applying Eq. (1.4.27) yields

$$\frac{\partial p}{\partial r} \mathbf{i}_r + \frac{1}{r} \frac{\partial p}{\partial \theta} \mathbf{i}_\theta + \frac{\partial p}{\partial z} \mathbf{i}_z = \rho \omega^2 r \mathbf{i}_r - \rho g \mathbf{i}_z \quad (1.5.24)$$

The three components of Eq. (1.5.24) must be satisfied individually,

$$\frac{\partial p}{\partial r} = \rho \omega^2 r, \quad \frac{\partial p}{\partial \theta} = 0, \quad \frac{\partial p}{\partial z} = -\rho g \quad (1.5.25)$$

Integrating these equations independently produces three relations for $p(r, \theta, z)$

$$p = \frac{1}{2} \rho \omega^2 r^2 + f_1(\theta, z), \quad p = f_2(r, z), \quad p = -\rho g z + f_3(r, \theta) \quad (1.5.26)$$

Applying the condition $p(0, \theta, 0) = p_0$ to Eq. (1.5.26) yields the complete solution

$$p = \frac{1}{2} \rho \omega^2 r^2 - \rho g z + p_0 \quad (1.5.26)$$

Thus, we see that the pressure is parabolic in r , linear in z , and independent of θ . The surfaces of constant pressure for this solution are paraboloids of revolution,

$$z = \frac{\omega^2}{2g} r^2 - \frac{p - p_0}{\rho g} \quad (1.5.26)$$

1.6. The Standard Atmosphere

The lift and drag acting on objects moving through the atmosphere are strong functions of air density. Thus, to predict the forces acting on an aircraft or some other object moving through the atmosphere, we must be able to determine the density of the air at a known altitude. For this purpose, the atmosphere can be regarded as a static fluid.

We have seen from Eq. (1.5.14) that in a static fluid, the change in pressure, p , with respect to geometric altitude, H , is

$$\frac{dp}{dH} = -\rho g \quad (1.6.1)$$

where ρ is the fluid density and g is the local acceleration of gravity. For all practical purposes, the atmosphere can be assumed to behave as an ideal gas, which gives

$$\rho = \frac{p}{RT} \quad (1.6.2)$$

where R is the ideal gas constant for the gas and T is the absolute temperature. Thus, the pressure variation in the atmosphere is governed by

$$\frac{dp}{dH} = -\frac{g}{R} \frac{p}{T} \quad (1.6.3)$$

From Newton's law of gravitation, g varies with altitude according to Eq. (1.5.11)

$$g = g_o \left(\frac{R_E}{R_E + H} \right)^2$$

where g_o is the standard acceleration of gravity at mean sea level, $g_o = 9.806645 \text{ m/s}^2$, and R_E is the radius of the Earth at mean sea level, $R_E = 6,356,766 \text{ m}$. Using Newton's law of gravitation in Eq. (1.6.3) gives

$$\frac{dp}{dH} = -\frac{g_o}{R} \left(\frac{R_E}{R_E + H} \right)^2 \frac{p}{T} \quad (1.6.4)$$

The integration of Eq. (1.6.4) is simplified by introducing the change of variables

$$Z \equiv \frac{R_E H}{R_E + H} \quad (1.6.5)$$

Differentiating Eq. (1.6.5) with respect to H yields

$$\frac{dZ}{dH} = \frac{(R_E + H)R_E - R_E H}{(R_E + H)^2} = \left(\frac{R_E}{R_E + H} \right)^2$$

With this change of variables, Eq. (1.6.4) can be written as

$$\frac{dp}{dH} = \frac{dp}{dZ} \frac{dZ}{dH} = \frac{dp}{dZ} \left(\frac{R_E}{R_E + H} \right)^2 = -\frac{g_o}{R} \left(\frac{R_E}{R_E + H} \right)^2 \frac{p}{T}$$

Thus, applying the change of variables defined in Eq. (1.6.5) to Eq. (1.6.4), we can write **the differential equation governing the pressure variation in a static ideal gas** as

$$\frac{dp}{dZ} = -\frac{g_o}{R} \frac{p}{T} \quad (1.6.6)$$

The new variable Z that is defined in Eq. (1.6.5) is called the *geopotential altitude*. The difference between the geopotential and geometric altitudes is small in the first several thousand feet above sea level. However, at higher altitudes the difference is significant.

Since the temperature and pressure variations in the atmosphere are different from day to day, aircraft instruments make use of the concept of a *standard atmosphere*. Many different “standard” atmospheres have been defined. However, most are essentially indistinguishable below 100,000 feet, which encloses the domain of most aircraft.

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The standard atmosphere commonly used in the calibration of aircraft instruments is divided into several regions, each of which is assumed to have a constant temperature gradient with respect to geopotential altitude. The temperature gradients defined for this standard atmosphere are given in Table 1.6.1. This information completely defines the temperature throughout the standard atmosphere. Other atmosphere definitions are quite commonly used for special area environments. These include the *polar atmosphere*, the *desert atmosphere*, and the *10 percent hot day*.

Since the temperature gradient is constant in each of the altitude ranges defined in Table 1.6.1, the temperature in each range is linear,

$$T(Z) = T(Z_i) + T'_i(Z - Z_i) \quad (1.6.7)$$

where Z_i is the minimum geopotential altitude in the range and T'_i is the temperature gradient for the range, defined as the change in temperature with respect to geopotential altitude, dT/dZ . The negative of this temperature gradient is commonly referred to as the *lapse rate*. Applying Eq. (1.6.7), the integration of Eq. (1.6.6) subject to the boundary condition, $p(Z_i) = p_i$, yields

$$p(Z) = \begin{cases} p_i \exp\left[-\frac{g_o(Z - Z_i)}{RT_i}\right], & T'_i = 0 \\ p_i \left[\frac{T_i + T'_i(Z - Z_i)}{T_i}\right]^{\frac{-g_o}{RT'_i}}, & T'_i \neq 0 \end{cases} \quad (1.6.8)$$

From Eqs. (1.6.7) and (1.6.8), combined with the information given in Table 1.6.1, the temperature and pressure throughout the atmosphere can be determined from the ideal gas constant and the temperature and pressure at standard sea level. The gas constant for the standard atmosphere is defined to be $R = 287.0528 \text{ N}\cdot\text{m}/\text{kg}\cdot\text{K} = 287.0528 \text{ m}^2/\text{s}^2\cdot\text{K}$. The standard atmosphere is defined to have a temperature at sea level of 288.150 K, and standard atmospheric pressure at sea level is defined as $101,325 \text{ N}/\text{m}^2$.

Geopotential Altitude Range		Initial Temperature	Temperature Gradient
Z_i (m)	Z_{i+1} (m)	T_i (K)	T'_i (K/km)
0	11,000	288.150	−6.5
11,000	20,000	216.650	0.0
20,000	32,000	216.650	1.0
32,000	47,000	228.650	2.8
47,000	52,000	270.650	0.0
52,000	61,000	270.650	−2.0
61,000	79,000	252.650	−4.0
79,000	~90,000	180.650	0.0

Table 1.6.1. Temperature gradients with respect to geopotential altitude for the standard atmosphere commonly used in the calibration of aircraft instruments.

Once the temperature and pressure are known at any altitude, the density can be determined from Eq. (1.6.2) and the speed of sound can be computed from

$$a = \sqrt{\gamma RT} \quad (1.6.9)$$

where γ is 1.4 for the standard atmosphere. From the known temperature, the dynamic viscosity is obtained from Sutherland's equation, Eq. (1.2.23),

$$\mu = 1.7153 \times 10^{-5} (\text{kg/s}\cdot\text{m}) \left(\frac{T}{273 \text{ K}} \right)^{3/2} \frac{383.4 \text{ K}}{T + 110.4 \text{ K}} \quad (1.6.10)$$

and the kinematic viscosity is obtained from the definition $\nu \equiv \mu/\rho$.

Once the speed of sound and the kinematic viscosity are known at any altitude, the molecular mean free path can be determined from Eq. (1.2.26). After rearranging, this gives

$$\lambda = \sqrt{\frac{\pi\gamma}{2}} \frac{\nu}{a} \quad (1.6.11)$$

For a given geometry and airspeed, the molecular mean free path and the speed of sound can be used to evaluate the Knudsen number and Mach number. The result can be used with Eq. (1.1.1) or Fig. 1.1.2 to assess the validity of continuum theory.

EXAMPLE 1.6.1. Compute the absolute temperature, pressure, density, speed of sound, kinematic viscosity, and the molecular mean free path for the standard atmosphere defined in Table 1.6.1 at a geometric altitude of 30,000 meters.

Solution. We start by determining geopotential altitude from the known geometric altitude. From Eq. (1.6.5)

$$Z \equiv \frac{R_E H}{R_E + H} = \frac{6,356,766 \text{ m}(30,000 \text{ m})}{6,356,766 \text{ m} + 30,000 \text{ m}} = 29,859 \text{ m}$$

Since this is in the third of altitude ranges given in Table 1.6.1, we must first integrate across the first two ranges. At the boundary between the first and the second range ($Z_1 = 11 \text{ km}$), from Eqs. (1.6.7) and (1.6.8) we obtain

$$T_1 = T_0 + T'_0(Z_1 - Z_0) = 288.150 \text{ K} + (-6.5 \text{ K/km})(11 - 0) \text{ km} = 216.650 \text{ K}$$

$$\begin{aligned} p_1 &= p_0 \left[\frac{T_0 + T'_0(Z_1 - Z_0)}{T_0} \right]^{\frac{-g_0}{RT'_0}} = p_0 \left(\frac{T_1}{T_0} \right)^{\frac{-g_0}{RT'_0}} \\ &= 101,325 \text{ N/m}^2 \left(\frac{216.650 \text{ K}}{288.150 \text{ K}} \right)^{\frac{-9.806645 \text{ m/s}^2}{287.0528 \text{ m}^2/\text{s}^2 \cdot \text{K}(-0.0065 \text{ K/m})}} = 22,632 \text{ N/m}^2 \end{aligned}$$

and at the boundary between the second and the third range ($Z_2 = 20$ km),

$$T_2 = T_1 + T_1'(Z_2 - Z_1) = 216.650 \text{ K} + (0.0 \text{ K/km})(20 - 11) \text{ km} = 216.650 \text{ K}$$

$$\begin{aligned} p_2 &= p_1 \exp\left[-\frac{g_o(Z_2 - Z_1)}{RT_1}\right] = 22,632 \text{ N/m}^2 \exp\left[-\frac{9.806645(20 - 11) \times 10^3}{287.0528(216.650)}\right] \\ &= 5,474.9 \text{ N/m}^2 \end{aligned}$$

Integrating from this point to the specified geopotential altitude of 29,859 meters, Eqs. (1.6.7) and (1.6.8) yield

$$T = T_2 + T_2'(Z - Z_2) = 216.650 \text{ K} + (1.0 \text{ K/km})(29.859 - 20) \text{ km} = \underline{226.509 \text{ K}}$$

$$\begin{aligned} p &= p_2 \left[\frac{T_2 + T_2'(Z - Z_2)}{T_2} \right]^{\frac{-g_o}{RT_2^2}} = p_2 \left(\frac{T}{T_2} \right)^{\frac{-g_o}{RT_2^2}} \\ &= 5,474.9 \text{ N/m}^2 \left(\frac{226.509 \text{ K}}{216.650 \text{ K}} \right)^{\frac{-9.806645 \text{ m/s}^2}{287.0528 \text{ m}^2/\text{s}^2 \cdot \text{K}(0.0010 \text{ K/m})}} = \underline{1,197.0 \text{ N/m}^2} \end{aligned}$$

From Eq. (1.6.2), the density is

$$\rho = \frac{p}{RT} = \frac{1,197.0 \text{ N/m}^2}{287.0528 \text{ N} \cdot \text{m/kg} \cdot \text{K}(226.509 \text{ K})} = \underline{0.018410 \text{ kg/m}^3}$$

and from Eq. (1.6.9), the speed of sound is

$$a = \sqrt{\gamma RT} = \sqrt{1.4(287.0528 \text{ m}^2/\text{s}^2 \cdot \text{K})226.509 \text{ K}} = \underline{301.71 \text{ m/s}}$$

From Eq. (1.6.10), the kinematic viscosity is

$$\begin{aligned} \nu &= \frac{1.7153 \times 10^{-5} (\text{kg/s} \cdot \text{m})}{\rho} \left(\frac{T}{273 \text{ K}} \right)^{3/2} \frac{383.4 \text{ K}}{T + 110.4 \text{ K}} \\ &= \frac{1.7153 \times 10^{-5} (\text{kg/s} \cdot \text{m})}{0.018410 \text{ kg/m}^3} \left(\frac{226.509 \text{ K}}{273 \text{ K}} \right)^{3/2} \frac{383.4 \text{ K}}{226.509 \text{ K} + 110.4 \text{ K}} \\ &= \underline{8.0132 \times 10^{-4} \text{ m}^2/\text{s}} \end{aligned}$$

and from Eq. (1.6.11), the molecular mean free path is

$$\lambda = \sqrt{\frac{\pi \gamma}{2}} \frac{\nu}{a} = \sqrt{\frac{\pi(1.4)}{2}} \frac{8.0132 \times 10^{-4} \text{ m}^2/\text{s}}{301.71 \text{ m/s}} = \underline{3.94 \times 10^{-6} \text{ m}}$$

EXAMPLE 1.6.2. Compute the absolute temperature, pressure, density, speed of sound, kinematic viscosity, and molecular mean free path, in English units, for the standard atmosphere defined in Table 1.6.1 at a geometric altitude of 100,000 ft.

Solution. Since this standard atmosphere is defined in SI units, we begin by converting the given geometric altitude to meters:

$$H = 100,000 \text{ ft} (0.304800 \text{ m/ft}) = 30,480 \text{ m}$$

Following the procedure used in Example 1.6.1, we obtain

$$Z = 30,335 \text{ m} / 0.3048 \text{ m/ft} = 99,523 \text{ ft}$$

$$T = 226.985 \text{ K} (1.80000^\circ \text{R/K}) = \underline{408.572^\circ \text{R}}$$

$$p = 1,114.3 \text{ N/m}^2 (0.02088543 \text{ lbf/ft}^2 / \text{N/m}^2) = 23.272 \text{ lbf/ft}^2 = \underline{0.16161 \text{ psi}}$$

$$\rho = 0.017102 \text{ kg/m}^3 (0.001940320 \text{ slug/ft}^3 / \text{kg/m}^3) = \underline{0.000033182 \text{ slug/ft}^3}$$

$$a = 302.03 \text{ m/s} / 0.3048 \text{ m/ft} = \underline{990.90 \text{ ft/s}}$$

$$\nu = 8.6414 \times 10^{-4} \text{ m}^2/\text{s} / (0.3048 \text{ m/ft})^2 = \underline{9.3015 \times 10^{-3} \text{ ft}^2/\text{s}}$$

$$\lambda = 4.24 \times 10^{-6} \text{ m} / 0.3048 \text{ m/ft} = \underline{1.39 \times 10^{-5} \text{ ft}}$$

Atmospheric temperature and pressure can vary substantially with time and location on the Earth. Nevertheless, for the correlation of test data taken at different times and locations and for the calibration of aircraft flight instruments, it is important to have an agreed-upon standard for atmospheric properties. From the assumption of an ideal gas, pressure, density, and the speed of sound can be determined directly from a defined temperature profile. The standard atmosphere, based on the temperature profile defined in Table 1.6.1 and tabulated in Appendices A and B, is commonly used for the purposes mentioned above. Slightly different temperature profiles have also been used to define other standard atmospheres, but the variation is small below 100,000 feet.

A few words on terminology may be in order here. So far we have used the explicit term *geometric altitude* to designate a vertical height above the Earth's mean-sea-level surface. This is commonly referred to simply as *altitude*, which is formally defined as:

altitude – the vertical height above a given datum level, usually measured above mean sea level (MSL) or above ground level (AGL).

When referring to geographic locations or the height of points near the Earth's surface, it is common to use the term *elevation*, which is formally defined as:

elevation – the vertical height above the Earth's mean-sea-level surface.

Thus, **the term elevation is technically synonymous with MSL altitude.**

1.7. Manometers and Barometers

When gravity is the only body force, under the approximations of constant density and constant gravitational acceleration, the pressure difference between two points in a static fluid can be determined from Eq. (1.5.16). This predicts that the pressure difference is linearly proportional to the difference in elevation. We have seen that the approximations used to obtain Eq. (1.5.16) are accurate for both liquids and gases provided that the change in elevation between the two points is on the order of a few meters or less. This suggests that a pressure difference could be determined by measuring an elevation difference between two points in a static fluid. Devices that make use of this principle for determining pressure differences are called *manometers*.

Consider, for example, the conventional U-tube manometer shown in Fig. 1.7.1. The fluid in chamber *A* and the upper portion of the right leg has density ρ_1 . The fluid in chamber *D* and the upper portion of the left leg has density ρ_3 . The fluid on the right is separated from the fluid on the left by a heavier liquid of density ρ_2 . Either or both of the fluids in chambers *A* and *D* could be gases or liquids. The only requirements for the conventional U-tube manometer shown in Fig. 1.7.1 are that ρ_2 must be greater than both ρ_1 and ρ_3 , and the separating liquid must be immiscible with both of the other fluids.

From Eq. (1.5.16), the pressure increase from point *A* to point *B* is the product of the fluid density, the acceleration of gravity, and the increase in depth,

$$p_B - p_A = \rho_1 g (H_1 - H_2) \quad (1.7.1)$$

Similarly, the pressure difference between points *B* and *C* is

$$p_C - p_B = \rho_2 g (H_2 - H_3) \quad (1.7.2)$$

and the pressure difference between points *C* and *D* is

$$p_D - p_C = \rho_3 g (H_3 - H_4) \quad (1.7.3)$$

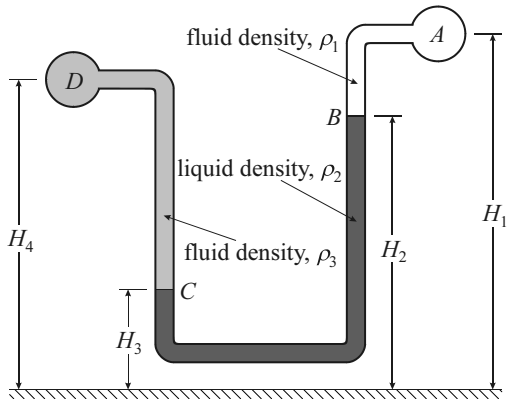


Figure 1.7.1. U-tube manometer for measuring the pressure difference between points *A* and *B*.

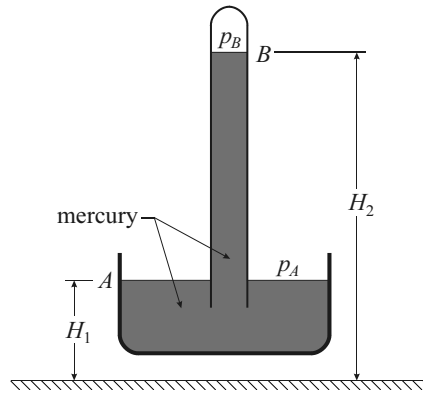


Figure 1.7.2. Mercury Barometer used for measuring atmospheric pressure.

Adding Eqs. (1.7.1), (1.7.2), and (1.7.3) yields

$$\begin{aligned} (p_B - p_A) + (p_C - p_B) + (p_D - p_C) &= \\ p_D - p_A &= \rho_1 g (H_1 - H_2) + \rho_2 g (H_2 - H_3) + \rho_3 g (H_3 - H_4) \end{aligned} \quad (1.7.4)$$

As shown in Fig. 1.7.1, H_1 is greater than H_2 , so the pressure increases as we move down from point A to point B . Similarly, because H_2 is greater than H_3 , there is another increase in pressure as we move down from point B to the bottom of the tube and back up to point C . Because H_4 is greater than H_3 , the pressure decreases as we move up from point C to point D . **Pressure always increases as we move down in a fluid, and pressure always decreases as we move up in a fluid.**

A device that makes use of this same principle for determining absolute atmospheric pressure is called a *barometer*. A mercury barometer is obtained by using a long tube, open at one end and closed at the other. The tube is filled with mercury and its open end is submerged in mercury contained in an open vessel as shown in Fig. 1.7.2. The mercury in the tube will rise to some level above that of the mercury in the vessel. The difference between the level of the mercury in the vessel A and the level of the mercury in the tube B depends on the atmospheric pressure.

From Eq. (1.5.16) and the geometry shown in Fig. 1.7.2, the pressure increase from point B to point A is the product of the mercury density, the acceleration of gravity, and the increase in depth,

$$p_A - p_B = \rho_{Hg} g (H_2 - H_1) \quad (1.7.5)$$

Because no air was admitted to the tube, the space above the mercury column contains only mercury vapor. Mercury was chosen for such barometers because, compared with atmospheric pressure, the vapor pressure of mercury at normal ambient temperatures is negligible. The vapor pressure of mercury at 20 °C is 0.16 N/m², which would support only 0.0012 millimeters of mercury. Thus, for all practical purposes, the space above the mercury in the tube is a perfect vacuum, i.e., $p_B = 0$. This gives a very simple relation for the atmospheric pressure in terms of the height of the mercury column in the tube,

$$p_A = \rho_{Hg} g (H_2 - H_1) \quad (1.7.6)$$

For standard conditions at mean sea level, atmospheric pressure is 101,325 N/m², the density of mercury is 13,558.2 kg/m³, and the acceleration of gravity is 9.806645 m/s². Thus, at standard mean-sea-level temperature and pressure, the atmosphere will support a column of mercury 762.07 mm (30.003 in) high.

A few decades ago, any laboratory conducting experiments that require knowledge of the ambient pressure would be equipped with a mercury barometer. Today, modern instrumentation makes use of resonating silicon crystals and other electronic technology to provide ambient temperature and pressure as digital readouts. Nevertheless, ambient pressures are still commonly expressed in millimeters or inches of mercury. Thus, it is important for students to understand the basic principle behind the mercury barometer and to be able to convert pressure readings in millimeters or inches of mercury to consistent SI and English units.

Because the density of mercury varies somewhat with temperature and the acceleration of gravity varies slightly with location on the Earth, accurate ambient pressure measurements made with a traditional mercury barometer require corrections for local gravitational acceleration and ambient temperature. Because barometric pressure is still commonly reported in millimeters or inches of mercury, for the correlation of data taken at different times and locations, and for the calibration of aircraft flight instruments, it is important to have a standard for defining a millimeter of mercury (mm Hg) relative to a newton per square meter (N/m^2). For this purpose we use the standard density of mercury at 0.000 C and the standard acceleration of gravity at mean sea level. This convention leads to the conversions

$$760.000 \text{ mm Hg} = 101,325 \text{ N/m}^2 \quad (1.7.7)$$

$$29.9213 \text{ in Hg} = 2,116.22 \text{ lbf/ft}^2 = 14.6959 \text{ psi} \quad (1.7.8)$$

EXAMPLE 1.7.1. Consider the multi-liquid manometer shown in Fig. 1.7.3. The pipe on the left at F contains water ($\rho_{\text{H}_2\text{O}} = 1.94 \text{ slug/ft}^3$) and the manometer is open to the atmosphere on the right at A , where the temperature is 70 F and the corrected barometric pressure is 29.00 in Hg. The liquid between points B and C , as well as that between points D and E , is mercury ($\rho_{\text{Hg}} = 26.3 \text{ slug/ft}^3$). The liquid between points C and D is oil ($\rho_{\text{oil}} = 1.55 \text{ slug/ft}^3$). Determine the pressure of the water in the pipe at F in psi.

Solution. We begin by converting the given corrected barometric pressure at A to consistent English units,

$$29.00 \text{ in Hg} \frac{2,116.22 \text{ lbf/ft}^2}{29.9213 \text{ in Hg}} = 2,051 \text{ lbf/ft}^2 \text{ (14.24 psi)}$$

The ideal gas constant in consistent English units is

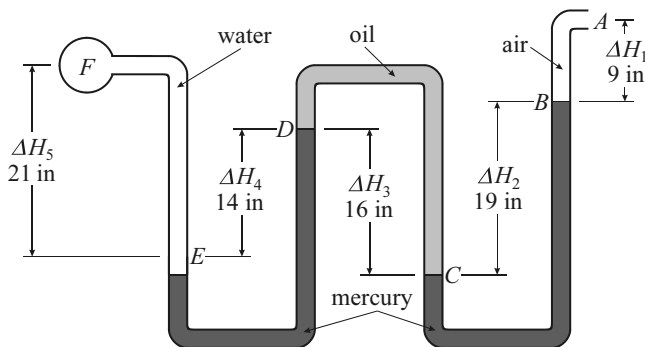


Figure 1.7.3. Manometer geometry for Example 1.7.1.

$$R = \frac{287.0528 \text{ N} \cdot \text{m/kg} \cdot \text{K} \cdot 14.5939029372 \text{ kg/slug}}{4.44822161526 \text{ N/lbf} \cdot 0.3048 \text{ m/ft} \cdot 1.8 \text{ R/K}} = 1716.561 \text{ lbf} \cdot \text{ft/slug} \cdot \text{R}$$

The air density in consistent English units is

$$\rho_{\text{air}} = \frac{p}{RT} = \frac{2,051 \text{ lbf/ft}^2}{1716.561 \text{ lbf} \cdot \text{ft/slug} \cdot \text{R} (70 + 459.67) \text{ R}} = 0.0022558 \text{ slug/ft}^3$$

Because point B is below point A , the pressure increases in going from A to B . Using 32.2 ft/s^2 for the local acceleration of gravity, we have

$$\begin{aligned} p_B &= p_A + \rho_{\text{air}} g \Delta H_1 = 2,051 \text{ lbf/ft}^2 + 0.0022558 \text{ slug/ft}^3 (32.2 \text{ ft/s}^2)(9/12) \text{ ft} \\ &= 2,051 \text{ lbf/ft}^2 + 0.054 \text{ lbf/ft}^2 = 2,051 \text{ lbf/ft}^2 \end{aligned}$$

Thus, we see that there is no significant change in pressure through the air in right leg. Because point C is below point B , the pressure increases as we move down from point B through the mercury to the bottom of the tube and up to point C ,

$$\begin{aligned} p_C &= p_B + \rho_{\text{Hg}} g \Delta H_2 = 2,051 \text{ lbf/ft}^2 + 26.3 \text{ slug/ft}^3 (32.2 \text{ ft/s}^2)(19/12) \text{ ft} \\ &= 2,051 \text{ lbf/ft}^2 + 1,341 \text{ lbf/ft}^2 = 3,392 \text{ lbf/ft}^2 \end{aligned}$$

Because point D is above point C , the pressure decreases as we move up through the oil from point C to point D ,

$$\begin{aligned} p_D &= p_C - \rho_{\text{oil}} g \Delta H_3 = 3,392 \text{ lbf/ft}^2 - 1.55 \text{ slug/ft}^3 (32.2 \text{ ft/s}^2)(16/12) \text{ ft} \\ &= 3,392 \text{ lbf/ft}^2 - 67 \text{ lbf/ft}^2 = 3,325 \text{ lbf/ft}^2 \end{aligned}$$

Because point E is below point D , the pressure increases in going from D to E ,

$$\begin{aligned} p_E &= p_D + \rho_{\text{Hg}} g \Delta H_4 = 3,325 \text{ lbf/ft}^2 + 26.3 \text{ slug/ft}^3 (32.2 \text{ ft/s}^2)(14/12) \text{ ft} \\ &= 3,325 \text{ lbf/ft}^2 + 988 \text{ lbf/ft}^2 = 4,313 \text{ lbf/ft}^2 \end{aligned}$$

Because point F is above point E , the pressure decreases in going from E to F ,

$$\begin{aligned} p_F &= p_E - \rho_{\text{H}_2\text{O}} g \Delta H_4 = 4,313 \text{ lbf/ft}^2 - 1.94 \text{ slug/ft}^3 (32.2 \text{ ft/s}^2)(21/12) \text{ ft} \\ &= 4,313 \text{ lbf/ft}^2 - 109 \text{ lbf/ft}^2 = 4,204 \text{ lbf/ft}^2 \quad (29.19 \text{ psi}) \end{aligned}$$

Note that the pressure change through the air in the previous example was negligible. Even in a column of air 10 ft high, the pressure change is less than 0.8 lbf/ft^2 (0.006 psi). The density of air at standard sea level is $0.0023769 \text{ slugs/ft}^3$. Thus, the density of air is negligible compared with the density of typical manometer fluids. This is true for gases in general. **For gases at normal ambient temperatures and pressures, the pressure change over the heights encountered in manometers can be neglected.**

A word of caution with regard to standards may be in order. Standard atmospheric temperature and pressure at mean sea level (MSL) are 15.0 C and 101,325 N/m², respectively. However, barometric pressure is conventionally reported in millimeters or inches of mercury, corrected to the standard density of mercury at 0.0 C. Because mercury expands slightly when warmed from 0.0 C to 15.0 C, at standard MSL temperature and pressure, the atmosphere will support a column of mercury 762.1 mm high. However, before this result is reported on television or the internet, it is corrected for the difference between the measured temperature (15.0 C in this case) and the standard temperature used to define the standard density of mercury (0.0 C). Thus, a measured column of mercury 762.1 mm high at MSL and 15.0 C would be corrected for temperature and reported as a barometric pressure of 760.0 mm (29.92 in) of mercury.

1.8. Hydrostatic Forces

Now that we know how to determine the pressure distribution in a static fluid, we shall turn our attention to determining the total resultant force acting on a solid surface as a result of the contact pressure between the surface and a static fluid. The pressure in a static fluid is commonly referred to as the *hydrostatic pressure* and a force that results only from hydrostatic pressure integrated over a finite surface area is commonly called a *hydrostatic force*.

Because force is a vector, a hydrostatic force is conveniently described in terms of its components relative to some coordinate system. Thus, one of the first steps associated with formulating an engineering problem that requires evaluating a hydrostatic force is choosing a coordinate system. The choice of a coordinate system is an important part of the formulation process, because the ease or difficulty of evaluating the resulting surface integral often depends on our choice of the coordinate system.

Consider the surface shown in Fig. 1.8.1, which has a total surface area of $\mathcal{S}$. This surface is completely submerged in a fluid that has the density ρ , which in general could vary with position. The unit vector normal to this surface at an arbitrary point on the surface is denoted as $\mathbf{n}$ and called the *outward normal*, because it points **outward from the surface and into the fluid**. Assuming that gravity is the only body force, we wish to evaluate the resultant hydrostatic force acting on this surface.

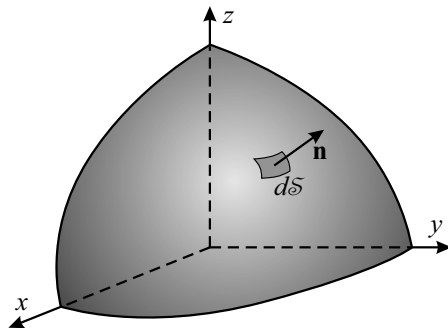


Figure 1.8.1. Geometry for evaluating the hydrostatic force on a curved submerged surface.

We begin by considering the differential hydrostatic force $d\mathbf{F}$ that is acting on the differential surface area $d\mathcal{S}$, which is shown in Fig. 1.8.1. Because there can be no shear stresses acting in a static fluid, the hydrostatic force acting on any differential element of the surface area must act normal to the surface and in a direction opposite to the direction of $\mathbf{n}$. Thus, we can write the vector relation

$$d\mathbf{F} = -p \mathbf{n} d\mathcal{S} \quad (1.8.1)$$

where p is the hydrostatic pressure. The positive direction for the unit normal vector $\mathbf{n}$ is outward from the surface and into the fluid. The negative sign on the right-hand side of Eq. (1.8.1) indicates that the hydrostatic force on $d\mathcal{S}$ acts inward against the surface and away from the fluid, i.e., in the direction opposite to that of $\mathbf{n}$.

The total resultant force $\mathbf{F}$ acting on the complete surface is found by integrating the contributions from the differential hydrostatic forces acting on all of the differential surface area elements that make up the entire surface area $\mathcal{S}$. This gives the **resultant hydrostatic force acting on a surface submerged in a static fluid**

$$\mathbf{F} = - \iint_{\mathcal{S}} p \mathbf{n} d\mathcal{S} \quad (1.8.2)$$

Before the surface integral shown symbolically on the right-hand side of Eq. (1.8.2) can be evaluated, the hydrostatic pressure p , the outward normal $\mathbf{n}$, the differential surface area $d\mathcal{S}$, and the boundaries of the complete surface $\mathcal{S}$ must be expressed in terms of the chosen coordinates; in this case x , y , and z . This procedure is best demonstrated by example.

EXAMPLE 1.8.1. The inside dimensions of a glass-walled aquarium are length l , width w , and height h . The aquarium is setting on a perfectly level surface and is filled completely to the top with water. Assume that the water has a constant density ρ and that the ambient pressure at the free surface of the water is p_a . If gravity is the only body force, and the acceleration of gravity is constant, determine the total resultant hydrostatic force vector acting on each inside face of the aquarium. Also find the total net force acting on the aquarium.

Solution. We begin by choosing the Cartesian coordinate system that is shown in Fig. 1.8.2. The coordinate system origin is located at the back-left-top inside corner of the aquarium. The x - y plane is horizontal and the z axis points vertically upward. Thus, the elevation at any point is given by $H = z + H_0$, where H_0 is the elevation at $z = 0$. From Eq. (1.5.16) the hydrostatic pressure in the water is

$$p = p_a - \rho g z \quad (1.8.3)$$

For the back inside surface of the aquarium (i.e., the gray surface shown in Fig. 1.8.2), the unit outward normal points in the positive x direction. Remember, we are evaluating the hydrostatic force on the inside surface of the aquarium glass. Thus, the outward unit normal points outward from the glass and into the water.

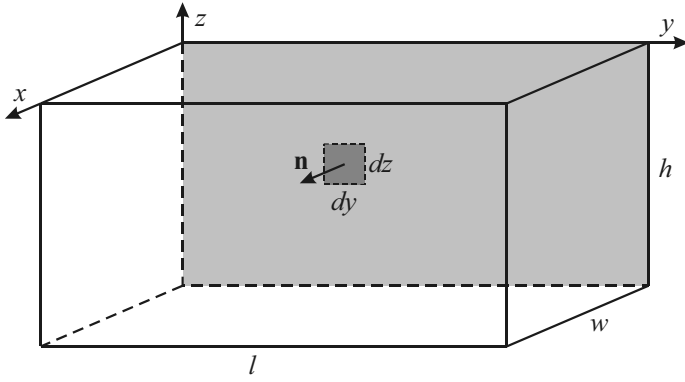


Figure 1.8.2. Geometry of the aquarium described in Example 1.8.1.

From the geometry of the back surface shown in Fig. 1.8.2, the surface geometry, outward unit normal, and differential area are described in terms of the chosen coordinates as

$$x = 0, \quad 0 \leq y \leq l, \quad -h \leq z \leq 0, \quad \mathbf{n} = \mathbf{i}_x, \quad d\mathcal{S} = dy dz \quad (1.8.4)$$

Using Eqs. (1.8.3) and (1.8.4) in Eq. (1.8.2) yields

$$\begin{aligned} \mathbf{F}_{\text{back}} &= - \iint_{\mathcal{S}} p \mathbf{n} d\mathcal{S} = - \int_{z=-h}^0 \int_{y=0}^l (p_a - \rho g z) \mathbf{i}_x dy dz \\ &= - \int_{z=-h}^0 (p_a - \rho g z) l \mathbf{i}_x dz = -(p_a h + \frac{1}{2} \rho g h^2) l \mathbf{i}_x \\ &= \underline{-(p_a + \frac{1}{2} \rho g h) l h \mathbf{i}_x} \end{aligned}$$

Likewise, on the inside surface of the front glass, the surface geometry, outward unit normal, differential area, and hydrostatic force are

$$\begin{aligned} x = w, \quad 0 \leq y \leq l, \quad -h \leq z \leq 0, \quad \mathbf{n} = -\mathbf{i}_x, \quad d\mathcal{S} = dy dz \\ \mathbf{F}_{\text{front}} &= - \iint_{\mathcal{S}} p \mathbf{n} d\mathcal{S} = - \int_{z=-h}^0 \int_{y=0}^l (p_a - \rho g z) (-\mathbf{i}_x) dy dz \\ &= \underline{+(p_a + \frac{1}{2} \rho g h) l h \mathbf{i}_x} \end{aligned}$$

Similarly, for the left surface

$$\begin{aligned} y = 0, \quad 0 \leq x \leq w, \quad -h \leq z \leq 0, \quad \mathbf{n} = \mathbf{i}_y, \quad d\mathcal{S} = dx dz \\ \mathbf{F}_{\text{left}} &= - \iint_{\mathcal{S}} p \mathbf{n} d\mathcal{S} = - \int_{z=-h}^0 \int_{x=0}^w (p_a - \rho g z) \mathbf{i}_y dx dz \\ &= \underline{-(p_a + \frac{1}{2} \rho g h) w h \mathbf{i}_y} \end{aligned}$$

For the right surface

$$y = l, \quad 0 \leq x \leq w, \quad -h \leq z \leq 0, \quad \mathbf{n} = -\mathbf{i}_y, \quad d\mathcal{S} = dx dz$$

$$\begin{aligned} \mathbf{F}_{\text{right}} &= - \iint_{\mathcal{S}} p \mathbf{n} d\mathcal{S} = - \int_{z=-h}^0 \int_{x=0}^w (p_a - \rho g z)(-\mathbf{i}_y) dx dz \\ &= \underline{+(p_a + \frac{1}{2} \rho g h) w h \mathbf{i}_y} \end{aligned}$$

For the bottom surface

$$z = -h, \quad 0 \leq x \leq w, \quad 0 \leq y \leq l, \quad \mathbf{n} = \mathbf{i}_z, \quad d\mathcal{S} = dx dy$$

$$\begin{aligned} \mathbf{F}_{\text{bottom}} &= - \iint_{\mathcal{S}} p \mathbf{n} d\mathcal{S} = - \int_{y=0}^l \int_{x=0}^w (p_a + \rho g h) \mathbf{i}_z dx dy \\ &= \underline{-(p_a + \rho g h) w l \mathbf{i}_z} \end{aligned}$$

Ambient pressure acts on the outside surface of each face of the aquarium. Because the variation in ambient air pressure with elevation is small compared with that in the water, the ambient air pressure can be approximated as constant over the entire outside surface of the aquarium. On each outside surface, the outward normal is directly opposite to that on the inside surface of the same face. Thus, the **net hydrostatic force on each face of the aquarium resulting from the water and air combined** is

$$\begin{aligned} \mathbf{F}_{\text{back}} &= -(p_a + \frac{1}{2} \rho g h) l h \mathbf{i}_x + p_a l h \mathbf{i}_x = -\frac{1}{2} \rho g l h^2 \mathbf{i}_x \\ \mathbf{F}_{\text{front}} &= +(p_a + \frac{1}{2} \rho g h) l h \mathbf{i}_x - p_a l h \mathbf{i}_x = +\frac{1}{2} \rho g l h^2 \mathbf{i}_x \\ \mathbf{F}_{\text{left}} &= -(p_a + \frac{1}{2} \rho g h) w h \mathbf{i}_y + p_a w h \mathbf{i}_y = -\frac{1}{2} \rho g w h^2 \mathbf{i}_y \\ \mathbf{F}_{\text{right}} &= +(p_a + \frac{1}{2} \rho g h) w h \mathbf{i}_y - p_a w h \mathbf{i}_y = +\frac{1}{2} \rho g w h^2 \mathbf{i}_y \\ \mathbf{F}_{\text{bottom}} &= -(p_a + \rho g h) w l \mathbf{i}_z + p_a w l \mathbf{i}_z = -\rho g h w l \mathbf{i}_z \end{aligned}$$

Adding the net hydrostatic force on all faces of the aquarium yields the **total net force acting on the aquarium as a result of contact with the water and air**,

$$\mathbf{F}_{\text{total}} = \underline{-\rho g h w l \mathbf{i}_z}$$

Notice that the magnitude of this net force is the product of the water density, the water volume, and the acceleration of gravity, which is simply the weight of the water in the aquarium. This force acts in the $-\mathbf{i}_z$ direction, i.e., down.

It is important to remember that in the previous example the ambient air pressure was approximated as constant over the entire outside surface of the aquarium. In reality, the air pressure at the bottom of the aquarium is slightly greater than that at the top. A more precise solution would predict a total net force on the aquarium that is slightly less than the weight of the water. We will return to this point shortly.

EXAMPLE 1.8.2. The aquarium described in Example 1.8.1 is modified as shown on Fig. 1.8.3. The volume of the aquarium is decreased by one sixth as a result of cutting the front, right, and bottom faces on the diagonal and sealing the opening with a triangular glass plane. The aquarium is setting on a perfectly level surface and is filled completely to the top with water. Assume that the water has a constant density ρ and that the ambient air pressure acting on all outside surfaces is constant at p_a . If gravity is the only body force, and the acceleration of gravity is constant, find the net hydrostatic force acting on each face of the aquarium resulting from the water and air combined. Also find the total net force acting on the aquarium.

Solution. Using the same coordinate system that was used in Example 1.8.1, the water pressure is given by Eq. (1.8.3), which can be rearranged as

$$p - p_a = -\rho g z$$

Because water pressure p acts on the inside surface of each face and air pressure p_a acts on the outside of each face, the **net hydrostatic force on any face of the aquarium resulting from the water and air combined** is

$$\mathbf{F} = - \iint_{\mathcal{S}} (p - p_a) \mathbf{n} d\mathcal{S} = \iint_{\mathcal{S}} \rho g z \mathbf{n} d\mathcal{S} \quad (1.8.5)$$

where $\mathbf{n}$ is the outward normal on the glass surface in contact with the water.

The back and left faces are unchanged from Example 1.8.1. Thus, the net forces on these two faces are

$$\begin{aligned} \mathbf{F}_{\text{back}} &= -\frac{1}{2} \rho g l h^2 \mathbf{i}_x \\ \mathbf{F}_{\text{left}} &= -\frac{1}{2} \rho g w h^2 \mathbf{i}_y \end{aligned}$$

The diagonal edge of the front face passes through the points having (x, y, z) coordinates of $(w, 0, -h)$ and $(w, l, 0)$. Any two non-coincident points can be used

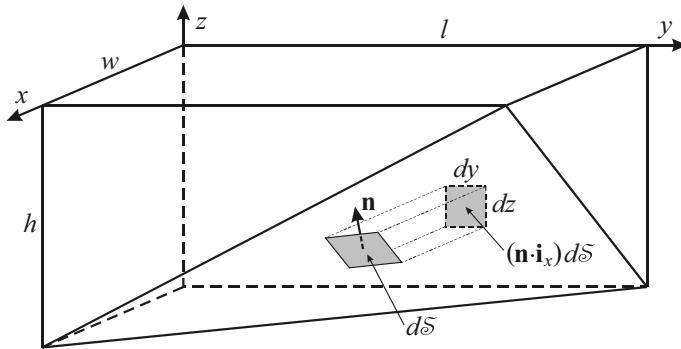


Figure 1.8.3. Geometry of the aquarium described in Example 1.8.2.

to define a straight line. This diagonal line lies in the plane $x = w$ and it can be defined by an equation of the form $y = C_0 + C_1 z$, where C_0 and C_1 are constants. Forcing this line to pass through the points $(w, 0, -h)$ and $(w, l, 0)$ yields the two equations

$$\begin{aligned} 0 &= C_0 + C_1(-h) \\ l &= C_0 + C_1(0) \end{aligned}$$

which result in $C_0 = l$, $C_1 = l/h$, and $y = l + (l/h)z$. Thus, the surface geometry, outward normal, and differential area are

$$x = w, \quad 0 \leq y \leq l + (l/h)z, \quad -h \leq z \leq 0, \quad \mathbf{n} = -\mathbf{i}_x, \quad d\mathcal{S} = dy dz \quad (1.8.6)$$

Using Eqs. (1.8.6) in Eq. (1.8.5) yields the net force on the front face

$$\begin{aligned} \mathbf{F}_{\text{front}} &= \iint_{\mathcal{S}} \rho g z \mathbf{n} d\mathcal{S} = \int_{z=-h}^0 \int_{y=0}^{l+(l/h)z} \rho g z (-\mathbf{i}_x) dy dz \\ &= -\rho g \mathbf{i}_x \int_{z=-h}^0 z [l + (l/h)z] dz = -\rho g \mathbf{i}_x \int_{z=-h}^0 [lz + (l/h)z^2] dz \\ &= -\rho g \mathbf{i}_x \left[\frac{1}{2} l z^2 + \frac{1}{3} (l/h) z^3 \right]_{-h}^0 = -\rho g \mathbf{i}_x \left[0 - \frac{1}{2} l (-h)^2 - \frac{1}{3} (l/h) (-h)^3 \right] \\ &= \underline{\underline{\frac{1}{6} \rho g l h^2 \mathbf{i}_x}} \end{aligned}$$

Similarly, on the right face, the equation for the diagonal edge is $x = w + (w/h)z$. Thus, the surface geometry, outward normal, and differential area are

$$y = l, \quad 0 \leq x \leq w + (w/h)z, \quad -h \leq z \leq 0, \quad \mathbf{n} = -\mathbf{i}_y, \quad d\mathcal{S} = dx dz$$

and the net force on the right face is

$$\begin{aligned} \mathbf{F}_{\text{right}} &= \iint_{\mathcal{S}} \rho g z \mathbf{n} d\mathcal{S} = \int_{z=-h}^0 \int_{x=0}^{w+(w/h)z} \rho g z (-\mathbf{i}_y) dx dz \\ &= -\rho g \mathbf{i}_y \int_{z=-h}^0 [wz + (w/h)z^2] dz = -\rho g \mathbf{i}_y \left[\frac{1}{2} w z^2 + \frac{1}{3} (w/h) z^3 \right]_{-h}^0 \\ &= \underline{\underline{\frac{1}{6} \rho g w h^2 \mathbf{i}_y}} \end{aligned}$$

On the horizontal bottom face, the equation for the diagonal edge is $x = w - (w/l)y$ and

$$z = -h, \quad 0 \leq x \leq w - (w/l)y, \quad 0 \leq y \leq l, \quad \mathbf{n} = \mathbf{i}_z, \quad d\mathcal{S} = dx dy$$

$$\begin{aligned} \mathbf{F}_{\text{bottom}} &= \iint_{\mathcal{S}} \rho g z \mathbf{n} d\mathcal{S} = \int_{y=0}^l \int_{x=0}^{w-(w/l)y} \rho g (-h) \mathbf{i}_z dx dy \\ &= -\rho g h \mathbf{i}_z \int_{y=0}^l [w - (w/l)y] dy = -\rho g h \mathbf{i}_z \left[wy - \frac{1}{2} (w/l) y^2 \right]_0^l \\ &= \underline{\underline{-\frac{1}{2} \rho g h w l \mathbf{i}_z}} \end{aligned}$$

On the remaining diagonal face, the resultant force has components in all three coordinate directions. The x component of any vector is its dot product with the unit vector $\mathbf{i}_x$. Similarly, the y and z components are obtained from the vector's dot product with $\mathbf{i}_y$ and $\mathbf{i}_z$, respectively. Thus, Eq. (1.8.5) can be written conveniently in terms of its three components

$$\mathbf{F} = \mathbf{i}_x \iint_S \rho g z (\mathbf{n} \cdot \mathbf{i}_x) d\mathcal{S} + \mathbf{i}_y \iint_S \rho g z (\mathbf{n} \cdot \mathbf{i}_y) d\mathcal{S} + \mathbf{i}_z \iint_S \rho g z (\mathbf{n} \cdot \mathbf{i}_z) d\mathcal{S}$$

It is important to recognize that the magnitude of $(\mathbf{n} \cdot \mathbf{i}_x) d\mathcal{S}$ is the projection of $d\mathcal{S}$ on the y - z plane as shown in Fig. 1.8.3. In general, the sign of $(\mathbf{n} \cdot \mathbf{i}_x) d\mathcal{S}$ could be either positive or negative, depending on the direction of $\mathbf{n}$. For the particular case shown in Fig. 1.8.3, $\mathbf{n}$ points outward from the inside surface of the glass and into the water (i.e., toward the back face), which gives $(\mathbf{n} \cdot \mathbf{i}_x) d\mathcal{S} = -dy dz$. Similarly, $(\mathbf{n} \cdot \mathbf{i}_y) d\mathcal{S}$ and $(\mathbf{n} \cdot \mathbf{i}_z) d\mathcal{S}$ are the signed projections of $d\mathcal{S}$ on the x - z and x - y planes, respectively. From the geometry shown in Fig. 1.8.3, we have

$$\mathbf{F}_{\text{diagonal}} = -\mathbf{i}_x \iint_S \rho g z dy dz - \mathbf{i}_y \iint_S \rho g z dx dz + \mathbf{i}_z \iint_S \rho g z dx dy \quad (1.8.7)$$

The equation for any plane can be written in the form $z = C_0 + C_1 x + C_2 y$, where C_0 , C_1 , and C_2 are constants. This particular diagonal plane passes through the points $(w, 0, -h)$, $(w, l, 0)$ and $(0, l, -h)$, which provides three equations in the three unknown constants C_0 , C_1 , and C_2 . The solution yields

$$z = -2h + (h/w)x + (h/l)y \quad (1.8.8)$$

As previously discussed, the equations for the three lines that form the boundaries of the diagonal face are

$$y = l + (l/h)z, \quad x = w + (w/h)z, \quad x = w - (w/l)y \quad (1.8.9)$$

Using Eqs. (1.8.8) and (1.8.9) in Eq. (1.8.7) results in

$$\begin{aligned} \mathbf{F}_{\text{diagonal}} &= -\mathbf{i}_x \int_{z=-h}^0 \int_{y=l+(l/h)z}^l \rho g z dy dz - \mathbf{i}_y \int_{z=-h}^0 \int_{x=w+(w/h)z}^w \rho g z dx dz \\ &\quad + \mathbf{i}_z \int_{y=0}^l \int_{x=w-(w/l)y}^w \rho g [-2h + (h/w)x + (h/l)y] dx dy \\ &= -\mathbf{i}_x \int_{z=-h}^0 \rho g z [-(l/h)z] dz - \mathbf{i}_y \int_{z=-h}^0 \rho g z [-(w/h)z] dz \\ &\quad + \mathbf{i}_z \int_{y=0}^l \rho g [-(hw/l)y + \frac{1}{2}(hw/l^2)y^2] dy \\ &= \underline{\underline{\frac{1}{3} \rho g l h^2 \mathbf{i}_x + \frac{1}{3} \rho g w h^2 \mathbf{i}_y - \frac{1}{3} \rho g h w l \mathbf{i}_z}} \end{aligned}$$

The vector sum of the net hydrostatic forces on all six faces of the aquarium yields the **total net force acting on the aquarium as a result of contact with the water and air**,

$$\begin{aligned}
\mathbf{F}_{\text{total}} &= \mathbf{F}_{\text{back}} + \mathbf{F}_{\text{left}} + \mathbf{F}_{\text{front}} + \mathbf{F}_{\text{right}} + \mathbf{F}_{\text{bottom}} + \mathbf{F}_{\text{diagonal}} \\
&= -\frac{1}{2}\rho g l h^2 \mathbf{i}_x - \frac{1}{2}\rho g w h^2 \mathbf{i}_y + \frac{1}{6}\rho g l h^2 \mathbf{i}_x + \frac{1}{6}\rho g w h^2 \mathbf{i}_y - \frac{1}{2}\rho g h w l \mathbf{i}_z \\
&\quad + \frac{1}{3}\rho g l h^2 \mathbf{i}_x + \frac{1}{3}\rho g w h^2 \mathbf{i}_y - \frac{1}{3}\rho g h w l \mathbf{i}_z \\
&= \left(-\frac{1}{2} + \frac{1}{6} + \frac{1}{3}\right)\rho g l h^2 \mathbf{i}_x + \left(-\frac{1}{2} + \frac{1}{6} + \frac{1}{3}\right)\rho g w h^2 \mathbf{i}_y + \left(-\frac{1}{2} - \frac{1}{3}\right)\rho g h w l \mathbf{i}_z \\
&= \underline{-\frac{5}{6}\rho g h w l \mathbf{i}_z}
\end{aligned}$$

This is a downward force equal to the product of the water density, water volume, and acceleration of gravity, which is the weight of the water in the aquarium.

1.9. Hydrostatic Moments and Center of Pressure

In the previous section we learned how to determine the resultant hydrostatic force acting on a solid surface as a result of the contact pressure between the surface and a static fluid. The resultant force given by Eq. (1.8.2) is a vector, which gives us both the magnitude and direction of this pressure force. However, Eq. (1.8.2) says nothing about the *point of application*, which is required for the resultant force to produce the same mechanical reactions as the distributed pressure load. The desire to define a point of application for any resultant force is related to the fact that the mechanical reactions required to hold a surface in place against a distributed load may include moments as well as forces.

*The **center of pressure** is a point of application for the resultant pressure force that will produce the same mechanical reactions as the distributed pressure load acting on a submerged surface.*

In most fluid mechanics textbooks, the concepts of pressure-induced moments and center of pressure are introduced through applications that involve the containment of liquids behind dams and in liquid storage tanks, such as those used for Examples 1.8.1 and 1.8.2. The presentation in this textbook will be no exception. However, because the title of this textbook is *Aerodynamics of Flight*, a student might rightfully ask, “What do dams and aquariums have to do with flight?” Hence, a few words on this topic may be in order before we proceed to our consideration of hydrostatic moments and center of pressure.

The weight of an airplane in flight is supported primarily by the air pressure acting on its wings. In addition to carrying the weight of the airplane, this air pressure produces a resultant bending moment about the points where the wings are attached to the fuselage. Furthermore, the air pressure acting on the wings of an airplane can produce a resultant moment about the airplane’s center of gravity. Any bending moments generated by air pressure must be resisted by the airplane’s structure, and the airplane must be designed so that the summation of all moments acting about its center of gravity is zero when the airplane is in equilibrium flight. Thus, the concepts of pressure-induced moments and center of pressure are of critical importance to understanding the aerodynamics of flight.

The flow field and pressure distribution about the curved surface of an airplane wing are quite complex. In the interest of building a thorough understanding of the underlying concepts of flight, most students will find it easier to begin with simple problems and

build toward the more complex. To this end, we will continue for a while with the simple analysis of the resultant forces and moments that are produced on flat surfaces as a result of the contact pressure between a surface and a static fluid. Problems that involve the containment of liquids behind dams and in liquid storage tanks are among the simplest problems of this type.

EXAMPLE 1.9.1. A dam is holding back the water in a reservoir, as shown in Fig. 1.9.1. The depth of the water is h , the width of the dam perpendicular to the page is w , and the air pressure at the free surface is p_a . Determine the y coordinate of the center of pressure for the resultant hydrostatic force $\mathbf{F}$ that is exerted on the dam by the water, i.e., the point of application that will produce the same reaction at the origin of the coordinate system shown in Fig. 1.9.1 as that of the distributed pressure load.

Solution. The water pressure at the surface is p_a , that at the bottom of the dam is $p_a + \rho gh$, and the vertical pressure distribution acting on the dam is linear. Thus, following Example 1.8.1, or simply by inspection of the *pressure trapezoid* shown in Fig. 1.9.1, it is easily shown that the resultant hydrostatic force is

$$\mathbf{F} = (p_a + \frac{1}{2} \rho gh)wh \mathbf{i}_x \quad (1.9.1)$$

Consider a differential strip of the dam's wetted surface, which is located at the height y , extends across the full width of the dam w , and has a thickness in the vertical direction of dy . The moment about the origin of the coordinate system shown in Fig. 1.9.1 that is produced by the water pressure acting on this strip has a magnitude equal to the product of the water pressure p , the area $w dy$, and the vertical moment arm y . Since the water pressure on the dam acts in the x - y plane, the moment produced by this pressure is about the z axis (pointing out of the page).

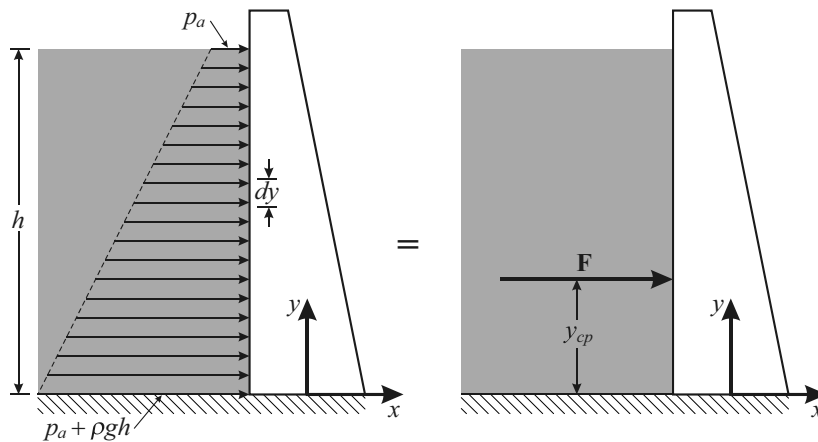


Figure 1.9.1. Center of pressure for the resultant hydrostatic force that is associated with the water pressure acting on a dam.

Inspection of Fig. 1.9.1 reveals that the moment about the z axis produced by the water pressure acting on the dam is a negative moment as defined by the right-hand rule. Thus, the moment about the origin of the coordinate system shown in Fig. 1.9.1 that is produced by the water pressure acting on a differential strip of the dam's wetted surface is

$$d\mathbf{M}_0 = -pwydy \mathbf{i}_z \quad (1.9.2)$$

The linear pressure distribution in the water is given by

$$p(y) = p_a + \rho gh - \rho gy \quad (1.9.3)$$

Therefore, using Eq. (1.9.3) in Eq. (1.9.2) and integrating over the wetted area, the moment about the origin that is produced by the distributed pressure load is

$$\begin{aligned} \mathbf{M}_0 &= - \int_{y=0}^h pwydy \mathbf{i}_z = - \int_{y=0}^h [(p_a + \rho gh)y - \rho gy^2] w dy \mathbf{i}_z \\ &= - \left[\frac{1}{2}(p_a + \rho gh)h^2 - \frac{1}{3}\rho gh^3 \right] w \mathbf{i}_z = - \frac{1}{2}(p_a + \frac{1}{3}\rho gh)wh^2 \mathbf{i}_z \end{aligned} \quad (1.9.4)$$

Similarly, in view of Eq. (1.9.1), the moment about the origin that is produced by the resultant hydrostatic force shown in Fig. 1.9.1 is

$$\mathbf{M}_0 = -y_{cp}|\mathbf{F}| \mathbf{i}_z = -y_{cp}(p_a + \frac{1}{2}\rho gh)wh \mathbf{i}_z \quad (1.9.5)$$

Equating the moments from Eqs. (1.9.4) and (1.9.5), and then solving the resulting equation for the y coordinate of the center of pressure yields

$$y_{cp} = \frac{3p_a + \rho gh}{6p_a + 3\rho gh} h \quad (1.9.6)$$

Note that the water pressure acting on the dam in Example 1.9.1 is the sum of the air pressure and the gage pressure. By setting the water density in Eq. (1.9.6) to zero, it can be seen that the *center of air pressure* on the wetted surface is $h/2$, i.e., at the midpoint between the bottom of the dam and the surface of the water. Similarly, by setting the air pressure in Eq. (1.9.6) to zero, it can be seen that the *center of gage pressure* is $h/3$, i.e., one third of the distance from the bottom of the dam to the surface of the water. The center of pressure on the wetted surface of a dam is closer to the bottom of the dam than it is to the free surface of the water, because the pressure is greater at the bottom of the dam than it is at the free surface.

Because the dry side of a dam is exposed to the atmosphere, air pressure acts on both sides of a dam, yielding a zero resultant. Hence, problems like the one considered in Example 1.9.1 are commonly worked using only gage pressure. In this case, the net resultant hydrostatic force from the pressure on both the wet and dry sides of the dam is $\mathbf{F} = (\rho gh^2 w/2) \mathbf{i}_x$ and the vertical coordinate for the point of application is the center of gage pressure, which is a distance of $h/3$ above the bottom of the dam.

The method demonstrated in Example 1.9.1 for locating the center of pressure on a flat surface does not depend on our choice of the coordinate system. Furthermore, the method can be generalized to three dimensions, which allows for evaluation of both the horizontal and vertical locations of the center of pressure on an arbitrary flat surface.

Consider the arbitrary force $\mathbf{F}$ applied at the arbitrary point (x, y, z) in the arbitrary Cartesian coordinate system, which is shown in Fig. 1.9.2. This figure also shows the projections of $\mathbf{F}$ and its components on the x - y , y - z , and z - x planes. We wish to determine the moment vector $\mathbf{M}_0$ generated about the origin by this force. The x component of $\mathbf{F}$ is parallel to the x axis, so only the y and z components of $\mathbf{F}$ contribute to the x component of $\mathbf{M}_0$. From Fig. 1.9.2, we see that a positive z component of $\mathbf{F}$ acting at a positive value of y produces a right-hand moment about the x axis. Figure 1.9.2 also shows that a positive y component of $\mathbf{F}$ acting at a positive value of z produces a left-hand moment about the x axis. Thus, the x component of $\mathbf{M}_0$ is $yF_z - zF_y$. Similarly, Fig. 1.9.2 also shows that the y component of $\mathbf{M}_0$ is $zF_x - xF_z$ and the z component of $\mathbf{M}_0$ is $xF_y - yF_x$, which yields

$$\mathbf{M}_0 = (yF_z - zF_y)\mathbf{i}_x + (zF_x - xF_z)\mathbf{i}_y + (xF_y - yF_x)\mathbf{i}_z \quad (1.9.7)$$

Comparing Eq. (1.9.7) to the definition of the cross product in Eq. (1.4.15), we see that Eq. (1.9.7) can be written as

$$\mathbf{M}_0 = (x\mathbf{i}_x + y\mathbf{i}_y + z\mathbf{i}_z) \times (F_x\mathbf{i}_x + F_y\mathbf{i}_y + F_z\mathbf{i}_z) \quad (1.9.8)$$

The vector $x\mathbf{i}_x + y\mathbf{i}_y + z\mathbf{i}_z$ is commonly called the position vector and will be denoted here as $\mathbf{r}$. Hence, the moment about the origin $\mathbf{M}_0$ that is generated by the force $\mathbf{F}$ acting at the position $\mathbf{r}$ is given by

$$\mathbf{M}_0 = \mathbf{r} \times \mathbf{F} \quad (1.9.9)$$

Consider the application of Eq. (1.9.9) to the moment generated by the differential hydrostatic force $d\mathbf{F}$ that is acting on the differential surface area $d\mathcal{S}$ shown in Fig. 1.8.1.

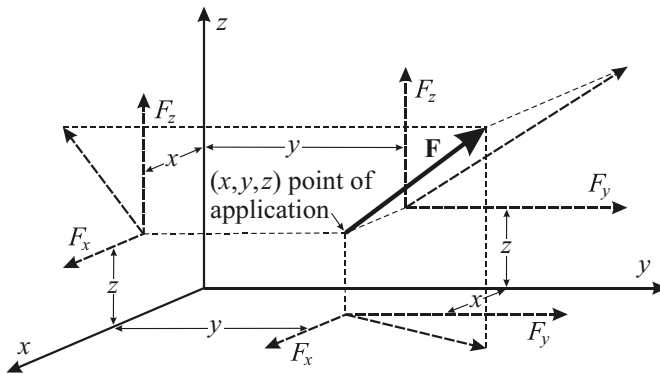


Figure 1.9.2. Geometry for computing the moment generated about the origin by an arbitrary force.

This differential hydrostatic force is related to the pressure p and the outward normal $\mathbf{n}$ according to Eq. (1.8.1). Consequently, we have

$$d\mathbf{M}_0 = \mathbf{r} \times d\mathbf{F} = -p\mathbf{r} \times \mathbf{n} d\mathcal{S} \quad (1.9.10)$$

The total resultant moment about the origin $\mathbf{M}_0$ generated by the pressure p acting on the complete surface is found by integrating the contributions from all of the differential surface area elements that make up the entire surface area $\mathcal{S}$. This gives **the resultant hydrostatic moment about the origin acting on a surface submerged in a static fluid**

$$\mathbf{M}_0 = - \iint_{\mathcal{S}} p\mathbf{r} \times \mathbf{n} d\mathcal{S} \quad (1.9.11)$$

Because the center of pressure is a point of application for the resultant hydrostatic force that will produce the same mechanical reactions as the distributed pressure load, the total resultant moment about the origin $\mathbf{M}_0$ can also be expressed using the relation given in Eq. (1.9.9) combined with Eq. (1.8.2),

$$\mathbf{M}_0 = \mathbf{r}_{cp} \times \mathbf{F} = -\mathbf{r}_{cp} \times \iint_{\mathcal{S}} p\mathbf{n} d\mathcal{S} \quad (1.9.12)$$

where $\mathbf{r}_{cp}$ is the position vector for the center of pressure, $\mathbf{r}_{cp} = x_{cp}\mathbf{i}_x + y_{cp}\mathbf{i}_y + z_{cp}\mathbf{i}_z$, and $\mathbf{F}$ is the resultant hydrostatic force. **The center of pressure is a point that will force the moment obtained from Eq. (1.9.11) to match that obtained from Eq. (1.9.12).**

The position vector from an arbitrary reference point $\mathbf{r}_{ref} = x_{ref}\mathbf{i}_x + y_{ref}\mathbf{i}_y + z_{ref}\mathbf{i}_z$ to another arbitrary point $\mathbf{r} = x\mathbf{i}_x + y\mathbf{i}_y + z\mathbf{i}_z$ is simply $\mathbf{r} - \mathbf{r}_{ref}$. Hence, from a development similar to that used to obtain Eq. (1.9.11), **the resultant hydrostatic moment about any fixed reference point acting on a surface submerged in a static fluid is**

$$\mathbf{M}_{ref} = - \iint_{\mathcal{S}} p(\mathbf{r} - \mathbf{r}_{ref}) \times \mathbf{n} d\mathcal{S} = - \iint_{\mathcal{S}} p\mathbf{r} \times \mathbf{n} d\mathcal{S} + \mathbf{r}_{ref} \times \iint_{\mathcal{S}} p\mathbf{n} d\mathcal{S} \quad (1.9.13)$$

or after applying Eqs. (1.9.11) and (1.8.2)

$$\mathbf{M}_{ref} = \mathbf{M}_0 + \mathbf{r}_{ref} \times \iint_{\mathcal{S}} p\mathbf{n} d\mathcal{S} = \mathbf{M}_0 - \mathbf{r}_{ref} \times \mathbf{F} \quad (1.9.14)$$

For the special case where the reference point in Eq. (1.9.14) is chosen to be the center of pressure, combining the relations given in Eqs. (1.9.12) and (1.9.14), we find that **the resultant hydrostatic moment about the center of pressure is**

$$\mathbf{M}_{cp} = \mathbf{M}_0 + \mathbf{r}_{cp} \times \iint_{\mathcal{S}} p\mathbf{n} d\mathcal{S} = \mathbf{M}_0 - \mathbf{r}_{cp} \times \mathbf{F} = 0 \quad (1.9.15)$$

Equation (1.9.15) provides an alternate definition for the center of pressure, which is very commonly used in fluid mechanics and aerodynamics:

*The **center of pressure** is a point about which the resultant moment produced by the distributed pressure load acting on a submerged surface is zero.*

Although the center of pressure is usually defined from Eq. (1.9.15), it is most easily located by evaluating the resultant moment from both Eq. (1.9.11) and Eq. (1.9.12) and forcing the two results to match. Before the surface integral shown symbolically on the right-hand side of Eq. (1.9.11) can be evaluated, the hydrostatic pressure p , the position vector $\mathbf{r}$, the outward normal $\mathbf{n}$, the differential surface area $d\mathcal{S}$, and the boundaries of the complete surface $\mathcal{S}$ must be expressed in terms of the chosen coordinate system. This procedure is best demonstrated by example.

EXAMPLE 1.9.2. Determine the center of pressure for the hydrostatic force on the left inside face of the aquarium described in Example 1.8.1. Assume that the water has a constant density ρ , that the ambient pressure at the free surface of the water is p_a , and that constant acceleration of gravity provides the only body force.

Solution. Continuing with the Cartesian coordinate system shown in Fig. 1.8.2, the coordinate system origin is located at the back-left-top inside corner of the aquarium. The hydrostatic pressure distribution in the water in contact with the left inside face of the aquarium is shown in Fig. 1.9.3. From Example 1.8.1, this pressure distribution is given by

$$p = p_a - \rho g z \quad (1.9.16)$$

For the left inside surface of the aquarium, the unit outward normal points in the positive y direction. From Example 1.8.1, the surface geometry, outward unit normal, differential area, and resultant hydrostatic force are described in terms of the chosen coordinate system as

$$y = 0, \quad 0 \leq x \leq w, \quad -h \leq z \leq 0, \quad \mathbf{n} = \mathbf{i}_y, \quad d\mathcal{S} = dx dz \quad (1.9.17)$$

$$\begin{aligned} \mathbf{F} &= - \iint_{\mathcal{S}} p \mathbf{n} d\mathcal{S} = - \int_{z=-h}^0 \int_{x=0}^w (p_a - \rho g z) \mathbf{i}_y dx dz \\ &= -(p_a + \frac{1}{2} \rho g h) w h \mathbf{i}_y \end{aligned} \quad (1.9.18)$$

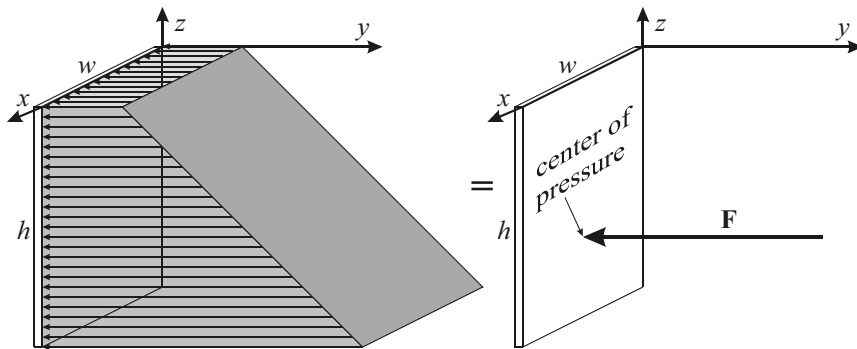


Figure 1.9.3. The pressure prism and center of pressure for the surface in Example 1.9.2.

Because y is zero at all points on this surface, the position vector for an arbitrary point on the surface is $\mathbf{r} = x\mathbf{i}_x + z\mathbf{i}_z$, and the cross product of the position vector $\mathbf{r}$ with the unit outward normal $\mathbf{n} = \mathbf{i}_y$ is $\mathbf{r} \times \mathbf{n} = (x\mathbf{i}_x + z\mathbf{i}_z) \times \mathbf{i}_y$. Hence, from the vector relations in Eqs. (1.4.17), (1.4.19) and (1.4.21), we obtain

$$\begin{aligned}\mathbf{r} \times \mathbf{n} &= (x\mathbf{i}_x + z\mathbf{i}_z) \times \mathbf{i}_y = x\mathbf{i}_x \times \mathbf{i}_y + z\mathbf{i}_z \times \mathbf{i}_y \\ &= x\mathbf{i}_x \times \mathbf{i}_y - z\mathbf{i}_y \times \mathbf{i}_z = x\mathbf{i}_z - z\mathbf{i}_x\end{aligned}\quad (1.9.19)$$

Using Eqs. (1.9.16), (1.9.17), and (1.9.19) in Eq. (1.9.11) yields

$$\begin{aligned}\mathbf{M}_0 &= - \iint_S p \mathbf{r} \times \mathbf{n} dS \\ &= - \int_{z=-h}^0 \int_{x=0}^w (p_a - \rho g z) (x\mathbf{i}_z - z\mathbf{i}_x) dx dz \\ &= \int_{z=0}^{-h} (p_a - \rho g z) \left(\frac{1}{2} w^2 \mathbf{i}_z - wz \mathbf{i}_x \right) dz \\ &= \frac{1}{2} [p_a(-h) - \frac{1}{2} \rho g (-h)^2] w^2 \mathbf{i}_z \\ &\quad - \left[\frac{1}{2} p_a (-h)^2 - \frac{1}{3} \rho g (-h)^3 \right] w \mathbf{i}_x \\ &= -\frac{1}{2} (p_a + \frac{1}{2} \rho g h) w^2 h \mathbf{i}_z - \frac{1}{2} (p_a + \frac{2}{3} \rho g h) w h^2 \mathbf{i}_x\end{aligned}\quad (1.9.20)$$

Similarly, the position vector for the center of pressure is $\mathbf{r}_{cp} = x_{cp} \mathbf{i}_x + z_{cp} \mathbf{i}_z$, and using this with Eq. (1.9.18) in Eq. (1.9.12), the moment about the origin that is produced by the resultant hydrostatic force shown in Fig. 1.9.3 is

$$\begin{aligned}\mathbf{M}_0 &= \mathbf{r}_{cp} \times \mathbf{F} = (x_{cp} \mathbf{i}_x + z_{cp} \mathbf{i}_z) \times [-(p_a + \frac{1}{2} \rho g h) w h \mathbf{i}_y] \\ &= -x_{cp} (p_a + \frac{1}{2} \rho g h) w h \mathbf{i}_x \times \mathbf{i}_y - z_{cp} (p_a + \frac{1}{2} \rho g h) w h \mathbf{i}_z \times \mathbf{i}_y \\ &= -x_{cp} (p_a + \frac{1}{2} \rho g h) w h \mathbf{i}_z + z_{cp} (p_a + \frac{1}{2} \rho g h) w h \mathbf{i}_x\end{aligned}\quad (1.9.21)$$

Equating the moments from Eqs. (1.9.20) and (1.9.21), and solving the resulting vector equation for the x and z coordinate of the center of pressure yields

$$x_{cp} = \frac{1}{2} w, \quad z_{cp} = -\frac{3p_a + 2\rho g h}{6p_a + 3\rho g h} h \quad (1.9.22)$$

The mathematical problem that was solved in Example 1.9.2 is identical to that solved in Example 1.9.1. The only difference in these two examples is the method of solution and the chosen coordinate system. In both examples, we were computing the center of pressure on a vertical surface of width w and height h . The solution obtained in Example 1.9.2 could have been deduced directly from the solution to Example 1.9.1 and inspection of Fig. 1.9.3.

Because the pressure shown in Fig. 1.9.3 does not vary with x , we should expect the center of pressure to be located at the horizontal midpoint of this rectangular surface, i.e., $x_{cp} = w/2$. The vertical position of the center of pressure obtained in Example 1.9.2 is identical to that obtained in Example 1.9.1. The only difference between the solution

given by Eq. (1.9.6) and that given by Eq. (1.9.22) is the chosen coordinate system. From Eq. (1.9.6), we found that the vertical center of air pressure on the wetted surface is the midpoint ($y = h/2$). Setting the water density in Eq. (1.9.22) to zero also reveals that the vertical center of air pressure is the midpoint ($z = -h/2$). Similarly, Eq. (1.9.6) places the center of gage pressure one third of the distance up from the bottom ($y = h/3$), and setting the air pressure in Eq. (1.9.22) to zero places the center of gage pressure two thirds of the distance down from the top ($z = -2h/3$). These two solutions are identical.

The fact that we obtained identical solutions in Examples 1.9.1 and 1.9.2 by taking moments about two different points is the result of an important principle that students should remember from their introductory course in *statics*:

Two force systems are equivalent if they have the same magnitude, are parallel, and produce the same moment about any point.

Hence, when locating the center of pressure as demonstrated in Examples 1.9.1 and 1.9.2 the choice of the point about which the moments are summed is arbitrary. The resulting mathematics will be simplified if the origin of the reference coordinate system is used. However, our choice of this coordinate system is also arbitrary.

For the simple problem of locating the center of pressure on a vertical rectangular surface, the solution presented in Example 1.9.1 may seem simpler and more intuitive, while that presented in Example 1.9.2 may seem unnecessarily complex. The systematic procedure, which uses the cross product of the position vector and the force vector, has the advantage of more general applicability to surfaces of arbitrary geometry.

EXAMPLE 1.9.3. Locate the center of gage pressure for the net hydrostatic force acting on the combined wet and dry sides of the diagonal triangular glass plane of the aquarium that is described in Example 1.8.2.

Solution. From Example 1.8.2, the equation for this diagonal plane is

$$z = -2h + (h/w)x + (h/l)y \quad (1.9.23)$$

and the equations for the three lines in this plane that form the surface boundaries were found to be

$$y = l + (l/h)z, \quad x = w + (w/h)z, \quad x = w - (w/l)y \quad (1.9.24)$$

Also from Example 1.8.2, the hydrostatic gage pressure distribution in the water in contact with the inside of the aquarium is

$$p - p_a = -\rho g z \quad (1.9.25)$$

and the net hydrostatic force on this diagonal face of the aquarium resulting from the water and air combined was found to be

$$\begin{aligned} \mathbf{F} &= - \iint_{\mathcal{S}} (p - p_a) \mathbf{n} d\mathcal{S} = \iint_{\mathcal{S}} \rho g z \mathbf{n} d\mathcal{S} \\ &= \frac{1}{3} \rho g l h^2 \mathbf{i}_x + \frac{1}{3} \rho g w h^2 \mathbf{i}_y - \frac{1}{3} \rho g h w l \mathbf{i}_z \end{aligned} \quad (1.9.26)$$

Because water pressure p acts on the inside of this face and air pressure p_a acts on the outside, following the development of Eq. (1.9.11), **the resultant hydrostatic moment about the origin generated on this face by water and air combined is**

$$\mathbf{M}_0 = - \iint_{\mathcal{S}} (p - p_a) \mathbf{r} \times \mathbf{n} d\mathcal{S} = \iint_{\mathcal{S}} \rho g z \mathbf{r} \times \mathbf{n} d\mathcal{S} \quad (1.9.27)$$

where $\mathbf{n}$ is the outward normal on the glass surface in contact with the water. The x component of any vector is its dot product with the unit vector $\mathbf{i}_x$. Similarly, the y and z components are obtained from the vector's dot product with $\mathbf{i}_y$ and $\mathbf{i}_z$, respectively. Thus, Eq. (1.9.27) can be written conveniently as

$$\begin{aligned} \mathbf{M}_0 = & \iint_{\mathcal{S}} \rho g z \mathbf{r} \times \mathbf{i}_x (\mathbf{n} \cdot \mathbf{i}_x) d\mathcal{S} + \iint_{\mathcal{S}} \rho g z \mathbf{r} \times \mathbf{i}_y (\mathbf{n} \cdot \mathbf{i}_y) d\mathcal{S} \\ & + \iint_{\mathcal{S}} \rho g z \mathbf{r} \times \mathbf{i}_z (\mathbf{n} \cdot \mathbf{i}_z) d\mathcal{S} \end{aligned} \quad (1.9.28)$$

In Example 1.8.2 it was shown that

$$(\mathbf{n} \cdot \mathbf{i}_x) d\mathcal{S} = -dy dz, \quad (\mathbf{n} \cdot \mathbf{i}_y) d\mathcal{S} = -dx dz, \quad (\mathbf{n} \cdot \mathbf{i}_z) d\mathcal{S} = dx dy \quad (1.9.29)$$

From the vector relations in Eqs. (1.4.17), (1.4.19), (1.4.20), and (1.4.21)

$$\mathbf{r} \times \mathbf{i}_x = (x\mathbf{i}_x + y\mathbf{i}_y + z\mathbf{i}_z) \times \mathbf{i}_x = -y\mathbf{i}_z + z\mathbf{i}_y \quad (1.9.30)$$

$$\mathbf{r} \times \mathbf{i}_y = (x\mathbf{i}_x + y\mathbf{i}_y + z\mathbf{i}_z) \times \mathbf{i}_y = +x\mathbf{i}_z - z\mathbf{i}_x \quad (1.9.31)$$

$$\mathbf{r} \times \mathbf{i}_z = (x\mathbf{i}_x + y\mathbf{i}_y + z\mathbf{i}_z) \times \mathbf{i}_z = -x\mathbf{i}_y + y\mathbf{i}_x \quad (1.9.32)$$

Using Eqs. (1.9.24) and (1.9.29)–(1.9.32) in Eq. (1.9.28) yields

$$\begin{aligned} \mathbf{M}_0 = & \int_{z=-h}^0 \int_{y=l+(l/h)z}^l \rho g z (y\mathbf{i}_z - z\mathbf{i}_y) dy dz \\ & + \int_{z=-h}^0 \int_{x=w+(w/h)z}^w \rho g z (z\mathbf{i}_x - x\mathbf{i}_z) dx dz \\ & + \int_{y=0}^l \int_{x=w-(w/l)y}^w \rho g z (y\mathbf{i}_x - x\mathbf{i}_y) dx dy \end{aligned} \quad (1.9.33)$$

The first surface integral in Eq. (1.9.33) can be evaluated directly to give

$$\begin{aligned} & \int_{z=-h}^0 \int_{y=l+(l/h)z}^l \rho g z (y\mathbf{i}_z - z\mathbf{i}_y) dy dz \\ & = \int_{z=-h}^0 \rho g z \left\{ \frac{1}{2} l^2 \mathbf{i}_z - \frac{1}{2} [l + (l/h)z]^2 \mathbf{i}_z - lz\mathbf{i}_y + [l + (l/h)z]z\mathbf{i}_y \right\} dz \\ & = \int_{z=-h}^0 \rho g \left\{ -[(l^2/h)z^2 + \frac{1}{2}(l/h)^2 z^3] \mathbf{i}_z + (l/h)z^3 \mathbf{i}_y \right\} dz \\ & = \rho g \left\{ [(l^2/h) \frac{1}{3} (-h)^3 + \frac{1}{2} (l/h)^2 \frac{1}{4} (-h)^4] \mathbf{i}_z - (l/h) \frac{1}{4} (-h)^4 \mathbf{i}_y \right\} \\ & = -\frac{5}{24} \rho g l^2 h^2 \mathbf{i}_z - \frac{1}{4} \rho g l h^3 \mathbf{i}_y \end{aligned} \quad (1.9.34)$$

Similarly, the second surface integral in Eq. (1.9.33) gives

$$\begin{aligned}
 & \int_{z=-h}^0 \int_{x=w+(w/h)z}^w \rho g z (z \mathbf{i}_x - x \mathbf{i}_z) dx dz \\
 &= \int_{z=-h}^0 \rho g z \{ w z \mathbf{i}_x - [w + (w/h)z] z \mathbf{i}_x - \frac{1}{2} w^2 \mathbf{i}_z + \frac{1}{2} [w + (w/h)z]^2 \mathbf{i}_z \} dz \\
 &= \int_{z=-h}^0 \rho g \{ -(w/h) z^3 \mathbf{i}_x + [(w^2/h) z^2 + \frac{1}{2} (w/h)^2 z^3] \mathbf{i}_z \} dz \quad (1.9.35) \\
 &= \rho g \{ (w/h) \frac{1}{4} (-h)^4 \mathbf{i}_x - [(w^2/h) \frac{1}{3} (-h)^3 + \frac{1}{2} (w/h)^2 \frac{1}{4} (-h)^4] \mathbf{i}_z \} \\
 &= \frac{1}{4} \rho g w h^3 \mathbf{i}_x + \frac{5}{24} \rho g w^2 h^2 \mathbf{i}_z
 \end{aligned}$$

In Eq. (1.9.34) we were integrating a function of only y and z with respect to y and z , and in Eq. (1.9.35) we were integrating a function of only x and z with respect to x and z . However, to evaluate the third surface integral in Eq. (1.9.33) we must first use Eq. (1.9.23) to express z in terms of x and y , which gives

$$\begin{aligned}
 & \int_{y=0}^l \int_{x=w-(w/l)y}^w \rho g z (y \mathbf{i}_x - x \mathbf{i}_y) dx dy \\
 &= \int_{y=0}^l \int_{x=w-(w/l)y}^w \rho g [-2h + (h/w)x + (h/l)y] (y \mathbf{i}_x - x \mathbf{i}_y) dx dy \\
 &= \int_{y=0}^l \rho g \left[\frac{1}{2} (wh/l^2) y^3 - (wh/l) y^2 \right] \mathbf{i}_x dy \quad (1.9.36) \\
 &\quad + \int_{y=0}^l \rho g \left[(w^2 h/l) y - (w^2 h/l^2) y^2 + \frac{1}{6} (w^2 h/l^3) y^3 \right] \mathbf{i}_y dy \\
 &= -\frac{5}{24} \rho g w l^2 h \mathbf{i}_x + \frac{5}{24} \rho g w^2 h l \mathbf{i}_y
 \end{aligned}$$

Using Eqs. (1.9.34)–(1.9.36) in Eq. (1.9.33) yields

$$\begin{aligned}
 \mathbf{M}_0 = & \rho g w h \left(\frac{1}{4} h^2 - \frac{5}{24} l^2 \right) \mathbf{i}_x + \rho g l h \left(\frac{5}{24} w^2 - \frac{1}{4} h^2 \right) \mathbf{i}_y \\
 & + \rho g h^2 \left(\frac{5}{24} w^2 - \frac{5}{24} l^2 \right) \mathbf{i}_z \quad (1.9.37)
 \end{aligned}$$

Likewise, as specified in Eq. (1.9.12), the resultant moment about the origin can be written as the cross product of the position vector for the center of gage pressure $\mathbf{r}_{cp} = x_{cp} \mathbf{i}_x + y_{cp} \mathbf{i}_y + z_{cp} \mathbf{i}_z$ and the net hydrostatic force resulting from the water and air combined, which is given in Eq. (1.9.26),

$$\begin{aligned}
 \mathbf{M}_0 = & \mathbf{r}_{cp} \times \mathbf{F} = (x_{cp} \mathbf{i}_x + y_{cp} \mathbf{i}_y + z_{cp} \mathbf{i}_z) \\
 & \times \left(\frac{1}{3} \rho g l h^2 \mathbf{i}_x + \frac{1}{3} \rho g w h^2 \mathbf{i}_y - \frac{1}{3} \rho g h w l \mathbf{i}_z \right) \\
 = & \frac{1}{3} \rho g h \begin{vmatrix} \mathbf{i}_x & \mathbf{i}_y & \mathbf{i}_z \\ x_{cp} & y_{cp} & z_{cp} \\ lh & wh & -wl \end{vmatrix} \quad (1.9.38) \\
 = & \frac{1}{3} \rho g h [w(-l y_{cp} - h z_{cp}) \mathbf{i}_x + l(h z_{cp} + w x_{cp}) \mathbf{i}_y + h(w x_{cp} - l y_{cp}) \mathbf{i}_z]
 \end{aligned}$$

Equating moments from Eqs. (1.9.37) and (1.9.38) yields a vector equation with three scalar components that must be satisfied independently,

$$\begin{aligned} & \frac{1}{3} \rho g w h (-l y_{cp} - h z_{cp}) \mathbf{i}_x + \frac{1}{3} \rho g l h (h z_{cp} + w x_{cp}) \mathbf{i}_y + \frac{1}{3} \rho g h^2 (w x_{cp} - l y_{cp}) \mathbf{i}_z \\ & = \rho g w h \left(\frac{1}{4} h^2 - \frac{5}{24} l^2 \right) \mathbf{i}_x + \rho g l h \left(\frac{5}{24} w^2 - \frac{1}{4} h^2 \right) \mathbf{i}_y + \rho g h^2 \left(\frac{5}{24} w^2 - \frac{5}{24} l^2 \right) \mathbf{i}_z \end{aligned}$$

These three scalar equations can be simplified to the linear system

$$\begin{bmatrix} 0 & l & h \\ w & 0 & h \\ w & -l & 0 \end{bmatrix} \begin{bmatrix} x_{cp} \\ y_{cp} \\ z_{cp} \end{bmatrix} = \begin{bmatrix} \frac{5}{8} l^2 - \frac{3}{4} h^2 \\ \frac{5}{8} w^2 - \frac{3}{4} h^2 \\ \frac{5}{8} w^2 - \frac{5}{8} l^2 \end{bmatrix} \quad (1.9.39)$$

At first thought, one might think that the three scalar equations in Eq. (1.9.39) could be solved for the three scalar coordinates of the center of pressure x_{cp} , y_{cp} , and z_{cp} . Each of the three scalar equations in Eq. (1.9.39) is the equation of a plane. In general, we should expect any two nonparallel planes to intersect in a straight line. The intersection of this straight line with a third plane that is not parallel with the line would be a point. However, it should be noted that the third equation in Eq. (1.9.39) can be obtained by simply subtracting the first equation from the second. Hence, the three equations represented by Eq. (1.9.39) are not linearly independent and this linear system has an infinite number of solutions. Because the two linearly independent equations in Eq. (1.9.39) are the equations for two nonparallel planes, which intersect in a straight line, **the locus of points satisfying Eq. (1.9.39) is a straight line traditionally called the line of action**. Every point on this line of action satisfies our definition for the center of pressure.

To isolate a center of pressure as a single point along the line of action, some additional piece of information is required. For any *flat surface*, it is customary to define the center of pressure to be the intersection of the line of action with the plane of the surface. Thus, with this additional defined constraint, the three scalar coordinates of the center of pressure x_{cp} , y_{cp} , and z_{cp} can be located by replacing the third equation in Eq. (1.9.39) with the equation for the plane of the surface, which in this case is given by Eq. (1.9.23). The resulting system simplifies to

$$\begin{bmatrix} 0 & l & h \\ w & 0 & h \\ 1/w & 1/l & -1/h \end{bmatrix} \begin{bmatrix} x_{cp} \\ y_{cp} \\ z_{cp} \end{bmatrix} = \begin{bmatrix} \frac{5}{8} l^2 - \frac{3}{4} h^2 \\ \frac{5}{8} w^2 - \frac{3}{4} h^2 \\ 2 \end{bmatrix} \quad (1.9.40)$$

The solution to Eq. (1.9.40) is readily obtained by Gauss elimination

$$x_{cp} = \frac{5}{8} w, \quad y_{cp} = \frac{5}{8} l, \quad z_{cp} = -\frac{3}{4} h \quad (1.9.41)$$

Example 1.9.3 demonstrates that **the classical definition for the center of pressure mathematically defines a line not a point**. Formally, we have the following definition:

The **line of action** is the locus of points about which the resultant moment produced by the distributed pressure load acting on a submerged surface is zero.

Solutions to the two 3×3 linear systems of equations that are given in Eqs. (1.9.39) and (1.9.40) are displayed graphically in Fig. 1.9.4. The plane labeled $M_x = 0$ is the locus of points about which the x component of the resultant moment is zero. Similarly, the planes labeled $M_y = 0$ and $M_z = 0$ are the loci of points about which the y and z components of the resultant moment are zero, respectively. The three components of Eq. (1.9.39) are the equations for these three planes. Notice that all three of these planes intersect along the same line, which is the line of action. On the other hand, the three planes described in Eq. (1.9.40) have only a single point in common, the center of pressure.

The center-of-pressure concept is important in the study of fluid dynamics as well as in fluid statics. The relations given in Eqs. (1.9.11)–(1.9.15), which are commonly used to locate the center of pressure, apply to fluids in motion as well as to fluids at rest. **For the curved surfaces typically encountered on an aircraft, the center of pressure is not traditionally defined to be the intersection of the line of action with aircraft's surface.** For the pressure distributions encountered on an aircraft in flight, the line of action will commonly intersect the aircraft's surface at more than one point, and in some cases the line of action may not intersect the aircraft's surface at all. Two such cases are shown in Fig. 1.9.5 for flow over a commonly used airfoil.

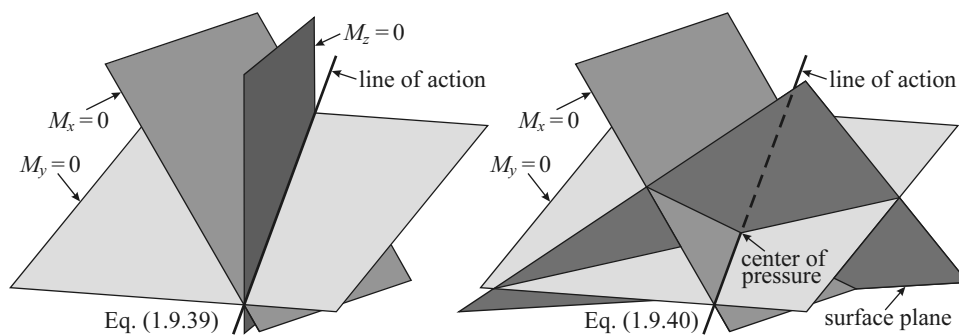


Figure 1.9.4. Geometric illustrations of the two linear systems given in Eqs. (1.9.39) and (1.9.40).

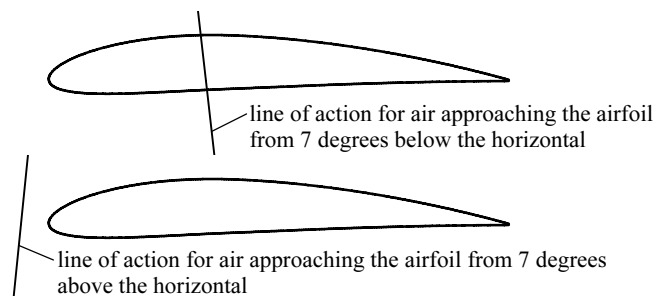


Figure 1.9.5. Line of action for the pressure acting on an airfoil at two different flow conditions.

Summary: The *center of pressure* is defined to be a point about which the resultant moment produced by the distributed pressure load acting on a submerged surface is zero. There are always an infinite number of points that satisfy this definition for any pressure distribution about any surface. The locus of points about which this resultant moment is zero is always a straight line called the *line of action*. For a planar surface, the center of pressure is traditionally chosen to be the intersection of the line of action with the plane of the surface. For a curved surface, there is no universally accepted method for isolating a center of pressure as a single point on the line of action. For a curved surface, the line of action could intersect the surface at several points. For a closed surface, the line of action for *hydrostatic pressure* will intersect the surface at no less than two points. For fluids in motion, the line of action for the resultant pressure force may not intersect the surface at any point. ***The only universal requirement for choosing a single-point center of pressure for a curved surface is that it must fall on the line of action.***

1.10. The Buoyant Force

In Examples 1.8.1 and 1.8.2, the ambient air pressure was approximated as constant over the entire outside surface of the aquarium. In reality, the air pressure at the bottom of the aquarium is slightly greater than that at the top. Neglecting this pressure difference is a reasonable approximation for these particular examples, because the density of air is negligible compared with that of water. For both of these examples, the total net force acting on the aquarium as a result of contact with the water and air was found to be equal to the weight of the water in the aquarium. A more precise solution would predict a total net force on the aquarium that is slightly less than the weight of the water.

EXAMPLE 1.10.1. For the aquarium described in Example 1.8.1, determine the total resultant hydrostatic force exerted by the air on all faces of the aquarium and the free surface of the water combined. Assume that the constant acceleration of gravity provides the only body force, that the air has a constant density ρ_a , and that the ambient pressure at the free surface of the water is p_0 . Do not assume that the air pressure is constant as was done in Example 1.8.1.

Solution. Using the same coordinate system that was used in Example 1.8.1 and shown in Fig. 1.8.2, from Eq. (1.5.16) the hydrostatic pressure in the air is

$$p_a = p_0 - \rho_a g z$$

For the back outside surface of the aquarium, the unit outward normal points in the negative x direction. From Eq. (1.8.2) and the geometry shown in Fig. 1.8.2, the surface geometry, outward unit normal, differential area, and hydrostatic force are

$$x = 0, \quad 0 \leq y \leq l, \quad -h \leq z \leq 0, \quad \mathbf{n} = -\mathbf{i}_x, \quad d\mathcal{S} = dy dz$$

$$\begin{aligned} \mathbf{F}_{\text{back}} &= - \iint_{\mathcal{S}} p_a \mathbf{n} d\mathcal{S} = - \int_{z=-h}^0 \int_{y=0}^l (p_0 - \rho_a g z) (-\mathbf{i}_x) dy dz \\ &= (p_0 + \frac{1}{2} \rho_a g h) l h \mathbf{i}_x \end{aligned}$$

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Similarly, on the outside surface of the front glass, the surface geometry, outward unit normal, differential area, and hydrostatic force are

$$x = w, \quad 0 \leq y \leq l, \quad -h \leq z \leq 0, \quad \mathbf{n} = \mathbf{i}_x, \quad d\mathcal{S} = dy dz$$

$$\begin{aligned} \mathbf{F}_{\text{front}} &= - \iint_{\mathcal{S}} p_a \mathbf{n} d\mathcal{S} = - \int_{z=-h}^0 \int_{y=0}^l (p_0 - \rho_a g z) \mathbf{i}_x dy dz \\ &= -(p_0 + \frac{1}{2} \rho_a g h) l h \mathbf{i}_x \end{aligned}$$

For the outside left surface

$$y = 0, \quad 0 \leq x \leq w, \quad -h \leq z \leq 0, \quad \mathbf{n} = -\mathbf{i}_y, \quad d\mathcal{S} = dx dz$$

$$\begin{aligned} \mathbf{F}_{\text{left}} &= - \iint_{\mathcal{S}} p_a \mathbf{n} d\mathcal{S} = - \int_{z=-h}^0 \int_{x=0}^w (p_0 - \rho_a g z) (-\mathbf{i}_y) dx dz \\ &= (p_0 + \frac{1}{2} \rho_a g h) w h \mathbf{i}_y \end{aligned}$$

For the outside right surface

$$y = l, \quad 0 \leq x \leq w, \quad -h \leq z \leq 0, \quad \mathbf{n} = \mathbf{i}_y, \quad d\mathcal{S} = dx dz$$

$$\begin{aligned} \mathbf{F}_{\text{right}} &= - \iint_{\mathcal{S}} p_a \mathbf{n} d\mathcal{S} = - \int_{z=-h}^0 \int_{x=0}^w (p_0 - \rho_a g z) \mathbf{i}_y dx dz \\ &= -(p_0 + \frac{1}{2} \rho_a g h) w h \mathbf{i}_y \end{aligned}$$

For the outside bottom surface

$$z = -h, \quad 0 \leq x \leq w, \quad 0 \leq y \leq l, \quad \mathbf{n} = -\mathbf{i}_z, \quad d\mathcal{S} = dx dy$$

$$\begin{aligned} \mathbf{F}_{\text{bottom}} &= - \iint_{\mathcal{S}} p_a \mathbf{n} d\mathcal{S} = - \int_{y=0}^l \int_{x=0}^w (p_0 + \rho_a g h) (-\mathbf{i}_z) dx dy \\ &= (p_0 + \rho_a g h) w l \mathbf{i}_z \end{aligned}$$

For the top free surface of the water

$$z = 0, \quad 0 \leq x \leq w, \quad 0 \leq y \leq l, \quad \mathbf{n} = \mathbf{i}_z, \quad d\mathcal{S} = dx dy$$

$$\begin{aligned} \mathbf{F}_{\text{top}} &= - \iint_{\mathcal{S}} p_a \mathbf{n} d\mathcal{S} = - \int_{y=0}^l \int_{x=0}^w p_0 \mathbf{i}_z dx dy \\ &= -p_0 w l \mathbf{i}_z \end{aligned}$$

Adding the hydrostatic forces acting on all five outside surfaces of the aquarium and the free surface of the water yields the **total hydrostatic force acting on the aquarium and water as a result of contact with the air**,

$$\mathbf{F}_{\text{total}} = \underline{\rho_a g h w l \mathbf{i}_z}$$

Notice that the magnitude of this net force is the product of the air density, the aquarium volume, and the acceleration of gravity. This is the weight of the air that is displaced by the aquarium filled with water. This force acts in the $+\mathbf{i}_z$ direction, i.e., up.

Combining this result with the result obtained in Example 1.8.1, we see that the water exerts a downward force equal to the weight of the water, and when the pressure variation in the air is included in the analysis, the air exerts an upward force equal to the weight of the air displaced. This upward force exerted by the surrounding air is called the *buoyant force*.

For this particular example, the buoyant force is negligible compared with the weight of the water. One cubic foot of water weighs about 62.4 lbf and one cubic foot of air at standard sea level weighs only 0.0765 lbf. Thus, the buoyant force exerted by the air is only 0.12% of the hydrostatic force exerted by the water.

Although the buoyant force is negligible in the previous example, this is not always the case. If the surrounding fluid had a higher density or the liquid in the aquarium had a lower density, the buoyant force could be important. The buoyant force is what keeps ships afloat and hot-air balloons aloft.

For the solution obtained in Example 1.10.1, the air density and the acceleration of gravity were approximated as being constant. Within these approximations, the buoyant force acting on a closed surface submerged in a static fluid was found to be equal to the weight of the fluid displaced. It can be shown that this result is more general.

Consider the arbitrary closed surface $\mathcal{S}$ enclosing the volume $\mathcal{V}$ within a static fluid, which is shown in Fig. 1.10.1. The same static fluid both fills and surrounds this closed surface. Because the fluid is static, the vector sum of the body forces and the surface forces must be zero. If gravity is the only body force, from Eq. (1.5.8) the body force acting on the differential volume $d\mathcal{V}$ is

$$d\mathbf{F}_{\text{gravity}} = -\rho g \nabla H d\mathcal{V} \quad (1.10.1)$$

and the total gravitational force acting on the fluid enclosed in the surface $\mathcal{S}$ is

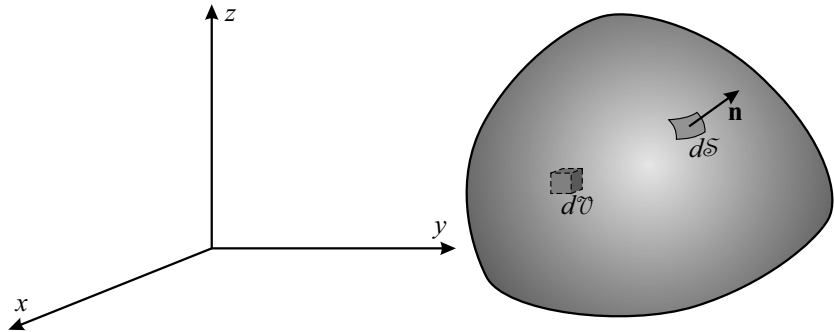


Figure 1.10.1. Geometry for evaluating the buoyant force on a closed submerged surface.

$$\mathbf{F}_{\text{gravity}} = -\nabla H \iiint_{\mathcal{V}} \rho g d\mathcal{V} \quad (1.10.2)$$

From Eq. (1.8.2), the total hydrostatic buoyant force exerted on the surface $\mathcal{S}$ by the surrounding static fluid is

$$\mathbf{F}_{\text{buoyant}} = - \iint_{\mathcal{S}} p \mathbf{n} d\mathcal{S} \quad (1.10.3)$$

Because the fluid is static, the vector sum of the gravitational force and the buoyant force must be zero, which gives

$$\mathbf{F}_{\text{buoyant}} = -\mathbf{F}_{\text{gravity}} \quad (1.10.4)$$

Combining Eqs. (1.10.2), (1.10.3), and (1.10.4) shows that **the buoyant force acting on any closed surface submerged in a static fluid is an upward force equal to the weight of the fluid displaced**

$$\mathbf{F}_{\text{buoyant}} = - \iint_{\mathcal{S}} p \mathbf{n} d\mathcal{S} = \nabla H \iiint_{\mathcal{V}} \rho g d\mathcal{V} = -\mathbf{W}_{\text{displaced}} \quad (1.10.5)$$

where $\mathbf{W}_{\text{displaced}}$ is the weight vector for the fluid displaced by the surface $\mathcal{S}$ enclosing the volume $\mathcal{V}$. This is commonly referred to as *Archimedes' principle*, which is named for the famous Greek mathematician and physicist who lived 287–212 BC.

It should be noted that Eq. (1.10.5) was developed for the special case where the surface $\mathcal{S}$ encloses the same static fluid that surrounds it. However, Eq. (1.10.3) shows that the buoyant force acting on this surface depends only on the surface geometry and the fluid pressure. Because the pressure in a static fluid depends on only the elevation, Eq. (1.10.3) shows that the buoyant force acting on any closed surface does not depend on what the surface encloses. Equation (1.10.5) could be applied to the surface enclosing a submarine, a weather balloon, or any other object submerged in any static fluid. It should be emphasized that Eq. (1.10.5) does not require that either the fluid density or the acceleration of gravity remain constant over the closed surface $\mathcal{S}$.

EXAMPLE 1.10.2. An atmospheric weather station is supported by a balloon that contains 17,000 ft³ of helium at standard temperature and pressure (288.150 K and 101,325 N/m²), which is enclosed by a thin membrane. Determine the magnitude of the weight and buoyant force acting on the helium at standard sea level. The ideal gas constant for air is 287.0528 J/kg·K = 287.0528 N·m/kg·K and that for helium is 2077.0602 J/kg·K = 2077.0602 N·m/kg·K.

Solution. Using the ideal gas equation of state given by Eq. (1.2.6), the density and mass of the helium are

$$\begin{aligned} \rho_{\text{He}} &= \frac{p}{R_{\text{He}} T} = \frac{101,325 \text{ N/m}^2}{(2077.0602 \text{ N·m/kg·K})(288.150 \text{ K})} = 0.16930 \text{ kg/m}^3 \\ &= 0.16930 \text{ kg/m}^3 (0.001940320 \text{ slug/ft}^3 / \text{kg/m}^3) = 0.00032849 \text{ slug/ft}^3 \end{aligned}$$

$$m_{\text{He}} = \rho_{\text{He}} \mathcal{V}_{\text{He}} = (0.00032849 \text{ slug/ft}^3)(17,000 \text{ ft}^3) = 5.5843 \text{ slugs}$$

The acceleration of gravity at standard sea level is

$$g = g_o = 9.806645 \text{ m/s}^2 = \frac{9.806645 \text{ m/s}^2}{0.3048 \text{ m/ft}} = 32.174 \text{ ft/s}^2$$

Thus, the weight of the helium at standard sea level is

$$W_{\text{He}} = m_{\text{He}} g = (5.5843 \text{ slugs})(32.174 \text{ ft/s}^2) = \underline{179.7 \text{ lbf}}$$

Similarly, the buoyant force is obtained from the density, mass, and weight of the displaced air

$$\begin{aligned} \rho_{\text{air}} &= \frac{p}{R_{\text{air}} T} = \frac{101,325 \text{ N/m}^2}{(287.0528 \text{ N}\cdot\text{m/kg}\cdot\text{K})(288.150 \text{ K})} = 1.2250 \text{ kg/m}^3 \\ &= 1.2250 \text{ kg/m}^3 (0.001940320 \text{ slug/ft}^3 / \text{kg/m}^3) = 0.0023769 \text{ slug/ft}^3 \end{aligned}$$

$$m_{\text{air displaced}} = \rho_{\text{air}} \mathcal{V}_{\text{He}} = (0.0023769 \text{ slug/ft}^3)(17,000 \text{ ft}^3) = 40.41 \text{ slugs}$$

$$F_{\text{buoyant}} = W_{\text{air displaced}} = m_{\text{air}} g = (40.41 \text{ slugs})(32.174 \text{ ft/s}^2) = \underline{1300 \text{ lbf}}$$

Hence, the helium will support about 1120 lbf, in addition to its own weight.

EXAMPLE 1.10.3. For the balloon described in Example 1.10.2, determine the magnitude of the weight and buoyant force that would act on the helium when submerged in the standard atmosphere at an altitude of 100,000 ft above mean sea level, if the helium could expand to the same temperature and pressure as the atmosphere (408.572 R = 226.985 K and 23.272 lbf/ft² = 1114.27 N/m²).

Solution. From Example 1.10.2, the mass of the helium is 5.5843 slugs. Using the ideal gas equation of state, the density and volume of the helium are

$$\begin{aligned} \rho_{\text{He}} &= \frac{p}{R_{\text{He}} T} = \frac{1114.27 \text{ N/m}^2}{(2077.0602 \text{ N}\cdot\text{m/kg}\cdot\text{K})(226.985 \text{ K})} = 0.0023634 \text{ kg/m}^3 \\ &= 0.0023634 \text{ kg/m}^3 (0.001940320 \text{ slug/ft}^3 / \text{kg/m}^3) = 4.5858 \times 10^{-6} \text{ slug/ft}^3 \end{aligned}$$

$$\mathcal{V}_{\text{He}} = m_{\text{He}} / \rho_{\text{He}} = 5.5843 \text{ slugs} / 4.5858 \times 10^{-6} \text{ slug/ft}^3 = 1.2177 \times 10^6 \text{ ft}^3$$

The acceleration of gravity at 100,000 ft MSL is

$$g = g_o \left(\frac{R_E}{R_E + H} \right)^2 = \frac{9.806645 \text{ m/s}^2}{0.3048 \text{ m/ft}} \left(\frac{6,356,766 \text{ m}}{6,356,766 \text{ m} + 30,480 \text{ m}} \right)^2 = 31.868 \text{ ft/s}^2$$

Thus, the weight of the helium at 100,000 ft MSL is

$$W_{\text{He}} = m_{\text{He}}g = (5.5843 \text{ slugs})(31.868 \text{ ft/s}^2) = \underline{178.0 \text{ lbf}}$$

Similarly, the buoyant force is obtained from the density, mass, and weight of the displaced air

$$\begin{aligned} \rho_{\text{air}} &= \frac{p}{R_{\text{air}}T} = \frac{1114.27 \text{ N/m}^2}{(287.0528 \text{ N}\cdot\text{m/kg}\cdot\text{K})(226.985 \text{ K})} = 0.017101 \text{ kg/m}^3 \\ &= 0.017101 \text{ kg/m}^3 (0.001940320 \text{ slug/ft}^3 / \text{kg/m}^3) = 3.3182 \times 10^{-5} \text{ slug/ft}^3 \end{aligned}$$

$$m_{\text{air displaced}} = \rho_{\text{air}} \mathcal{V}_{\text{He}} = (3.3182 \times 10^{-5} \text{ slug/ft}^3)(1.2177 \times 10^6 \text{ ft}^3) = 40.41 \text{ slugs}$$

$$F_{\text{buoyant}} = W_{\text{air displaced}} = m_{\text{air}}g = (40.41 \text{ slugs})(31.868 \text{ ft/s}^2) = \underline{1288 \text{ lbf}}$$

Note that the helium displaces the same mass of air at 100,000 ft as it does at sea level. The 12-lbf reduction in the buoyant force results entirely from a decrease in the acceleration of gravity.

EXAMPLE 1.10.4. A 1/32-scale model of a transport airplane weighs 87.61 lbf in air at standard sea level (0.0023769 slugs/ft³) and it weighs 55.19 lbf when immersed in fresh water at standard sea level (1.94032 slugs/ft³). Estimate the buoyant force acting on the full-scale airplane at standard sea level.

Solution. The apparent weight of any object immersed in a fluid is the actual weight less the buoyant force. Hence, we can write the following relations for the weight of the model in air and water, respectively,

$$W_{\text{model}} - \mathcal{V}_{\text{model}} \rho_{\text{air}} g = W_{\text{in air}} \quad (1.10.6)$$

$$W_{\text{model}} - \mathcal{V}_{\text{model}} \rho_{\text{water}} g = W_{\text{in water}} \quad (1.10.7)$$

Subtracting Eq. (1.10.7) from Eq. (1.10.6) and solving the result for the volume of the model yields

$$\begin{aligned} \mathcal{V}_{\text{model}} &= \frac{W_{\text{in air}} - W_{\text{in water}}}{(\rho_{\text{water}} - \rho_{\text{air}})g} = \frac{87.61 \text{ lbf} - 55.19 \text{ lbf}}{(1.94032 \text{ slug/ft}^3 - 0.0023769 \text{ slug/ft}^3)32.174 \text{ ft/s}^2} \\ &= 0.519956 \text{ ft}^3 \end{aligned}$$

Thus, the volume and buoyant force for the full-scale airplane are

$$\mathcal{V}_{\text{airplane}} = (32)^3 \mathcal{V}_{\text{model}} = (32)^3 0.519956 \text{ ft}^3 = 17,038 \text{ ft}^3$$

$$F_{\text{buoyant}} = \mathcal{V}_{\text{airplane}} \rho_{\text{air}} g = 17,038 \text{ ft}^3 (0.0023769 \text{ slug/ft}^3) 32.174 \text{ ft/s}^2 = \underline{1303 \text{ lbf}}$$

1.11. Buoyancy of Gases in the Atmosphere

In Examples 1.10.2 and 1.10.3, we found that 17,000 ft³ of helium at the temperature and pressure of the standard atmosphere at sea level will occupy more than 1.2 million ft³ when expanded to the temperature and pressure of the standard atmosphere 100,000 ft above mean sea level. As the helium expands from sea level to 100,000 ft, it displaces an increasing volume of atmospheric air. A significant amount of work is required to displace this air. The energy required to perform the work necessary to displace the air must come from the helium and/or the surrounding air. In this section we shall examine how work is done by an expanding homogeneous gas.

Consider first a fixed mass of some gas contained by a cylinder and piston, as shown in Fig. 1.11.1. Let x denote the coordinate distance measured from the closed end of the cylinder parallel with the piston's travel, as shown in Fig. 1.11.1. We wish to determine the work done on the piston by the gas as the piston moves slowly from x_1 to x_2 . The work done by the gas as the piston moves a differential distance dx is the product of the force on the piston and the distance traveled by the piston. The total force exerted on the piston by the gas is pS_c , where p is the gas pressure and S_c is the cross-sectional area of the cylinder (i.e., the contact area between the piston and gas). Hence, the differential work $d\tilde{w}$ that is done by the gas as the piston moves a differential distance dx is

$$d\tilde{w} = pS_c dx \quad (1.11.1)$$

However, the product $S_c dx$ is equal to the differential change in the volume of the gas $d\tilde{v}$. Thus, Eq. (1.11.1) can be written equivalently as

$$d\tilde{w} = p d\tilde{v} \quad (1.11.2)$$

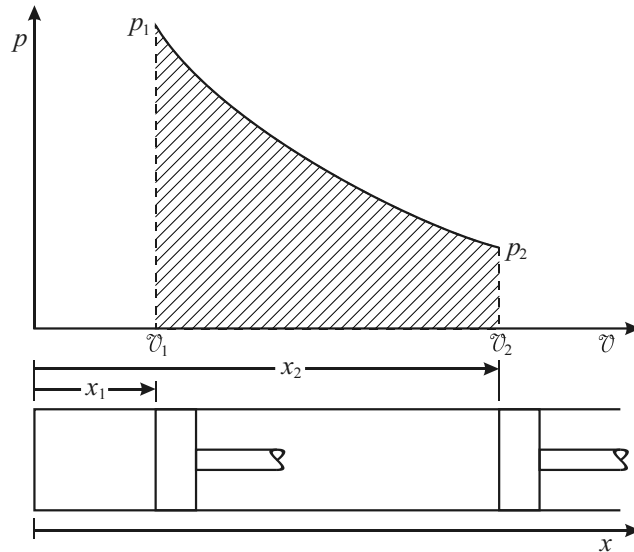


Figure 1.11.1. Pressure-volume diagram showing the work done by an expanding gas.

The volume of any homogeneous substance is the product of its mass m and its specific volume v . Hence, a differential change in the volume of a fixed mass of any homogeneous fluid can be expressed as the product of its mass and a differential change in its specific volume, $d\mathcal{V} = m dv$. Using this relation in Eq. (1.11.2) yields

$$d\tilde{w} = p d\mathcal{V} = m p dv \quad (1.11.3)$$

If the fluid is an ideal gas, $v = RT/p$, and

$$dv = \frac{R}{p} dT - \frac{RT}{p^2} dp \quad (1.11.4)$$

where R is the ideal gas constant and T is the absolute temperature. Using Eq. (1.11.4) in Eq. (1.11.3) yields

$$d\tilde{w} = p d\mathcal{V} = m R dT - m \frac{RT}{p} dp \quad (1.11.5)$$

The total work done on the piston by the gas as the piston moves slowly from x_1 to x_2 is found by integrating Eq. (1.11.5)

$${}_1\tilde{w}_2 = \int_{\mathcal{V}_1}^{\mathcal{V}_2} p d\mathcal{V} = m R \int_{T_1}^{T_2} dT - m R \int_{p_1}^{p_2} \frac{T}{p} dp \quad (1.11.6)$$

where $\mathcal{V}_1$ and $\mathcal{V}_2$ are the gas volumes with the piston located at x_1 and x_2 , respectively. The volume integral in Eq. (1.11.6) is the crosshatched area under the curve in the p - $\mathcal{V}$ diagram shown in Fig. 1.11.1. However, this integral cannot be evaluated analytically unless the relationship between p and $\mathcal{V}$ during the process is known. The temperature integral in Eq. (1.11.6) can be evaluated independent of the process. Hence, **the work done by an expanding or compressing homogeneous ideal gas is**

$${}_1\tilde{w}_2 = m R (T_2 - T_1) - m R \int_{p_1}^{p_2} \frac{T}{p} dp \quad (1.11.7)$$

To evaluate the remaining integral in Eq. (1.11.7) the relation between T and p during the process must be known.

Once the work done by the expanding gas is computed from Eq. (1.11.7), the heat transferred to the gas ${}_1\mathcal{Q}_2$ can be obtained from the first law of thermodynamics. For a fixed mass, the differential heat transfer $d\mathcal{Q}$ that is transferred to the gas as the piston moves a differential distance dx is

$$d\mathcal{Q} = m du + d\tilde{w} \quad (1.11.8)$$

where u is the specific internal energy. After applying Eq. (1.2.7), this yields

$$d\mathcal{Q} = m C_v dT + d\tilde{w} \quad (1.11.9)$$

Integrating Eq. (1.11.9) for the case of constant specific heat, the heat transfer to an expanding or compressing homogeneous ideal gas is

$${}_1Q_2 = mC_v(T_2 - T_1) + {}_1\tilde{w}_2 \quad (1.11.10)$$

EXAMPLE 1.11.1. The weather balloon described in Examples 1.10.2 and 1.10.3 is released at sea level and rises slowly through the standard atmosphere to an altitude of 100,000 ft above mean sea level. Assuming that the helium expands so as to always remain at the same temperature and pressure as the atmosphere, determine the work done by the helium on the atmosphere and the heat transferred to the helium from the atmosphere. The constant volume specific heat for helium is 3115.5903 J/kg·K.

Solution. From Examples 1.6.1 and 1.6.2, the geopotential altitude at 100,000 ft is

$$Z = \frac{R_E H}{R_E + H} = \frac{6,356,766 \text{ m}[100,000 \text{ ft}(0.3048 \text{ m/ft})]}{6,356,766 \text{ m} + 100,000 \text{ ft}(0.3048 \text{ m/ft})} = 30,335 \text{ m}$$

which is in the third of the altitude ranges given for the standard atmosphere in Table 1.6.1. In the first altitude range, temperature and pressure are related by

$$p = p_0 \left[\frac{T_0 + T'_0(Z - Z_0)}{T_0} \right]^{-\frac{g_0}{RT'_0}} = p_0 \left(\frac{T}{T_0} \right)^{-\frac{g_0}{RT'_0}} = p_0 \left(\frac{T}{T_0} \right)^{\frac{-9.806645 \text{ m/s}^2}{287.0528 \text{ m}^2/\text{s}^2 \cdot \text{K}(-0.0065 \text{ K/m})}}$$

where $p_0 = 101,325 \text{ N/m}^2$ and $T_0 = 288.150 \text{ K}$. After rearranging, we have

$$T = T_0 \left(\frac{p}{p_0} \right)^{0.1903} \quad \text{for } p_0 \geq p \geq p_1$$

where $p_1 = 22,632 \text{ N/m}^2$. In the second altitude range of the standard atmosphere, the temperature remains constant

$$T = T_1 = 216.650 \text{ K} \quad \text{for } p_1 \geq p \geq p_2$$

where $p_2 = 5,474.9 \text{ N/m}^2$. Similarly, in the third altitude range, we obtain

$$T = T_2 \left(\frac{p}{p_2} \right)^{-0.0293} \quad \text{for } p_2 \geq p \geq 868.02 \text{ N/m}^2$$

where $T_2 = 216.650 \text{ K}$. Using these relations in Eq. (1.11.7), the integration from state 0 at sea level to state 3 at 100,000 ft MSL is carried out in three parts

$${}_0\tilde{w}_3 = mR(T_3 - T_0) - mR \int_{p_0}^{p_1} \frac{T}{p} dp - mR \int_{p_1}^{p_2} \frac{T}{p} dp - mR \int_{p_2}^{p_3} \frac{T}{p} dp$$

$$\begin{aligned}
{}_0\tilde{w}_3 &= mR(T_3 - T_0) - mRT_0 \int_{p_0}^{p_1} \left(\frac{p}{p_0}\right)^{0.1903} \frac{dp}{p} - mRT_1 \int_{p_1}^{p_2} \frac{dp}{p} - mRT_2 \int_{p_2}^{p_3} \left(\frac{p}{p_2}\right)^{-0.0293} \frac{dp}{p} \\
&= mR \left[T_3 - T_0 - T_0 \frac{(p_1/p_0)^{0.1903} - 1}{0.1902632} - T_1 \ln \frac{p_2}{p_1} - T_2 \frac{(p_3/p_2)^{-0.0293} - 1}{-0.0292713} \right]
\end{aligned}$$

where m and R are, respectively, the mass and ideal gas constant for the helium. From Example 1.10.2, the mass of the helium is 5.5843 slugs (81.497 kg) and the ideal gas constant for the helium is 2077.0602 J/kg·K. From Example 1.6.2, the pressure and temperature at 100,000 ft are $p_3 = 1,114.3$ N/m² and $T_3 = 226.985$ K. Thus, we have

$$\begin{aligned}
{}_0\tilde{w}_3 &= 81.497 \times 2077.0602 \\
&\times \left[226.985 - 288.150 - 288.150 \frac{(22632/101325)^{0.1903} - 1}{0.1902632} \right. \\
&\quad \left. - 216.650 \ln \frac{5474.9}{22632} - 216.650 \frac{(1114.3/5474.9)^{-0.0293} - 1}{-0.0292713} \right] \\
&= \underline{1.6507 \times 10^8 \text{ J}}
\end{aligned}$$

The heat transfer to the helium can now be calculated from Eq. (1.11.10),

$$\begin{aligned}
{}_0Q_3 &= mC_v(T_3 - T_0) + {}_0\tilde{w}_3 \\
&= 81.497 \times 3115.5903(226.985 - 288.150) + 1.6507 \times 10^8 = \underline{1.4954 \times 10^8 \text{ J}}
\end{aligned}$$

In the previous example, it was assumed that the helium within the balloon remained homogeneous and in constant temperature and pressure equilibrium with the surrounding atmosphere. As the balloon rises, the helium expands in response to the decreasing atmospheric pressure. The expanding helium displaces the surrounding atmosphere. This displacement of the atmosphere requires energy in the form of work done on the atmosphere by the helium. If no other energy transfer took place, the work done on the atmosphere by the helium would result in a significant reduction in the temperature of the helium. To maintain temperature equilibrium with the surrounding atmosphere, some of the energy lost from the helium as work must be returned to the helium as heat transfer. In this particular example, about 2025 kJ/kg are lost from the helium as work, but more than 90 percent of that energy must be returned to the helium as heat transfer to maintain temperature equilibrium with the surrounding atmosphere.

In reality, gases have low thermal conductivity and little heat is actually transferred from the atmospheric air to a rising balloon. A more realistic model for the expansion process in a rising balloon is obtained by treating the process as adiabatic. In the absence of heat transfer, Eq. (1.11.9) requires

$$d\tilde{w} = -mC_v dT \quad (1.11.11)$$

Using Eq. (1.11.11) in Eq. (1.11.5), rearranging, and applying Eq. (1.2.15) yields

$$\frac{dT}{T} = \frac{R}{C_v + R} \frac{dp}{p} = \frac{R/C_v}{1 + R/C_v} \frac{dp}{p} = \frac{\gamma - 1}{\gamma} \frac{dp}{p} \quad (1.11.12)$$

where γ is the specific heat ratio defined in Eq. (1.2.15). For constant γ , Eq. (1.11.12) is easily integrated to give

$$T = T_1 \left(\frac{p}{p_1} \right)^{\frac{\gamma-1}{\gamma}} \quad (1.11.13)$$

where T_1 and p_1 are, respectively, the temperature and pressure at the beginning of the process. Using Eq. (1.11.13) in Eq. (1.11.7) produces the result

$$\begin{aligned} {}_1\tilde{w}_2 &= mR(T_2 - T_1) - mR \int_{p_1}^{p_2} \frac{T}{p} dp = mR(T_2 - T_1) - mRT_1 \int_{p_1}^{p_2} \left(\frac{p}{p_1} \right)^{\frac{\gamma-1}{\gamma}} \frac{dp}{p} \\ &= mR(T_2 - T_1) - \frac{\gamma mRT_1 [(p_2/p_1)^{(\gamma-1)/\gamma} - 1]}{\gamma - 1} \end{aligned} \quad (1.11.14)$$

From Eq. (1.11.13), $T_1(p_2/p_1)^{(\gamma-1)/\gamma} = T_2$, and Eq. (1.11.14) can be simplified to yield **the work done by an expanding or compressing homogeneous adiabatic ideal gas**

$${}_1\tilde{w}_2 = \frac{mR(T_1 - T_2)}{\gamma - 1} \quad (1.11.15)$$

EXAMPLE 1.11.2. Rework Example 1.11.1, assuming that the helium expands in an adiabatic process as it rises from mean sea level to 100,000 ft above mean sea level. Also recompute the buoyant force acting on the helium at 100,000 ft under this same assumption. Compare this result with that obtained in Example 1.10.3. For helium use $\gamma = 5/3$.

Solution. From Example 1.6.2, the pressure at 100,000 ft is 1,114.3 N/m² and from Eq. (1.11.13), the temperature of the helium when it reaches an altitude of 100,000 ft above mean sea level would be

$$T_2 = T_1 \left(\frac{p_2}{p_1} \right)^{\frac{\gamma-1}{\gamma}} = 288.150 \text{ K} \left(\frac{1,114.3}{101,325} \right)^{0.4} = 47.439 \text{ K}$$

Using this result in Eq. (1.11.15), yields

$$\begin{aligned} {}_1\tilde{w}_2 &= \frac{mR(T_1 - T_2)}{\gamma - 1} = \frac{(81.497 \text{ kg})(2077.0602 \text{ J/kg}\cdot\text{K})(288.150 \text{ K} - 47.439 \text{ K})}{2/3} \\ &= \underline{6.1119 \times 10^7 \text{ J}} \end{aligned}$$

Using this result for the work in Eq. (1.11.10) will simply reiterate our original assumption that the process is adiabatic. Using Eq. (1.11.15) to eliminate the work from Eq. (1.11.10) yields

$${}_1Q_2 = mC_v(T_2 - T_1) + \frac{mR(T_1 - T_2)}{\gamma - 1} = m \frac{(\gamma - 1)C_v - R}{\gamma - 1}(T_2 - T_1)$$

From Eq. (1.2.15), it can be easily shown that $(\gamma - 1)C_v = R$. For helium we have $\gamma = 5/3$, $C_v = 3115.5903 \text{ J/kg}\cdot\text{K}$, and $R = 2077.0602 \text{ J/kg}\cdot\text{K}$, which exactly satisfies the required thermodynamic relation.

The buoyant force acting on the helium at 100,000 ft can now be determined from the known temperature and pressure. Following Example 1.10.3, we obtain

$$\begin{aligned}\rho_{\text{He}} &= \frac{p}{R_{\text{He}}T} = \frac{1114.27 \text{ N/m}^2}{(2077.0602 \text{ N}\cdot\text{m/kg}\cdot\text{K})(47.439 \text{ K})} = 0.0113085 \text{ kg/m}^3 \\ &= 0.0113085 \text{ kg/m}^3 (0.001940320 \text{ slug/ft}^3/\text{kg/m}^3) = 2.1942 \times 10^{-5} \text{ slug/ft}^3 \\ \mathcal{V}_{\text{He}} &= m_{\text{He}}/\rho_{\text{He}} = 5.5843 \text{ slugs}/2.1942 \times 10^{-5} \text{ slug/ft}^3 = 2.5450 \times 10^5 \text{ ft}^3\end{aligned}$$

The acceleration of gravity at 100,000 ft MSL is 31.868 ft/s^2 , and the buoyant force is obtained from the density, mass, and weight of the displaced air

$$\rho_{\text{air}} = 3.3182 \times 10^{-5} \text{ slug/ft}^3$$

$$m_{\text{air displaced}} = \rho_{\text{air}} \mathcal{V}_{\text{He}} = (3.3182 \times 10^{-5} \text{ slug/ft}^3)(2.5450 \times 10^5 \text{ ft}^3) = 8.445 \text{ slugs}$$

$$F_{\text{buoyant}} = W_{\text{air displaced}} = m_{\text{air}} g = (8.445 \text{ slugs})(31.868 \text{ ft/s}^2) = \underline{269 \text{ lbf}}$$

This is only about 20 percent of the buoyant force computed in Example 1.10.3 for the case where the helium was assumed to be at the same temperature as the air. Of course, if the balloon were to remain at 100,000 ft MSL for a long period of time, it would eventually come to temperature equilibrium with the atmosphere. However, for a large balloon, this could take a very long time.

EXAMPLE 1.11.3. Consider a hot-air balloon submerged in the atmosphere at a time and location where the atmospheric temperature and pressure are T_{a0} and p_{a0} , respectively. At this time and location, the atmospheric temperature varies linearly with geopotential altitude Z and the geopotential temperature gradient is T'_a . As the balloon rises through the atmosphere, the hot air in the balloon remains in pressure equilibrium with the atmosphere and it expands in an adiabatic process. If the initial geopotential altitude and temperature of the hot air in the balloon are Z_0 and T_{b0} , respectively, determine the change in the temperature of the air in the balloon with respect to geopotential altitude. Assume that the air in the balloon is chemically identical to that in the atmosphere.

Solution. Following the development presented in Sec. 1.6, if T'_a is nonzero, the temperature and pressure of the atmosphere are, respectively,

$$T_a = T_{a0} + T'_a(Z - Z_0) \quad (1.11.16)$$

$$p_a = p_{a0} \left[\frac{T_{a0} + T'_a(Z - Z_0)}{T_{a0}} \right]^{-\frac{g_o}{RT'_a}} \quad (1.11.17)$$

Because the air in the balloon expands in an adiabatic process, the temperature of the air in the balloon varies with pressure according to Eq. (1.11.13),

$$T_b = T_{b0} \left(\frac{p_b}{p_{b0}} \right)^{\frac{\gamma-1}{\gamma}} \quad (1.11.18)$$

Since the pressure in the balloon remains in equilibrium with the atmosphere, the pressure ratio in Eq. (1.11.18) can be replaced with that obtained from Eq. (1.11.17)

$$T_b = T_{b0} \left[\frac{T_{a0} + T'_a(Z - Z_0)}{T_{a0}} \right]^{-\frac{(\gamma-1)g_o}{\gamma RT'_a}} \quad (1.11.19)$$

Differentiating Eq. (1.11.19) with respect to Z yields

$$\begin{aligned} \frac{\partial T_b}{\partial Z} &= \frac{-(\gamma-1)g_o}{\gamma RT'_a} T_{b0} \left[\frac{T_{a0} + T'_a(Z - Z_0)}{T_{a0}} \right]^{-\frac{(\gamma-1)g_o}{\gamma RT'_a} - 1} \frac{T'_a}{T_{a0}} \\ &= -\frac{(\gamma-1)g_o}{\gamma R} \frac{T_{b0} \{ [T_{a0} + T'_a(Z - Z_0)] / T_{a0} \}^{-(\gamma-1)g_o/(\gamma RT'_a)}}{T_{a0} + T'_a(Z - Z_0)} \end{aligned} \quad (1.11.20)$$

From Eq. (1.11.19), we see that the numerator of the last term in Eq. (1.11.20) is the air temperature inside the balloon at the geopotential altitude Z . Similarly, we see from Eq. (1.11.16) that the denominator of this term is the atmospheric temperature at the geopotential altitude Z . Hence, the change in the temperature of the air in the balloon with respect to geopotential altitude can be written as

$$\frac{\partial T_b}{\partial Z} = -\frac{(\gamma-1)g_o}{\gamma R} \frac{T_b}{T_a} \quad (1.11.21)$$

The coefficient of the temperature ratio on the right-hand side of Eq. (1.11.21) is a temperature gradient, which is called the *dry adiabatic lapse rate* (DALR) for the standard atmosphere, and it has the numerical value

$$\begin{aligned} \text{DALR} &\equiv \frac{(\gamma-1)g_o}{\gamma R} = \frac{0.4(9.806645 \text{ m/s}^2)}{1.4(287.0528 \text{ m}^2/\text{s}^2 \cdot \text{K})} = 9.761 \text{ K/km} \\ &= 5.355 \text{ F/1000 ft} \end{aligned} \quad (1.11.22)$$

The dry adiabatic lapse rate, which was exposed in the previous example, is of critical importance in meteorology, particularly within the troposphere. In general, a **lapse rate is defined to be the negative of the rate of temperature change with altitude**, i.e., a decrease of temperature with altitude. Strictly speaking, a lapse rate is defined to be a decrease in air temperature with respect to geometric altitude. However, the thickness of the troposphere is only about 0.25% of the Earth's radius, so geometric altitude and geopotential altitude are nearly indistinguishable in the troposphere. There are three important lapse rates in the troposphere; the *environmental lapse rate* (ELR), the *dry adiabatic lapse rate* (DALR), and the saturated adiabatic lapse rate (SALR).

The environmental lapse rate is the actual rate of decrease of temperature with altitude in the atmosphere at a given time and location. From Table 1.6.1, note that the standard atmosphere has an environmental lapse rate of 6.5 K/km (3.6 F/1000 ft) from sea level to 11 km (36,089 ft). The standard atmosphere contains no moisture. Unlike the idealized standard atmosphere, temperature in the actual atmosphere below 11 km does not always decrease at a uniform rate with altitude. The actual environmental lapse rate can vary significantly with time and altitude. For example, at times there can be an *inversion layer* in the troposphere where the temperature actually increases with altitude.

The dry adiabatic lapse rate is commonly defined to be the rate of temperature decrease with altitude for a mass of dry or unsaturated air that is rising under adiabatic conditions. From Eq. (1.11.21), we see that the adiabatic rate of temperature decrease with altitude in rising air depends on the ratio of the temperature of the rising air to that of the surrounding atmosphere. Thus, strictly speaking, the dry adiabatic lapse rate that is defined in Eq. (1.11.22) is the rate of temperature decrease with altitude for dry air at the same temperature as the atmosphere, rising under adiabatic conditions. When air in the atmosphere rises by buoyant convection, the absolute temperature difference between the rising air and the surrounding atmosphere is typically small. Furthermore, air has low thermal conductivity, and the bodies of air involved in buoyant convection are very large, so heat transfer is negligibly small. Thus, the dry adiabatic lapse rate is very close to that observed in actual atmospheric buoyant convection. Note from Eq. (1.11.22) that the dry adiabatic lapse rate is independent of the local temperature and pressure. It depends only on the specific heat ratio, the ideal gas constant, and the acceleration of gravity.

The saturated adiabatic lapse rate is the rate of temperature decrease with altitude for a mass of air saturated with water vapor, which is rising under adiabatic conditions. When air is saturated with water vapor (at its dew point), any further reduction in temperature will cause water to condense. The saturated adiabatic lapse rate is less than the dry adiabatic lapse rate because latent heat is released when water condenses, thus decreasing the rate of temperature drop with increasing altitude. The saturated adiabatic lapse rate varies strongly with the air temperature but typical values are on the order of 5 K/km (2.7 F/1000 ft).

The three lapse rates discussed above are important in meteorology, because they determine if a mass of buoyant air will rise high enough for water vapor to condense to form clouds; and, having formed clouds, whether the air will continue to rise to form larger rain clouds; and whether these clouds will grow even larger to form cumulonimbus clouds (thunder clouds). As buoyant unsaturated air rises, its temperature drops at the dry adiabatic lapse rate. The dew point also drops as a result of decreasing air pressure,

but at a lower rate of about 2 K/km (1 F/1000 ft). If unsaturated air rises high enough, its temperature will reach the dew point and condensation will begin to form clouds. This altitude is commonly known as cloud base. The difference between the dry adiabatic lapse rate and the rate at which the dew point drops is about 8 K/km (4.4 F/1000 ft). From a known difference between the air temperature and the dew point temperature on the ground, the altitude at cloud base can be easily estimated.

If the environmental lapse rate is less than the saturated adiabatic lapse rate, the air is said to be *absolutely stable*. Under such conditions, rising air will always cool faster than the surrounding air and lose buoyancy. This typically occurs in the morning, when air near the ground has cooled overnight. Clouds are unlikely to form in such stable air.

If the environmental lapse rate is between the saturated and dry adiabatic lapse rates, the air is said to be *conditionally unstable*. Under these conditions, unsaturated air does not typically have sufficient buoyancy to rise to cloud base. If the air is saturated, it is unstable and buoyant air will rise, causing water vapor to condense.

If the environmental lapse rate is greater than the dry adiabatic lapse rate, it is said to be a *superadiabatic lapse rate*, and the air is said to be *absolutely unstable*. Under such conditions, buoyant rising air will gain buoyancy both below and above cloud base. This often occurs in the afternoon over warm land masses. Under these conditions, with sufficient moisture, cumulus clouds, rain clouds, and even thunderstorms are likely.

1.12. Introduction to Fluid Dynamics and Aerodynamics

To this point, we have considered the analysis of only static fluids and fluids that are undergoing slow homogeneous expansion or compression in an environment where the only body force is that of gravity. Because a fluid deforms continuously under the action of a shear stress, there can be no shear stresses acting in a static fluid. Similarly, because the shear stress in a Newtonian fluid is proportional to the rate of angular deformation as expressed in Eq. (1.2.21), there can be no shear stresses acting in a Newtonian fluid that is undergoing homogeneous expansion or compression. Furthermore, there is no relative acceleration in a static fluid, and when analyzing fluids undergoing slow homogeneous expansion or compression, the effects of fluid acceleration can be neglected. Hence, in the analysis of static fluids and fluids undergoing slow homogeneous expansion or compression, when the only body force is that of gravity, Newton's second law reduces to a simple balance between pressure forces and weight. Although there are important engineering problems in fluid mechanics that can be analyzed accurately using this simple relation, accurate analysis of many important problems in fluid mechanics requires including consideration of fluid shear stress and/or fluid acceleration.

In a general analysis of Newtonian fluid flow, we must include the effects of fluid acceleration, expansion and compression (called compressibility), pressure, shear stress, and weight. In some circumstances we must also include other body forces, such as the electromagnetic force and/or the centrifugal and Coriolis forces resulting from the Earth's rotation. In the most general case, the mathematical equations of motion that govern fluid flow are quite complex and we will eventually need to deal with this complexity. Nevertheless, there are many important problems in engineering and science that can be analyzed accurately by neglecting one or more of the associated physical phenomena.

The discipline of fluid mechanics is traditionally split into two divisions; *fluid statics* and *fluid dynamics*. As the names imply, fluid statics is the study of fluids at rest and fluid dynamics is the study of fluids in motion. As stated in Sec. 1.1, aerodynamics is the branch of fluid mechanics concerned with predicting and controlling the forces and moments acting on an aircraft or other object moving through the atmosphere.

For convenience, the total force $\mathbf{F}$ exerted on a moving object by the atmosphere can be divided into two parts; the *buoyant force* $\mathbf{F}_b$ and the *aerodynamic force* $\mathbf{F}_a$. As seen in Sec. 1.10, the buoyant force acts in the direction opposite to the gravitational force and has a magnitude equal to the weight of the fluid displaced by the object. Rigorously speaking, the aerodynamic force is simply defined to be the total force exerted on an object by the atmosphere, less the buoyant force $\mathbf{F}_a \equiv \mathbf{F} - \mathbf{F}_b$. Because the buoyant force is a *hydrostatic force*, the aerodynamic force can be thought of as a *dynamic force*, which results only from the relative motion between the object and the surrounding atmosphere. Although the buoyant force results only from a pressure distribution acting over the object's surface, in general, the aerodynamic force results from both pressure and shear stress distributions acting over the surface of the object.

The aerodynamic force is commonly divided into two components; the *lift force* and the *drag force*. The lift L and drag D are defined to be the scalar components of the resultant aerodynamic force perpendicular to and in the direction of the *relative wind*, respectively. The relative wind $\mathbf{V}_\infty$ is defined to be the velocity vector of the air far from the object, as measured relative to a coordinate system fixed to the object. For example, an aircraft moving through still air has a relative wind equal to the negative of the aircraft's velocity vector. The flow far from the object, as viewed from a reference frame fixed to the object, is commonly referred to as the *freestream*. Hence, the relative wind is also commonly called the *freestream velocity*. Using this notation, the lift and drag are by definition:

$$D \equiv \text{drag} \equiv \text{the scalar component of } \mathbf{F}_a \text{ parallel with } \mathbf{V}_\infty \quad (D = \mathbf{F}_a \cdot \mathbf{i}_\infty)$$

$$L \equiv \text{lift} \equiv \text{the scalar component of } \mathbf{F}_a \text{ perpendicular to } \mathbf{V}_\infty \quad (L = |\mathbf{F}_a - D\mathbf{i}_\infty|)$$

where $\mathbf{i}_\infty$ is the unit vector in the direction of the relative wind. The magnitude of the relative wind is commonly called the *airspeed*.

In this textbook, we shall be concerned primarily with aerodynamic flows about aircraft, rockets, and missiles. An *aircraft* is traditionally defined to be any *vehicle* that is able to *fly* having its weight supported primarily by air, or in general, the atmosphere of a planet. The weight of a flying aircraft is supported either by buoyant force, aerodynamic lift, or some combination of both. Although rockets and missiles also travel through the atmosphere, most are not typically classified as aircraft because they use rocket thrust instead of buoyancy or aerodynamic lift as the primary means of supporting weight. For the most part, rockets and missiles use aerodynamic lift only as a means of control and to carry a small fraction of their weight, whereas aircraft often use aerodynamic lift both for control and as the primary means of supporting their weight. Some missiles could be considered to be aircraft because they rely on small lifting wings to support a large fraction of their weight. Historically, aircraft have been divided into two categories; *aerostats* and *aerodynes*.

Aerostats are *lighter-than-air* aircraft, which use the buoyant force to support their weight. These aircraft *float* in air in much the same way that ships and submarines float in water. Aerostats typically have one or more large gasbags or canopies, filled with a relatively low density gas such as hydrogen, helium, or hot air, which is less dense than the surrounding atmosphere. To maintain constant altitude in a free-flying aerostat, the weight of the buoyant gas combined with the weight of the aircraft structure and payload must balance the weight of the air that is displaced by the aircraft. An aerostat that has no control or only altitude control is commonly called a balloon. Small hot air balloons date back to the third century BC. Hot air balloons were the second type of man-made aircraft to fly and the first to carry human passengers aloft. A powered aerostat with both steering and altitude control, like that shown in Fig. 1.12.1, is commonly referred to as a dirigible or blimp. Dirigibles often use aerodynamic lift as a means of control. The first aircraft of this type were the German *Zeppelins* like the well known *Hindenburg*.

Aerodynes are *heavier-than-air* aircraft, which use aerodynamic lift to support a large fraction of their weight. These aircraft produce aerodynamic lift as a result of relative motion between the atmosphere and some lifting surface or surfaces, such as the wings of an airplane or the rotor blades of a helicopter. Most aerodynes are commonly classified as either fixed-wing aircraft or rotorcraft. A fixed-wing aircraft, commonly called an airplane, produces lift as a result of the forward motion of the wing or wings relative to the atmosphere. A rotorcraft produces lift as a result of the rotational motion of the rotor blades relative to the atmosphere. To maintain constant altitude in a free-flying aerodyne, the aerodynamic lift developed by the wings or rotor blades combined with the weight of the air that is displaced by the aircraft must balance the weight of the aircraft structure and payload. For most aerodynes in common use today, the buoyant force is negligible compared with the aerodynamic lift and gross weight of the aircraft.



Figure 1.12.1. All aerostats are lighter than air and use the buoyant force to support their weight, although some use aerodynamic lift as a means of control. (U. S. Navy photograph)

However, a new class of very large aircraft (perhaps as large as ocean liners) has been considered in the early phases of design (see Rawdon and Hoisington 2003). Conceptual design studies have shown that these aircraft will likely support their weight using approximately equal fractions of aerodynamic lift and buoyancy. Aerodynes are the most important class of aircraft in use today. Furthermore, a very familiar type of aerodyne was actually the first man-made aircraft to fly. These small aircraft, which are still commonly flown today, are called kites.

The first controlled human flight in an aircraft heavier than air took place in 1891 with an unpowered fixed-wing aircraft, designed, built, and flown by Otto Lilienthal, one of the true giants in aviation history. A few of Lilienthal's many flights are shown in Fig. 1.12.2. Otto Lilienthal was truly the first aviator and the father of modern aviation.

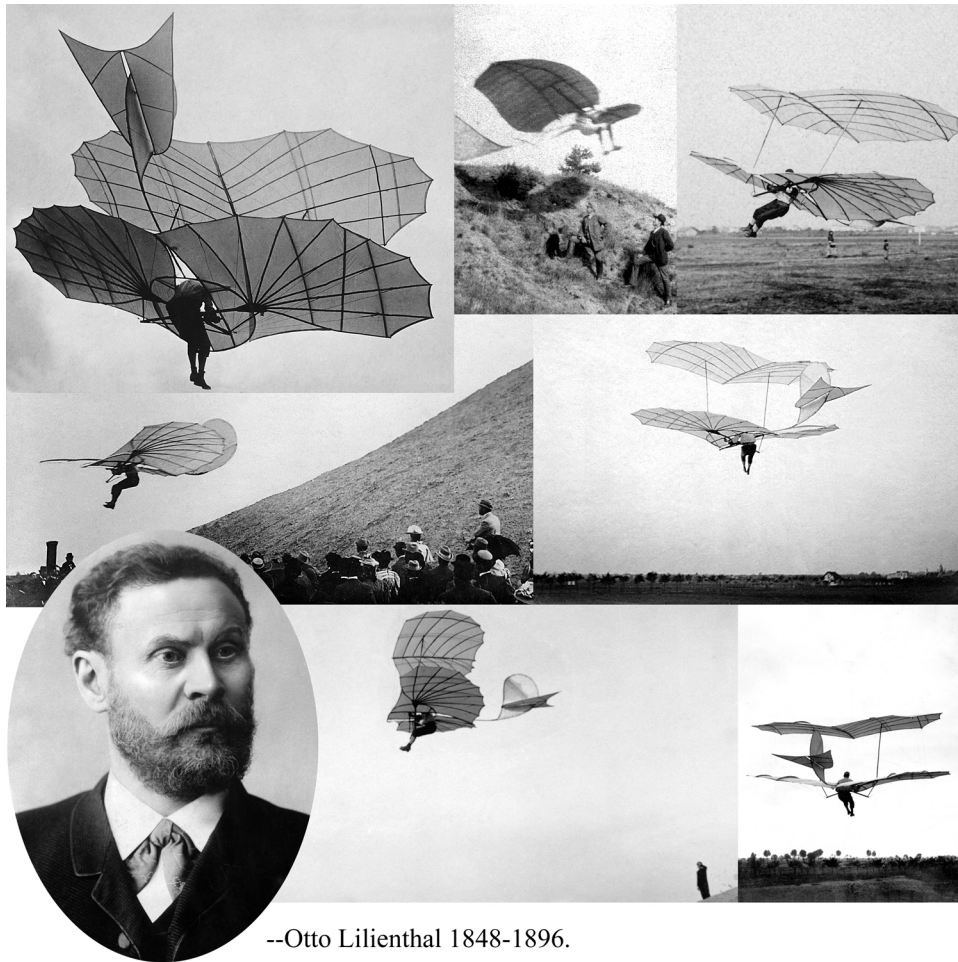


Figure 1.12.2. Otto Lilienthal, the true father of modern aviation, was the first human to pilot a fixed-wing aircraft in stable controlled flight. (Archives of the Otto-Lilienthal-Museum)

His nineteenth-century predecessors felt that the key to flight was to *build an engine with sufficient power and attach it to an airframe of sufficient strength and man could propel himself into aviation history through brute force alone*. By contrast, Lilienthal was the first to recognize the need to get in the air and fly with gliders before an engine could be used successfully for powered flight. The Wright brothers carried on his philosophy, which ultimately led to the control and propulsion solutions that made powered flight possible. During the period from 1891 to 1896, Lilienthal made more than 2,000 successful glider flights. He also published papers on flight that were read throughout the world, and these papers inspired other aviation pioneers, including the Wright brothers. At the time of his death, Lilienthal was working on powered flight and it is more than likely that had he lived a few more years, he would have achieved powered flight before the Wright brothers.

On Sunday, August 9, 1896, Otto Lilienthal was injured in a glider crash near Stollen, Germany. He died the next day in Berlin. According to a witness, Lilienthal did not know he was fatally injured. The witness gave Lilienthal's last words as: "Ich muss etwas ausruhen, dann machen wir weiter" (I must rest a little, then we continue). During his life, Lilienthal used the phrase "Opfer müssen gebracht werden" (Sacrifices must be made). This is the epitaph now carved on his tombstone in Lichterfelde cemetery.

The worldwide publicity surrounding Lilienthal's death spurred increased interest in human flight from two other aviation pioneers. Wilbur and Orville Wright had been following Lilienthal's progress with great interest. However, it was not until Lilienthal's death in 1896 that the Wright brothers began seriously pursuing their own journey toward powered human flight.

Although Lilienthal published much of his work on flight, the Wright brothers were not able to take up exactly where Lilienthal left off. It was not until October of 1900 that the Wrights flew their first full-size glider. This glider was flown from the ground with strings, usually unmanned. Only a few brief manned flights were ever made with this glider. Their second glider was more successful, and a series of manned flights were made during July and August of 1901. However, the Wright brothers were still not close to demonstrating the level of success that Lilienthal had demonstrated five years earlier. It was not until their third glider was flown in August and September of 1902 that the Wrights were able to approach the success demonstrated by Lilienthal in 1896. The Wright brothers made more than 1,000 successful flights during the of Fall 1902; but more than six years of aviation progress were likely lost as a result of Lilienthal's death.

As both Lilienthal and the Wright brothers believed, once the principles of flight were mastered in a glider, powered flight was not far behind. During the winter of 1903 the Wrights designed and built an engine that developed 12 horsepower and weighed about 200 pounds. They also designed and built the two propellers. During the summer of 1903 they built what was essentially a larger powered version of the 1902 glider. This aircraft made its first flight of 120 feet in about 12 seconds on December 17, 1903.

The major original contributions made by the Wright brothers, which made possible their first powered flight, were the development of a lighter internal combustion engine and the design of a more efficient propeller. The 1903 Wright Flyer, which is shown in Fig. 1.12.3, was powered by a four-cylinder, liquid-cooled internal-combustion engine having a power-to-weight ratio of only 0.06 hp/lb. Its first flight marked the beginning of



Figure 1.12.3. The 1903 Wright Flyer was powered by a four-cylinder, liquid-cooled internal-combustion engine that weighed 200 pounds and developed 12 horsepower, providing a power-to-weight ratio of 0.06 hp/lb.

a 40-year period of rapid development for fixed-wing aircraft powered by piston engine-propeller combinations. During this period the power-to-weight ratio for the piston engine was dramatically increased and propeller efficiency was significantly improved. The liquid-cooled piston engine continued to be used as a means of aircraft propulsion through the end of World War II, when power-to-weight ratios reached nearly 1.0 hp/lb.

1.13. Dimensional Analysis and Experimental Modeling

In Sections 1.5–1.11, our consideration of fluid statics was purely analytical in nature. We began by developing the differential formulation of Newton’s second law for a static fluid, which is presented in Eq. (1.5.10). This vector differential equation provides three scalar component equations, which can be used to determine the three components of the hydrostatic force vector acting on a surface submerged in a static fluid. In the following chapters, this analytical approach will be extended to fluids in motion. However, for the complex geometries that are encountered in most practical engineering applications, it is seldom possible to obtain an analytical solution to the equations of motion for fluid flow.

Because so few fluid flows of practical interest can be solved by analytical methods alone, the study of fluid dynamics and aerodynamics relies heavily on data obtained from experiments and *computational fluid dynamics* (CFD). However, obtaining experimental data and/or CFD results can be very time consuming. Hence, an engineer will always want to obtain the required information from the fewest possible number of experiments and CFD computations. A very powerful technique that is commonly used as an aid in accomplishing this is referred to as *dimensional analysis*.

When engineers need information related to the flow of a fluid about a very large object, it is often impossible or prohibitively expensive to perform experimental testing on a full-size prototype. In such cases, the only feasible experimental approach is to perform the testing on a smaller model of the full-size prototype. However, data taken

using the model are not useful for predicting results associated with flow about the full-size prototype, unless the model flow and the prototype flow are related through known scaling laws. Here again, dimensional analysis provides a powerful tool that can be used to design model experiments, which can be scaled to accurately predict the performance of a full-size prototype.

Introduction to Dimensional Analysis

Most phenomena associated with fluid mechanics and aerodynamics depend on several variables associated with the geometry, the fluid, and the flow. For example, suppose that we simply wish to predict the magnitude of the hydrostatic buoyant force acting on a very large spherical balloon that is to be deployed in the Martian atmosphere. Of course, for this simple problem we have already developed the analytical solution that is given in Eq. (1.10.5). Neglecting changes in the Martian atmospheric density and acceleration of gravity over the surface of the sphere, this yields

$$F_b = \frac{\pi}{6} d^3 \rho g \quad (1.13.1)$$

where F_b is the buoyant force acting on the sphere, d is the diameter of the sphere, ρ is the atmospheric density, and g is the acceleration of gravity in the Martian atmosphere at the altitude of the balloon. To demonstrate the utility of dimensional analysis, suppose that we had not been able to obtain this analytical solution from Eqs. (1.5.10) and (1.8.2). Instead we wish to conduct a series of experiments to determine how the buoyant force varies with the related dimensional variables.

Even if we had not been able to solve the governing equations analytically, we can see from examination of Eqs. (1.5.10) and (1.8.2) that the buoyant force should depend on the fluid density, the acceleration of gravity, and the diameter of the sphere. Hence, even without integrating the governing differential equation for static pressure, we should expect a functional relationship of the form

$$F_b = f(d, \rho, g) \quad (1.13.2)$$

To experimentally expose the functional relationship in this form, we would need to conduct a series of experiments in which the dependent variable F_b is measured with the independent variables d , ρ , and g varied systematically over the range of interest.

Imagine a series of experiments to determine the dependence of F_b on d , ρ , and g . To obtain the relation between F_b and d for fixed values of ρ and g , we might need to measure F_b on 5 spheres of different diameter. To explore the effects of fluid density, each series of 5 diameter tests could be repeated in 5 fluids of different density. If these 25 experiments were conducted in an accelerating laboratory like an elevator, they could be repeated 5 times to simulate 5 different values of g . This series of tests would require 125 measurements in an elaborate facility that would allow us to vary both the fluid density and the acceleration of gravity.

Fortunately, it is possible to obtain equivalent results with much less effort through the use of dimensional analysis. Essentially, dimensional analysis is a technique for reducing the number of mathematical variables that affect a given physical phenomenon,

by making use of the fact that the equations governing any physical phenomenon must be dimensionally homogeneous. Dimensions are measures of physical quantities like mass, length, time, and temperature. Dimensions are quantitatively expressed in units such as SI units or English units. For example, length could be expressed in units of feet (ft), meters (m), microns (μm), etc. **The principle of dimensional homogeneity** states: *In any equation that expresses a proper relationship among variables in a physical process, each of the additive terms on both sides of the equation must have the same dimensions.*

Buckingham's Pi Theorem

If a physical phenomenon is described as a relationship among n dimensional variables, the same phenomenon can be described as a relationship among $n - m$ dimensionless variables, where m is the total number of *fundamental dimensions* required to describe the n dimensional variables. In fluid mechanics, the four fundamental dimensions and corresponding SI units are usually mass (kg), length (m), time (s), and temperature (K). However, not all physical phenomena associated with fluid mechanics include all four of these fundamental dimensions. If temperature is included only in combination with a gas constant (RT), then the dimensions of the product are length squared per time squared (m^2/s^2 in SI units) and only the dimensions of mass, length, and time are required.

Mathematically, Buckingham's Pi theorem states: If a physical phenomenon depends on n dimensional parameters, say $p_1, p_2, \dots, p_n$, and a dimensional relationship among these parameters can be expressed as

$$f(p_1, p_2, \dots, p_n) = 0 \quad (1.13.3)$$

then the n dimensional parameters can be grouped into $n - m$ independent dimensionless ratios, $\Pi_1, \Pi_2, \dots, \Pi_{n-m}$, and the relationship among the parameters can be expressed as

$$F(\Pi_1, \Pi_2, \dots, \Pi_{n-m}) = 0 \quad (1.13.4)$$

Following Buckingham (1914) the $n - m$ independent dimensionless ratios are commonly referred to as the Π groups. Buckingham's theorem does not predict the form of either f or F . The functional relation among the dimensionless Π groups must be determined by some other means; usually analytically, experimentally, or numerically.

For example, consider the relation suggested by Eq. (1.13.2). This is a relationship among four dimensional variables; F_b ($\text{kg}\cdot\text{m}/\text{s}^2$), d (m), ρ (kg/m^3), and g (m/s^2). Note that force is expressed in terms of the fundamental dimensions; mass, length, and time. Because none of these dimensional variables contain the dimension of temperature, only three fundamental dimensions are included in the four dimensional variables. For this relation $n=4$, $m=3$, and the number of independent dimensionless ratios required to describe the physical relation suggested by Eq. (1.13.2) is $n - m = 1$. Hence, the physical relation suggested by Eq. (1.13.2) can be written in terms of a single dimensionless ratio

$$F(\Pi) = 0 \quad (1.13.5)$$

Clearly, the relation suggested by Eq. (1.13.5) is easier to expose experimentally than that suggested by Eq. (1.13.2). If Eq. (1.13.5) has a unique solution, the dimensionless ratio Π must be a constant for the physical relation that is suggested by Eq. (1.13.2).

Determining the Dimensionless Ratios

Regardless of the method used to determine the $n - m$ independent dimensionless ratios that describe a particular physical phenomenon, one must commence by listing all of the n dimensional variables that affect the phenomenon of interest. For this purpose, the best approach is to develop a complete mathematical model for the physical phenomenon to be studied. Even if the mathematical formulation is so complex that we are unable to obtain an analytical solution, the mathematical formulation will expose the dimensional variables that affect the phenomenon. All of these identified dimensional variables must be included in the set of $n - m$ independent dimensionless ratios that are to be used to describe the given physical phenomenon.

For the relation suggested by Eq. (1.13.2), $n = 4$, $m = 3$, and $n - m = 1$. Hence, there is only one dimensionless ratio that must include all four of the dimensional variables; F_b ($\text{kg} \cdot \text{m}/\text{s}^2$), d (m), ρ (kg/m^3), and g (m/s^2). With this knowledge, the dimensionless ratio to be identified could be written in the form

$$\Pi = F_b^{e_1} d^{e_2} \rho^{e_3} g^{e_4} \quad (1.13.6)$$

where e_1 , e_2 , e_3 , and e_4 are constant exponents, which are as yet unknown.

Once the list of dimensional variables to be included in any dimensionless ratio has been determined, it is a simple matter to define an appropriate dimensionless ratio from only the knowledge that the ratio must be dimensionless. In the case of the Π group to be determined from Eq. (1.13.6), the fact that Π is dimensionless requires

$$\left(\frac{\text{kg} \cdot \text{m}}{\text{s}^2} \right)^{e_1} \text{m}^{e_2} \left(\frac{\text{kg}}{\text{m}^3} \right)^{e_3} \left(\frac{\text{m}}{\text{s}^2} \right)^{e_4} = 1 \quad (1.13.7)$$

Here the dimensions are expressed in SI units, but this is arbitrary and has no effect on the final result. The dimensional relation in Eq. (1.13.7) can be rearranged as

$$\text{kg}^{e_1+e_3} \text{m}^{e_1+e_2-3e_3+e_4} \text{s}^{-2e_1-2e_4} = \text{kg}^0 \text{m}^0 \text{s}^0 \quad (1.13.8)$$

Equating exponents of the three fundamental dimensions of mass, length, and time on the two sides of Eq. (1.13.8) provides three equations in the four unknown exponents e_1 , e_2 , e_3 , and e_4 . This allows us to solve for three of the unknown exponents in terms of the fourth. For example, the three equations required by Eq. (1.13.8) could be written as

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & -3 & 1 \\ 0 & 0 & -2 \end{bmatrix} \begin{Bmatrix} e_2 \\ e_3 \\ e_4 \end{Bmatrix} = \begin{Bmatrix} -1 \\ -1 \\ 2 \end{Bmatrix} e_1 \quad (1.13.9)$$

The fourth exponent is arbitrary and can be chosen as we see fit.

To see that one of the exponents in Eq. (1.13.6) is arbitrary, we need to recognize that any dimensionless ratio can be raised to an arbitrary power and the result is still dimensionless. Hence, one of the exponents in Eq. (1.13.6) can be chosen simply for our convenience. Although it is not absolutely necessary, it will usually be most convenient to choose the exponent of the dependent variable to be 1.

For the relation suggested by Eq. (1.13.2), the dependent variable is F_b . Thus, if we choose $e_1 \equiv 1$ in Eq. (1.13.6), the dimensional requirements that are given in Eq. (1.13.9) can be solved by Gauss elimination to yield

$$e_1 \equiv 1, \quad e_2 = -3, \quad e_3 = -1, \quad e_4 = -1 \quad (1.13.10)$$

Using Eq. (1.13.10) in Eq. (1.13.6) easily reveals that the single dimensionless ratio that is required to express the relation suggested by Eq. (1.13.2) in the form of Eq. (1.13.5) is

$$\Pi = \frac{F_b}{d^3 \rho g} \quad (1.13.11)$$

If the relation suggested by Eq. (1.13.5) has a unique solution then the dimensionless ratio Π must be a constant and the physical relation that is suggested by Eq. (1.13.2) can be written equivalently as

$$\frac{F_b}{d^3 \rho g} = C \quad (1.13.12)$$

where at this point C is an unknown constant.

The unknown constant in Eq. (1.13.12) prevents us from using this relation alone to determine the hydrostatic force acting on a balloon deployed in the Martian atmosphere. However, the unknown constant C in Eq. (1.13.12) could be determined from a single experiment conducted on Earth in a common fluid. A carefully conducted experiment using a sphere of modest diameter in water would quickly reveal a result very close to $C=0.52$, which for all practical purposes is the same result given by Eq. (1.13.1).

In the text that follows Eq. (1.13.2), we considered a hypothetical series of 125 experimental measurements that could be made to reveal the functional relation that is suggested by Eq. (1.13.2). This was based on the assumption that 5 experiments would be required to reveal the dependence of F_b on each of the independent variables d , ρ , and g , with both of the other dimensional variables held constant. Because the dependent variable in this problem F_b is a function of three independent variables d , ρ , and g , this leads to a total of 5^3 or 125 experiments. Even if only three experiments were required to reveal the dependence of F_b on each of the independent variables, a total of 3^3 or 27 experiments would be required to expose the dimensional relation that is suggested by Eq. (1.13.2). Furthermore, this series of tests would require an elaborate facility that would allow us to vary both the fluid density and the acceleration of gravity. In contrast, dimensional analysis has allowed us to reduce the dimensional relation that is suggested by Eq. (1.13.2) to the much simpler dimensionless relation that is given by Eq. (1.13.12), which can be exposed by a single experiment conducted in a much less elaborate facility.

Although the problem of predicting the force acting on an object submerged in a static fluid is simple compared to problems involving fluids in motion, consideration of dimensional analysis applied to this problem has demonstrated the power of dimensional analysis as a tool for reducing the number and complexity of experiments or CFD computations that may be required to meet some engineering objective. As we begin to consider fluids in motion, dimensional analysis will prove to be even more beneficial.

When considering problems associated with fluids in motion, $n - m$ is typically greater than 1, and two or more dimensionless Π groups are required to mathematically describe the physical phenomenon. In such cases, the following steps outline a procedure that can be used for determining appropriate dimensionless Π groups:

1. *Find all n of the dimensional variables involved in the physical phenomenon.* This is best accomplished by developing a mathematical formulation for the physics of the phenomenon being studied. Do not include variables that can be obtained from other included variables. For example, if a rectangle has area S , length b , and width c ; you can include S and b , S and c , or b and c ; but you cannot include S , b , and c .
2. *List the variables found in step 1 with their fundamental dimensions and determine the total number of fundamental dimensions m .* Do not include any dimension that can be obtained from other included dimensions. For example, you can include the dimensions of mass, length, and time; or force, length, and time; but you cannot include dimensions of force, mass, length, and time.
3. *From the total list of n variables, select what is called the repeating subset of the m most important independent variables, which include all fundamental dimensions.* This repeating subset should not include any two variables that have exactly the same dimensions raised to any power. For example, if the list of n variables includes an area S and a length b , you cannot include both S and b in the repeating subset, because S has dimensions of length squared and b has dimensions of length. The repeating subset should not include the dependent variable.
4. *Develop a set of $n - m$ dimensionless Π groups by successively combining each of the variables not included in the repeating subset with all variables included in the repeating subset.* In each Π group, the single variable not included in the repeating subset is given the exponent 1 and each variable included in the repeating subset is given an unknown exponent $e_1, e_2, \dots, e_m$. The unknown exponents are evaluated by setting the net exponent for each fundamental dimension in the Π group to zero.
5. *Examine and rearrange your $n - m$ dimensionless Π groups so that, to the extent possible, you are using traditional dimensionless groups as defined in the existing literature.* Although any consistent set of independent dimensionless Π groups can be used to correlate experimental data and/or CFD results, traditional dimensionless groups should be used unless there is cause to do otherwise.

If this procedure is followed correctly, the Π groups so obtained will be dimensionless. Nevertheless, it is always wise to check your computations by verifying that all of your Π groups are in fact dimensionless.

The first step in the above process is the most difficult, and it is also the most important. If any important variable is omitted in step 1, the entire process is flawed and the final results obtained from a very costly set of experiments could prove to be almost useless. On the other hand, if an additional unnecessary variable is included in step 1, the final experimental results will show that this variable should not have been included. However, including this unnecessary variable in the experimental design process could increase the cost of conducting the experiments by an order of magnitude or more.

At this point, most students will not possess the knowledge and analytical skills needed to implement step 1 of the preceding process effectively for problems involving fluids in motion. Much of the material presented in this textbook is devoted to providing students with the required knowledge and analytical skills. Hence, for the examples and homework problems presented in this chapter, results that would normally be obtained from step 1 of the preceding process will be given a priori.

In a subsequent development of the equations of motion for fluid flow, it will be shown that the total surface force exerted on an object of given shape and orientation moving through an ideal gas depends on six dimensional variables; the density of the freestream fluid ρ_∞ (kg/m^3), the dynamic viscosity of the freestream fluid μ_∞ ($\text{kg/s}\cdot\text{m}$), the velocity of the object relative to the freestream fluid V_∞ (m/s), the speed of sound in the freestream fluid a_∞ (m/s), the characteristic length of the object d (m), and the local acceleration of gravity g (m/s^2). The total surface force acting on an object moving through any fluid can be divided into two parts; the *buoyant force* and the *aerodynamic force*. A significant advantage of this division is a reduction in the number of variables affecting each part of the surface force. If the aerodynamic force is defined to be the total force exerted on the object by the fluid less the buoyant force, which is computed from Eq. (1.10.5), the buoyant force does not depend on V_∞ , μ_∞ , or a_∞ , and as will be shown in a subsequent chapter, the aerodynamic force does not depend on g .

EXAMPLE 1.13.1. It is known from the equations of motion for fluid flow that the aerodynamic drag force D acting on a smooth sphere moving through a fluid depends on five dimensional variables; the freestream density ρ_∞ , the freestream dynamic viscosity μ_∞ , the relative freestream velocity V_∞ , the freestream speed of sound a_∞ , and the sphere's diameter d . Use the five-step dimensional analysis procedure presented in this section to obtain appropriate dimensionless ratios to be used in an experimental investigation of fluid flow about a sphere.

Solution. In this problem statement the results of step 1 are given a priori. Hence, we begin step 2 by listing all of the dimensional variables with their fundamental dimensions. For example, expressing the dimensions in SI units we have

$$D \text{ (kg}\cdot\text{m/s}^2\text{)}, \rho_\infty \text{ (kg/m}^3\text{)}, \mu_\infty \text{ (kg/s}\cdot\text{m)}, V_\infty \text{ (m/s)}, a_\infty \text{ (m/s)}, d \text{ (m)}$$

Examination of the variables reveals that there are three fundamental dimensions; mass (kg), length (m), and time (s). So we see that for this example, $n=6$, $m=3$, and $n-m=3$. Thus, we must obtain three dimensionless Π groups.

Because this particular problem involves three fundamental dimensions, step 3 suggests that we choose three of the most important independent variables as our repeating subset. The repeating subset should not include the dependent variable, so we should not include D in the repeating subset. Furthermore, the repeating subset cannot include any two variables that have exactly the same dimensions raised to any power, so both V_∞ and a_∞ cannot be included in the repeating subset. Even the inexperienced reader will most likely discern that the relative velocity is more important than the speed of sound. Hence, we should select our repeating

subset of the three most important independent variables from the four variables ρ_∞ , μ_∞ , V_∞ , and d . Although any three of these four variables could be chosen as our repeating subset, to minimize the effort required in step 5, it helps to know something about tradition. Here we will choose:

$$\text{repeating subset} \equiv \rho_\infty (\text{kg/m}^3), V_\infty (\text{m/s}), d (\text{m})$$

We shall subsequently consider an alternate choice, examine the consequences of that choice, and discuss the motivation for our present choice.

Proceeding with step 4, our first Π group shall include only the drag D raised to the power 1 combined with the three variables chosen as our repeating subset, each raised to an unknown exponent

$$\Pi_1 = D \rho_\infty^{e_1} V_\infty^{e_2} d^{e_3}$$

The fact that this Π group must be dimensionless requires that the net exponent for each fundamental dimension must be zero. Hence, expressing the dimensions in SI units we obtain

$$\frac{\text{kg} \cdot \text{m}}{\text{s}^2} \left(\frac{\text{kg}}{\text{m}^3} \right)^{e_1} \left(\frac{\text{m}}{\text{s}} \right)^{e_2} \text{m}^{e_3} = 1$$

$$\text{kg}^{1+e_1} \text{m}^{1-3e_1+e_2+e_3} \text{s}^{-2-e_2} = \text{kg}^0 \text{m}^0 \text{s}^0$$

which yields

$$\begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 1 \\ 0 & -1 & 0 \end{bmatrix} \begin{Bmatrix} e_1 \\ e_2 \\ e_3 \end{Bmatrix} = \begin{Bmatrix} -1 \\ -1 \\ 2 \end{Bmatrix}, \quad \begin{Bmatrix} e_1 \\ e_2 \\ e_3 \end{Bmatrix} = \begin{Bmatrix} -1 \\ -2 \\ -2 \end{Bmatrix}, \quad \Pi_1 = \frac{D}{\rho_\infty V_\infty^2 d^2}$$

Similarly, our second Π group will include only the second dimensional variable not used in our repeating subset μ_∞ raised to the power 1 and the three repeating variables, each raised to an unknown exponent

$$\Pi_2 = \mu_\infty \rho_\infty^{e_1} V_\infty^{e_2} d^{e_3}$$

$$\frac{\text{kg}}{\text{s} \cdot \text{m}} \left(\frac{\text{kg}}{\text{m}^3} \right)^{e_1} \left(\frac{\text{m}}{\text{s}} \right)^{e_2} \text{m}^{e_3} = 1$$

$$\text{kg}^{1+e_1} \text{m}^{-1-3e_1+e_2+e_3} \text{s}^{-1-e_2} = \text{kg}^0 \text{m}^0 \text{s}^0$$

$$\begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 1 \\ 0 & -1 & 0 \end{bmatrix} \begin{Bmatrix} e_1 \\ e_2 \\ e_3 \end{Bmatrix} = \begin{Bmatrix} -1 \\ 1 \\ 1 \end{Bmatrix}, \quad \begin{Bmatrix} e_1 \\ e_2 \\ e_3 \end{Bmatrix} = \begin{Bmatrix} -1 \\ -1 \\ -1 \end{Bmatrix}, \quad \Pi_2 = \frac{\mu_\infty}{\rho_\infty V_\infty d}$$

Finally, our third Π group is obtained by combining the last dimensional variable a_∞ with the three repeating variables in a similar manner

$$\Pi_3 = a_\infty \rho_\infty^{e_1} V_\infty^{e_2} d^{e_3}$$

$$\frac{\text{m}}{\text{s}} \left(\frac{\text{kg}}{\text{m}^3} \right)^{e_1} \left(\frac{\text{m}}{\text{s}} \right)^{e_2} \text{m}^{e_3} = 1$$

$$\text{kg}^{e_1} \text{m}^{1-3e_1+e_2+e_3} \text{s}^{-1-e_2} = \text{kg}^0 \text{m}^0 \text{s}^0$$

$$\begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 1 \\ 0 & -1 & 0 \end{bmatrix} \begin{Bmatrix} e_1 \\ e_2 \\ e_3 \end{Bmatrix} = \begin{Bmatrix} 0 \\ -1 \\ 1 \end{Bmatrix}, \quad \begin{Bmatrix} e_1 \\ e_2 \\ e_3 \end{Bmatrix} = \begin{Bmatrix} 0 \\ -1 \\ 0 \end{Bmatrix}, \quad \Pi_3 = \frac{a_\infty}{V_\infty}$$

Now that we have obtained the proper number of independent dimensionless Π groups, which contain all of the dimensional variables involved in the physical phenomenon, we can proceed to step 5 and rearrange our Π groups in a more traditional form. Of course, this always requires some familiarity with the existing literature. In the present case, step 5 is easy. We should immediately recognize the dimensionless Π_2 group as the inverse of the well-known Reynolds number, which is defined in Eq. (1.3.3). Furthermore, Π_3 is the inverse of the familiar Mach number defined in Eq. (1.3.1). With only a little more insight, it will be noticed that Π_1 differs only by a constant from the traditional drag coefficient defined in Eq. (1.3.8). The traditional reference area used to define the drag coefficient for a sphere is the cross-sectional area $S_c = \pi d^2/4$. Hence, our final selection for the three required independent dimensionless ratios should be

$$C_D = \frac{8\Pi_1}{\pi} = \frac{D}{\frac{1}{2}\rho_\infty V_\infty^2 S_c}, \quad R_\infty = \frac{1}{\Pi_2} = \frac{\rho_\infty V_\infty d}{\mu_\infty}, \quad M_\infty = \frac{1}{\Pi_3} = \frac{V_\infty}{a_\infty}$$

Accordingly, dimensional analysis has shown that the drag coefficient for a sphere depends on the Reynolds number and Mach number.

In step 3 of the preceding example, we had to select three of the “*most important*” variables from among the four variables that were found to qualify for a repeating subset. The four qualifying variables were ρ_∞ , μ_∞ , V_∞ , and d . Because we are considering the drag acting on an object moving through a fluid, common experiences like holding a hand out the window of a moving automobile or riding a bicycle tell us that the relative velocity between the object and the fluid is of paramount importance. Similarly, our common experiences also tell us that an object’s size is a critical factor in determining the drag acting on the object. Furthermore, because dimensional analysis is commonly used to design scale model experiments, the object’s size is critical to our intended purpose. With this reasoning, the reader should be quite comfortable with choosing a characteristic velocity and a characteristic length for inclusion in the repeating subset. However, the

remaining choice between fluid density and fluid viscosity is not so obvious. In fact, suppose that one were to ask 100 beginning aerodynamics students, “Which has the greater effect on airfoil drag, fluid density or fluid viscosity?” It is likely that more than 50 would pick fluid viscosity. Furthermore, if that same question were asked to 100 experienced aerodynamicists; it is more than likely that all 100 would pick the fluid viscosity. Especially in view of this last statement, the choice made in Example 1.13.1 may be in need of some elaboration.

The advantage of using the particular repeating subset that was chosen in the previous example is best viewed in the light of an alternate choice. If Example 1.13.1 is reworked using the repeating subset composed of μ_∞ (kg/s·m), V_∞ (m/s), and d (m), the associated dimensionless Π groups are readily found to be

$$\Pi_1 = \frac{D}{\mu_\infty V_\infty d}, \quad \Pi_2 = \frac{\rho_\infty V_\infty d}{\mu_\infty}, \quad \Pi_3 = \frac{a_\infty}{V_\infty} \quad (1.13.13)$$

The development of these Π groups is left as an exercise for the student.

From the Π groups presented in Eq. (1.13.13), we should immediately recognize Π_2 and Π_3 as the Reynolds number and the inverse Mach number, respectively. The relation between these Π groups and the traditional drag coefficient may not be quite so obvious. However, careful examination reveals that the traditional drag coefficient can be obtained from the Π groups presented in Eq. (1.13.13) according to the relation

$$C_D = \frac{8\Pi_1}{\pi\Pi_2} = \frac{D}{\frac{1}{2}\rho_\infty V_\infty^2 (\frac{1}{4}\pi d^2)} = \frac{D}{\frac{1}{2}\rho_\infty V_\infty^2 S_c} \quad (1.13.14)$$

Clearly the choice between fluid density and fluid viscosity as the third member of our repeating subset was not critical. With either choice we are able to reproduce the three classical dimensionless ratios that have long been used to correlate experimental data for the drag acting on an object moving through a fluid; i.e., the drag coefficient, the Reynolds number, and the Mach number. With either choice the Reynolds number and Mach number were directly exposed. However, the traditional drag coefficient was more directly revealed when the repeating subset ρ_∞ , V_∞ , and d was selected.

Even a very superficial examination of the literature on fluid mechanics reveals that many traditional dimensionless ratios include the fluid density, whereas fewer include the fluid viscosity. This is because high-speed flows in low-viscosity fluids like air and water have traditionally been of greatest interest. In such flows, the fluid acceleration and dynamic pressure forces in the equations of motion are more important than the viscous forces. ***As a general guideline, traditional dimensionless ratios are usually more directly exposed if a characteristic density, velocity, and length are included in the repeating subset.*** However, dimensional analysis has value beyond that associated with simply reproducing traditional dimensionless ratios. It is often enlightening and useful to view experimental data and CFD results from a nontraditional perspective.

Simple inspection of Eq. (1.13.14) reveals that the traditional drag coefficient is a ratio of the drag force to a dynamic pressure force. On the other hand, the nontraditional drag coefficient exposed in Eq. (1.13.13) is a ratio of the drag to a viscous shear force.

The dynamic pressure effects are proportional to the product $\rho_\infty V_\infty^2 d^2$ and the viscous shear stress effects are proportional to the product $\mu_\infty V_\infty d$. Hence, the Reynolds number could be thought of as a ratio of the dynamic pressure effects to the viscous shear stress effects, $R_\infty \equiv \rho_\infty V_\infty^2 d^2 / (\mu_\infty V_\infty d)$.

Experimental results show that when the Mach number is less than about 0.3, both pressure and shear stress are independent of Mach number and the aerodynamic drag force exerted on an object of given shape and orientation moving through a Newtonian fluid depends only on some combination of dynamic pressure effects and viscous shear stress effects. This means that any appropriately nondimensionalized drag coefficient should depend only on the Reynolds number. For the case of a sphere, Fig. 1.13.1 shows how the traditionally defined drag coefficient varies over a range of Reynolds numbers encompassing eight orders of magnitude. The smooth curve shown in this figure is an empirical fit to experimental data from a number of sources. These data are also shown in Fig. 1.13.1. Note that for Reynolds numbers less than 1,000 the traditional drag coefficient decreases substantially with increasing Reynolds number. Over the range of Reynolds numbers from 1,000 to almost 300,000 this drag coefficient remains nearly constant. The abrupt drop in drag coefficient that occurs at a Reynolds number of about 300,000 is the result of an important fluid flow phenomenon that will be discussed at a later point in this chapter.

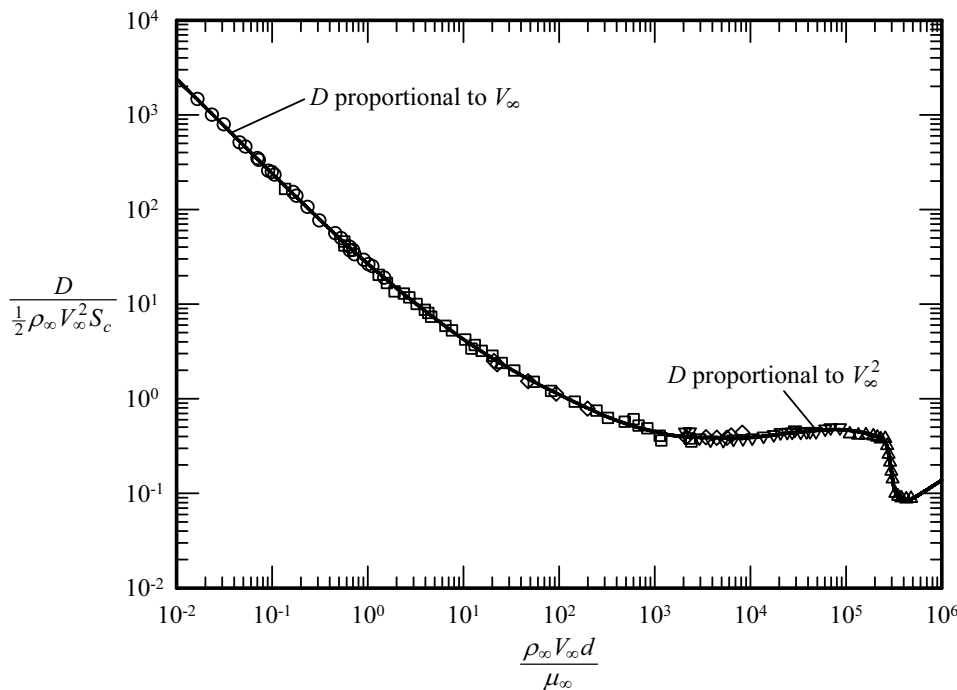


Figure 1.13.1. The traditional dynamic-pressure-based drag coefficient as a function of Reynolds number for flow about smooth spheres at low Mach number.

Although the alternate drag coefficient exposed in Eq. (1.13.13) is not typically used in the fluid mechanics literature, it could be properly called the *Stokes drag coefficient*, after G. G. Stokes (1851) who was the first to obtain a closed-form analytical solution for the drag on a smooth sphere at low Mach number and low Reynolds number. The same experimental data that are shown in Fig. 1.13.1 are plotted in Fig. 1.13.2, as viewed from the perspective of the nontraditional Stokes viscosity-based drag coefficient. Note that for Reynolds numbers less than about 1, this ratio of the drag force to the viscous shear force is constant, which demonstrates that the drag force is not affected by the dynamic pressure at low Reynolds numbers. This fact is not so obvious from the dimensionless perspective of Fig. 1.13.1, because defining the traditional drag coefficient to be the ratio of the drag force to a dynamic pressure force introduces an artificial dynamic pressure effect where none exists in the physics.

Neither Fig. 1.13.1 nor Fig. 1.13.2 portrays a true picture of the relation between drag and velocity, because in both figures the drag force has been nondimensionalized with respect to a velocity-dependent reference force. To obtain a dimensionless view of the relation between the drag and the velocity, we can return to our dimensional analysis procedure and use yet a different repeating subset. In this case, we want a dimensionless representation where the drag and velocity are each included in only one dimensionless ratio. Therefore, we choose a repeating subset that does not include the relative velocity.

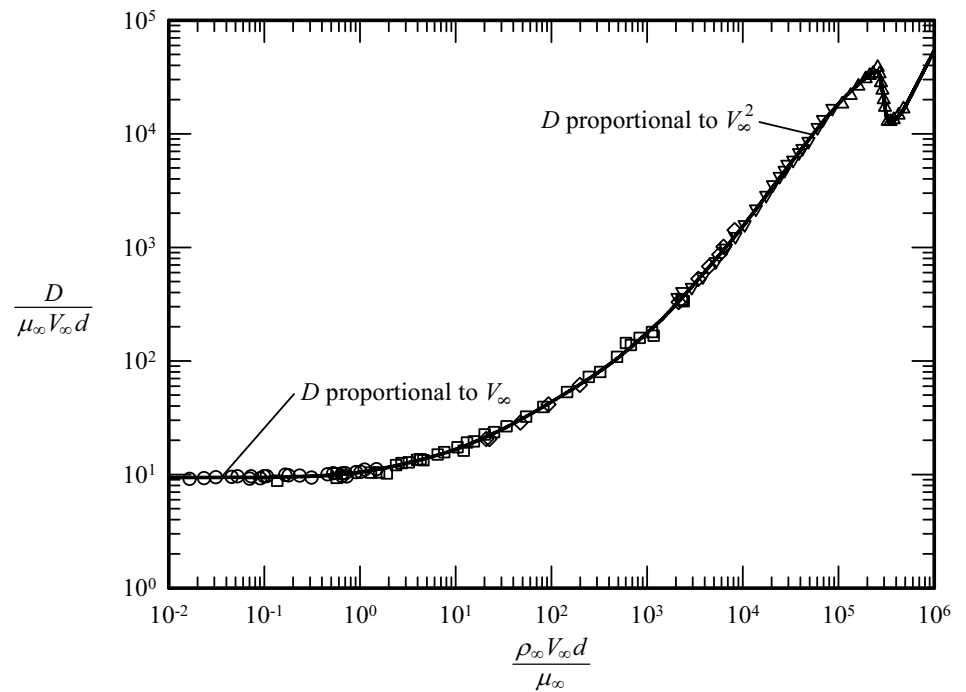


Figure 1.13.2. The nontraditional Stokes viscosity-based drag coefficient as a function of Reynolds number for flow about smooth spheres at low Mach number.

If Example 1.13.1 is reworked using a repeating subset that is composed of ρ_∞ (kg/m³), μ_∞ (kg/s·m), and d (m), the associated dimensionless Π groups are found to be

$$\Pi_1 = \frac{\rho_\infty D}{\mu_\infty^2}, \quad \Pi_2 = \frac{\rho_\infty V_\infty d}{\mu_\infty}, \quad \Pi_3 = \frac{\rho_\infty a_\infty d}{\mu_\infty} \quad (1.13.15)$$

The development of these Π groups is also left as an exercise for the student.

In the dimensionless Π groups presented in Eq. (1.13.15), the Reynolds number once again appears directly as Π_2 , which is our nondimensional velocity. The Mach number is also seen as the ratio Π_2/Π_3 . The dimensionless ratio Π_1 is yet another nontraditional drag coefficient, which was developed here specifically to be independent of velocity. Experimental data recorded in terms of the traditional drag coefficient can be reformatted easily in terms of the nontraditional drag coefficient specified in Eq. (1.13.15),

$$\frac{\rho_\infty D}{\mu_\infty^2} = \frac{\pi}{8} \frac{D}{\frac{1}{2} \rho_\infty V_\infty^2 (\frac{1}{4} \pi d^2)} \left(\frac{\rho_\infty V_\infty d}{\mu_\infty} \right)^2 = \frac{\pi}{8} C_D R_\infty^2 \quad (1.13.16)$$

The same experimental data shown in Fig. 1.13.1 are plotted in Fig. 1.13.3 in terms of the nontraditional drag coefficient defined in Eq. (1.13.15).

The ability to apply the principles of dimensional analysis to rearrange experimental data and/or CFD results in nontraditional dimensionless forms is often useful for solving engineering problems. This is typically the case when we are required to solve for one of the dimensional variables, which was included in the repeating subset used to develop the traditional dimensionless groups. For example, experimental data and CFD results for drag are traditionally arranged in the form $C_D = f(R_\infty, M_\infty)$. However, because the drag coefficient, Reynolds number, and Mach number all include the velocity, this form is not convenient for solving problems that involve an unknown velocity. The solution of such problems is often simplified by using nontraditional dimensionless groups.

EXAMPLE 1.13.2. Use the experimental data presented in Fig. 1.13.1 to estimate the velocity of a 0.5-mm-diameter spherical rain drop falling at standard sea level. Assume that the rain drop has reached its *terminal velocity*, i.e., the velocity for which the acceleration is zero.

Solution. At zero acceleration, the summation of forces acting on the rain drop must be zero. Hence, the aerodynamic drag acting on the rain drop must equal its weight less the buoyant force exerted on the rain drop by the air

$$D = \frac{\pi}{6} d^3 (\rho_w - \rho_\infty) g$$

where ρ_w is the density of water and ρ_∞ is the density of the air. In this case, we have all the information required to determine the drag on the rain drop. However, because the velocity is unknown, we cannot evaluate either the traditional drag coefficient or the Reynolds number, which were used to correlate the experimental data shown in Fig. 1.13.1. Thus, using the experimental data in the format of Fig. 1.13.1 requires an iterative trial-and-error process to determine the velocity.

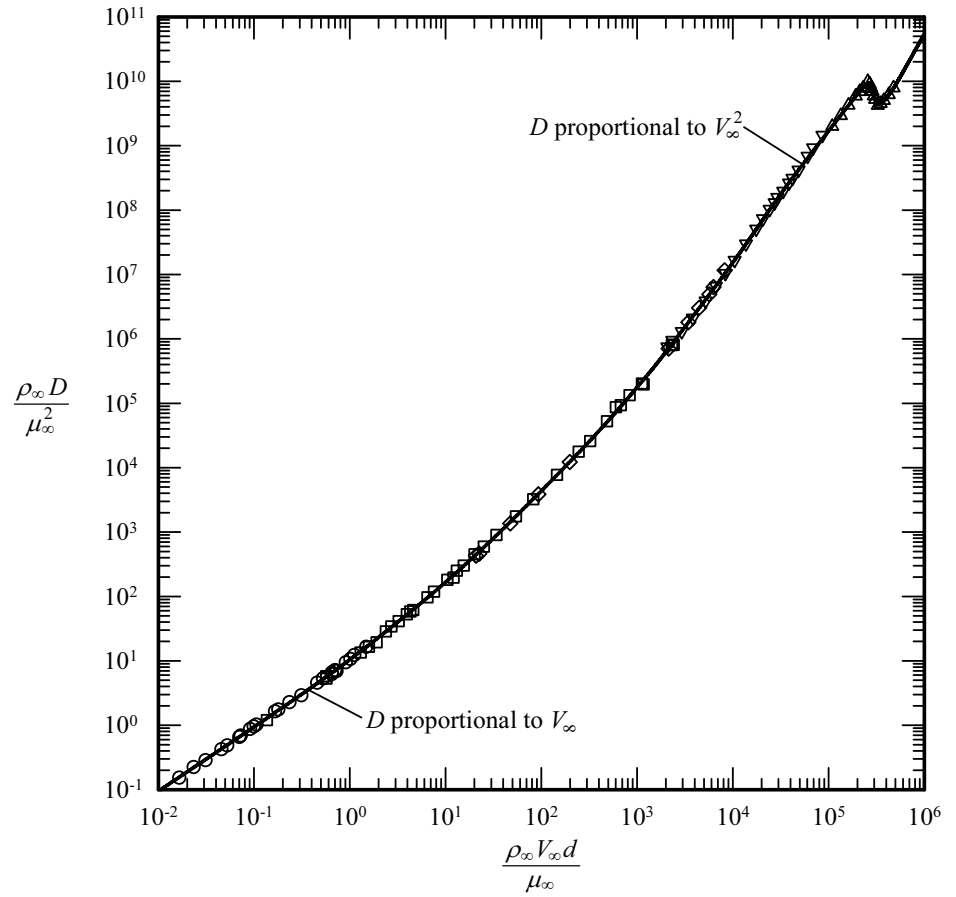


Figure 1.13.3. The nontraditional drag coefficient that is defined in Eq. (1.13.15) as a function of Reynolds number for flow about smooth spheres at low Mach number.

On the other hand, the nontraditional drag coefficient that was used to plot the same data in Fig. 1.13.3 can be evaluated directly,

$$\begin{aligned}
 \frac{\rho_{\infty} D}{\mu_{\infty}^2} &= \frac{\pi d^3 (\rho_w - \rho_{\infty}) \rho_{\infty} g}{6 \mu_{\infty}^2} \\
 &= \frac{\pi (0.0005 \text{ m})^3 (999.0 \text{ kg/m}^3 - 1.225 \text{ kg/m}^3) (1.225 \text{ kg/m}^3) (9.8066 \text{ m/s}^2)}{6 (1.7893 \times 10^{-5} \text{ kg/s} \cdot \text{m})^2} \\
 &= 2,450
 \end{aligned}$$

Comparing this result with the data in Fig. 1.13.3, we see that the corresponding Reynolds number is somewhat less than 100 but significantly above 10. The data in this range of Reynolds number are shown on an expanded scale in Fig. 1.13.4.

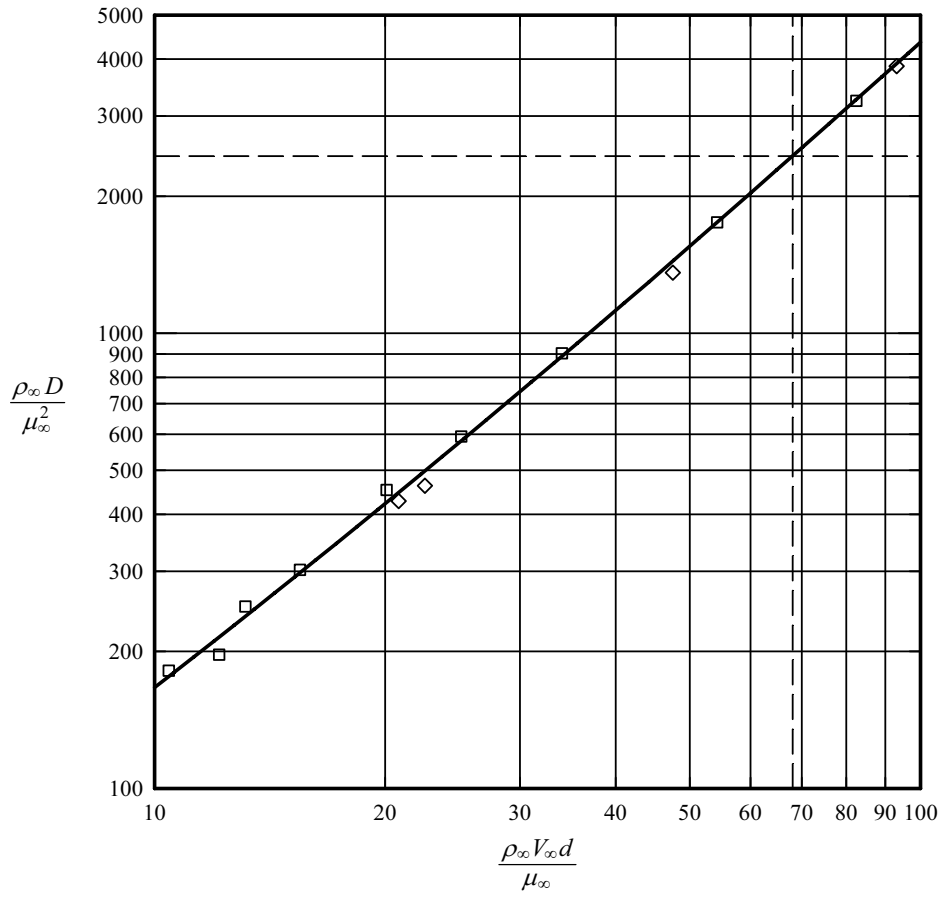


Figure 1.13.4. The nontraditional drag coefficient that is defined in Eq. (1.13.15) as a function of Reynolds number plotted on an expanded scale.

The horizontal dashed line in this figure corresponds to $\rho_{\infty} D / \mu_{\infty}^2 = 2,450$ and the vertical dashed line corresponds to a Reynolds number of

$$R_{\infty} \equiv \frac{\rho_{\infty} V_{\infty} d}{\mu_{\infty}} \cong 68$$

Using this value for the Reynolds number together with the known properties of atmospheric air at standard sea level and the given diameter of the rain drop, the terminal velocity is found to be

$$V_{\infty} = R_{\infty} \frac{\mu_{\infty}}{\rho_{\infty} d} \cong 68 \frac{1.7893 \times 10^{-5} \text{ kg/s}\cdot\text{m}}{(1.225 \text{ kg/m}^3)(0.0005 \text{ m})} \cong 2.0 \text{ m/s}$$

This yields a Mach number of 0.006, which is well below the limit of 0.3.

Summary of Mach Number and Reynolds Number Effects

The aerodynamic lift and drag forces that act on an aircraft, or some other object moving through the atmosphere, are traditionally nondimensionalized in terms of the lift and drag coefficients, which are defined in Eqs. (1.3.7) and (1.3.8), respectively. For continuum flows, the lift and drag coefficients are found to be functions of the Mach number and the Reynolds number, which are defined in Eqs. (1.3.1) and (1.3.3), respectively. The magnitudes of the Mach number and Reynolds number associated with aerodynamic flows determine the relative significance of fluid compressibility, acceleration, shear stress, and pressure in the equations of fluid motion that are used for the analytical prediction of lift and drag coefficients. In certain Mach number and Reynolds number regimes one or more of these associated physical phenomena can be neglected in the equations of motion and the prediction of lift and drag can be significantly simplified with little loss in accuracy.

For example, the Mach number for an aerodynamic flow characterizes the relative importance of fluid compressibility. If the Mach number is less than approximately 0.3, the effects of compressibility are negligible and atmospheric air can be approximated as being an incompressible Newtonian fluid, for the purpose of computing the aerodynamic lift and drag acting on an aircraft or some other object moving through the atmosphere. On the other hand, at high supersonic Mach numbers, compressibility effects dominate, and other simplified mathematical models can be applied.

Similarly, the Reynolds number characterizes the relative importance of fluid inertia (i.e., the product of mass and acceleration) as compared to the viscous forces acting on the fluid. Fluid inertia effects are characterized by the dynamic pressure $\frac{1}{2}\rho_\infty V_\infty^2$ acting over some reference area. Hence, inertia effects are proportional to the product $\rho_\infty V_\infty^2 d^2$, where d is some characteristic length. The viscous forces are proportional to the product $\mu_\infty V_\infty d$, so the Reynolds number is a ratio of the dynamic pressure effects to the viscous shear stress effects, $R_\infty \equiv \rho_\infty V_\infty^2 d^2 / (\mu_\infty V_\infty d) = \rho_\infty V_\infty d / \mu_\infty$. For Reynolds numbers less than about 1, the viscous effects dominate and the acceleration terms in the general equations of fluid motion can be neglected. This Reynolds number regime is commonly referred to as *creeping flow*. On the other hand, for Reynolds numbers that are greater than about 2000, the viscous forces become small compared to the inertia terms in the equations of motion, and accurate predictions for aerodynamic lift can be made using a simplified mathematical model that neglects the viscous shear stresses.

1.14. Streamlines, Streaklines, Pathlines, and Timelines

The quantitative study of fluid dynamics and aerodynamics requires the application and comprehension of advanced mathematics. The output obtained from both *experimental fluid dynamics* and *computational fluid dynamics* is usually a very large set of numerical data. The typical human brain is ill equipped for quickly extracting the essence from a large numerical data set. In contrast, most humans can very rapidly process an enormous amount of visual information. This is likely the basis for the well-known statement, “*A picture is worth a thousand words.*” Although the mathematical equations that govern fluid flow contain considerable information that cannot be processed visually, images can aid significantly in our comprehension of this underlying mathematics.

In the study of fluid dynamics and aerodynamics, it is frequently helpful to be able to visualize a flow field associated with fluid motion. There are many ways to visualize the flow patterns that occur in moving fluids, and much can be learned from examining photographs and computer generated images that display these flow patterns.

The four fundamental types of curved lines shown in Figs. 1.14.1 are commonly used in flow visualization:

1. A **streamline** is a curve that is tangent to the instantaneous local fluid velocity vector at all points along its length.
2. A **streakline** is a curve comprising the instantaneous locus of points occupied by all fluid particles that have formerly passed through a given point in the flow field.
3. A **pathline** is a curve comprising the locus of points traced by a single fluid particle as it moves with the flow over a period of time.
4. A **timeline** is a curve comprising the instantaneous locus of points occupied by a set of adjacent fluid particles that were all marked at the same previous instant in time.

Note that streamlines, streaklines, and timelines are all instantaneous images or *snapshots* displaying some feature of the flow field at one particular instant in time. On the other hand, a pathline is a historical trace or *time exposure* displaying the path followed by a single marked fluid particle as it moves through the flow field with the passing of time. Streamlines are flow snapshots that depend only on the instantaneous state of the fluid velocity field. They are completely independent of the flow history associated with how the fluid came to be in a particular state of motion. Like streamlines, streaklines and timelines are also snapshots that display some instantaneous feature of the flow field at one instant in time. However, like pathlines, streaklines and timelines have a shape that depends significantly on the history of the flow field.

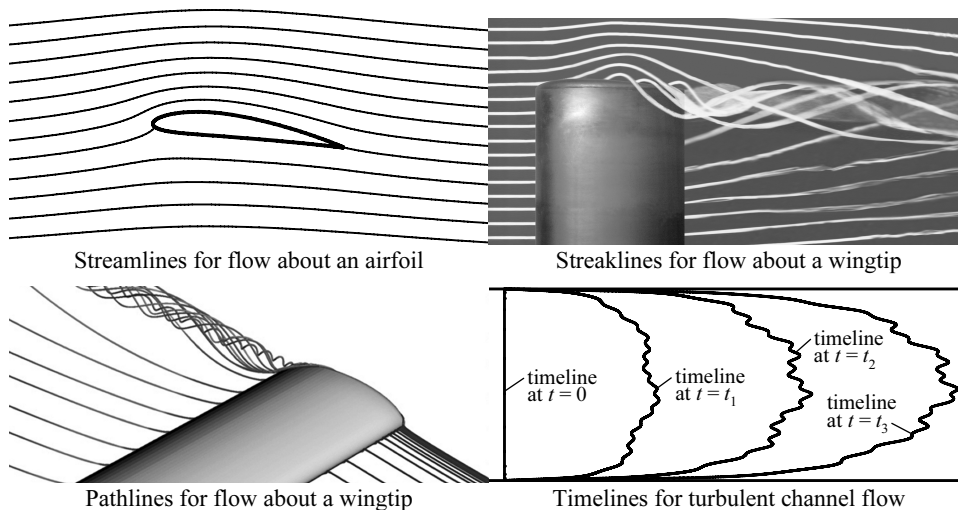


Figure 1.14.1. Streamlines, streaklines, pathlines, and timelines.

In any *steady fluid flow field*, the velocity at each point remains constant with time. Accordingly, the streamlines in a steady flow field do not change from one instant in time to another. As a result, any fluid particle that is located on one particular streamline at one instant in time will remain on that same streamline for all time, simply traveling along the length of the streamline with the passing of time. Similarly, all fluid particles that have passed through a fixed point on some particular streamline will still be on the same streamline at any later instant in time. Hence, ***in a steady flow field, streamlines, streaklines, and pathlines are identical curves throughout the flow field.***

Streamlines are the most useful indicator of the instantaneous direction of the fluid velocity throughout a flow field. Streamlines are very difficult to observe experimentally except in the case of steady flow fields, where they are coincident with streaklines and pathlines. However, streamlines are relatively easy to define mathematically.

Consider a differential vector segment of some streamline $\mathbf{ds} = dx\mathbf{i}_x + dy\mathbf{i}_y + dz\mathbf{i}_z$. Because a streamline is always tangent to the local velocity vector, $\mathbf{ds}$ must be parallel to the velocity vector $\mathbf{V} = V_x\mathbf{i}_x + V_y\mathbf{i}_y + V_z\mathbf{i}_z$. From the simple geometric relations shown in Fig. 1.14.2, the components of $\mathbf{ds}$ must be proportional to the components of $\mathbf{V}$. Thus, **a streamline can be defined in Cartesian coordinates from the differential relations**

$$\frac{dx}{ds} = \frac{V_x}{V}, \quad \frac{dy}{ds} = \frac{V_y}{V}, \quad \frac{dz}{ds} = \frac{V_z}{V} \quad (1.14.1)$$

where ds and V are the magnitudes of $\mathbf{ds}$ and $\mathbf{V}$, respectively. If the velocity field is a known function of space and time, the x , y , and z coordinates along a streamline can be obtained as a function of the arc length s by integrating Eq. (1.14.1) numerically. It is sometimes convenient to eliminate the arc length from Eq. (1.14.1), which yields

$$\frac{dy}{dx} = \frac{V_y}{V_x}, \quad \frac{dz}{dx} = \frac{V_z}{V_x} \quad (1.14.2)$$

The general solution to Eq. (1.14.2) includes two arbitrary constants of integration. Each set of values chosen for these constants represents a different streamline. The family of

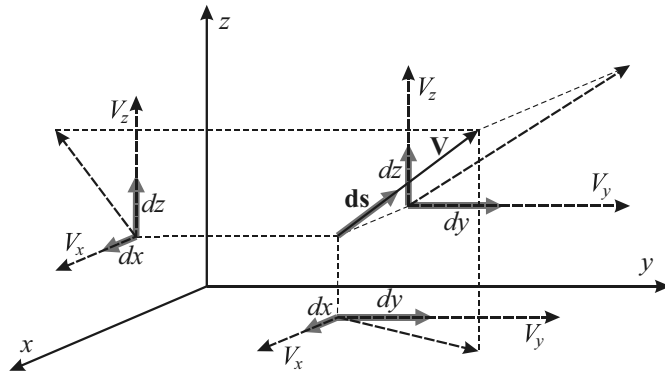


Figure 1.14.2. Geometric relations used for mathematically defining a streamline.

curves described by the general solution to Eq. (1.14.2) is the family of streamlines for the flow field. From Eq. (1.4.32), the differential relations in Eq. (1.14.2) can be written in **cylindrical coordinates**

$$\frac{1}{r} \frac{dr}{d\theta} = \frac{V_r}{V_\theta}, \quad \frac{1}{r} \frac{dz}{d\theta} = \frac{V_z}{V_\theta} \quad (1.14.3)$$

and from Eq. (1.4.38), these relations can be written in **spherical coordinates**

$$\frac{1}{r} \frac{dr}{d\theta} = \frac{V_r}{V_\theta}, \quad \sin \theta \frac{d\varphi}{d\theta} = \frac{V_\varphi}{V_\theta} \quad (1.14.4)$$

where θ is the co-latitude and φ is the longitude.

EXAMPLE 1.14.1. Flow without viscous shear stress is called *inviscid flow*. As will be shown in a following chapter, the fluid velocity field for steady inviscid flow normal to a long circular cylinder is expressed in cylindrical coordinates as

$$\mathbf{V} = V_r \mathbf{i}_r + V_\theta \mathbf{i}_\theta + V_z \mathbf{i}_z = V_\infty (1 - R^2/r^2) \cos \theta \mathbf{i}_r - V_\infty (1 + R^2/r^2) \sin \theta \mathbf{i}_\theta + 0 \mathbf{i}_z$$

where r is the radial coordinate measured from the axis of the cylinder, θ is the angular coordinate measured from the downstream vector aligned with the free-stream, z is the axial coordinate measured along the axis of the cylinder, V_∞ is the freestream airspeed, and R is the radius of the cylinder. Obtain the equations for the family of streamlines associated with this velocity field.

Solution. Using the given velocity field in Eq. (1.14.3), the differential equations that describe this family of streamlines are

$$\frac{1}{r} \frac{dr}{d\theta} = - \frac{(r^2 - R^2) \cos \theta}{(r^2 + R^2) \sin \theta} \quad (1.14.5)$$

$$\frac{dz}{d\theta} = 0 \quad (1.14.6)$$

Separating variables and integrating Eq. (1.14.5) yields

$$\begin{aligned} \int \frac{r dr}{r^2 - R^2} + R^2 \int \frac{dr}{r(r^2 - R^2)} &= - \int \frac{\cos \theta d\theta}{\sin \theta} \\ \frac{1}{2} \ln[(r^2 - R^2)] - \frac{1}{2} \ln[r^2/(r^2 - R^2)] &= -\ln(\sin \theta) + \ln(C_1) \\ (1 - R^2/r^2) r \sin \theta &= C_1 \end{aligned}$$

Transforming to Cartesian coordinates by using the relations $x = r \cos \theta$, $y = r \sin \theta$, and combining this result with the solution to Eq. (1.14.6) yields

$$[1 - R^2/(x^2 + y^2)]y = C_1, \quad z = C_2 \quad (1.14.7)$$

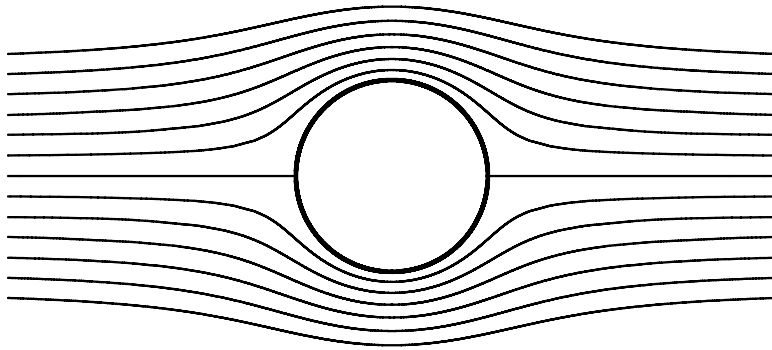


Figure 1.14.3. Streamlines for inviscid flow about a circular cylinder.

The streamlines obtained from Example 1.14.1 for inviscid flow about a cylinder in cross flow are shown graphically in Fig. 1.14.3 with the constant of integration set to $C_1 = 0.2iR$, where $i = -6, -5, \dots, 0, 1, \dots, 6$. Note the perfect fore-and-aft symmetry of these streamlines. Also notice from Eq. (1.14.7) that in the limit as x and/or y approach $\pm$ infinity, the equation for these streamlines reduces to $y = C_1$. Hence, far from the cylinder, the streamlines shown in Fig. 1.14.3 become equally spaced straight lines, parallel with the x axis, and separated by a distance of $0.2R$.

Because streamlines are everywhere tangent to the local fluid velocity vector, the component of velocity normal to a streamline is always zero, and by definition, **fluid cannot cross a streamline**. Thus, the gaps between streamlines such as those shown in Fig. 1.14.3 can be viewed as channels that confine the flow. Like in a river with no tributaries, the flow must remain constant along the length of each of these channels; so the fluid moves faster in regions where the channels are narrow and it moves slower in regions where the channels are wide. Hence, a quick glance at Fig. 1.14.3 reveals regions of high velocity directly above and below the cylinder and regions of low velocity directly fore and aft of the cylinder.

As will be shown in a following chapter, the pressure in an inviscid flow field depends primarily on the fluid velocity. In regions of high velocity the pressure is low, and in regions of low velocity the pressure is high. Maximum pressure in a flow field occurs at points where the velocity is zero. A point of zero velocity is commonly called a *stagnation point* and the pressure at a stagnation point is called the *stagnation pressure*. From the velocity field presented in Example 1.14.1, the velocity is zero on the surface of the cylinder ($r = R$) at the leading and trailing edges ($\theta = \pi$ and $\theta = 0$, respectively). The pressure distribution associated with the inviscid flow field presented in Example 1.14.1 exhibits fore-and-aft symmetry similar to the streamlines. This predicts zero net drag acting on the cylinder in the absence of viscous shear forces.

In summary, more can be gleaned from a visual inspection of streamlines than simply the instantaneous direction of the fluid velocity throughout a flow field. The closely spaced streamlines directly above and below the cylinder in Fig. 1.14.3 indicate regions of high velocity and low pressure. In contrast, the widely spaced streamlines fore and aft of the cylinder reveal regions of low velocity and high pressure.

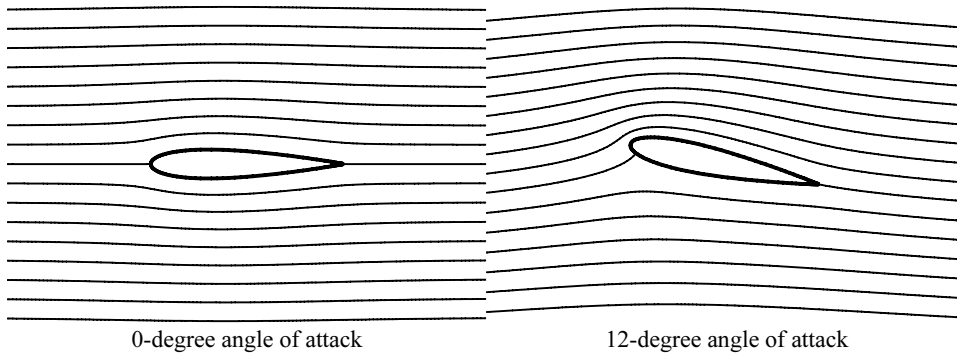


Figure 1.14.4. Streamlines for flow about a symmetric airfoil at two different angles of attack.

Similar observations can be made from Fig. 1.14.4, which shows the streamlines for flow about a symmetric airfoil at two different angles of attack. At zero angle of attack, the streamlines are symmetric above and below the airfoil. This reveals that the velocity and pressure fields are also symmetric, so no lift is produced on the airfoil. On the other hand, for the 12-degree angle of attack, streamline spacing is decreased above the airfoil and increased below the airfoil. This reveals high velocity and low pressure above the airfoil and low velocity and high pressure below the airfoil. This pressure difference between the upper and lower surfaces is the origin of the lift produced on an airfoil.

1.15. The Boundary Layer Concept

When an aircraft or some other solid object is moving through the atmosphere under conditions that allow the air to be modeled accurately as a continuum, the air in direct contact with the solid surface will always move with the same velocity as the surface. This is commonly referred to as the *no-slip boundary condition*, because it requires zero relative motion between the surface and the adjacent air. Since the air in contact with an object moves with the velocity of the object and the air far from the object moves with the velocity of the atmosphere, whenever there is relative motion between an object and the atmosphere, there will be shear deformation in the air near the object.

For example, consider the two-dimensional shear deformation shown in Fig. 1.2.1. The angular deformation rate is given by Eq. (1.2.20), and for a Newtonian fluid, the shear stress in the fluid is given by Eq. (1.2.21),

$$\text{angular deformation rate} = \frac{\partial V_x}{\partial y} + \frac{\partial V_y}{\partial x} \quad (1.15.1)$$

$$\text{shear stress} \equiv \tau_{yx} = \tau_{xy} = \mu \left(\frac{\partial V_x}{\partial y} + \frac{\partial V_y}{\partial x} \right) \quad (1.15.2)$$

where μ is the dynamic viscosity. If we use a coordinate system attached to some object and let x denote the coordinate direction measured tangent to a flat surface on the object,

with y denoting the normal coordinate measured outward from the surface, then the no-slip boundary condition at the surface requires $V_x(x,0)=0$ and $V_y(x,0)=0$. Hence, the change in V_y with respect to x evaluated at the surface is zero, and the angular deformation rate and shear stress in the fluid at the surface of the object are

$$\text{angular deformation rate at the surface} = \left(\frac{\partial V_x}{\partial y} \right)_{y=0} \quad (1.15.3)$$

$$\text{shear stress at the surface} \equiv \tau_w = \mu \left(\frac{\partial V_x}{\partial y} \right)_{y=0} \quad (1.15.4)$$

The shear stress in a fluid at the surface of an object τ_w is commonly referred to as the *wall shear stress*.

The dynamic viscosity of atmospheric air is extremely low. Hence, a large velocity gradient is required to produce a significant shear stress. For example, atmospheric air at standard sea level has a dynamic viscosity of $\mu = 1.7893 \times 10^{-5}$ kg/s·m. In English units this is $\mu = 3.7371 \times 10^{-7}$ slugs/s·ft = 3.7371×10^{-7} lbf·s/ft². Hence, we see from Eq. (1.15.4) that to produce a wall shear stress of 0.0001 psi (0.0144 lbf/ft²) in air at standard sea level, a velocity gradient at the wall of $38,533 \text{ s}^{-1}$ is required. To place this value in perspective, note that if this velocity gradient were maintained over a distance of 1 inch, the velocity 1 inch from the surface would be 3,211 ft/s (2,189 mph), which is about Mach 2.9. From this example, we see that to produce a significant shear stress on the surface of an object moving through the atmosphere, either the velocity of the object relative to the atmosphere must be very large or the viscous shear layer between the no-slip boundary and the surrounding flow field must be very thin.

Because the dynamic viscosity of atmospheric air is extremely low, for the airspeeds and vehicle dimensions typically encountered in human flight, the Reynolds numbers are high and viscous forces are usually small compared to pressure forces and inertia effects. Under such conditions, both experimental and analytical results show that, in flow over a *streamlined body* with its long axis aligned closely with the relative wind, the effects of viscosity are essentially confined to a thin layer adjacent to the surface of the body, as shown in Fig. 1.15.1. This thin layer is called the *boundary layer* and the angle that the long axis of a streamlined body makes with the relative wind is called the *angle of attack*.

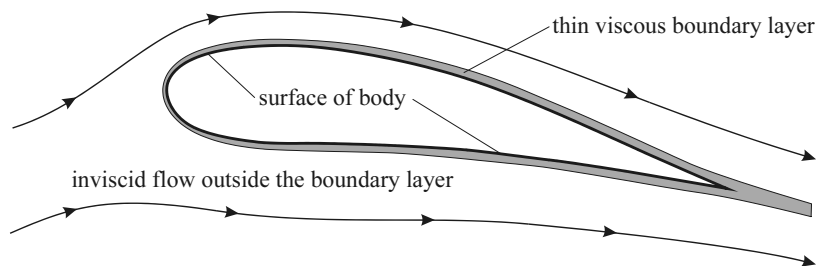


Figure 1.15.1. Boundary layer flow over a streamlined body at a low angle of attack.

Inside the boundary layer, the velocity gradient normal to the surface is very high and viscous shear stresses are important. Outside the boundary layer, the viscous shear stress can be neglected. Since the boundary layer is thin, the change in pressure across the thickness of this layer is insignificant. With this flow model, the pressure forces can be determined from the inviscid flow outside the boundary layer, and the shear forces can be obtained from a solution to the boundary layer equations.

While boundary layer theory provides a tremendous simplification over the general equations of motion for a Newtonian fluid, solutions to the boundary layer equations are far from trivial, especially for the complex geometry that is often encountered in an aircraft. A complete presentation of boundary layer theory is beyond the intended scope of this chapter and will be postponed to a later point. However, there are some important results of boundary layer theory that the reader should know and understand.

1. For the high-Reynolds-number flows typically encountered in flight, the viscous shear forces are small compared to the pressure forces.
2. For flow over streamlined bodies at high Reynolds numbers and low angles of attack, the pressure forces acting on a body can be closely approximated from an inviscid flow analysis, outside the boundary layer.
3. For high-Reynolds-number flow over bluff bodies and streamlined bodies at high angles of attack, boundary layer separation occurs, as shown in Fig. 1.15.2, and the pressure forces dominate the drag.

Boundary layer separation in the flow over airfoils and wings is commonly referred to as *stall*. An understanding of stall and its effect on lift and drag is critical in the study of aerodynamics. As the angle of attack for a wing is increased from zero, at first the boundary layer remains attached as shown in Fig. 1.15.1; the lift coefficient increases as a nearly linear function of angle of attack; and the drag coefficient increases approximately with the angle of attack squared. As angle of attack continues to increase, the positive pressure gradient on the aft portion of the upper surface of the wing also increases. At some angle of attack this *adverse pressure gradient* may result in local boundary layer separation and the increase in lift with angle of attack will begin to diminish. At a slightly higher angle of attack, boundary layer separation becomes complete, as shown in Fig. 1.15.2, and the lift rapidly decreases as the angle of attack is increased further. Typical variations in airfoil and wing lift coefficients with angle of attack are shown in Fig. 1.15.3. Boundary layer separation also greatly accelerates the increase in drag coefficient with angle of attack. The maximum lift coefficient and the exact shape of the lift curve for angles of attack near stall depend substantially on the airfoil section geometry and Reynolds number. As the angle of attack increases beyond stall, the lift and drag coefficients become less sensitive to airfoil section geometry and for angles of attack beyond about 25 degrees, the lift and drag coefficients are nearly independent of the section geometry.

The maximum lift coefficient that can be attained on a given wing before boundary layer separation is very important in aerodynamics. This parameter not only determines the maximum weight that can be carried by a given wing at a particular airspeed, it also affects takeoff distance, turning radius, and other measures of airplane performance.

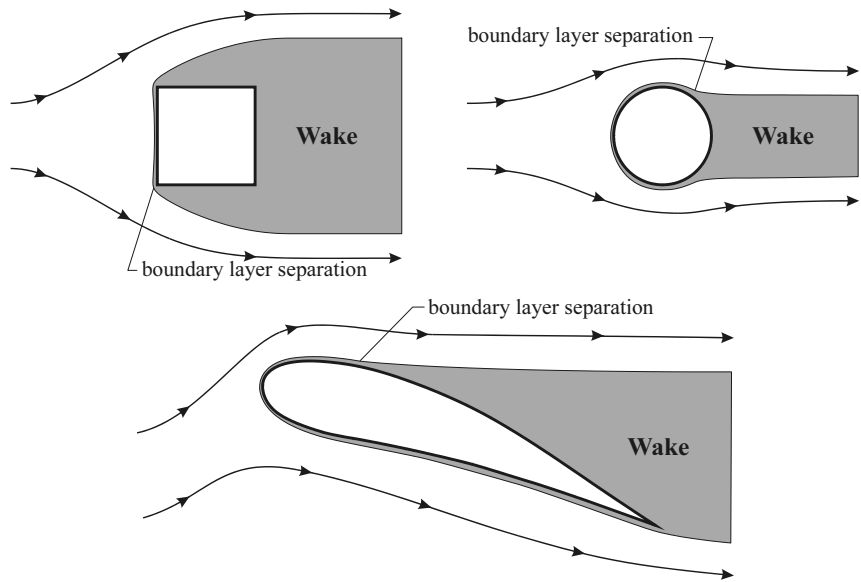


Figure 1.15.2. Examples of boundary layer separation.

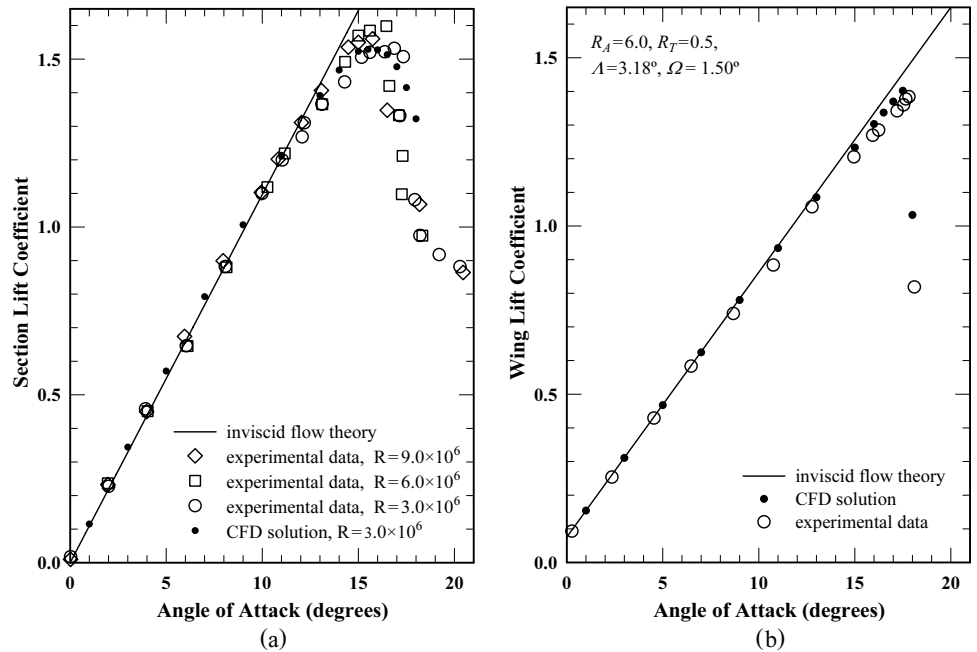


Figure 1.15.3. Comparisons among experimental data, CFD results, and inviscid flow theory for: (a) A NACA 0012 airfoil section at a Reynolds number of 3.0×10^6 and a Mach number of 0.25; (b) A tapered wing of NACA 64-210 airfoil section at a Reynolds number of 3.15×10^6 and a Mach number of 0.10; (from Alley, Phillips and Spall 2007).

To better understand the important phenomenon of boundary layer separation, we shall return to our consideration of the steady inviscid flow normal to a long circular cylinder, which was introduced in Example 1.14.1. From the velocity field presented in that example, the predicted velocity at the surface of the cylinder ($r = R$) is

$$\mathbf{V}(R, \theta, z) = 0\mathbf{i}_r - 2V_\infty \sin \theta \mathbf{i}_\theta + 0\mathbf{i}_z \quad (1.15.5)$$

This velocity is tangent to the cylinder and has perfect fore-and-aft symmetry, as noted from the streamlines plotted in Fig. 1.14.3. The surface velocity is zero at the leading and trailing edges ($\theta = \pi$ and $\theta = 0$, respectively) and it has a maximum value of twice the freestream velocity at the points of maximum thickness normal to the flow ($\theta = \pm \pi/2$). The surface pressure distribution predicted from this same inviscid flow solution is

$$p(R, \theta, z) = p_\infty + \frac{1}{2} \rho_\infty V_\infty^2 (1 - 4 \sin^2 \theta) \quad (1.15.6)$$

This surface pressure distribution is shown in Fig. 1.15.4 compared with experimental data taken at three different Reynolds numbers and Mach numbers less than 0.3.

From Fig. 1.15.4, we see that experimental data for flow about a cylinder deviate significantly from the predictions of inviscid flow theory, particularly on the downstream side of the cylinder. Also notice that there is little difference between the dimensionless pressure distributions at the two lower Reynolds numbers. However, there is substantial difference between these pressure distributions and that for the higher Reynolds number. Streamlines and drag coefficients for flow over a cylinder are shown in Fig. 1.15.5 for a broad range of Reynolds numbers.

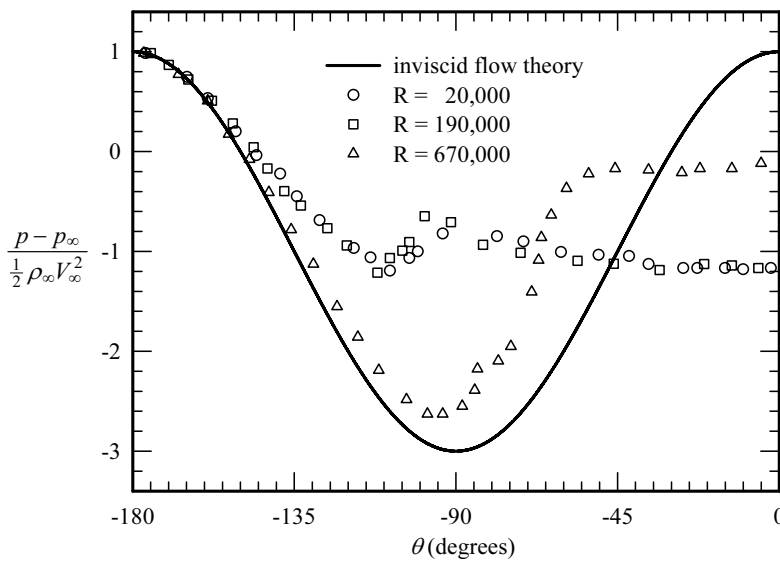


Figure 1.15.4. Comparison between surface pressure from inviscid flow theory and experimental data for flow about a circular cylinder at Reynolds numbers of 2.0×10^4 , 1.9×10^5 , and 6.7×10^5 .

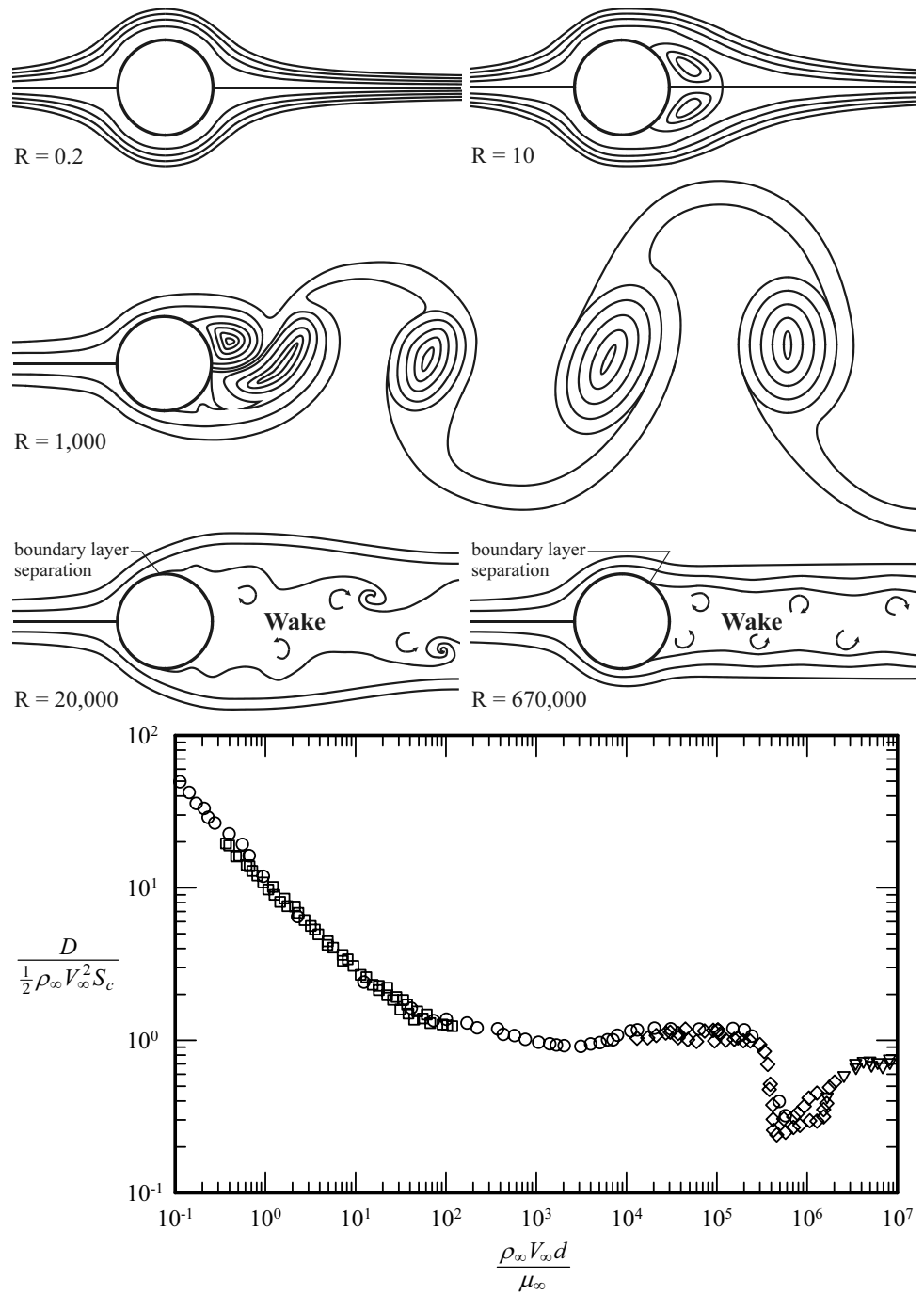


Figure 1.15.5. Streamlines and experimental data for drag coefficient as a function of Reynolds number for flow about smooth circular cylinders at low Mach number.

For Reynolds numbers less than about 1.0, inertia effects are negligible compared to the viscous effects and the flow about a cylinder is laminar with perfect fore-and-aft symmetry, as shown for a Reynolds number of 0.2 in Fig. 1.15.5. These streamlines look quite similar to the streamlines for inviscid flow, which are shown in Fig. 1.14.3. However, the velocity field for creeping flow is very different from that for inviscid flow. Notice from Fig. 1.14.3 that the streamlines for inviscid flow are spaced closer directly above and below the center of the cylinder. This spacing increases as we move away from the cylinder. The creeping flow streamlines shown in Fig. 1.15.5 have the largest spacing next to the cylinder and this spacing decreases as we move away from the cylinder. For inviscid flow, the velocity is highest at the surface of the cylinder directly above and below its center and the velocity *decreases monotonically* as we move outward from these points. For creeping flow, the velocity is zero at the surface of the cylinder and it *increases monotonically* as we move outward from the surface.

At Reynolds numbers slightly higher than unity the flow remains laminar, but as a result of fluid inertia, the mainstream flow does not follow the contour of the cylinder on its downstream side. Instead, two attached counter-rotating eddies form in the lee of the cylinder, as shown for a Reynolds number of 10 in Fig. 1.15.5. Such a circulating eddy is commonly called a *vortex*. The vortex pair that forms in the lee of a cylinder in cross flow is stable only for a very small range of Reynolds numbers.

At a Reynolds number of about 60, the flow breaks away or *separates* from the downstream surface of the cylinder. The vortices are alternately shed from first one side of the cylinder and then the other. As each vortex is shed and moves downstream with the flow, another forms in its place. This results in a repeating or *periodic vortex pattern* in the flow field downstream from the cylinder, as shown for a Reynolds number of 1,000 in Fig. 1.15.5. This wake consisting of pairs of vortices shed alternately from the two sides of the aft surface is commonly called a *von Kármán vortex street*. A similar flow pattern on a much larger scale is seen in the satellite image shown in Fig. 1.15.6, which displays a vortex street outlined in a cloud bank downwind from an island. A periodic von Kármán vortex street is observed downstream from a cylinder in cross flow only in the range of Reynolds numbers from about 60 to 5,000.

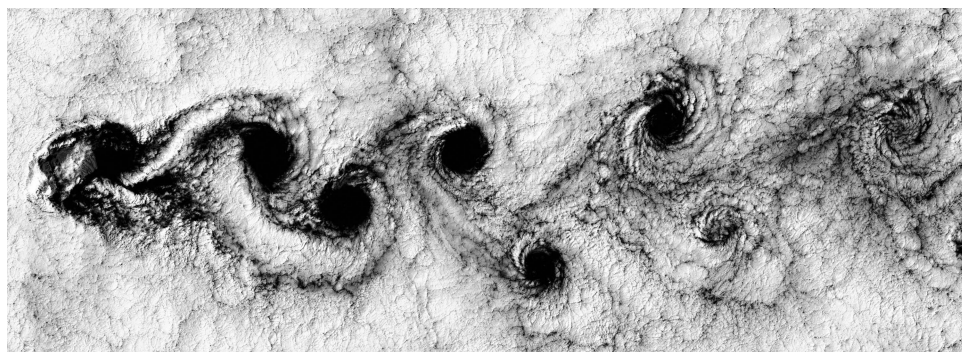


Figure 1.15.6. Example of a von Kármán vortex street, which is visible in the clouds downwind from Alexander Selkirk Island (NASA photograph).

For Reynolds numbers greater than about 5,000 there is nearly complete turbulent mixing in the wake downstream from the cylinder and a thin boundary layer develops on the upstream side. For Reynolds numbers in this range, inviscid flow theory accurately predicts the velocity and pressure fields outside the boundary layer and turbulent wake. However, the wake downstream from the cylinder significantly changes the shape of the inviscid region, which in turn changes the velocity and pressure fields in the region upstream from the wake. For Reynolds numbers less than about 300,000, the boundary layer remains laminar, although the wake may be completely turbulent. Inertia effects cause the laminar boundary layer to separate at a point approximately 82 degrees aft of the forward stagnation point, as shown for a Reynolds number of 20,000 in Fig. 1.15.5. The pressure in the wake downstream from the point of separation is nearly constant and less than the average pressure on the upstream side of the cylinder (see Fig. 1.15.4). This produces a large pressure drag, which is directly proportional to the dynamic pressure and completely overshadows the viscous drag. Thus, in the range of Reynolds numbers from about 5,000 to 300,000, the traditional drag coefficient for a smooth cylinder in cross flow remains nearly constant between about 1.0 and 1.2 (see Fig. 1.15.5).

For Reynolds numbers greater than about 300,000, the laminar boundary layer on a smooth cylinder in cross flow becomes unstable and undergoes transition to turbulence. A remarkable phenomenon linked to the transition from a laminar to turbulent boundary layer occurs for flow about bluff bodies, such as cylinders and spheres. Notice from the experimental data plotted in Fig. 1.15.5 that there is an abrupt drop in drag coefficient between Reynolds numbers of about 300,000 and 500,000. This is because a turbulent boundary layer can overcome a larger adverse pressure gradient than can a laminar boundary layer. As a consequence, transition from a laminar to turbulent boundary layer causes the separation point to move a considerable distance downstream. For the case of a smooth circular cylinder in cross flow, the point of boundary layer separation, which lies about 82 degrees aft of the forward stagnation point for a laminar boundary layer, moves downstream to a point about 125 degrees aft of the forward stagnation point, as shown in Fig. 1.15.5 for a Reynolds number of 670,000. This produces a thinner wake and a pressure distribution closer to that predicted from inviscid theory (see Fig. 1.15.4). This same phenomenon occurs in flow about spheres (see Fig. 1.15.7) and is responsible for the abrupt drop in drag coefficient seen in Fig. 1.13.1. As the Reynolds number is increased beyond about 1,000,000, the separation point moves gradually back upstream, causing the drag coefficient to increase.

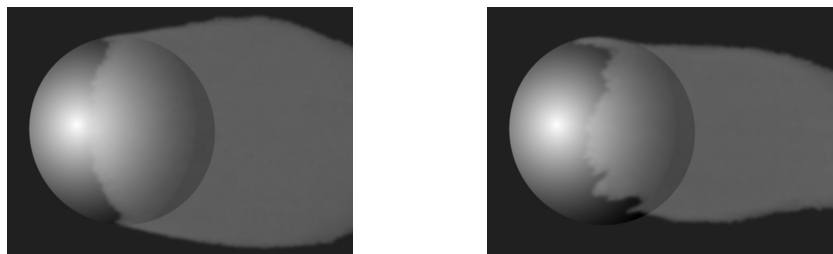


Figure 1.15.7. Laminar (left) and turbulent (right) boundary layer separation on a sphere.

It should be emphasized that results shown in Fig. 1.15.5 are for smooth cylinders in a freestream with very low levels of ambient turbulence. The transition from a laminar to turbulent boundary layer is strongly affected by both surface roughness and ambient turbulence. An increase in either will decrease the critical Reynolds number at which transition to turbulence takes place. This shifts the region of lower drag coefficient, which occurs between Reynolds numbers of 300,000 and 3,000,000 in Fig. 1.15.5, to lower Reynolds numbers. Thus, in a certain Reynolds-number range, adding surface roughness to a bluff body can actually decrease the drag, whereas in other ranges of Reynolds number, surface roughness will increase the drag. This fact is exploited by making golf and tennis balls rough, whereas table-tennis balls are smooth. For example, the dimples on a golf ball reduce the critical Reynolds number for boundary layer transition from nearly 300,000 to around 40,000. This reduces the drag by a factor of about two in the Reynolds-number range 40,000 to 280,000 (see Fig. 1.15.8).

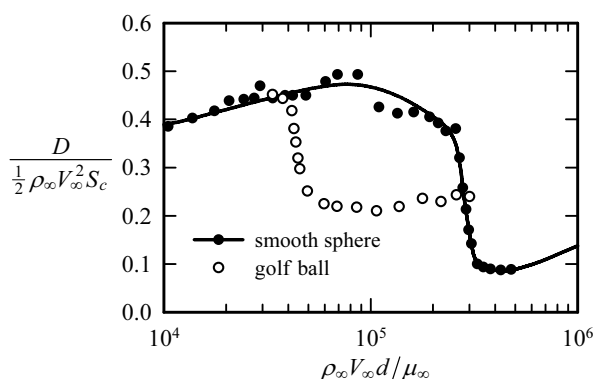


Figure 1.15.8. Drag coefficient for flow about smooth spheres and golf balls.

The agreement between inviscid flow theory and experimental data is considerably better for high-Reynolds-number flows about streamlined bodies. The experimentally measured pressure distribution on the surface of a slender streamlined body at high Reynolds numbers and low angles of attack differs very little from that predicted from inviscid flow theory, as seen for the case of an airfoil in Fig. 1.15.9. This is because the pressure increase downstream from the point of maximum velocity is so gradual that there is no separation. As a consequence, there is no significant pressure drag, so the total drag is primarily viscous drag, which is relatively small.

Boundary layer separation is still important in the study of flow about airfoils and wings. As the angle of attack is increased, there is increased danger of separation on the upper surface of an airfoil or wing, because the pressure increase becomes steeper as is seen in Fig. 1.15.9. At some angle of attack this adverse pressure gradient becomes too steep and separation occurs. The separation point is typically located fairly close to the leading edge (see Fig. 1.15.10), so the decrease in lift and increase in drag are usually quite abrupt (see Fig. 1.15.3). Surface roughness can postpone boundary layer separation to a higher angle of attack, but it also increases drag at lower angles of attack.

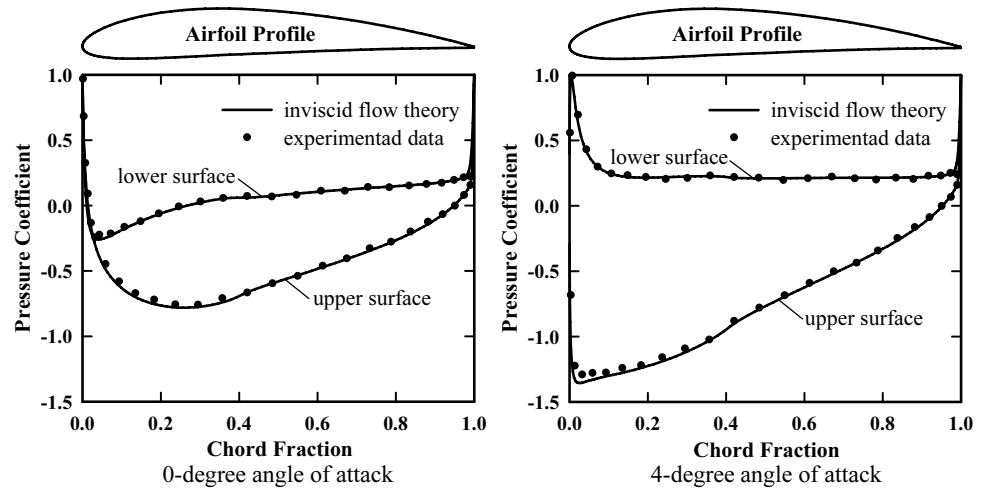


Figure 1.15.9. Comparison between surface pressure from inviscid flow theory and experimental data for flow about a NACA 4412 airfoil at a Reynolds number of 3.0×10^6 .

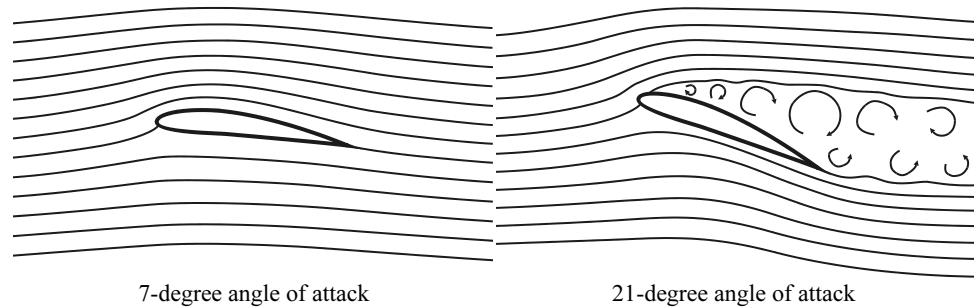


Figure 1.15.10. Streamlines for flow about a NACA 4412 airfoil at low and high angles of attack.

1.16. Aerodynamic Forces and Moments

The aerodynamic forces and moments acting on any body moving through the atmosphere originate from only two sources, *the pressure distribution over the body surface* and *the shear stress distribution over the body surface*. The resultant aerodynamic force, $\mathbf{F}_a$, and the resultant aerodynamic moment, $\mathbf{M}_a$, are the net effects of the pressure and shear stress distributions integrated over the entire surface of the body. To express these two vectors in terms of components, we must define a coordinate system. While several different coordinate systems are often used in the study of aeronautics, the coordinate system most commonly used in the study of aerodynamics will be referred to here as *Cartesian aerodynamic coordinates*. When considering flow over a body such as an airfoil, wing, or airplane, the x axis of this particular coordinate system is aligned with the body axis or *chord line*, pointing in the general direction of relative airflow. The origin is typically located at the front of the body or *leading edge*. The y axis is chosen

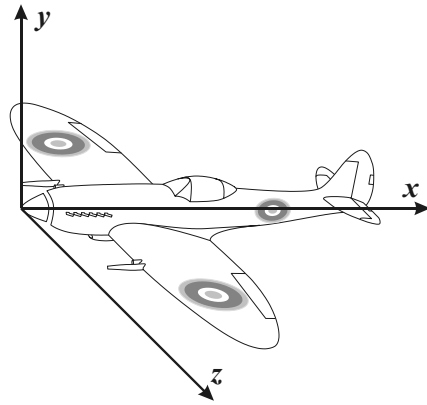


Figure 1.16.1. Cartesian aerodynamic coordinate system used in the study of aerodynamics.

normal to the x axis in an upward direction. Choosing a conventional right-handed coordinate system requires the z axis to be pointing in the spanwise direction from right to left, as shown in Fig. 1.16.1. Here, the components of the resultant aerodynamic force and moment, described in this particular coordinate system, are denoted as

$$\begin{aligned}\mathbf{F}_a &= A\mathbf{i}_x + N\mathbf{i}_y + B\mathbf{i}_z \\ \mathbf{M}_a &= -\ell\mathbf{i}_x - n\mathbf{i}_y - m\mathbf{i}_z\end{aligned}$$

where $\mathbf{i}_x$, $\mathbf{i}_y$, and $\mathbf{i}_z$ are the unit vectors in the x , y , and z directions, respectively. The terminology that describes these components is

- A $\equiv$ aftward axial force $\equiv x$ component of $\mathbf{F}_a$ (parallel to the chord)
- N $\equiv$ upward normal force $\equiv y$ component of $\mathbf{F}_a$ (normal to the chord and span)
- B $\equiv$ leftward side force $\equiv z$ component of $\mathbf{F}_a$ (parallel with the span)
- ℓ $\equiv$ rolling moment (positive right wing down)
- n $\equiv$ yawing moment (positive nose right)
- m $\equiv$ pitching moment (positive nose up)

The traditional definitions for the moments in roll, pitch, and yaw do not follow the right-hand rule in this coordinate system. It is often convenient to split the resultant aerodynamic force into only two components,

$$\begin{aligned}D &\equiv \text{drag} \equiv \text{the component of } \mathbf{F}_a \text{ parallel to } \mathbf{V}_\infty \quad (D = \mathbf{F}_a \cdot \mathbf{i}_\infty) \\ L &\equiv \text{lift} \equiv \text{the component of } \mathbf{F}_a \text{ perpendicular to } \mathbf{V}_\infty \quad (L = |\mathbf{F}_a - D\mathbf{i}_\infty|)\end{aligned}$$

where $\mathbf{V}_\infty$ is the freestream velocity or relative wind far from the body and $\mathbf{i}_\infty$ is the unit vector in the direction of the freestream.

For two-dimensional flow, it is often advantageous to define the *section force* and *section moment* to be the force and moment per unit span. For these definitions the notation used in this book will be

$\tilde{D}$ $\equiv$ section drag $\equiv$ drag force per unit span (parallel to $\mathbf{V}_\infty$)

$\tilde{L}$ $\equiv$ section lift $\equiv$ lift force per unit span (perpendicular to $\mathbf{V}_\infty$)

$\tilde{A}$ $\equiv$ section axial force $\equiv$ axial force per unit span (parallel to chord)

$\tilde{N}$ $\equiv$ section normal force $\equiv$ normal force per unit span (perpendicular to chord)

$\tilde{m}$ $\equiv$ section moment $\equiv$ pitching moment per unit span (positive nose up)

where the chord is a line extending from the leading edge to the trailing edge of the body. The chord length, c , is the length of this chord line.

The aerodynamic forces and moments are usually expressed in terms of dimensionless force and moment coefficients. For example,

$$C_D \equiv \text{drag coefficient} \equiv \frac{D}{\frac{1}{2} \rho_\infty V_\infty^2 S_{\text{ref}}}$$

$$C_L \equiv \text{lift coefficient} \equiv \frac{L}{\frac{1}{2} \rho_\infty V_\infty^2 S_{\text{ref}}}$$

$$C_A \equiv \text{axial force coefficient} \equiv \frac{A}{\frac{1}{2} \rho_\infty V_\infty^2 S_{\text{ref}}}$$

$$C_N \equiv \text{normal force coefficient} \equiv \frac{N}{\frac{1}{2} \rho_\infty V_\infty^2 S_{\text{ref}}}$$

$$C_m \equiv \text{pitching moment coefficient} \equiv \frac{m}{\frac{1}{2} \rho_\infty V_\infty^2 S_{\text{ref}} l_{\text{ref}}}$$

where ρ_∞ is the freestream density, S_{ref} is the reference area, and l_{ref} is the reference length. For a streamlined body such as a wing, S_{ref} is the planform area and l_{ref} is a reference chord length. For a bluff body, the maximum cross-sectional area, which is commonly called the *frontal area*, is used as the reference area. For two-dimensional flow, the section aerodynamic coefficients per unit span are defined:

$$\tilde{C}_D \equiv \text{section drag coefficient} \equiv \frac{\tilde{D}}{\frac{1}{2} \rho_\infty V_\infty^2 c}$$

$$\tilde{C}_L \equiv \text{section lift coefficient} \equiv \frac{\tilde{L}}{\frac{1}{2} \rho_\infty V_\infty^2 c}$$

$$\tilde{C}_A \equiv \text{section axial force coefficient} \equiv \frac{\tilde{A}}{\frac{1}{2} \rho_\infty V_\infty^2 c}$$

$$\tilde{C}_N \equiv \text{section normal force coefficient} \equiv \frac{\tilde{N}}{\frac{1}{2} \rho_\infty V_\infty^2 c}$$

$$\tilde{C}_m \equiv \text{section moment coefficient} \equiv \frac{\tilde{m}}{\frac{1}{2} \rho_\infty V_\infty^2 c^2}$$

where c is the section chord length defined in Fig. 1.16.2.

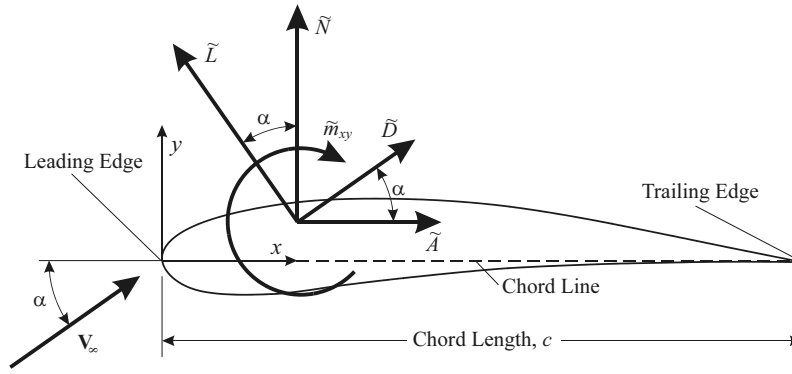


Figure 1.16.2. Section forces and moment.

The resultant aerodynamic force acting on a two-dimensional airfoil section is completely specified in terms of either lift and drag or axial and normal force. These two equivalent descriptions of the resultant aerodynamic force are related to each other through the angle of attack, as shown in Fig. 1.16.2,

$\alpha \equiv \text{angle of attack} \equiv$ the angle from V_∞ to the chord line (positive nose up)

If the normal and axial coefficients are known, the lift and drag coefficients can be found from the relations

$$\tilde{C}_L = \tilde{C}_N \cos \alpha - \tilde{C}_A \sin \alpha \quad (1.16.1)$$

$$\tilde{C}_D = \tilde{C}_A \cos \alpha + \tilde{C}_N \sin \alpha \quad (1.16.2)$$

and when the lift and drag coefficients are known, the normal and axial coefficients are found from

$$\tilde{C}_N = \tilde{C}_L \cos \alpha + \tilde{C}_D \sin \alpha \quad (1.16.3)$$

$$\tilde{C}_A = \tilde{C}_D \cos \alpha - \tilde{C}_L \sin \alpha \quad (1.16.4)$$

Because the lift on a well-designed airfoil can be nearly two orders of magnitude larger than the drag, Eq. (1.16.4) shows why the axial force is often negative even though the angle of attack is small.

1.17. Center of Pressure and Aerodynamic Center

The resultant aerodynamic force and moment acting on a body must yield the same reactions as the distributed loads. Thus, the resultant moment will depend on where the resultant force is placed on the body. For example, let x be the coordinate measured along the chord line of an airfoil, from the leading edge toward the trailing edge; and let y be the normal coordinate measured positive above the chord line, as shown in Fig. 1.16.2.

If the resultant force and moment are placed at some arbitrary point in the x - y plane, the value of the resultant moment will depend on the x and y locations of the resultant force. Let $\tilde{m}_{xy}$ denote the resultant moment about some arbitrary point a distance x aft of the leading edge and a distance y above the chord line. From Fig. 1.16.2, this moment is related to the resultant moment about the leading edge, $\tilde{m}_{le}$, according to

$$\tilde{m}_{le} = \tilde{m}_{xy} - \tilde{N}x + \tilde{A}y$$

or in terms of dimensionless coefficients,

$$\tilde{C}_{m_{le}} = \tilde{C}_{m_{xy}} - \tilde{C}_N \frac{x}{c} + \tilde{C}_A \frac{y}{c} \quad (1.17.1)$$

If the normal and axial force coefficients as well as the aerodynamic moment coefficient about any point in the x - y plane are determined either experimentally or analytically, the moment coefficient about the leading edge can be evaluated from Eq. (1.17.1).

Another location of special interest in fluid mechanics was introduced in our study of fluid statics,

$x_{cp}, y_{cp} \equiv$ **center of pressure** $\equiv$ a point about which the resultant aerodynamic moment is zero.

Using the definition of the center of pressure in Eq. (1.17.1) and rearranging, yields an important relation that must be satisfied at the center of pressure

$$\tilde{C}_N \frac{x_{cp}}{c} - \tilde{C}_A \frac{y_{cp}}{c} = -\tilde{C}_{m_{le}} \quad (1.17.2)$$

If the normal and axial force coefficients as well as the moment coefficient about any point in the x - y plane are known, using Eq. (1.17.1) in Eq. (1.17.2) provides a relation between the two coordinates of the center of pressure, x_{cp} and y_{cp} . Note that the relation between x_{cp} and y_{cp} given in Eq. (1.17.2) is the equation of a straight line. **The locus of points about which the resultant aerodynamic pitching moment acting on an airfoil is zero will always be a straight line called the *line of action*. In general, the position and orientation of the line of action for the resultant aerodynamic force acting on an airfoil can vary significantly with the angle of attack.**

Every point on the line of action defined by Eq. (1.17.2) satisfies our definition for the center of pressure. Isolating one center of pressure as a single point along this line of action requires an additional piece of information. Recall that the center of hydrostatic pressure acting on a flat surface was defined to be the intersection of the line of action with the plane of the surface. This is not practical for the curved surface of an airfoil. Note that if $\tilde{C}_A$ is zero, Eq. (1.17.2) defines a vertical line (i.e., x_{cp} is independent of y_{cp}). For the operating conditions that are most commonly encountered in flight, $\tilde{C}_A$ is small compared to $\tilde{C}_N$. In such cases, the line of action defined by Eq. (1.17.2) is nearly vertical and intersects the airfoil surface at two points; once on the upper surface and once on the lower surface. However, this is not the case for all operating conditions. Notice that when $\tilde{C}_N$ is zero, the line of action defined by Eq. (1.17.2) is horizontal.

EXAMPLE 1.17.1. For reasons that will become clear as we study airfoil theory in more detail, airfoil pitching moments are typically reported about a point on the chord line that is one quarter of the distance aft from the leading edge toward the trailing edge. This point on an airfoil section is called the *quarter chord*. For instance, consider a particular airfoil that has section lift, drag, and quarter-chord moment coefficients given by

$$\tilde{C}_L = 6.0\alpha + 0.2, \quad \tilde{C}_D = 0.2\alpha^2 + 0.006, \quad \tilde{C}_{m_{c/4}} = -0.05 - 0.01\alpha$$

where α is the angle of attack in radians. For this airfoil, find the equations for the line of action at angles of attack of -5 , 0 , 5 , and 10 degrees. Also find the equations for the line of action at the angles of attack for which the line of action is vertical and horizontal.

Solution. At the quarter chord, $x/c = 0.25$ and $y/c = 0.00$. Thus, after applying Eqs. (1.16.3), (1.16.4), and (1.17.1), the normal and axial force coefficients as well as the leading-edge moment coefficient can be expressed in terms of the lift, drag, and quarter-chord moment coefficients,

$$\begin{aligned} \tilde{C}_N &= \tilde{C}_L \cos \alpha + \tilde{C}_D \sin \alpha \\ &= (6.0\alpha + 0.2)\cos \alpha + (0.2\alpha^2 + 0.006)\sin \alpha \\ \tilde{C}_A &= \tilde{C}_D \cos \alpha - \tilde{C}_L \sin \alpha \\ &= (0.2\alpha^2 + 0.006)\cos \alpha - (6.0\alpha + 0.2)\sin \alpha \\ \tilde{C}_{m_{le}} &= \tilde{C}_{m_{c/4}} - (\tilde{C}_L \cos \alpha + \tilde{C}_D \sin \alpha)0.25 \\ &= -0.05 - 0.01\alpha - (1.5\alpha + 0.05)\cos \alpha - (0.05\alpha^2 + 0.0015)\sin \alpha \end{aligned}$$

From Eq. (1.17.2), the angles of attack for which the line of action is vertical and horizontal are simply the angles of attack that yield zero axial and normal force coefficients, respectively. There are two angles of attack that yield zero axial force. Hence, for this particular airfoil we obtain

α (deg)	$\tilde{C}_N$	$\tilde{C}_A$	$\tilde{C}_{m_{le}}$
-5.0000	-0.32302	-0.020709	0.031628
-3.0779	-0.12249	0.000000	-0.018839
-1.9079	0.00000	0.006225	-0.049666
0.0000	0.20000	0.006000	-0.100000
1.1029	0.31555	0.000000	-0.129081
5.0000	0.72150	-0.055571	-0.231248
10.0000	1.23035	-0.204665	-0.359333

These section force and moment coefficients are plotted as a function of angle of attack in Fig. 1.17.1. Note that the normal force coefficient and the leading-edge moment coefficient are nearly linear functions of angle of attack but the axial force coefficient is highly nonlinear.

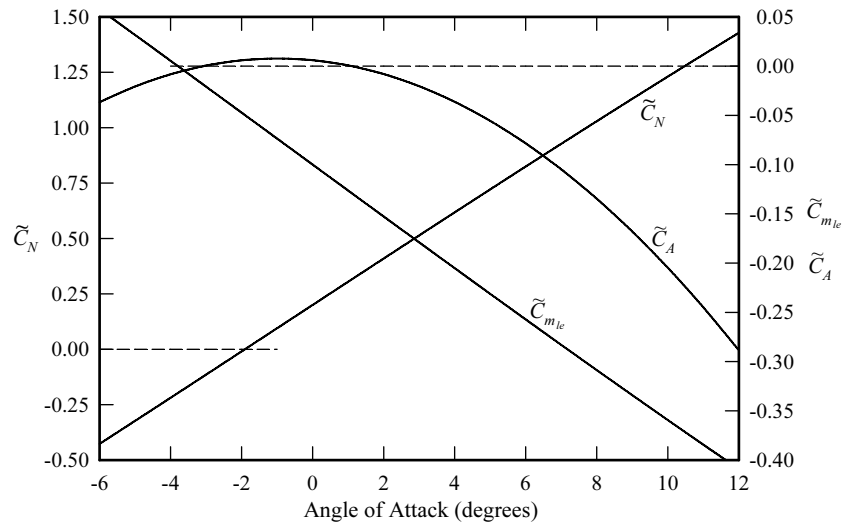


Figure 1.17.1. Section forces and moment coefficients from Example 1.17.1.

Using these coefficients in Eq. (1.17.2), the equations for the line of action at the specified angles of attack are

$$\frac{x_{cp}}{c} = -\frac{\tilde{C}_{m_{le}}}{\tilde{C}_N} + \frac{\tilde{C}_A}{\tilde{C}_N} \frac{y_{cp}}{c} \quad \text{or} \quad \frac{y_{cp}}{c} = \frac{\tilde{C}_{m_{le}}}{\tilde{C}_A} + \frac{\tilde{C}_N}{\tilde{C}_A} \frac{x_{cp}}{c}$$

$$\begin{aligned} \alpha = -5.0000, & \quad x_{cp}/c = 0.097914 + 0.064110 y_{cp}/c \\ \alpha = -3.0779, & \quad x_{cp}/c = -0.153799 \\ \alpha = -1.9079, & \quad y_{cp}/c = -7.978364 \\ \alpha = 0.0000, & \quad x_{cp}/c = 0.500000 + 0.030000 y_{cp}/c \\ \alpha = 1.1029, & \quad x_{cp}/c = 0.409059 \\ \alpha = 5.0000, & \quad x_{cp}/c = 0.320509 - 0.077022 y_{cp}/c \\ \alpha = 10.0000, & \quad x_{cp}/c = 0.292057 - 0.166347 y_{cp}/c \end{aligned}$$

Notice that at an angle of attack slightly less than -1.9 degrees, the line of action is horizontal and nearly 8 chord lengths below the airfoil. The lines of action for these other angles of attack are shown in Fig. 1.17.2.

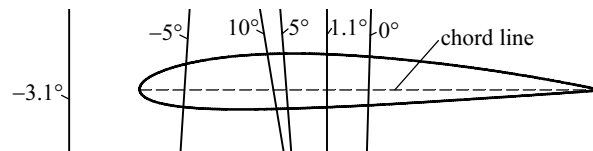


Figure 1.17.2. Line of action for the airfoil in Example 1.17.1 at several angles of attack.

To maintain steady equilibrium flight, the resultant force exerted on the surface of an aircraft by the surrounding air must be exactly balanced by the weight of the aircraft and the thrust developed by the engine. In addition, the resultant moment about an aircraft's center of gravity must be exactly zero. If the forces do not sum to zero, the aircraft will undergo translational acceleration; and if the resultant moment about the center of gravity is not zero, the aircraft will undergo rotational acceleration. This was recognized during the early development of aeronautics in the late nineteenth century.

Because the resultant moment about an aircraft's center of gravity must be zero for equilibrium flight, the understanding and prediction of an airfoil's center of pressure was thought to be of critical importance in the early years of aeronautics. The line of action at any given angle of attack is the locus of points about which the resultant aerodynamic moment is zero. Hence, to define a single fixed-point center of pressure for an airfoil, it would be necessary to locate a single point common to the lines of action at all angles of attack encountered during flight. A quick glance at Fig. 1.17.2 reveals why the aviation pioneers found a useful definition for the airfoil center of pressure to be a bit illusive.

The first practical airfoil theory was developed by Ludwig Prandtl and his students during the early part of the twentieth century. This theory combined with extensive wind-tunnel testing brought a fairly complete understanding of the center of pressure to the aeronautics community by the end of World War I. Because the line of action for the aerodynamic force acting on an airfoil moves so dramatically with angle of attack, the center of pressure has been nearly abandoned as a practical airfoil property.

Early analytical and experimental investigations introduced another location on an airfoil section that is of greater significance in aeronautics,

$x_{ac}, y_{ac} \equiv$ **aerodynamic center** $\equiv$ a point about which the change in the resultant aerodynamic moment with respect to the angle of attack is zero.

Using the aerodynamic center in Eq. (1.17.1) and rearranging we can write

$$\tilde{C}_N \frac{x_{ac}}{c} - \tilde{C}_A \frac{y_{ac}}{c} = \tilde{C}_{m_{ac}} - \tilde{C}_{m_{le}} \quad (1.17.3)$$

Differentiating Eq. (1.17.3) with respect to angle of attack while holding the position coordinates x_{ac} and y_{ac} constant yields

$$\frac{\partial \tilde{C}_N}{\partial \alpha} \frac{x_{ac}}{c} - \frac{\partial \tilde{C}_A}{\partial \alpha} \frac{y_{ac}}{c} = \frac{\partial \tilde{C}_{m_{ac}}}{\partial \alpha} - \frac{\partial \tilde{C}_{m_{le}}}{\partial \alpha}$$

By definition, the change in the moment about the aerodynamic center with respect to angle of attack is zero and we have

$$\frac{\partial \tilde{C}_N}{\partial \alpha} \frac{x_{ac}}{c} - \frac{\partial \tilde{C}_A}{\partial \alpha} \frac{y_{ac}}{c} = - \frac{\partial \tilde{C}_{m_{le}}}{\partial \alpha} \quad (1.17.4)$$

This provides an important relation between the coordinates of the aerodynamic center, x_{ac} and y_{ac} . Notice that the relation between x_{ac} and y_{ac} given in Eq. (1.17.4) is the

equation of a straight line, as was the case for the line of action. **The locus of points about which the change in the resultant aerodynamic pitching moment with respect to angle of attack is zero is always a straight line.** Here, we shall refer to this line as the *neutral axis* of the airfoil. At any given angle of attack, every point on the neutral axis defined by Eq. (1.17.4) satisfies our definition for the aerodynamic center. Isolating one point on this neutral axis as the aerodynamic center requires additional information.

The concept of an aerodynamic center is extremely important in aeronautics. We see from Eq. (1.17.4) that the location of the neutral axis does not depend on the magnitude of the aerodynamic coefficients, as the line of action does; rather, it depends only on derivatives of the aerodynamic coefficients with respect to angle of attack. To avoid the somewhat cumbersome notation used in Eq. (1.17.4), we shall use a shorthand notation for such derivatives. **In this textbook, a subscript preceded by a comma will be used to signify differentiation.** For example,

$$\tilde{C}_{L,\alpha} \equiv \frac{\partial \tilde{C}_L}{\partial \alpha}, \quad \tilde{C}_{D,\alpha} \equiv \frac{\partial \tilde{C}_D}{\partial \alpha}, \quad \tilde{C}_{N,\alpha} \equiv \frac{\partial \tilde{C}_N}{\partial \alpha}, \quad \tilde{C}_{m_{le},\alpha} \equiv \frac{\partial \tilde{C}_{m_{le}}}{\partial \alpha}, \quad \text{etc.}$$

Using this shorthand notation, the equation for the **neutral axis** defined by Eq. (1.17.4) is rearranged and written as

$$\frac{x_{ac}}{c} = -\frac{\tilde{C}_{m_{le},\alpha}}{\tilde{C}_{N,\alpha}} + \frac{\tilde{C}_{A,\alpha}}{\tilde{C}_{N,\alpha}} \frac{y_{ac}}{c} \quad (1.17.5)$$

EXAMPLE 1.17.2. For the airfoil in Example 1.17.1, determine the equations for the neutral axis at angles of attack of -5 , 0 , 5 , and 10 degrees.

Solution. From Eq. (1.16.3) and the relations for the lift and drag coefficients for this particular airfoil that are given in Example 1.17.1, the aerodynamic derivative $\tilde{C}_{N,\alpha}$ can be expressed as

$$\begin{aligned} \tilde{C}_{N,\alpha} &= \frac{\partial}{\partial \alpha} (\tilde{C}_L \cos \alpha + \tilde{C}_D \sin \alpha) \\ &= \tilde{C}_{L,\alpha} \cos \alpha - \tilde{C}_L \sin \alpha + \tilde{C}_{D,\alpha} \sin \alpha + \tilde{C}_D \cos \alpha \\ &= (\tilde{C}_{L,\alpha} + \tilde{C}_D) \cos \alpha - (\tilde{C}_L - \tilde{C}_{D,\alpha}) \sin \alpha \\ &= (6.006 + 0.2\alpha^2) \cos \alpha - (5.6\alpha + 0.2) \sin \alpha \end{aligned}$$

Similarly, from Eq. (1.16.4) and the relations in Example 1.17.1, $\tilde{C}_{A,\alpha}$ is written as

$$\begin{aligned} \tilde{C}_{A,\alpha} &= \frac{\partial}{\partial \alpha} (\tilde{C}_D \cos \alpha - \tilde{C}_L \sin \alpha) \\ &= -(\tilde{C}_L - \tilde{C}_{D,\alpha}) \cos \alpha - (\tilde{C}_{L,\alpha} + \tilde{C}_D) \sin \alpha \\ &= -(5.6\alpha + 0.2) \cos \alpha - (6.006 + 0.2\alpha^2) \sin \alpha \end{aligned}$$

and using Eqs. (1.16.3) and (1.17.1), $\tilde{C}_{m_{le},\alpha}$ is written

$$\begin{aligned}
\tilde{C}_{m_{le},\alpha} &= \frac{\partial}{\partial \alpha} [\tilde{C}_{m_{c/4}} - (\tilde{C}_L \cos \alpha + \tilde{C}_D \sin \alpha) 0.25] \\
&= \tilde{C}_{m_{c/4},\alpha} - [(\tilde{C}_{L,\alpha} + \tilde{C}_D) \cos \alpha - (\tilde{C}_L - \tilde{C}_{D,\alpha}) \sin \alpha] 0.25 \\
&= -0.01 - (1.5015 + 0.05 \alpha^2) \cos \alpha + (1.4 \alpha + 0.05) \sin \alpha
\end{aligned}$$

Hence, for this particular airfoil at the specified angles of attack we obtain

α (deg)	$\tilde{C}_{N,\alpha}$	$\tilde{C}_{A,\alpha}$	$\tilde{C}_{m_{le},\alpha}$
-5.0000	5.95950	0.81118	-1.49988
0.0000	6.00600	-0.20000	-1.51150
5.0000	5.92464	-1.20966	-1.49116
10.0000	5.71630	-2.20349	-1.43908

Using these aerodynamic derivatives in Eq. (1.17.5), the equations for the neutral axis at the specified angles of attack are

$$\begin{aligned}
\alpha = -5.0000, \quad x_{ac}/c &= 0.251678 + 0.136116 y_{ac}/c \\
\alpha = 0.0000, \quad x_{ac}/c &= 0.251665 - 0.033300 y_{ac}/c \\
\alpha = 5.0000, \quad x_{ac}/c &= 0.251688 - 0.204175 y_{ac}/c \\
\alpha = 10.0000, \quad x_{ac}/c &= 0.251749 - 0.385474 y_{ac}/c
\end{aligned}$$

The neutral axis for each of these angles of attack is shown in Fig. 1.17.3.

From the example results presented in Fig. 1.17.3, we see that the point where the neutral axis of the airfoil crosses the chord line does not vary significantly with angle of attack. This is typical of most airfoils at the operating conditions commonly encountered in flight. As a result, the aerodynamic center of an airfoil is traditionally defined to be located at the intersection of the neutral axis and the chord line.

The aerodynamic center of an airfoil is the point on the chord line about which the change in the resultant aerodynamic moment with respect to angle of attack is zero.

Using this definition in Eq. (1.17.5), the coordinates of the aerodynamic center are

$$\frac{x_{ac}}{c} = -\frac{\tilde{C}_{m_{le},\alpha}}{\tilde{C}_{N,\alpha}}, \quad \frac{y_{ac}}{c} = 0 \quad (1.17.6)$$

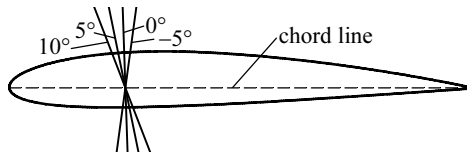


Figure 1.17.3. Neutral axes for the airfoil in Example 1.17.2 at several angles of attack.

Similarly, and perhaps simply for lack of a better option, the center of pressure for an airfoil is traditionally defined to be located at the intersection of the line of action and the chord line.

*The **center of pressure for an airfoil** is the point on the chord line about which the resultant aerodynamic moment is zero.*

Using this definition in Eq. (1.17.2), the coordinates of the center of pressure are

$$\frac{x_{cp}}{c} = -\frac{\tilde{C}_{m_{le}}}{\tilde{C}_N}, \quad \frac{y_{cp}}{c} = 0 \quad (1.17.7)$$

Although this definition provides a convention for choosing a single point on the line of action as the center of pressure for the airfoil, there is no physical justification for setting $y_{cp} = 0$. In general, the position of the center of pressure defined by Eq. (1.17.7) will vary significantly with angle of attack (see Fig. 1.17.2).

EXAMPLE 1.17.3. From the aerodynamic relations presented in Example 1.17.1, on a single graph, plot the location of the airfoil's center of pressure, x_{cp}/c , and the airfoil's aerodynamic center, x_{ac}/c , as a function of angle of attack from -6 to 12 degrees.

Solution. Using Eqs. (1.16.3) and (1.17.1) in Eq. (1.17.7), the center of pressure can be expressed in terms of the section lift, drag, and quarter-chord moment coefficients,

$$\begin{aligned} \frac{x_{cp}}{c} &= -\frac{\tilde{C}_{m_{le}}}{\tilde{C}_N} = -\frac{\tilde{C}_{m_{c/4}} - \tilde{C}_N 0.25}{\tilde{C}_N} = 0.25 - \frac{\tilde{C}_{m_{c/4}}}{\tilde{C}_N} \\ &= 0.25 - \frac{\tilde{C}_{m_{c/4}}}{\tilde{C}_L \cos \alpha + \tilde{C}_D \sin \alpha} \\ &= 0.25 - \frac{-0.05 - 0.01\alpha}{(6.0\alpha + 0.2)\cos \alpha + (0.2\alpha^2 + 0.006)\sin \alpha} \end{aligned}$$

Similarly, using Eqs. (1.16.3) and (1.17.1) in Eq. (1.17.6) yields

$$\begin{aligned} \frac{x_{ac}}{c} &= -\frac{\tilde{C}_{m_{le},\alpha}}{\tilde{C}_{N,\alpha}} = -\frac{\tilde{C}_{m_{c/4},\alpha} - \tilde{C}_{N,\alpha} 0.25}{\tilde{C}_{N,\alpha}} = 0.25 - \frac{\tilde{C}_{m_{c/4},\alpha}}{\tilde{C}_{N,\alpha}} \\ &= 0.25 - \frac{\tilde{C}_{m_{c/4},\alpha}}{\tilde{C}_{L,\alpha} \cos \alpha - \tilde{C}_L \sin \alpha + \tilde{C}_{D,\alpha} \sin \alpha + \tilde{C}_D \cos \alpha} \\ &= 0.25 - \frac{\tilde{C}_{m_{c/4},\alpha}}{(\tilde{C}_{L,\alpha} + \tilde{C}_D)\cos \alpha - (\tilde{C}_L - \tilde{C}_{D,\alpha})\sin \alpha} \\ &= 0.25 - \frac{-0.01}{(6.006 + 0.2\alpha^2)\cos \alpha - (5.6\alpha + 0.2)\sin \alpha} \end{aligned}$$

At angles of attack of -5 , 0 , 5 , and 10 degrees, these relations result in

α (deg)	α (rad)	x_{cp}/c	x_{ac}/c
-5.0	-0.08727	0.09791	0.25168
0.0	0.00000	0.50000	0.25167
5.0	0.08727	0.32051	0.25169
10.0	0.17453	0.29206	0.25175

For this particular airfoil, Fig. 1.17.4 shows the axial position of the center of pressure and the aerodynamic center for angles of attack from -6 to 12 degrees. Note that although the position of the center of pressure varies considerably with angle of attack, the position of the aerodynamic center is nearly independent of angle of attack. This result is typical for cambered airfoils, which are commonly used in the construction of aircraft wings.

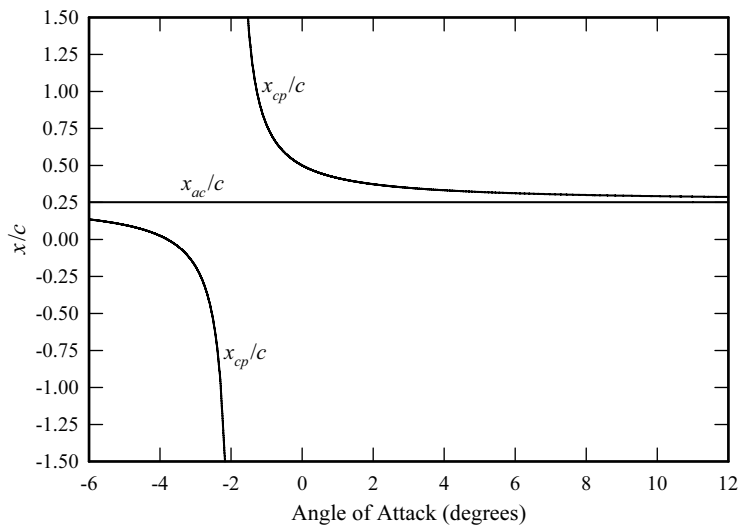


Figure 1.17.4. Airfoil center of pressure and aerodynamic center from Example 1.17.3.

1.18. Problems

Concepts and Definitions

- 1.1. What is the technical distinction between a fluid and a solid? How does a liquid differ from a gas? Name five substances each that can be treated as fluids, solids, liquids, and gases. What is the technical distinction between a Newtonian fluid and a non-Newtonian fluid?
- 1.2. Does atmospheric air precisely satisfy the engineering definition of a fluid? Why or why not?

- 1.3. In one or two sentences define a continuum as it applies to fluid mechanics and aerodynamics. Can atmospheric air always be treated as a continuum?
- 1.4. A sphere is moving through an ideal gas. Name four physical properties of the sphere and the gas that affect whether or not the flow about the sphere can be modeled accurately as a continuum flow.
- 1.5. A sphere is moving through an ideal gas. Name two independent dimensionless parameters that affect whether or not the flow about the sphere can be modeled accurately as a continuum flow.
- 1.6. What do the fields of *fluid mechanics*, *aerodynamics*, and the *kinetic theory of gases* have in common and how do they differ?
- 1.7. Name seven properties of a continuum fluid that are important in the study of fluid mechanics and aerodynamics. Are these properties independent?
- 1.8. Name four properties of a continuum ideal gas that depend only on temperature and are important in the study of fluid mechanics and aerodynamics.
- 1.9. Name two properties of a continuum ideal gas that depend on both temperature and pressure and are important in the study of fluid mechanics and aerodynamics.
- 1.10. Thermodynamic pressure is said to be isotropic. Look up the word *isotropic* in the dictionary and copy the definition. Is the normal stress in all substances always isotropic?
- 1.11. What is the difference between absolute pressure and gage pressure? Does a barometer read absolute pressure or gage pressure? Does the manometer shown in Fig. 1.6.3 read absolute pressure or gage pressure?
- 1.12. How does the density of an ideal gas depend on temperature and pressure? What is the relation between density and specific volume?
- 1.13. Does the specific internal energy of an ideal gas depend on temperature? Does the specific internal energy of an ideal gas depend on pressure? Does the specific internal energy of an ideal gas depend on density?
- 1.14. How is the property enthalpy defined? Does the enthalpy of an ideal gas depend on temperature? Does the enthalpy of an ideal gas depend on pressure? Does the enthalpy of an ideal gas depend on density?
- 1.15. How is the speed of sound defined? Does the speed of sound in an ideal gas depend on temperature? Does the speed of sound in an ideal gas depend on pressure? Does the speed of sound in an ideal gas depend on density?

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- 1.16. Does the dynamic viscosity of an ideal gas depend on temperature? Does the dynamic viscosity of an ideal gas depend on pressure? Does the dynamic viscosity of an ideal gas depend on density?
- 1.17. What is the relation between kinematic viscosity and dynamic viscosity? Does the kinematic viscosity of an ideal gas depend on temperature? Does the kinematic viscosity of an ideal gas depend on pressure? Does the kinematic viscosity of an ideal gas depend on density?
- 1.18. How is the Mach number defined? What is the accepted upper limit of the Mach number for which an aerodynamic flow can be considered to be incompressible?
- 1.19. How is the Knudsen number defined? At very low Mach numbers, what is the upper limit of the Knudsen number for which an aerodynamic flow can be considered to be continuum flow? What is the upper limit at Mach 1?
- 1.20. How is the Reynolds number defined? The Reynolds number characterizes the relative importance of what two effects in fluid flow? How does the Reynolds number change with altitude if the Mach number is held constant?
- 1.21. Does the creeping flow approximation apply to low Reynolds numbers or high Reynolds numbers? Does the boundary layer approximation apply to low Reynolds numbers or high Reynolds numbers?
- 1.22. Are the Mach number, Knudsen number, and Reynolds numbers in an ideal gas independent? Explain why or why not.
- 1.23. How is the magnitude of a vector related to its components? What is the magnitude of a unit vector?
- 1.24. How is the dot product of two vectors related to their magnitudes and the angle between the two vectors?
- 1.25. How is the cross product of two vectors related to their magnitudes and the angle between the two vectors?
- 1.26. Is the gradient of a scalar field a scalar or a vector? If a scalar field is a function only of the spatial coordinate x , what can be said about its gradient?
- 1.27. Is the divergence of a vector field a scalar or a vector? If a vector field is a function only of time, what can be said about its divergence?
- 1.28. Is the curl of a vector field a scalar or a vector? Consider a two-dimensional vector field with an x component that depends only on x and a y component that depends only on y . What can be said about the curl of this vector field?

- 1.29. What direction does the pressure gradient point in a static fluid when the only body force is that of gravity?
- 1.30. How does the shear stress in a static fluid vary in the vertical direction? How does this shear stress vary in the horizontal direction?
- 1.31. If the only body force is that associated with the constant acceleration of gravity, how does the pressure in a static liquid of constant density vary with depth below the surface? How does this pressure vary in the horizontal direction?
- 1.32. In a static isothermal ideal gas is the second derivative of pressure with respect to altitude positive or negative?
- 1.33. What is the distinction between geometric altitude and geopotential altitude, and which is greater?
- 1.34. What is the distinction between MSL altitude, AGL altitude, and elevation, and which is greater?
- 1.35. Define the hydrostatic pressure, the hydrostatic force, and the hydrostatic moment acting on a submerged surface. Is the hydrostatic pressure a scalar or a vector? Is the hydrostatic force a scalar or a vector? Is the hydrostatic moment a scalar or a vector?
- 1.36. State two different but equivalent definitions for a point to qualify as a center of pressure for the resultant pressure force acting on a submerged surface.
- 1.37. The procedure commonly used for locating the center of pressure for the resultant pressure force acting on a submerged surface involves summing moments about some point in space. How must this point be defined? State an important principle that you learned in your introductory course in statics that supports your answer.
- 1.38. State the definition for the line of action for the resultant pressure force acting on a submerged surface. How is the center of pressure traditionally related to the line of action for a planar (flat) surface? Is this same relation always applied to curved surfaces? Why or why not?
- 1.39. State Archimedes' principle both in words and in the form of a mathematical integral equation. Do your statements of Archimedes' principle depend on the fluid density or the acceleration of gravity remaining constant?
- 1.40. How is the buoyant force acting on a submerged object like a submarine or hot-air balloon related to the hydrostatic force? How is this buoyant force related to the volume of the object and properties of the surrounding fluid? How does the buoyant force depend on the properties of the submerged object?

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- 1.41. Consider two blocks of the same dimensions, one is made of aluminum and the other is made of lead. If both blocks are submerged in water at the same depth, will the buoyant force acting on the two blocks be the same or different? Explain your answer.
- 1.42. Consider two aluminum objects of the same mass, one is a sphere and the other is a rod having a length that is 10 times its diameter. If both objects are submerged in water at the same depth, will the buoyant force acting on the two objects be the same or different? Explain your answer.
- 1.43. Consider two identical rubber balls submerged in water at different depths. Will the buoyant force acting on the two balls be the same or different? Explain your answer.
- 1.44. As warm air or some other relatively light gas rises through the atmosphere, does its temperature increase or decrease? Explain your answer.
- 1.45. What is the distinction between the environmental lapse rate and the dry adiabatic lapse rate? Which is greater in the standard atmosphere below 36,000 ft?
- 1.46. Historically, aircraft have been divided into two broad categories. What are the technical names for these two classes of aircraft and how do they differ?
- 1.47. What is the common name for a fixed-wing aircraft? Explain how the weight of a fixed-wing aircraft is supported in flight? Who designed, built, and flew the first fixed-wing aircraft to carry a human in controlled flight and in what year did the first flight take place? Who designed, built, and flew the first powered fixed-wing aircraft to carry a human and in what year did that flight take place?
- 1.48. The aerodynamic force acting on an airplane in flight is usually divided into two components called the *lift* and the *drag*. These two components are defined in terms of what is called the *relative wind*. How are the lift, drag, and relative wind defined? How is the *airspeed* defined?
- 1.49. How are the dimensionless parameters known as the *lift coefficient* and the *drag coefficient* defined? What other two dimensionless parameters have the greatest effect on the lift and drag coefficients for an airplane in flight? How are these two dimensionless parameters defined?
- 1.50. The word *streamlined* has several meanings in the English language. Look up the word *streamlined* in the dictionary and copy the definition that best applies to fluid mechanics.
- 1.51. What is the distinction between *dimensions* and *units*? Give four examples of a dimension. For each dimension listed, give two examples of corresponding units.

- 1.52. List four *fundamental dimensions* that commonly occur in fluid mechanics and aerodynamics. For each fundamental dimension listed, give four examples of corresponding units. List the SI units first and for each of the other units, give their numerical value in SI units.
- 1.53. Write the definitions for *streamlines*, *streaklines*, *pathlines*, and *timelines*. Which of these are coincident in a steady flow field?
- 1.54. A very common means for visualizing flow patterns in a wind tunnel is to insert a small tube into the flow at some fixed location and introduce a continuous stream of visible tracer gas or smoke. When multiple streams of visible tracer gas are introduced along a fixed line in the flow, the device is commonly referred to as a *smoke rake*. Does a smoke rake allow us to visualize *streamlines*, *streaklines*, *pathlines*, or *timelines*? Explain your answer.
- 1.55. Flow patterns can be visualized in a water tunnel using what is called a *hydrogen bubble wire*. A very short pulse of electric current is sent through a fine cathode wire. As a result of electrolysis, small hydrogen bubbles form along this wire. Because these bubbles are extremely small, their weight and buoyant force are negligible compared to the drag. Thus, the bubbles follow the flow very closely. Does a hydrogen bubble wire allow us to visualize *streamlines*, *streaklines*, *pathlines*, or *timelines*? Explain your answer.
- 1.56. From the definitions of *streamlines*, *streaklines*, *pathlines*, and *timelines*, which are instantaneous images or snapshots displaying some feature of the flow field at one particular instant in time?
- 1.57. From the definitions of *streamlines*, *streaklines*, *pathlines*, and *timelines*, which are dependant only on the instantaneous state of the fluid velocity field and are completely independent of the flow history associated with how the fluid came to be in a particular state of motion?
- 1.58. The gap between two streamlines at point A in a two-dimensional inviscid flow field is 1 mm and the gap between the same two streamlines at point B is 2 mm. Is the fluid velocity at point B higher or lower than that at point A? Is the fluid pressure at point B higher or lower than that at point A?
- 1.59. What is the *no-slip boundary condition*? Does the velocity of a fluid in contact with a solid object depend on the fluid viscosity? Explain your answer.
- 1.60. What is an adverse pressure gradient? Is the dot product of the velocity vector and the pressure gradient positive or negative in an adverse pressure gradient?
- 1.61. What is another term commonly used in reference to boundary layer separation in the flow over airfoils and wings? Does this occur on the upper or lower surface?

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- 1.62. Would you expect the dot product of the velocity vector and the pressure gradient to be positive or negative at the point of boundary layer separation in the flow over an airfoil or wing?
- 1.63. In a certain Reynolds-number range, adding surface roughness to a bluff body can decrease the drag relative to that on a similar object of the same size with the same airspeed. Explain why.
- 1.64. State the definition for the line of action for the resultant aerodynamic force acting on an airfoil. How is the traditional definition for the center of pressure related to the line of action for an airfoil? Is the traditionally defined center of pressure for an airfoil always between the leading and trailing edges at small angles of attack?
- 1.65. State the definition for the neutral axis for the resultant aerodynamic force acting on an airfoil. How is the traditional definition for the aerodynamic center related to the neutral axis for an airfoil? Is the traditionally defined aerodynamic center for an airfoil always between the leading and trailing edges at small angles of attack?

Analysis and Design

- 1.66. A 1-cm-diameter sphere is moving through the atmosphere with a Mach number of 0.3. Using the atmospheric data that is given in Appendix A, determine the maximum altitude at which the flow can be treated as a continuum?
- 1.67. A 1-inch-diameter sphere is moving through the atmosphere with a Mach number of 0.3. Using the atmospheric data that is given in Appendix B, determine the maximum altitude at which the flow can be treated as a continuum?
- 1.68. At an altitude where the temperature and pressure of the air are -55°C and $1.2 \times 10^4 \text{ N/m}^2$, respectively, an object with a minimum significant characteristic length of 1.0 mm is moving through the air at 200 knots. Find the Mach number, Reynolds number, and Knudsen number. Do you expect continuum theory to be valid for this flow?
- 1.69. At an altitude where the temperature and pressure of the air are 6.0°F and 8.3 psi, respectively, an object with a minimum significant characteristic length of 0.1 inch is moving through the air at 200 mph. Find the Mach number, Reynolds number, and Knudsen number. Do you expect continuum theory to be valid for this flow?
- 1.70. A 1-cm-diameter sphere is moving through the atmosphere with a Mach number of 0.6 where the air temperature and pressure are -22.80°C and 287.14 N/m^2 , respectively. Determine the Knudsen number and the Reynolds number. Do you expect continuum theory to apply to this flow?

- 1.71. A 1-inch-diameter sphere is moving through the atmosphere with a Mach number of 0.6 where the air temperature and pressure are -69.70°F and 2.7301 psi , respectively. Determine the Knudsen number and Reynolds number. Do you expect continuum theory to apply to this flow?
- 1.72. If the parameters A , B , and C are constants, for each of the scalar fields given in Cartesian coordinates below, determine the gradient of ψ :
- (a) $\psi = Ax + Bxy + Cz^2$ (b) $\psi = Axy - Byz + Cx/z$
 (c) $\psi = Ax^2e^{-By} + C$ (d) $\psi = A(x^2 + z^2)^{1/2} + B/y^3$
 (e) $\psi = Ax^2y^2/z^4 + B$ (f) $\psi = Ax^2e^{-By}\cos(Cz)$
- 1.73. If the parameters A , B , and C are constants, for each of the scalar fields given in cylindrical coordinates below, determine the gradient of ψ :
- (a) $\psi = A\ln(r) + B\theta + C$ (b) $\psi = A\cos(B\theta)/r + C$
 (c) $\psi = Ar^2\cos^2(B\theta)/z^2 + C$ (d) $\psi = A\cos(B\theta)/r^2 - C/r^4$
- 1.74. If the parameters A , B , and C are constants, for each of the scalar fields given in spherical coordinates below, determine the gradient of ψ :
- (a) $\psi = (A/r - r/B + r^2/C)\sin^2\theta$ (b) $\psi = (Ar + B/r^2)\cos\theta + C$
- 1.75. If the parameters A , B , and C are constants, for each of the vector fields given in Cartesian coordinates below, determine the divergence of $\mathbf{V}$:
- (a) $\mathbf{V} = Ax\mathbf{i}_x + By\mathbf{i}_y + Cz\mathbf{i}_z$ (b) $\mathbf{V} = Axy\mathbf{i}_x - Byz\mathbf{i}_y + Cx/z\mathbf{i}_z$
 (c) $\mathbf{V} = Ax\mathbf{i}_x + Bxy\mathbf{i}_y + Cz^2\mathbf{i}_z$ (d) $\mathbf{V} = A(x^2 + y^2)^{1/2}\mathbf{i}_x + B/z^3\mathbf{i}_y$
 (e) $\mathbf{V} = (A + Bx)/(x^2 + y^2)\mathbf{i}_x + By/(x^2 + y^2)\mathbf{i}_y$
- 1.76. If the parameters A , B , and C are constants, for each of the vector fields given in cylindrical coordinates below, determine the divergence of $\mathbf{V}$:
- (a) $\mathbf{V} = (A/r)\mathbf{i}_r$ (b) $\mathbf{V} = -(A/r)\mathbf{i}_\theta$
 (c) $\mathbf{V} = A\cos\theta/r^2\mathbf{i}_r + A\sin\theta/r^2\mathbf{i}_\theta$ (d) $\mathbf{V} = (A\cos\theta + B/r)\mathbf{i}_r - A\sin\theta\mathbf{i}_\theta$
 (e) $\mathbf{V} = A(1 - B/r^2)\cos(\theta)\mathbf{i}_r - [A(1 + B/r^2)\sin(\theta) + C/r]\mathbf{i}_\theta$
- 1.77. If the parameters A and B are constants, for each of the vector fields given in spherical coordinates below, determine the divergence of $\mathbf{V}$:
- (a) $\mathbf{V} = (A - B/r^3)\cos(\theta)\mathbf{i}_r - (A + B/2r^3)\sin(\theta)\mathbf{i}_\theta$
 (b) $\mathbf{V} = A(1 - 3B/2r + B^3/2r^3)\cos(\theta)\mathbf{i}_r + A(3B/4r + B^3/4r^3 - 1)\sin(\theta)\mathbf{i}_\theta$
- 1.78. If the parameters A , B , and C are constants, for each of the vector fields given in Cartesian coordinates in problem 1.75, determine the curl of $\mathbf{V}$.
- 1.79. If the parameters A , B , and C are constants, for each of the vector fields given in cylindrical coordinates in problem 1.76, determine the curl of $\mathbf{V}$.
- 1.80. If the parameters A and B are constants, for each of the vector fields given in spherical coordinates in problem 1.77, determine the curl of $\mathbf{V}$.

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- 1.81. For each of the scalar fields given in Cartesian coordinates in problem 1.72, find the divergence of the gradient of ψ , i.e., $\nabla \cdot (\nabla \psi)$.
- 1.82. For each of the scalar fields given in cylindrical coordinates in problem 1.73, find the divergence of the gradient of ψ , i.e., $\nabla \cdot (\nabla \psi)$.
- 1.83. For each of the scalar fields given in spherical coordinates in problem 1.74, find the divergence of the gradient of ψ , i.e., $\nabla \cdot (\nabla \psi)$.
- 1.84. For each of the scalar fields given in Cartesian coordinates in problem 1.72, find the curl of the gradient of ψ , i.e., $\nabla \times (\nabla \psi)$.
- 1.85. For each of the scalar fields given in cylindrical coordinates in problem 1.73, find the curl of the gradient of ψ , i.e., $\nabla \times (\nabla \psi)$.
- 1.86. For each of the scalar fields given in spherical coordinates in problem 1.74, find the curl of the gradient of ψ , i.e., $\nabla \times (\nabla \psi)$.
- 1.87. The deepest known spot in the ocean is in the Mariana Trench and has a depth of 10,911 m. Neglecting all body forces except gravity, assuming seawater has a constant density of 1025 kg/m^3 , and assuming constant gravitational acceleration of 9.81 m/s^2 , estimate the pressure of water at this depth for a surface pressure of $101,325 \text{ N/m}^2$.
- 1.88. The deepest known spot in the ocean is in the Mariana Trench and has a depth of 35,797 ft. Neglecting all body forces except gravity, assuming seawater has a constant density of 64.0 lbm/ft^3 , and assuming constant gravitational acceleration of 32.2 ft/sec^2 , estimate the pressure of water at this depth for a surface pressure of 14.7 psi.
- 1.89. An approximate equation of state sometimes used for seawater is independent of temperature,

$$p/p_0 = (A+1)(\rho/\rho_0)^7 - A$$

where p is pressure, ρ is density, $A = 3000$, $p_0 = 14.7 \text{ psi}$, and $\rho_0 = 64.0 \text{ lbm/ft}^3$. The deepest known spot in the ocean is in the Mariana Trench and has a depth of 35,797 ft. Using the above equation of state and assuming the acceleration of gravity is constant at 32.2 ft/sec^2 , estimate the pressure and density of water at this depth for a surface pressure of 14.7 psi.

- 1.90. At 8,850 meters above sea level, Mount Everest is the highest mountain on Earth. Neglecting all body forces except gravity and assuming constant gravitational acceleration of 9.81 m/sec^2 , estimate the air pressure at this altitude for a sea-level pressure of $101,325 \text{ N/m}^2$, a sea-level temperature of 21°C , and a constant temperature gradient of -9.7°C/km .

- 1.91. At 29,035 feet above sea level, Mount Everest is the highest mountain on Earth. Neglecting all body forces except gravity and assuming constant gravitational acceleration of 32.2 ft/sec^2 , estimate the air pressure at this altitude for a sea-level pressure of 14.7 psi, a sea-level temperature of 70 F, and a constant temperature gradient of -5.3 F/1000 ft .
- 1.92. For the standard atmosphere defined in Sec. 1.6, compute the first and second derivatives of pressure with respect to geometric altitude at standard sea level.
- 1.93. Write a computer subroutine to compute the geopotential altitude, temperature, pressure, air density, speed of sound, dynamic viscosity, kinematic viscosity, and molecular mean free path as a function of geometric altitude for the standard atmosphere that is defined in Table 1.6.1. The only input variable should be the geometric altitude in meters. All output variables should be in standard SI units. Test this subroutine by comparing its output with the results that were obtained in Example 1.6.1 and those presented in Appendix A.
- 1.94. Write a computer subroutine to compute the geopotential altitude, temperature, pressure, air density, speed of sound, dynamic viscosity, kinematic viscosity, and molecular mean free path as a function of geometric altitude for the standard atmosphere that is defined in Table 1.6.1. The only input variable should be the geometric altitude in feet. All output variables should be in English engineering units. Test the subroutine by comparison with Example 1.6.2 and the results presented in Appendix B. Since this standard atmosphere is defined in SI units, this subroutine could be written using simple unit conversion and a single call to the subroutine written for problem 1.93.
- 1.95. A large jet transport has a wingtip chord length of 4.43 m. The airplane is flying in the standard atmosphere 12,000 m above mean sea level at 516 knots. Calculate the Mach number, Knudsen number, and Reynolds number based on the wingtip chord length.
- 1.96. The large jet transport in problem 1.95 is landing at standard sea level with an airspeed of 131 knots. Calculate the Mach number, Knudsen number, and Reynolds number based on the wingtip chord length.
- 1.97. A large jet transport has a wingtip chord length of 14.55 ft. The airplane is flying in the standard atmosphere 40,000 feet above mean sea level at 594 miles per hour. Calculate the Mach number, Knudsen number, and Reynolds number based on the wingtip chord length.
- 1.98. A small general aviation airplane has a constant wing chord length of 5.22 ft. The airplane is landing at standard sea level with an airspeed of 68 miles per hour. Calculate the Mach number, Knudsen number, and Reynolds number based on the wing chord length.

- 1.99. An airplane is flying in the standard atmosphere at a Mach number of 0.6 and the Reynolds number based on wing chord length is 6×10^6 . What is the Knudsen number based on wing chord length? If the wing chord length is 5.222 ft, what is the geometric altitude of the airplane?
- 1.100. The pressure in a water pipe is measured using the U-tube manometer shown in Fig. 1.18.1. If $\Delta H_1 = 21$ in, $\rho_{\text{H}_2\text{O}} = 1.94$ slug/ft³, $\Delta H_2 = 18$ in, $\rho_{\text{Hg}} = 26.3$ slug/ft³ and the local corrected barometric pressure is 31.2 in Hg, what is the absolute pressure in the water pipe in psi?
- 1.101. A U-tube manometer is partially filled with water ($\rho_{\text{H}_2\text{O}} = 999.0$ kg/m³) and oil as shown in Fig. 1.18.2. If both legs of the U-tube are open to the atmosphere, $\Delta H_1 = 243$ mm, and $\Delta H_2 = 145$ mm, what is the density of the oil?
- 1.102. The pressure in a water pipe is measured using the manometer that is shown in Fig. 1.18.3. If $\Delta H_1 = 738$ mm, $\rho_{\text{H}_2\text{O}} = 999.0$ kg/m³, $\Delta H_2 = 546$ mm, $\Delta H_3 = 642$ mm, $\rho_{\text{Hg}} = 13,558.2$ kg/m³ and the local corrected barometric pressure is 737 mm Hg, what is the absolute pressure in the water pipe at point A?
- 1.103. The open tank shown in Fig. 1.18.4 is perfectly level and filled with water to the depth h . The overall length of the tank is l and the width is w . Along the lower-left edge, the tank has a curved surface in the form of a quarter cylinder of radius r and length w , where r is less than h and l as shown. Atmospheric pressure acts on all outer surfaces of the tank. If the water has a constant density ρ , gravity is the only body force, and the acceleration of gravity is constant, determine the net resultant hydrostatic force vector acting on the front face of the tank, expressed in terms of the coordinate system shown.
- 1.104. For the tank shown in Fig. 1.18.4 and described in problem 1.103, determine the net resultant hydrostatic force vector acting on the curved cylindrical face of the tank, expressed in terms of the coordinate system shown.

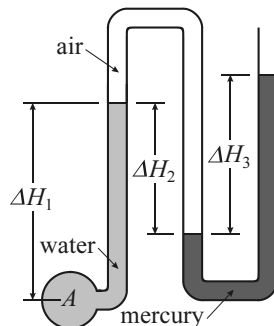
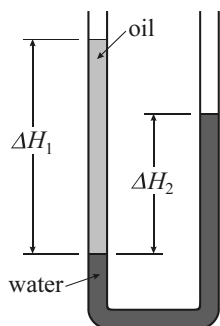
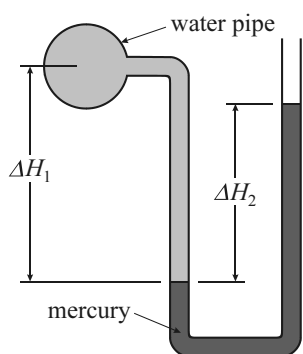


Figure 1.18.1. Problem 1.100. Figure 1.18.2. Problem 1.101. Figure 1.18.3. Problem 1.102.

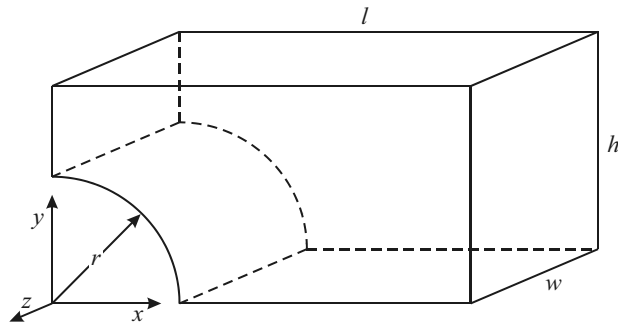


Figure 1.18.4. Problems 1.103–1.111.

- 1.105.** For the tank shown in Fig. 1.18.4 and described in problem 1.103, determine the resultant hydrostatic moment vector produced on the front face of the tank about the origin of the coordinate system shown.
- 1.106.** For the tank shown in Fig. 1.18.4 and described in problem 1.103, determine the resultant hydrostatic moment vector produced on the curved cylindrical face of the tank about the origin of the coordinate system shown.
- 1.107.** For the tank shown in Fig. 1.18.4 and described in problem 1.103, determine the line of action for the resultant hydrostatic force acting on the front face of the tank, expressed in terms of the coordinate system shown.
- 1.108.** For the tank shown in Fig. 1.18.4 and described in problem 1.103, determine the line of action for the resultant hydrostatic force acting on the curved cylindrical face of the tank, expressed in terms of the coordinate system shown.
- 1.109.** For the tank shown in Fig. 1.18.4 and described in problem 1.103, determine the center of pressure for the resultant hydrostatic force acting on the front face of the tank, expressed in terms of the coordinate system shown.
- 1.110.** For the tank shown in Fig. 1.18.4 and described in problem 1.103, determine the intersection of the line of action found in problem 1.108 with the cylindrical face of the tank, expressed in terms of the coordinate system shown.
- 1.111.** For the tank shown in Fig. 1.18.4 and described in problem 1.103, the depth h is 6.0 m, the length l is 11.5 m, the width is w is 4.5 m, and the radius is r is 3.5 m. Determine the combined resultant hydrostatic force exerted on all faces of the tank by the gage pressure in the water.
- 1.112.** For the tank shown in Fig. 1.18.4 and described in problems 1.103 and 1.111, at standard sea level, determine the buoyant force exerted by the air on all faces of the tank and the free surface of the water combined.

- 1.113.** The open tank shown in Fig. 1.18.5 is perfectly level and filled with water to the depth h . The overall length of the tank is l and the width is w . At the lower-left-front corner, the tank has a curved surface in the form of a one-quadrant spherical shell of radius r , where r is less than h , l , and w as shown. Atmospheric pressure acts on all outer surfaces of the tank. If the water has constant density ρ , gravity is the only body force, and the acceleration of gravity is constant, determine the net resultant hydrostatic force vector acting on the curved spherical face of the tank, expressed in terms of the coordinate system shown.
- 1.114.** For the tank shown in Fig. 1.18.5 and described in problem 1.113, determine the resultant hydrostatic moment vector produced on the curved spherical face of the tank about the origin of the coordinate system shown.
- 1.115.** For the tank shown in Fig. 1.18.5 and described in problem 1.113, determine the line of action for the resultant hydrostatic force acting on the curved spherical face of the tank, expressed in terms of the coordinate system shown.
- 1.116.** For the tank shown in Fig. 1.18.5 and described in problem 1.113, determine the intersection of the line of action found in problem 1.115 with the spherical face of the tank, expressed in terms of the coordinate system shown.
- 1.117.** For the tank shown in Fig. 1.18.5 and described in problem 1.113, the depth h is 6.0 m, the length l is 11.5 m, the width is w is 4.5 m, and the radius is r is 3.0 m. Determine the combined resultant hydrostatic force exerted on all faces of the tank by the gage pressure in the water.
- 1.118.** For the tank shown in Fig. 1.18.5 and described in problems 1.113 and 1.117, at standard sea level, determine the buoyant force exerted by the air on all faces of the tank and the free surface of the water combined.
- 1.119.** Determine the buoyant force exerted on a 1-m-diameter sphere immersed in water and in air at standard sea level.

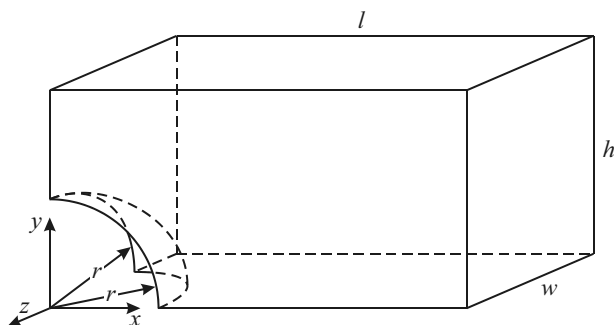


Figure 1.18.5. Problems 1.113–1.118.

- 1.120.** A ball made from lightweight closed-cell rigid foam weighs 1.0 ounce in air at standard sea level. A downward force of 18.9 lbf is required to submerge the ball in water (density 62.4 lbf/ft^3). Determine the diameter of the ball and the density of the foam.
- 1.121.** A cylindrical aluminum compressed-air tank used for scuba diving has an outside diameter of 7.25 inches and a 0.625-inch wall thickness. The tank has a flat end cap on the bottom and a hemispherical end cap on the top, which both have the same wall thickness as the sidewall. The overall length of the tank, including the end caps is 19.70 inches. The tank contains compressed air at 3,500 psi and 70 F. What is the apparent weight of the tank and compressed air submerged in water (density 62.4 lbf/ft^3)? What is the apparent weight of the empty tank submerged in water?
- 1.122.** The gasbag of a dirigible has the shape of a prolate spheroid, which is formed by rotating an ellipse about its major axis. The spheroid's major diameter is 83.4 m and the minor diameter is 18.5 m. If this gasbag is filled with pure hydrogen at standard atmospheric temperature and pressure (gas constant $4124.2421 \text{ J/kg}\cdot\text{K}$), what weight beyond that of the hydrogen will the dirigible support at 2,000 m above standard sea level?
- 1.123.** The mass of the gasbag, structure, and payload for a hot-air balloon is 450 kg. If the hot air is at ambient pressure and 150 F above ambient temperature, what volume of hot air is required to support the balloon at standard sea level? What volumes of helium and hydrogen at standard atmospheric conditions would be required to support this balloon?
- 1.124.** A spherical air bubble 1.0 cm in diameter at a temperature of 37 C is released by a scuba diver into seawater (density 1025 kg/m^3) at a depth of 50 m below the surface. If the surface is at standard atmospheric pressure, what is the buoyant force on the bubble at 50 m below the surface? Neglecting any heat transfer between the water and the air bubble, what is the diameter of the air bubble and the buoyant force on the bubble when it reaches the surface?
- 1.125.** A plume of hot air at 55 C is being released into the standard atmosphere at sea level. Neglecting heat transfer between the plume and the atmosphere, determine altitude to which the plume will rise.
- 1.126.** On a warm summer afternoon, you notice the formation of a large dark cloud with a very flat base, and you hear thunder coming from the direction of the cloud. The air temperature, relative humidity, and dew point at ground level are 90 F, 58% and 78 F, respectively. The change in the dew point temperature with respect to air pressure is 2.26 F/psi. Estimate the altitude above ground level at the base of the cloud. What is your best estimate for the minimum environmental lapse rate in the air around the cloud?

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- 1.127.** A 10-m-diameter weather balloon is rising through the standard atmosphere at an altitude of 10,000 m with an airspeed of 2.00 m/s. Under dynamically similar conditions, a 1/16 scale model of this balloon was tested in water at 15° C ($\rho = 999.1 \text{ kg/m}^3$, $\mu = 1.138 \times 10^{-3} \text{ kg/s}\cdot\text{m}$) and the model drag force was found to be 73.7 N. What velocity was used for the model and what is the drag force acting on the full-scale balloon? If the diameter of the model is cut in half, how would the model velocity and the model drag force change?
- 1.128.** An airplane is flying at 20,000 ft in the standard atmosphere with an airspeed of 170 knots. It is proposed that a 1/32 scale model of the airplane be tested in water at 68 F ($\rho = 1.937 \text{ slugs/ft}^3$, $\nu = 1.082 \times 10^{-5} \text{ ft}^2/\text{s}$) under dynamically similar conditions. What velocity is required for the model? What is the ratio of the drag on the model to that on the airplane? Is this proposed test reasonable?
- 1.129.** The aerodynamic drag force D acting on a smooth sphere moving through a fluid depends on the density ρ_∞ , the dynamic viscosity μ_∞ , the relative velocity V_∞ , the speed of sound a_∞ , and the sphere's diameter d . Use the dimensional analysis procedure presented in Sec. 1.13 with the repeating subset composed of μ_∞ , V_∞ , and d to obtain the associated dimensionless Π groups.
- 1.130.** Repeat problem 1.129 using the dimensional analysis procedure presented in Sec. 1.13 with the repeating subset composed of ρ_∞ , μ_∞ , and d .
- 1.131.** It is known from the kinetic theory of gases that the drag force D acting on a smooth sphere moving through an ideal gas depends on five dimensional variables; the freestream dynamic viscosity μ_∞ , the relative freestream velocity V_∞ , the freestream speed of sound a_∞ , the sphere's diameter d , and the freestream molecular mean free path λ_∞ . Use these variables with the five-step dimensional analysis procedure presented in Sec. 1.13 to obtain appropriate dimensionless ratios to be used in an experimental investigation of fluid flow about a sphere.
- 1.132.** From the kinetic theory of gases, an analytical solution has been obtained for the drag force D acting on a spherical aerosol particle moving through air,

$$\frac{D}{\mu_\infty V_\infty d} = \frac{\pi[15 - 6C_1 K + 4(8 + \pi\sigma)(C_1^2 + 2)C_2 K^2]}{5 + 8C_1 K + 12(C_1^2 + 1)K^2 + 48(C_1^2 + 2)C_2 K^3}$$

$$C_1 = \frac{2 - \sigma}{\sigma}, \quad C_2 = \frac{1}{2 - \sigma}, \quad \sigma = 0.895$$

where μ_∞ is the dynamic viscosity, V_∞ is the relative velocity, d is the diameter, and K is the Knudsen number. This solution is in agreement with experimental data for $M < 0.3$ and $M/K < 0.7$, where M is the Mach number. Using this relation for a 50- μm -diameter spherical ice particle ($\rho_s = 917.0 \text{ kg/m}^3$), plot the Reynolds number and Mach number corresponding to its terminal velocity in the standard atmosphere from sea level to a geometric altitude of 90,000 m.

- 1.133. Hail is produced as a result of ice particles being held aloft in the updraft of a thunderstorm. When the hail becomes large enough, the aerodynamic drag from the updraft cannot support the weight of the hailstones ($\rho_s = 917.0 \text{ kg/m}^3$), which fall from the thundercloud. The smooth curve that is shown in Fig. 1.13.1 is an empirical fit to experimental data from a number of sources. The following data are taken from this empirical fit:

R	C_D	R	C_D
1.000E-02	2.400E+03	1.491E+05	4.152E-01
3.500E-02	6.857E+02	2.252E+05	3.855E-01
1.000E-01	2.400E+02	2.653E+05	3.429E-01
3.508E-01	7.034E+01	2.734E+05	2.922E-01
8.496E-01	3.080E+01	2.835E+05	2.341E-01
2.160E+00	1.383E+01	2.939E+05	1.887E-01
8.865E+00	4.616E+00	3.029E+05	1.531E-01
4.417E+01	1.690E+00	3.122E+05	1.249E-01
2.857E+02	6.744E-01	3.237E+05	1.078E-01
7.446E+02	4.833E-01	3.588E+05	9.069E-02
1.433E+03	4.141E-01	3.978E+05	8.686E-02
2.660E+03	3.916E-01	4.491E+05	8.579E-02
7.641E+03	3.842E-01	5.070E+05	8.956E-02
1.563E+04	4.083E-01	5.759E+05	9.641E-02
2.249E+04	4.341E-01	6.908E+05	1.084E-01
4.517E+04	4.530E-01	1.000E+06	1.377E-01

Use this data in exactly this form to estimate the velocity of the updraft needed to produce a spherical hailstone 1.0 mm in diameter at standard sea level.

- 1.134. Express the data given in problem 1.133 in the form of Fig. 1.13.3. For this range of Reynolds numbers, use this data to plot the terminal velocity of spherical hailstones ($\rho_s = 917.0 \text{ kg/m}^3$) at standard sea level as a function of diameter. From the result, estimate the updraft velocity in miles per hour needed to produce a “golf-ball-sized” hailstone (1.68-inch diameter) at standard sea level.
- 1.135. The boundary layer thickness δ for incompressible flow over a flat plate depends on the distance downstream from the leading edge x , the relative freestream velocity V_∞ , the fluid density ρ , and the dynamic viscosity μ . Use the five-step dimensional analysis procedure presented in Sec. 1.13 to obtain appropriate dimensionless ratios to be used for this problem.
- 1.136. Because a streamline is tangent to the local velocity vector, any differential vector segment of a streamline $d\mathbf{s} = dx\mathbf{i}_x + dy\mathbf{i}_y + dz\mathbf{i}_z$ is parallel to the velocity vector $\mathbf{V} = V_x\mathbf{i}_x + V_y\mathbf{i}_y + V_z\mathbf{i}_z$. From Eq. (1.4.16), the cross product of any two parallel vectors is zero. Hence, the equations for the family of streamlines associated with any velocity field can be written in vector form as $\mathbf{V} \times d\mathbf{s} = 0$. Show that this vector equation yields the scalar equations given in Eq. (1.14.2).

- 1.137. A differential segment of a streamline can be written in cylindrical coordinates as $d\mathbf{s} = dr\mathbf{i}_r + r d\theta\mathbf{i}_\theta + dz\mathbf{i}_z$. Show that the vector equation $\mathbf{V} \times d\mathbf{s} = 0$ written in cylindrical coordinates yields the scalar equations given in Eq. (1.14.3).
- 1.138. A differential segment of a streamline can be written in spherical coordinates as $d\mathbf{s} = dr\mathbf{i}_r + r d\theta\mathbf{i}_\theta + r \sin \theta d\phi\mathbf{i}_\phi$. Show that the vector equation $\mathbf{V} \times d\mathbf{s} = 0$ written in spherical coordinates yields the scalar equations given in Eq. (1.14.4).
- 1.139. A steady two-dimensional flow field can be expressed in Cartesian coordinates as $\mathbf{V} = V_x\mathbf{i}_x + V_y\mathbf{i}_y + V_z\mathbf{i}_z = C_0 x\mathbf{i}_x - C_0 y\mathbf{i}_y + 0\mathbf{i}_z$, where C_0 is a positive constant. Obtain the equations for the streamlines associated with this velocity field. Over the domain $0 \leq x \leq 10\text{ m}$ and $0 \leq y \leq 10\text{ m}$, plot the six streamlines that pass through the (x, y, z) coordinates $(0, 10\text{ m}, 0)$, $(1\text{ m}, 10\text{ m}, 0)$, $(2\text{ m}, 10\text{ m}, 0)$, $(3\text{ m}, 10\text{ m}, 0)$, $(4\text{ m}, 10\text{ m}, 0)$, and $(5\text{ m}, 10\text{ m}, 0)$.
- 1.140. An unsteady two-dimensional flow field is expressed in Cartesian coordinates as $\mathbf{V} = A t\mathbf{i}_x + B\mathbf{i}_y + 0\mathbf{i}_z$, where A and B are positive constants. Find the equations for the streamline associated with this velocity field that passes through the point $(x_o, y_o, 0)$ at time t . For the case $A=0.5\text{ m/s}^2$, $B=1\text{ m/s}$, $x_o=1\text{ m}$, and $y_o=1\text{ m}$, plot the streamline at times $t = -1\text{ s}, 0, 1\text{ s}, 2\text{ s}, 3\text{ s}, 4\text{ s}, 5\text{ s}$ over the domain $0 \leq x \leq 10\text{ m}$ and $0 \leq y \leq 10\text{ m}$.
- 1.141. An unsteady two-dimensional flow field is expressed in Cartesian coordinates as $\mathbf{V} = A t\mathbf{i}_x + B\mathbf{i}_y + 0\mathbf{i}_z$, where A and B are positive constants. Find the equations for the time dependent coordinates of the pathline traced by the particle that passes through the point $(x_o, y_o, 0)$ at time $t = t_o$. For the case $A=0.5\text{ m/s}^2$, $B=1\text{ m/s}$, $x_o=1\text{ m}$, $y_o=1\text{ m}$, and $t_o=0$, plot the pathline traced by this particle over the domain $0 \leq x \leq 10\text{ m}$ and $0 \leq y \leq 10\text{ m}$. Compare this pathline with the streamline that passes through the point $(x_o, y_o, 0)$ at time $t = 0$.
- 1.142. Creeping flow about a sphere is expressed in spherical coordinates as

$$\begin{aligned}\mathbf{V} &= V_r\mathbf{i}_r + V_\theta\mathbf{i}_\theta + V_\phi\mathbf{i}_\phi \\ &= V_\infty \left[\frac{1}{2} \left(\frac{R}{r} \right)^3 - \frac{3}{2} \left(\frac{R}{r} \right) + 1 \right] \cos \theta \mathbf{i}_r + V_\infty \left[\frac{1}{4} \left(\frac{R}{r} \right)^3 + \frac{3}{4} \left(\frac{R}{r} \right) - 1 \right] \sin \theta \mathbf{i}_\theta + 0\mathbf{i}_\phi\end{aligned}$$

where r is the radial coordinate measured from the center of the sphere, θ is the co-latitude measured from the downstream vector aligned with the freestream, ϕ is the longitude, V_∞ is the freestream airspeed, and R is the radius of the sphere. Obtain the equations for the streamlines associated with this velocity field.

- 1.143. In a certain Reynolds-number range, adding surface roughness to a bluff body can decrease the drag relative to that on a similar object of the same size with the same airspeed. The diameter of a golf ball is 1.68 inches. From the data plotted in Fig. 1.15.8, determine the range of airspeed over which the dimples on a golf ball are expected to reduce the drag at standard sea level.

- 1.144.** From the data plotted in Fig. 1.15.8, estimate the drag on a golf ball and a smooth sphere of the same diameter (1.68 inches) at standard sea level and 100 ft/s.
- 1.145.** The following is a tabulation of the section lift, drag, and quarter-chord moment coefficients taken from test data for a particular airfoil section.

α (deg)	$\tilde{C}_L$	$\tilde{C}_D$	$\tilde{C}_{m_{c/4}}$
-6.0	-0.38	0.0077	-0.0446
-5.0	-0.28	0.0071	-0.0440
-4.0	-0.18	0.0066	-0.0434
-3.0	-0.07	0.0063	-0.0428
-2.0	0.04	0.0060	-0.0422
-1.0	0.14	0.0059	-0.0416
0.0	0.24	0.0058	-0.0410
1.0	0.35	0.0058	-0.0404
2.0	0.46	0.0060	-0.0398
3.0	0.56	0.0062	-0.0392
4.0	0.65	0.0066	-0.0386
5.0	0.76	0.0070	-0.0380
6.0	0.87	0.0075	-0.0374
7.0	0.97	0.0081	-0.0368
8.0	1.07	0.0088	-0.0362
9.0	1.16	0.0096	-0.0356
10.0	1.26	0.0105	-0.0350
11.0	1.35	0.0115	-0.0344
12.0	1.44	0.0126	-0.0338

On one graph, plot both the section lift coefficient and the section normal force coefficient as a function of angle of attack. On another graph, plot both the section drag coefficient and the section axial force coefficient as a function of angle of attack.

- 1.146.** From the data presented in problem 1.145, on one graph, plot the location of the center of pressure, x_{cp}/c , and the aerodynamic center, x_{ac}/c , as a function of angle of attack.
- 1.147.** As predicted from inviscid thin airfoil theory, the section lift, drag, and leading-edge moment coefficients for a NACA 2412 airfoil section are given by

$$\tilde{C}_L = 2\pi(\alpha + 0.03625), \quad \tilde{C}_D = 0, \quad \text{and} \quad \tilde{C}_{m_{le}} = -\frac{\pi}{2}(\alpha + 0.07007)$$

where α is in radians. On one graph, plot both the section lift and normal force coefficients as a function of angle of attack, from -6 to $+12$ degrees. On another graph, plot both the section drag and axial force coefficients as a function of angle of attack, from -6 to $+12$ degrees.

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- 1.148.** From the thin airfoil relations for the airfoil section presented in problem 1.147, determine the equations for the line of action for the resultant aerodynamic force at angles of attack of -5 , 0 , 5 , and 10 degrees.
- 1.149.** From the thin airfoil relations for the airfoil section presented in problem 1.147, determine the equations for the neutral axis for the resultant aerodynamic force at angles of attack of -5 , 0 , 5 , and 10 degrees.
- 1.150.** From the thin airfoil relations for the airfoil section presented in problem 1.147, on one graph plot the location of the center of pressure, x_{cp}/c , and the aerodynamic center, x_{ac}/c , as a function of angle of attack, from -6 to $+12$ degrees.